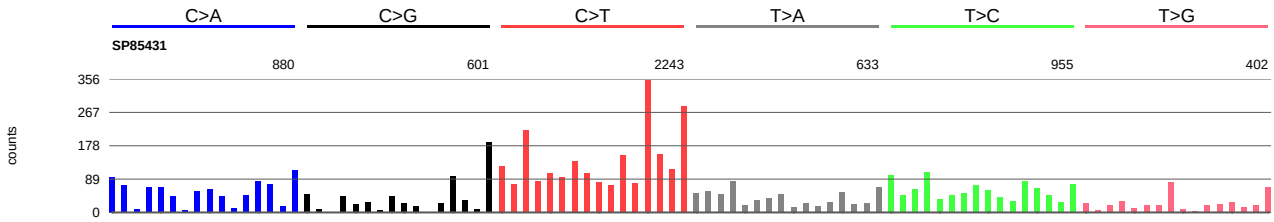
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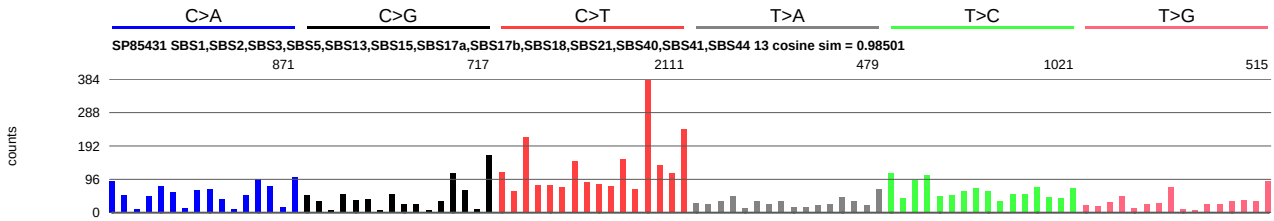
SP85431
880
601
2243
633
955
402
counts
356
267
178
89
0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SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 13 cosine sim = 0.98501
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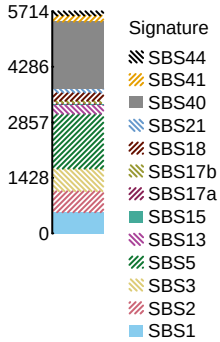
717
2111
479
1021
515
counts
384
288
192
96
0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4286
2857
1428
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Signature
SBS44
SBS41
SBS40
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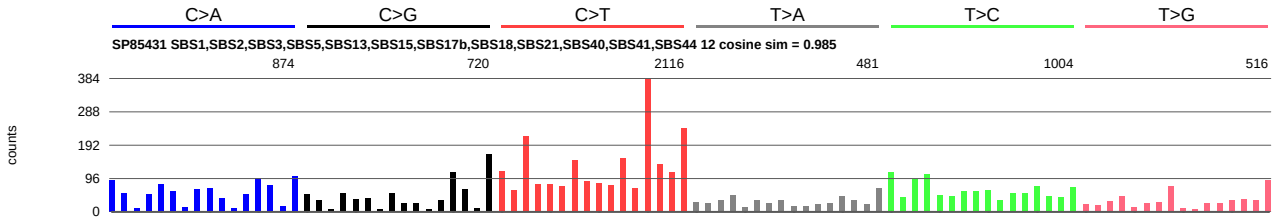
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.985
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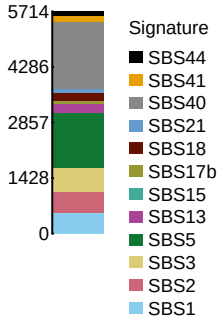
720
2116
481
1004
516
counts
384
288
192
96
0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4286
2857
1428
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Signature
SBS44
SBS41
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SBS21
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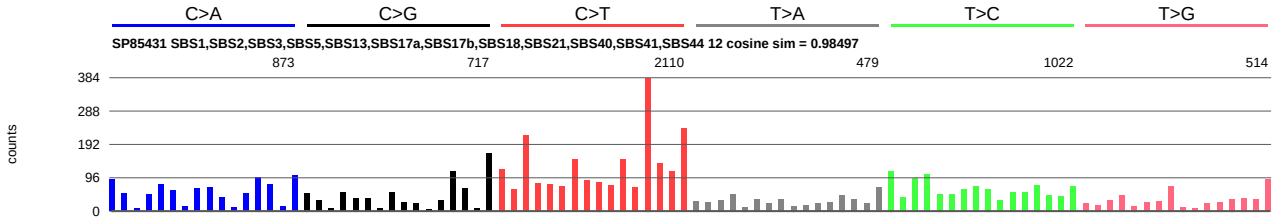
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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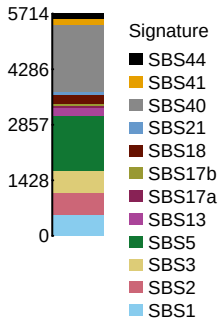
717
2110
479
1022
514
counts
384
288
192
96
0


5714
4286
2857
1428
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Signature
SBS44
SBS41
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SBS21
SBS18
SBS17b
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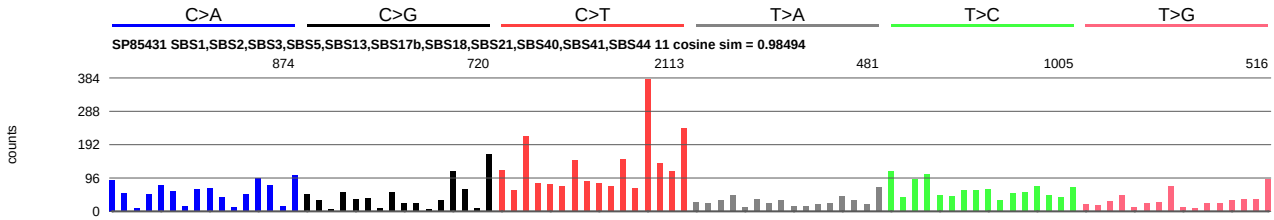
SBS2
SBS1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C>T
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T>C
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SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98494
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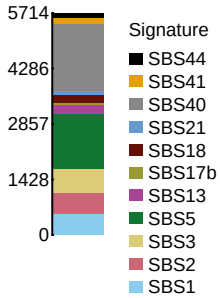
720
2113
481
1005
516
counts
384
288
192
96
0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4286
2857
1428
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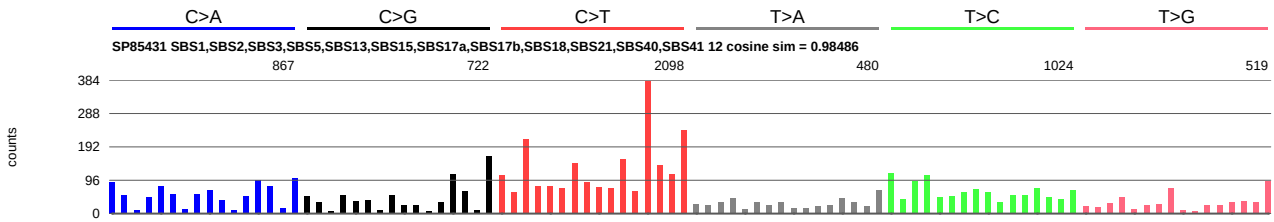
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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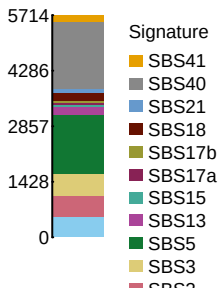
722
2098
480
1024
519
counts
384
288
192
96
0


5714
4286
2857
1428
0
Signature
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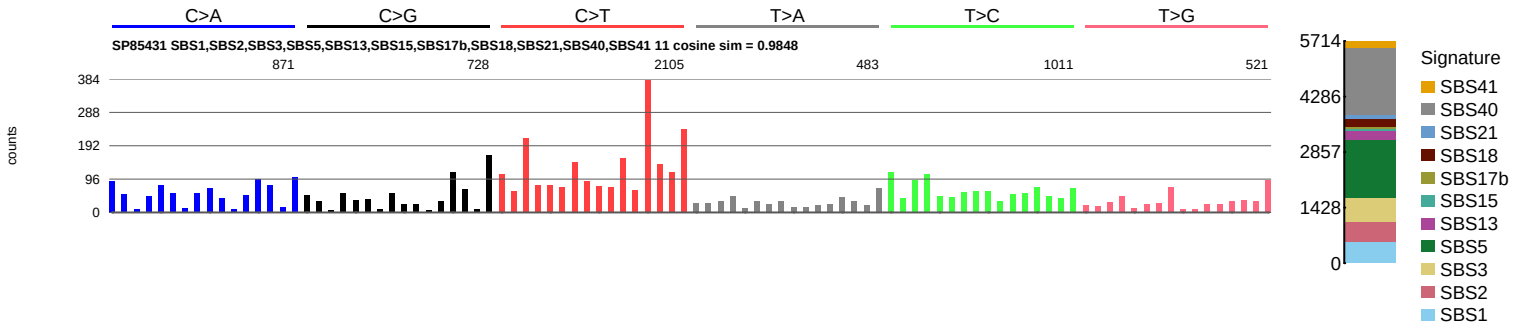
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3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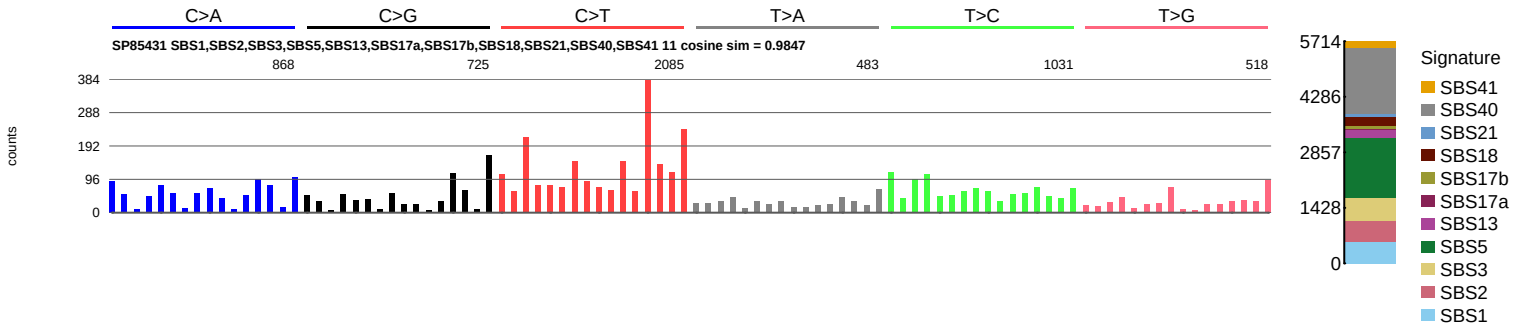
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.9848
871
728
2105
483
1011
521
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.9847
868
725
2085
483
1031
518
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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SBS18
SBS17b
SBS17a
SBS13
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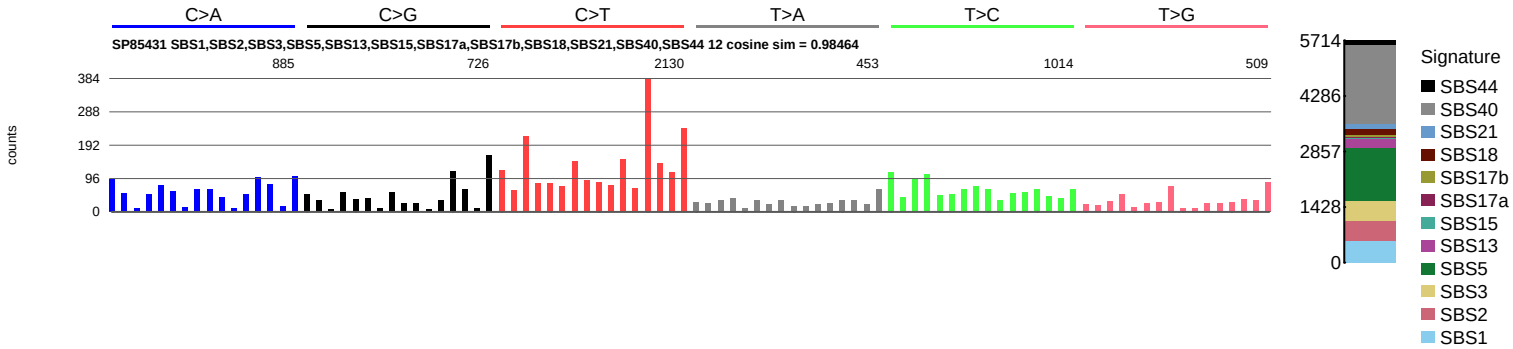
SBS2
SBS1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C>G
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SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 12 cosine sim = 0.98464
885
726
2130
453
1014
509
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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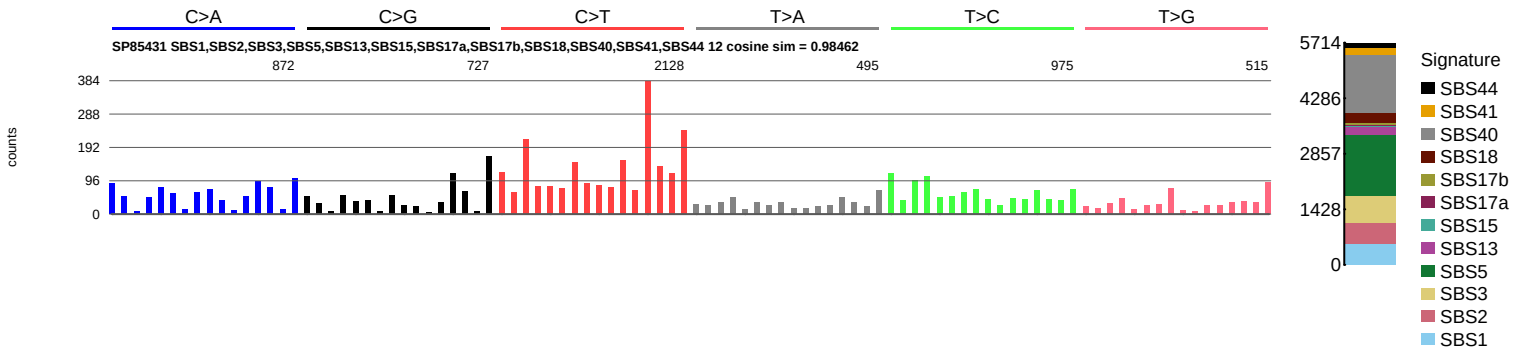
SBS2
SBS1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T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 12 cosine sim = 0.98462
872
727
2128
495
975
515
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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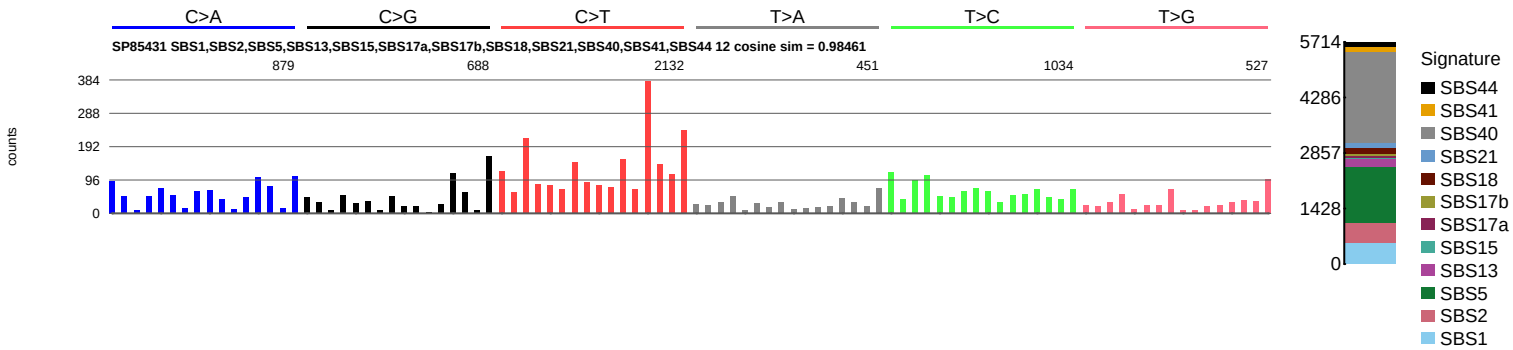
SBS2
SBS1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T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98461
879
688
2132
451
1034
527
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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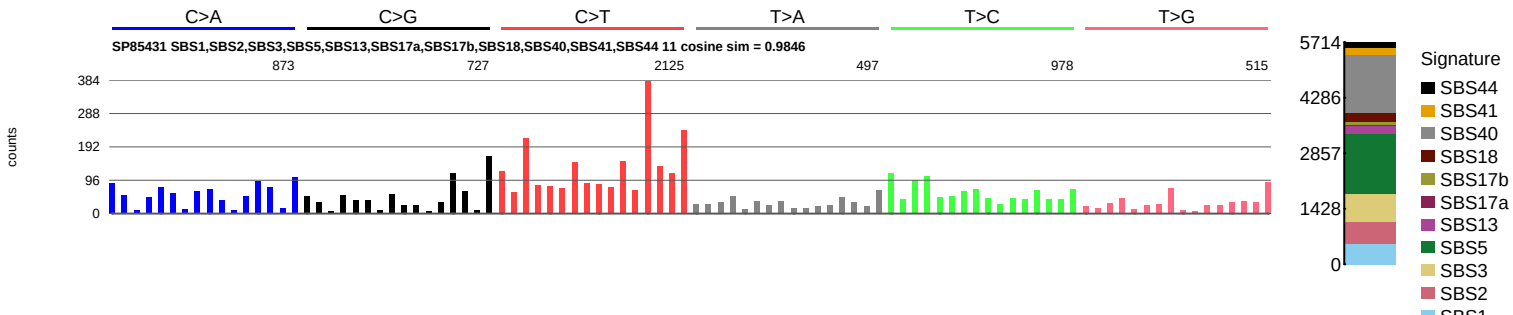
SBS2
SBS1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T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.9846
873
727
2125
497
978
515
counts
384
288
192
96
0
5714
4286
2857
1428
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Signature
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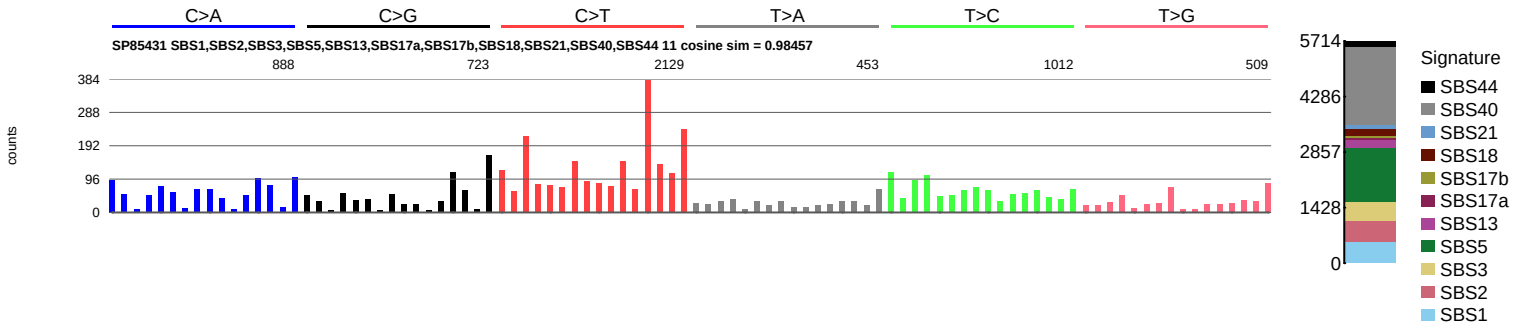
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2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T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98457
888
723
2129
453
1012
509
counts
384
288
192
96
0
5714
4286
2857
1428
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Signature
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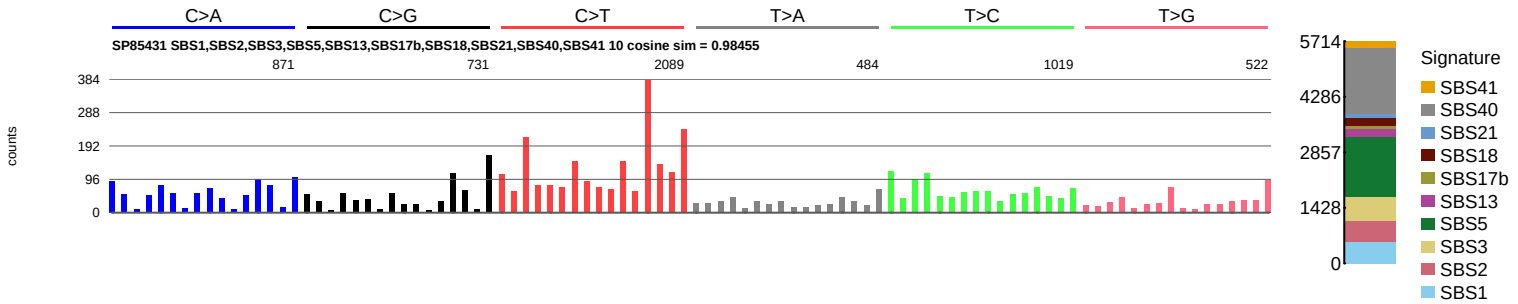
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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T>C
T>G
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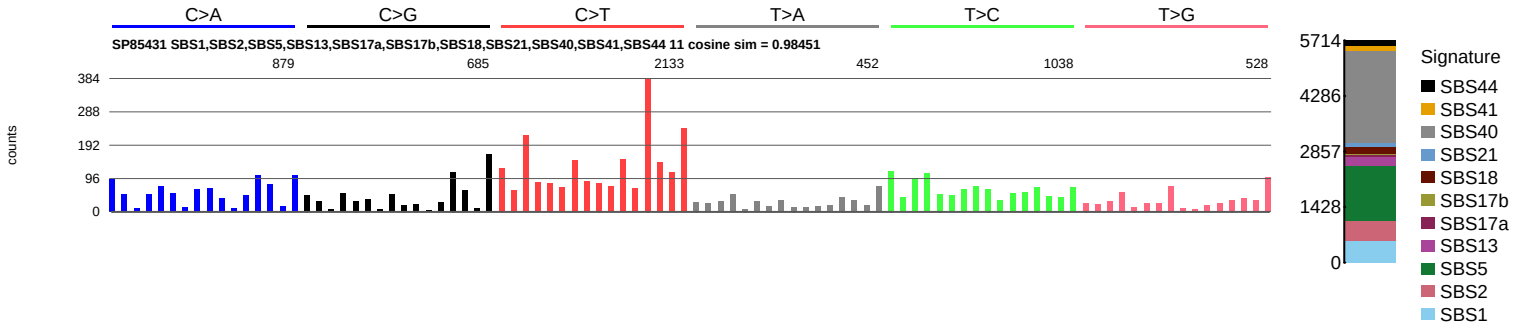
871
731
2089
484
1019
522
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98451
879
685
2133
452
1038
528
counts
384
288
192
96
0
5714
4286
2857
1428
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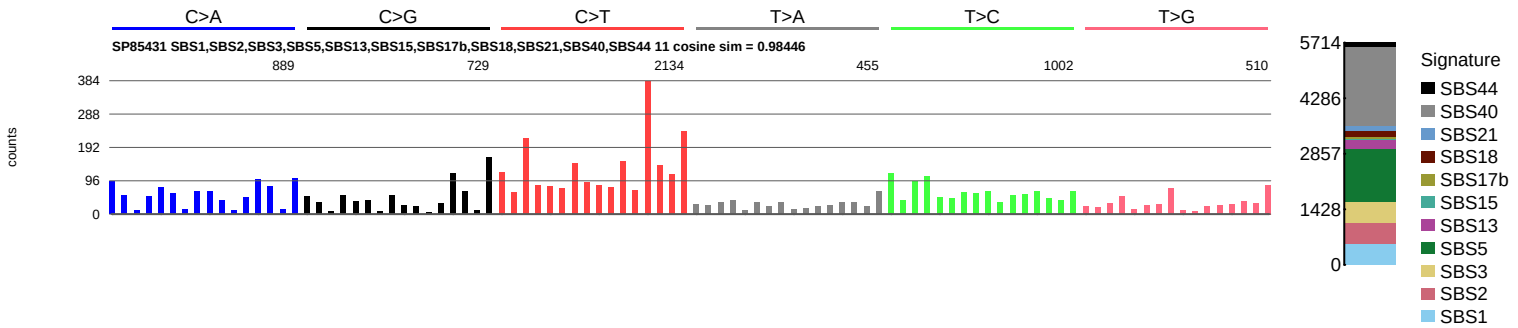
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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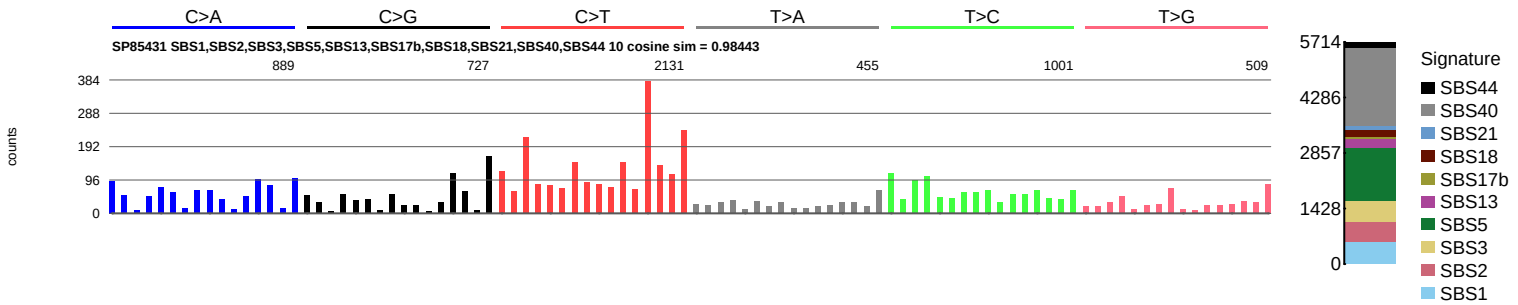
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98446
889
729
2134
455
1002
510
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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T>C
T>G
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889
727
2131
455
1001
509
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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SBS40
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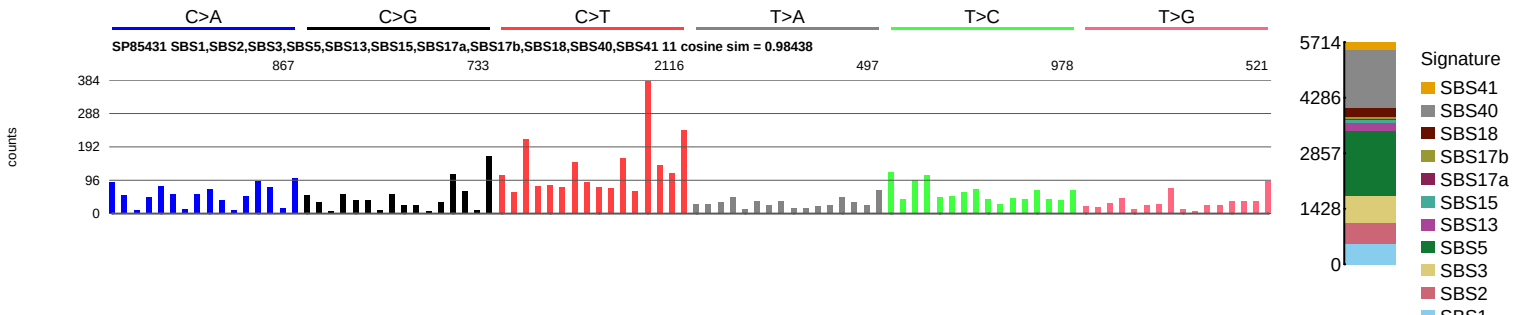
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
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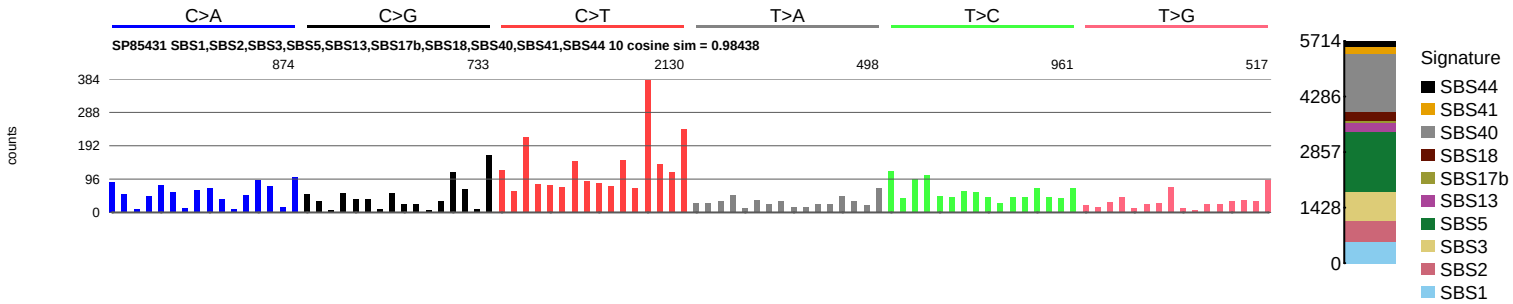
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 11 cosine sim = 0.98438
867
733
2116
497
978
521
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2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C>T
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T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98438
874
733
2130
498
961
517
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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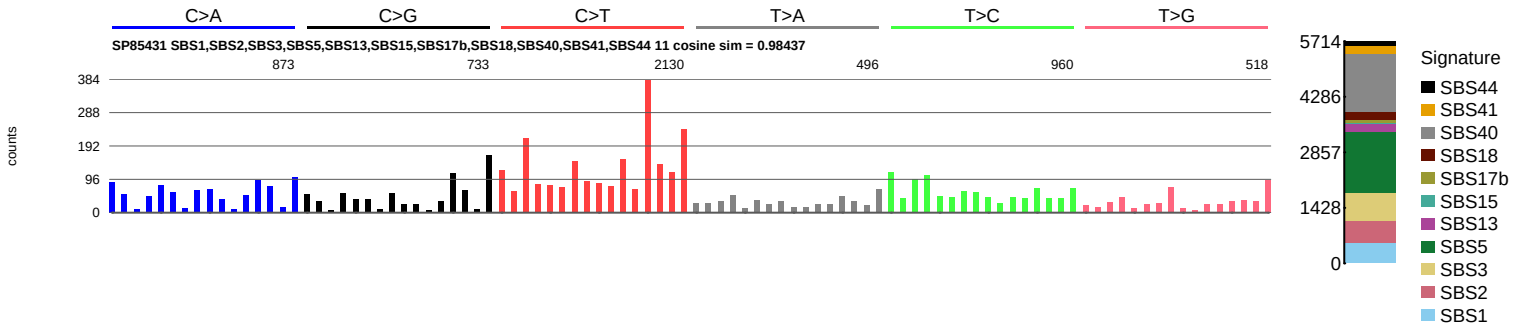
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98437
873
733
2130
496
960
518
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
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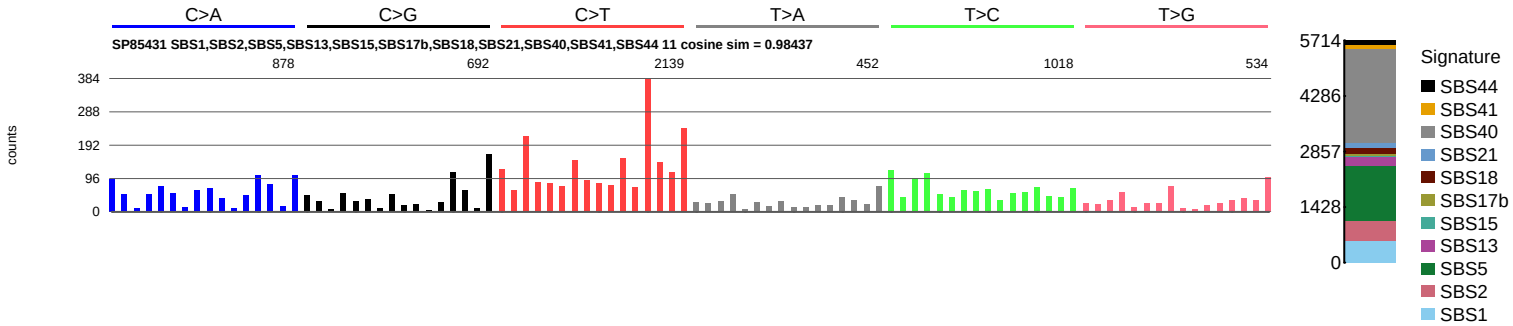
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98437
878
692
2139
452
1018
534
counts
384
288
192
96
0
5714
4286
2857
1428
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Signature
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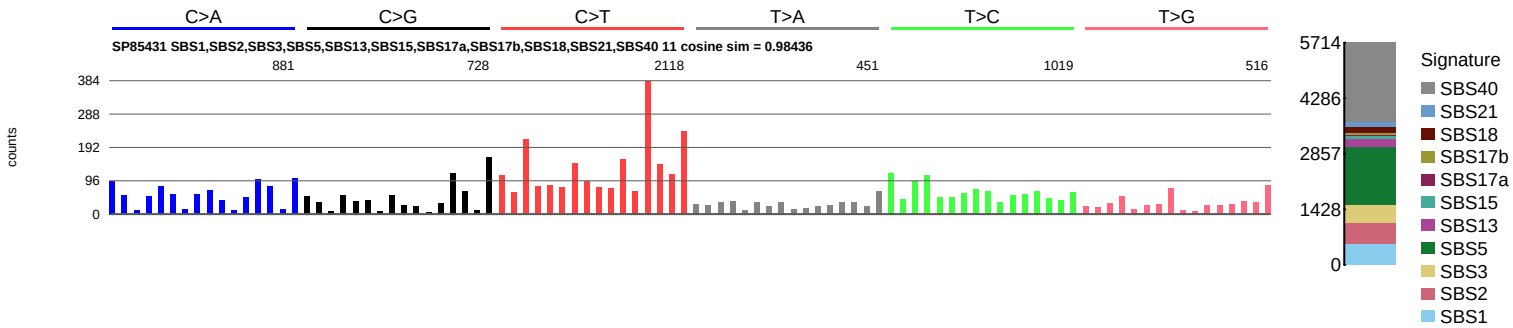
SBS2
SBS1


C>A
C>G
C>T
T>A
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T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 11 cosine sim = 0.98436
881
728
2118
451
1019
516
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
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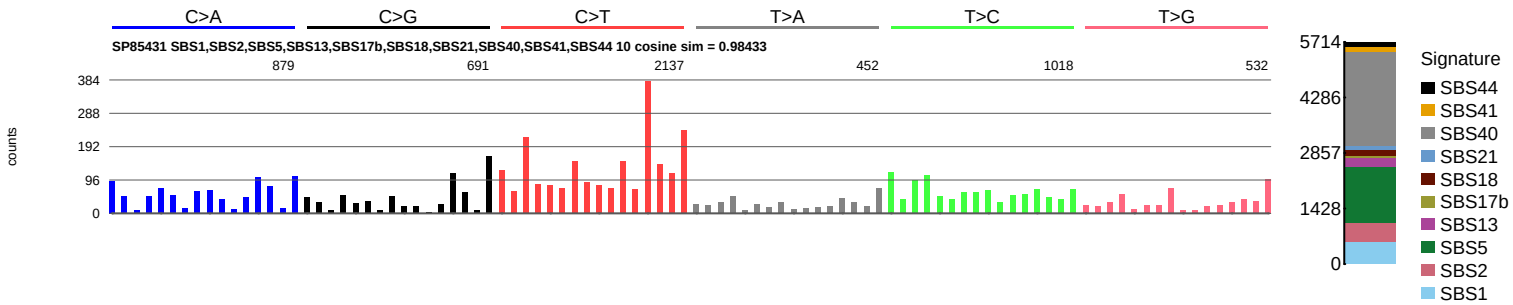
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
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T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98433
879
691
2137
452
1018
532
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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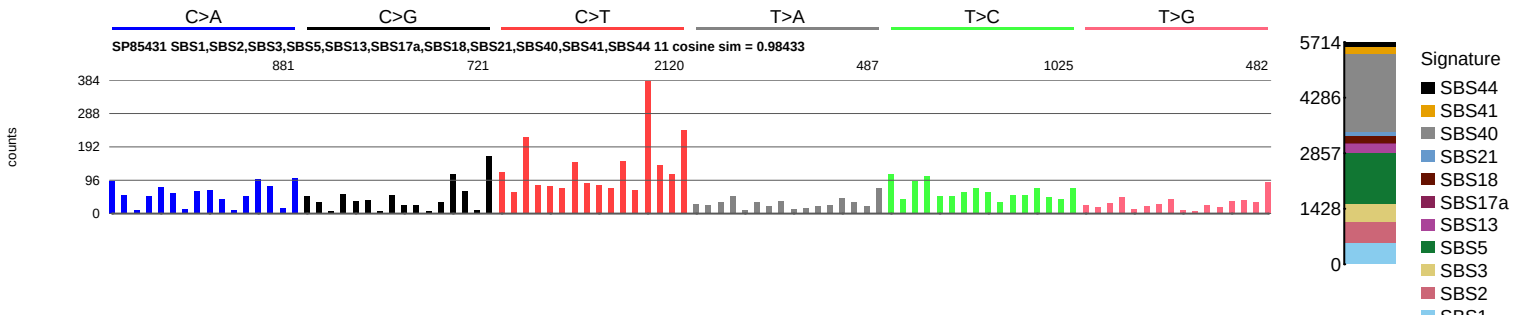
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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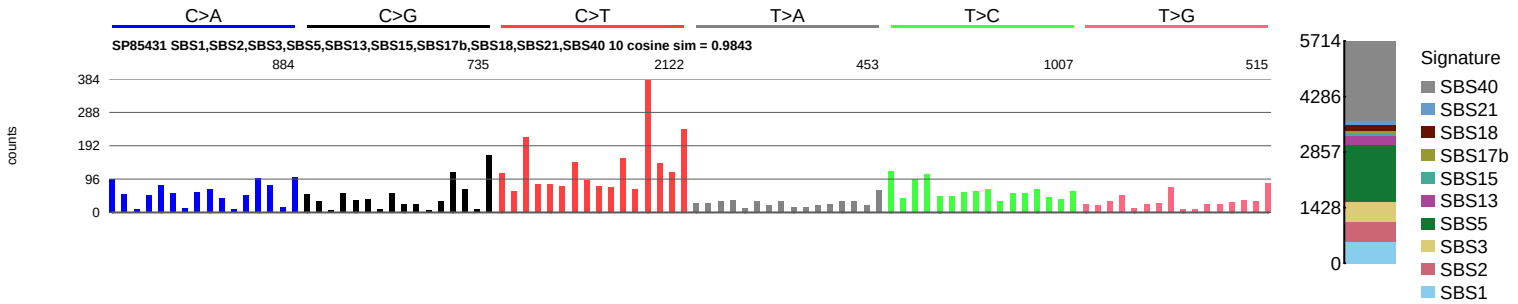
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98433
881
721
2120
487
1025
482
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2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C>G
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T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.9843
884
735
2122
453
1007
515
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
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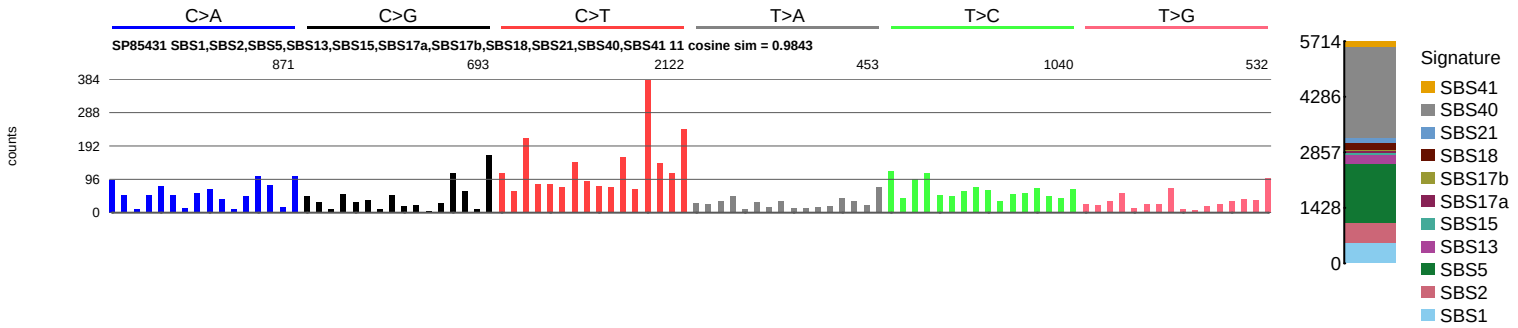
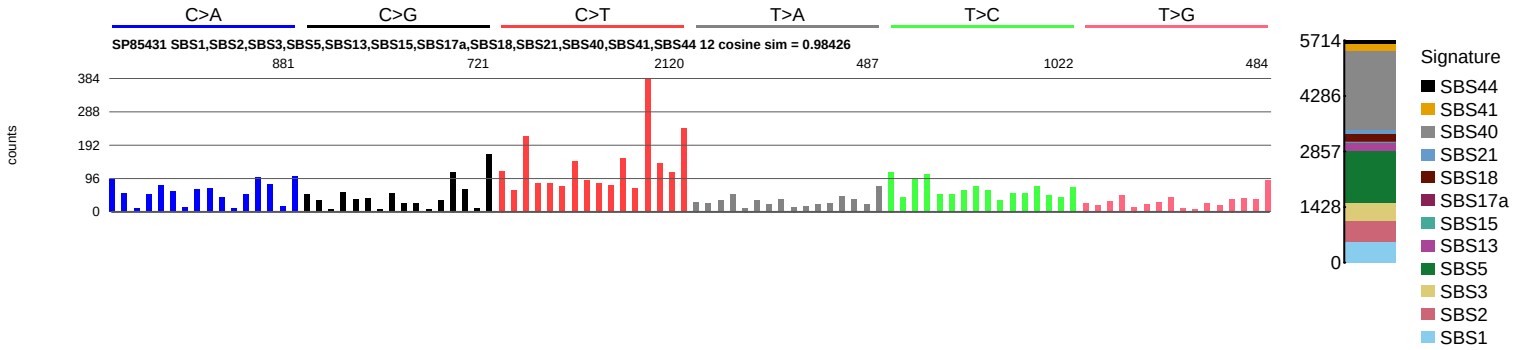
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.9843
871
693
2122
453
1040
532
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98426
881
721
2120
487
1022
484
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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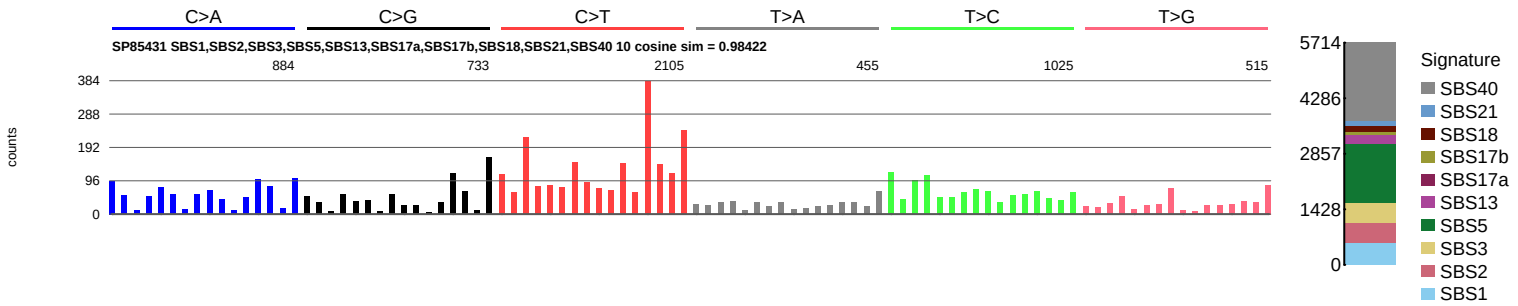
SBS2
SBS1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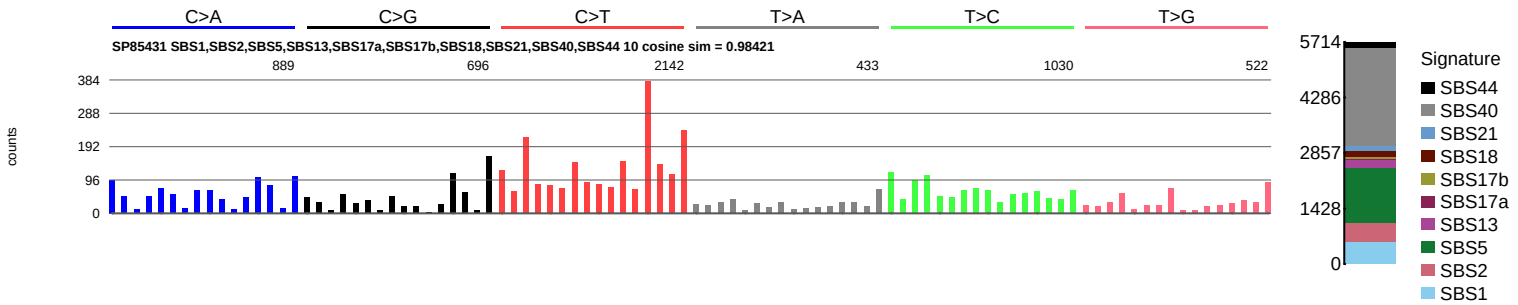
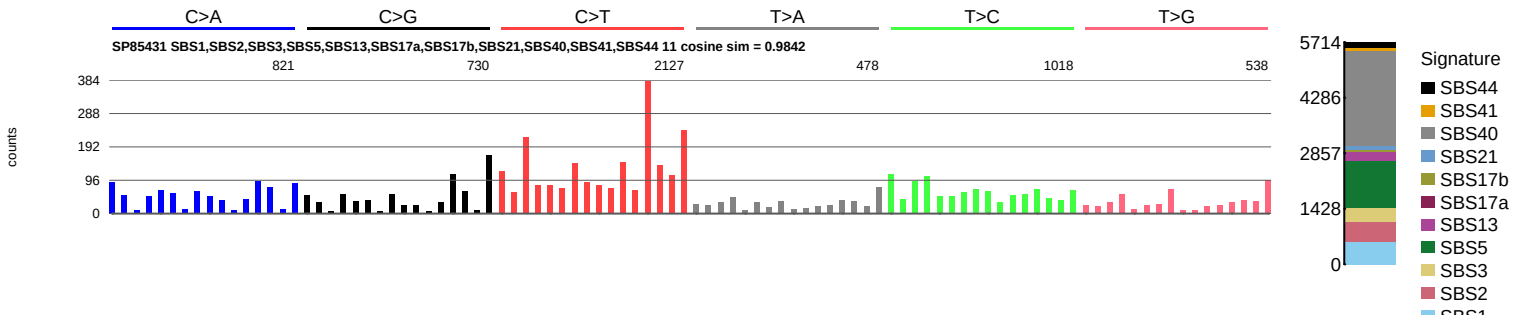
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98422
884
733
2105
455
1025
515
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98421
889
696
2142
433
1030
522
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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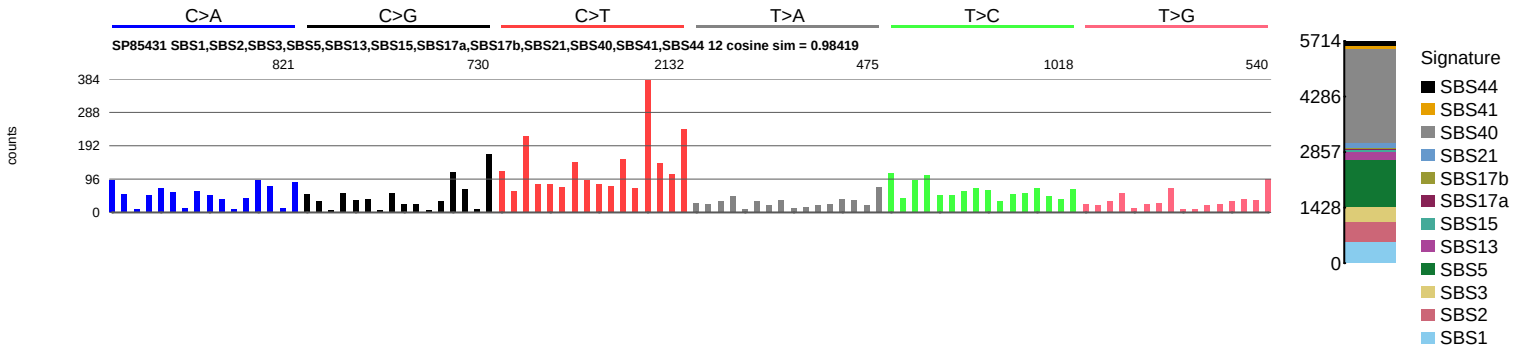
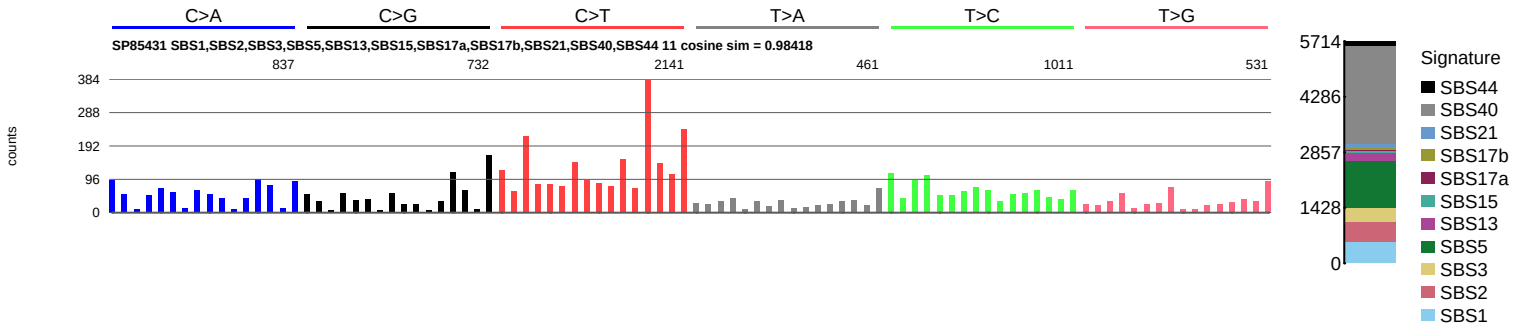
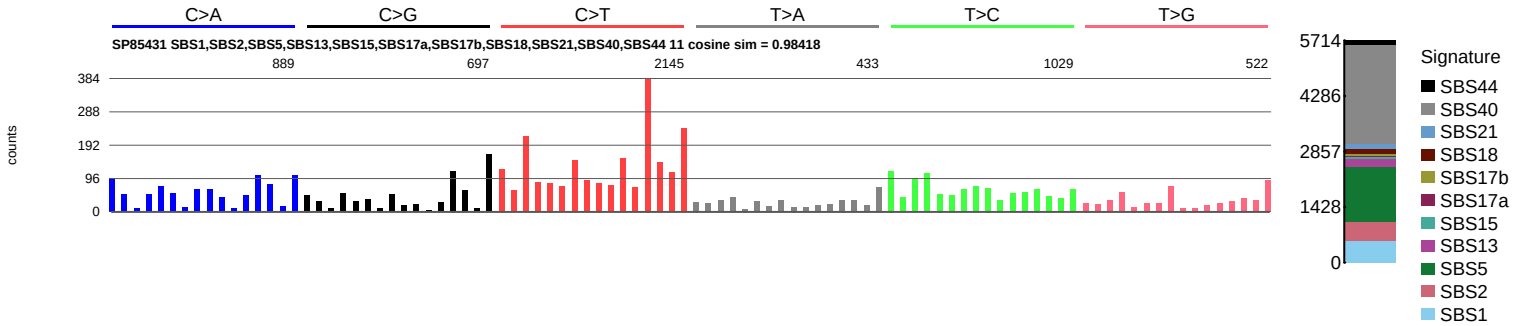
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.9842
821
730
2127
478
1018
538
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98419
821
730
2132
475
1018
540
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 11 cosine sim = 0.98418
837
732
2141
461
1011
531
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98418
889
697
2145
433
1029
522
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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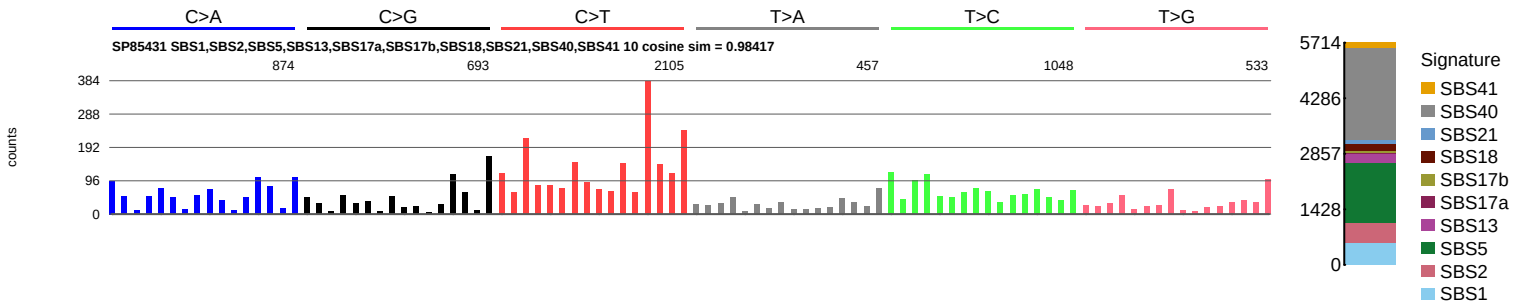
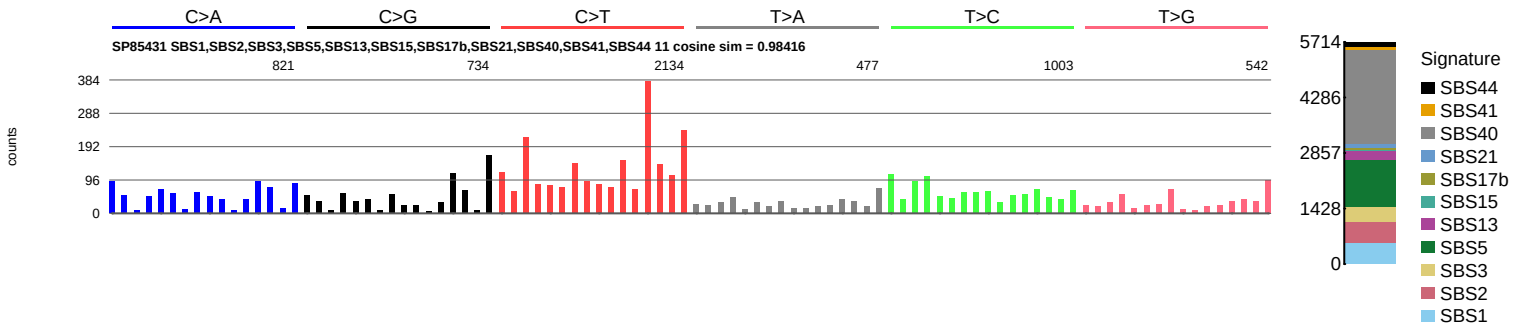
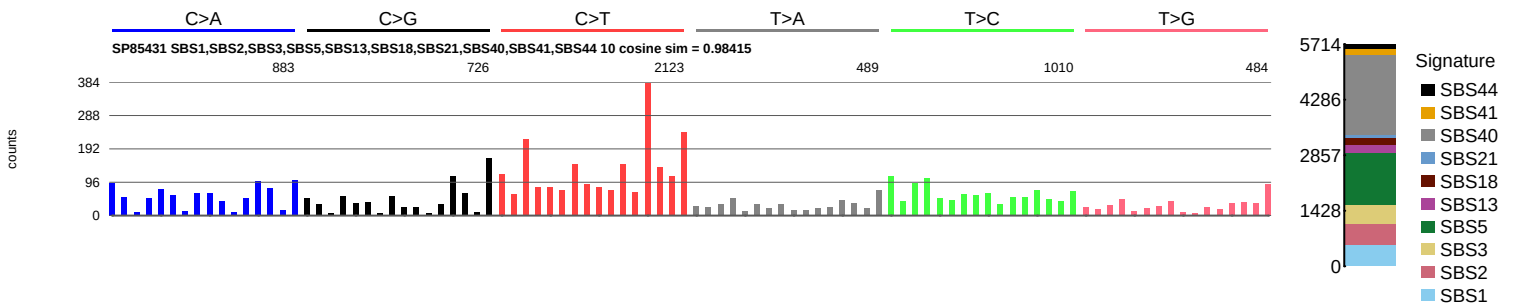
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98417
874
693
2105
457
1048
533
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98416
821
734
2134
477
1003
542
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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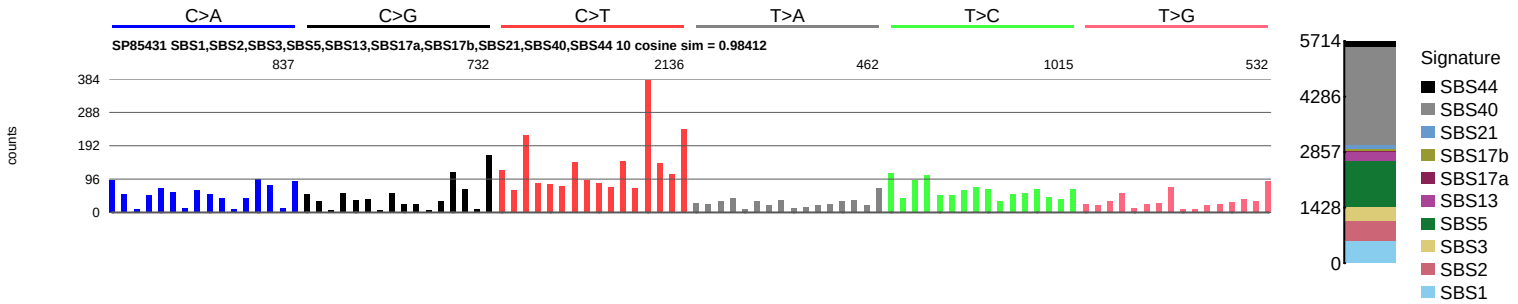
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98415
883
726
2123
489
1010
484
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 10 cosine sim = 0.98412
837
732
2136
462
1015
532
counts
384
288
192
96
0
5714
4286
2857
1428
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Signature
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SBS21
SBS17b
SBS17a
SBS13
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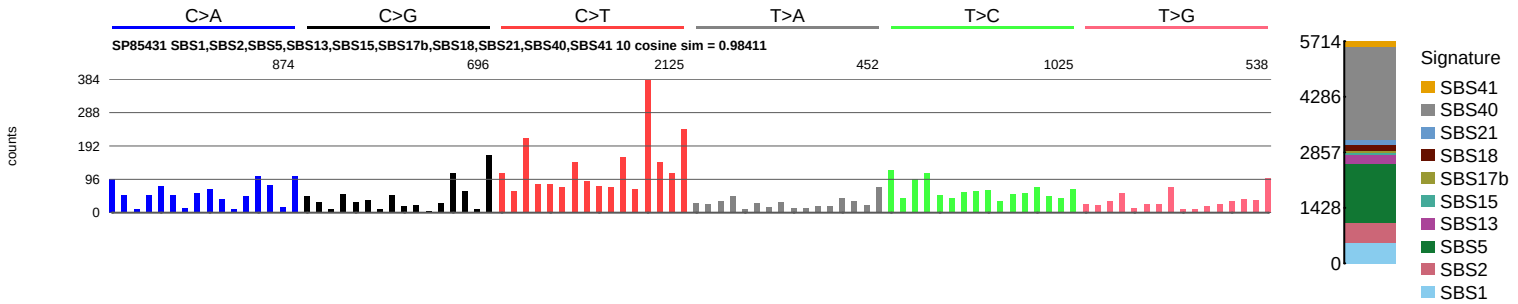
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98411
874
696
2125
452
1025
538
counts
384
288
192
96
0
5714
4286
2857
1428
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Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
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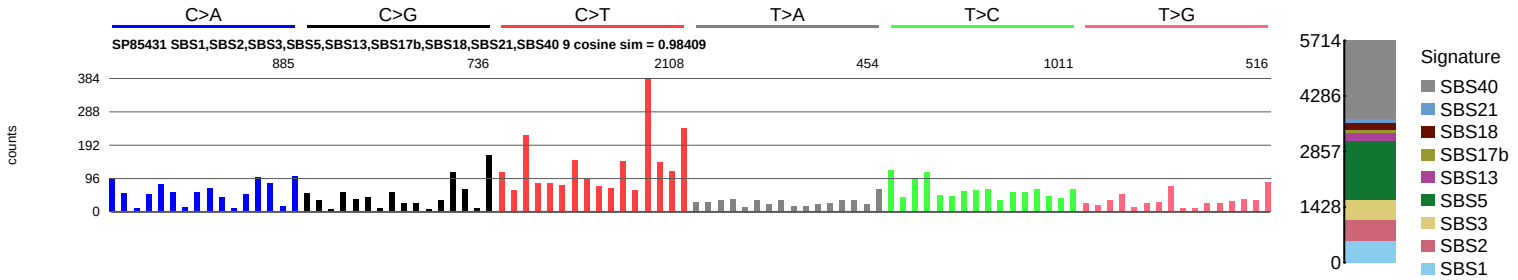
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98409
885
736
2108
454
1011
516
counts
384
288
192
96
0
5714
4286
2857
1428
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Signature
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SBS21
SBS18
SBS17b
SBS13
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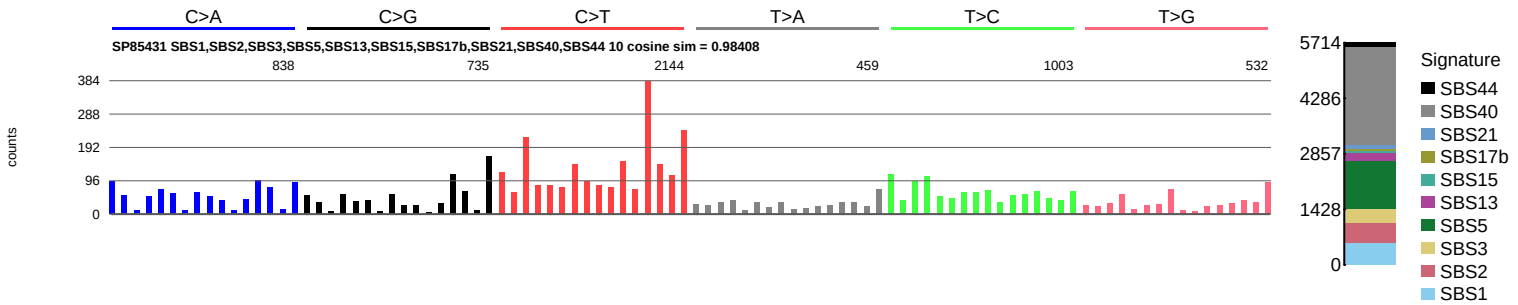
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS21,SBS40,SBS44 10 cosine sim = 0.98408
838
735
2144
459
1003
532
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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SBS17b
SBS15
SBS13
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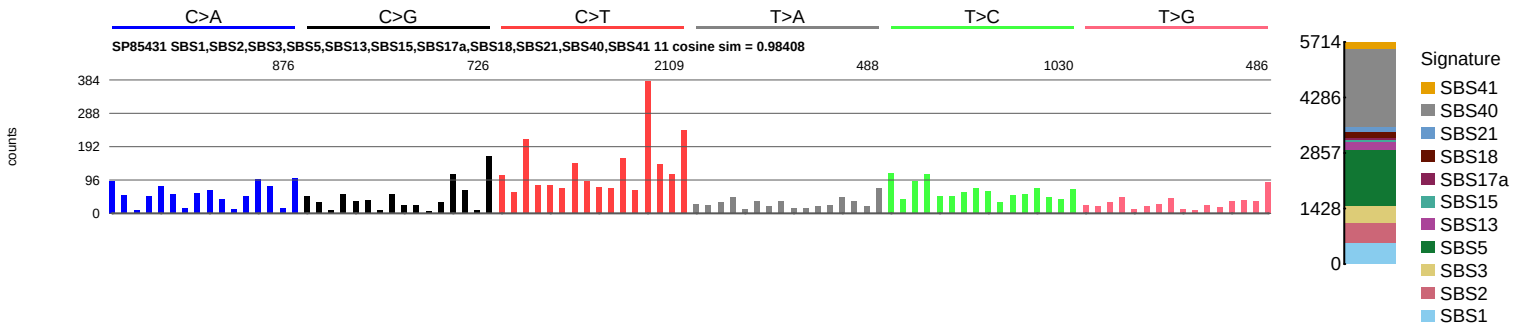
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98408
876
726
2109
488
1030
486
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
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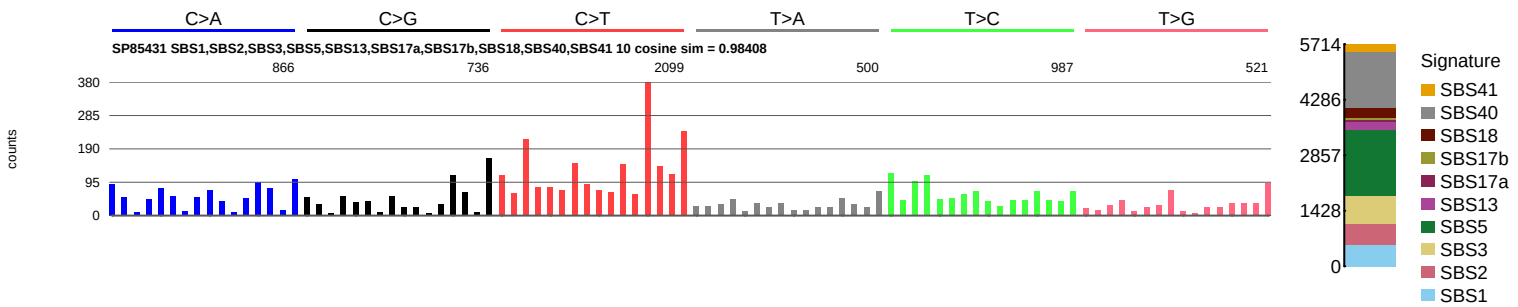
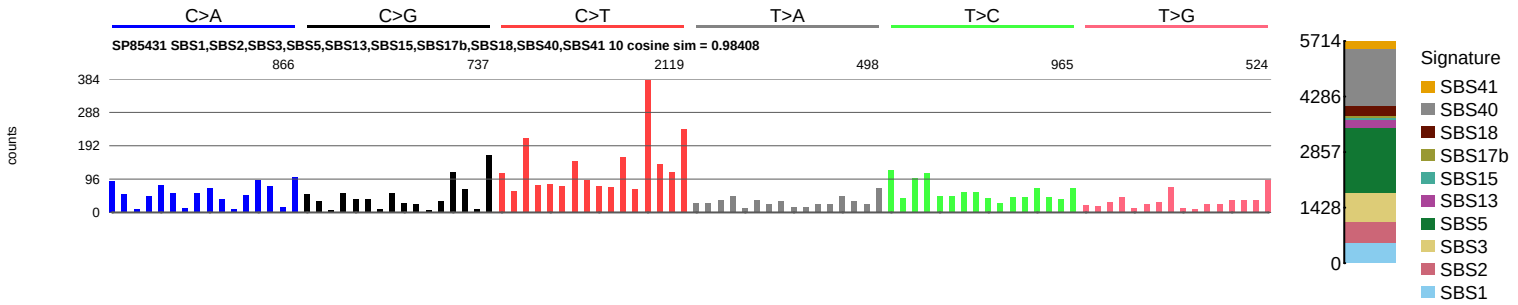
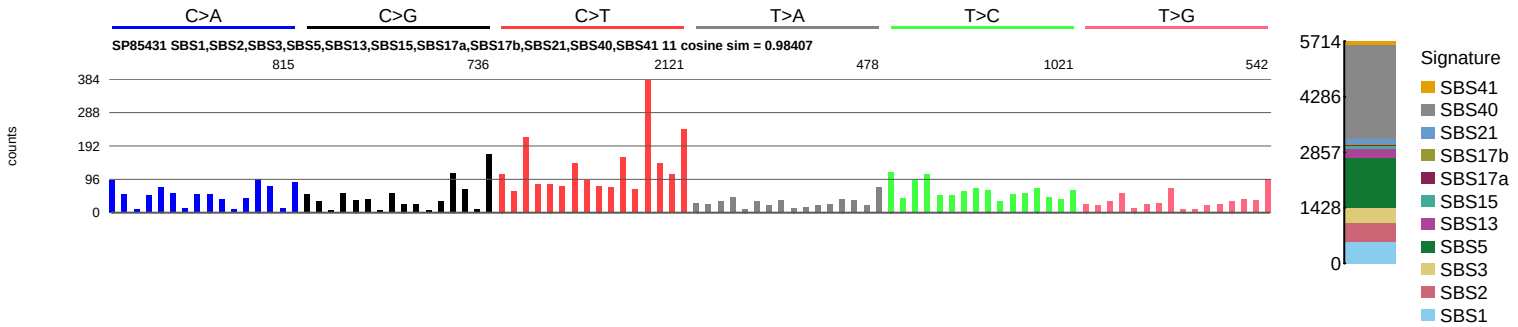
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98408
866
736
2099
500
987
521
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98408
866
737
2119
498
965
524
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 11 cosine sim = 0.98407
815
736
2121
478
1021
542
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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SBS17b
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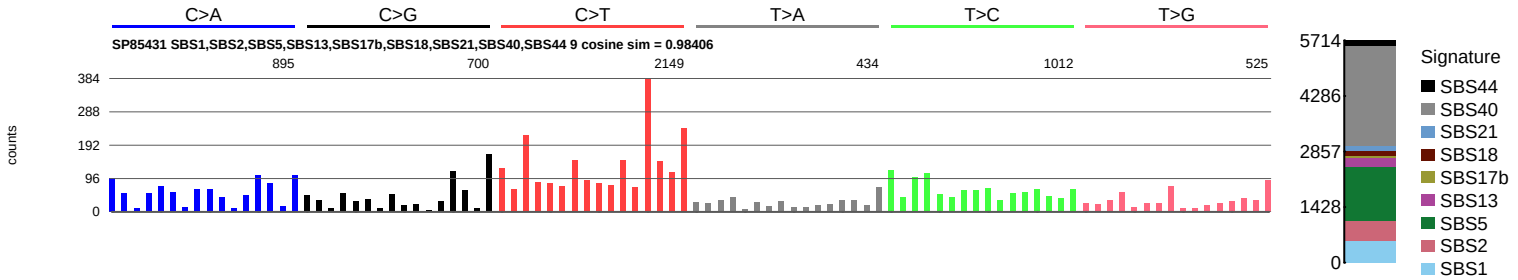
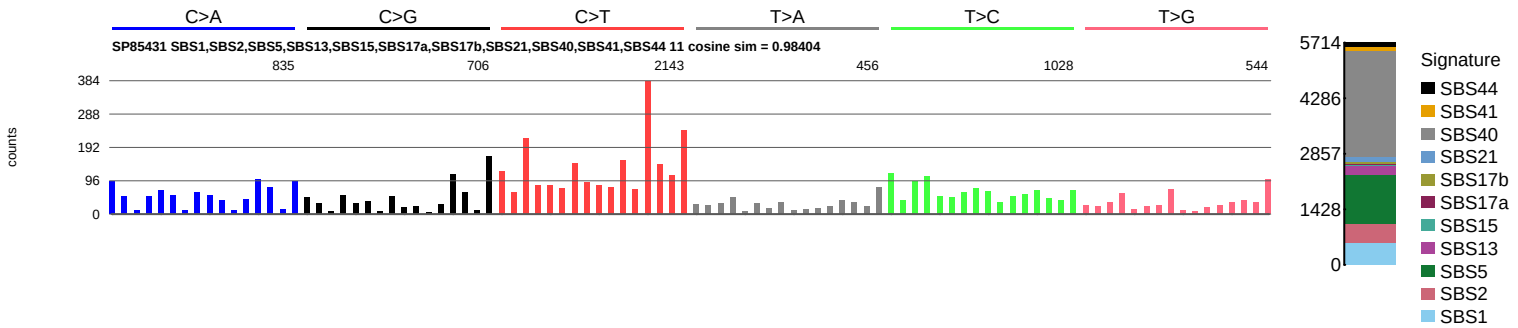
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98406
895
700
2149
434
1012
525
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98404
835
706
2143
456
1028
544
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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SBS40
SBS21
SBS17b
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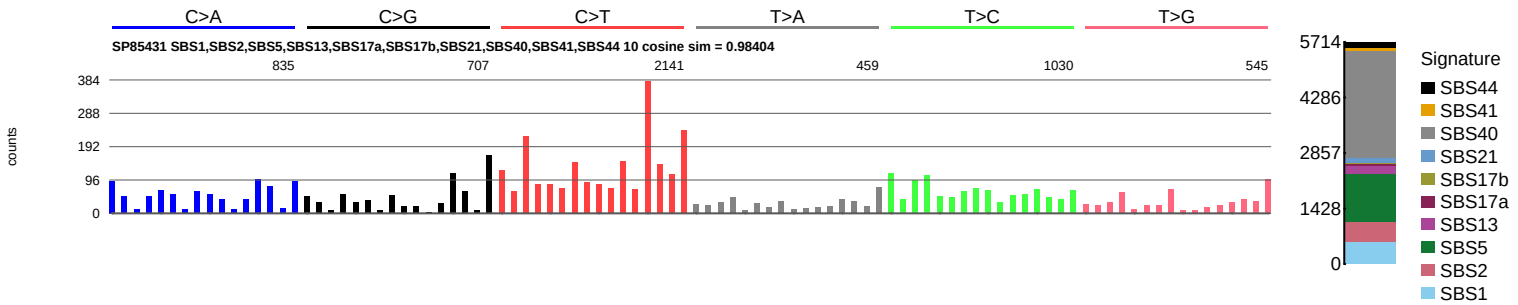
SBS1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SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98404
835
707
2141
459
1030
545
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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SBS40
SBS21
SBS17b
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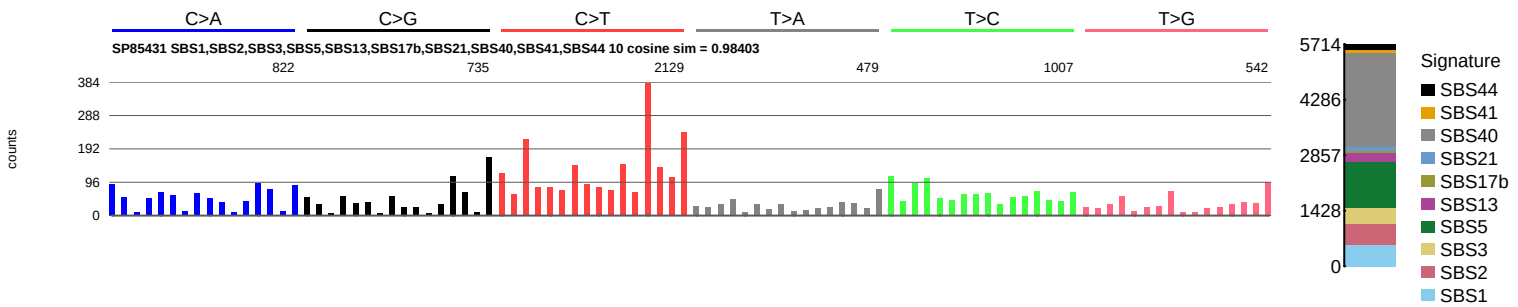
SBS2
SBS1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SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98403
822
735
2129
479
1007
542
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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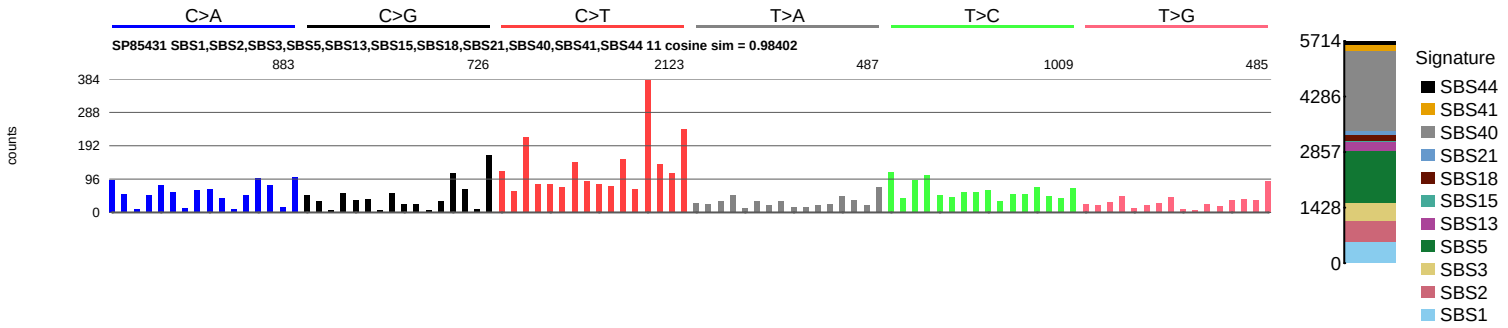
SBS2
SBS1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SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98402
883
726
2123
487
1009
485
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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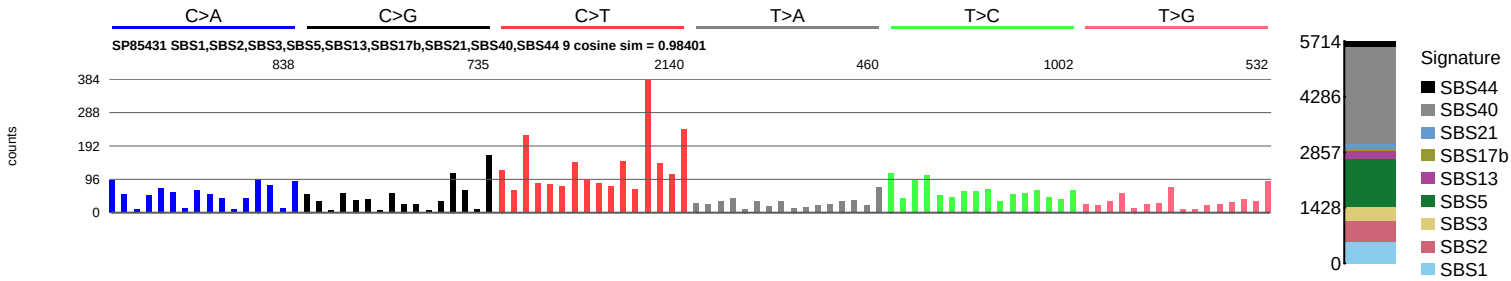
SBS1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SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98401
838
735
2140
460
1002
532
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
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SBS17b
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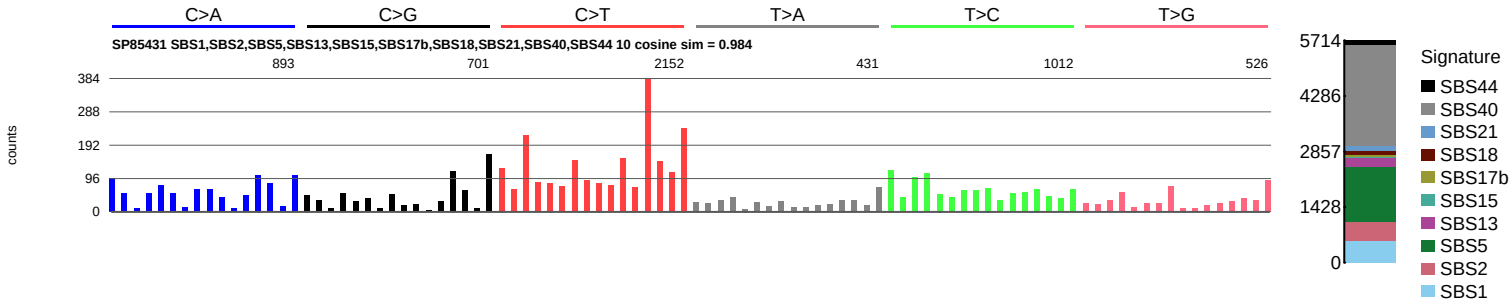
SBS2
SBS1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SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.984
893
701
2152
431
1012
526
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
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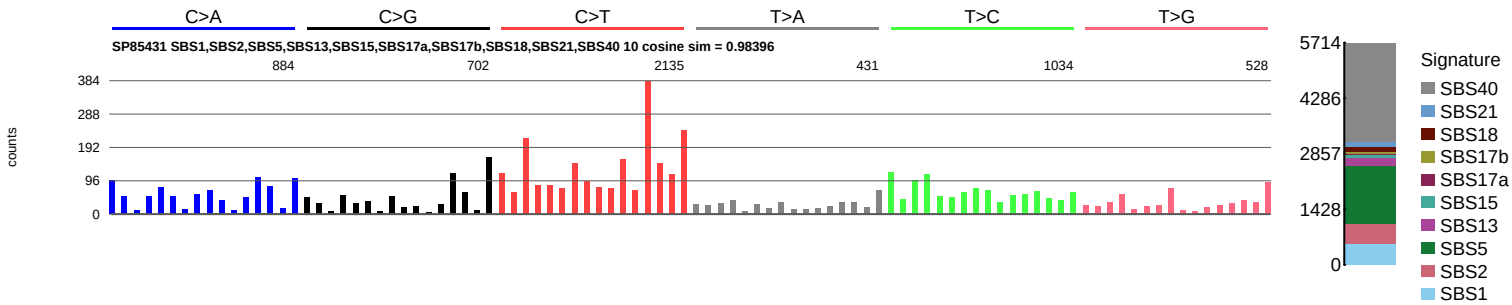
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98396
884
702
2135
431
1034
528
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
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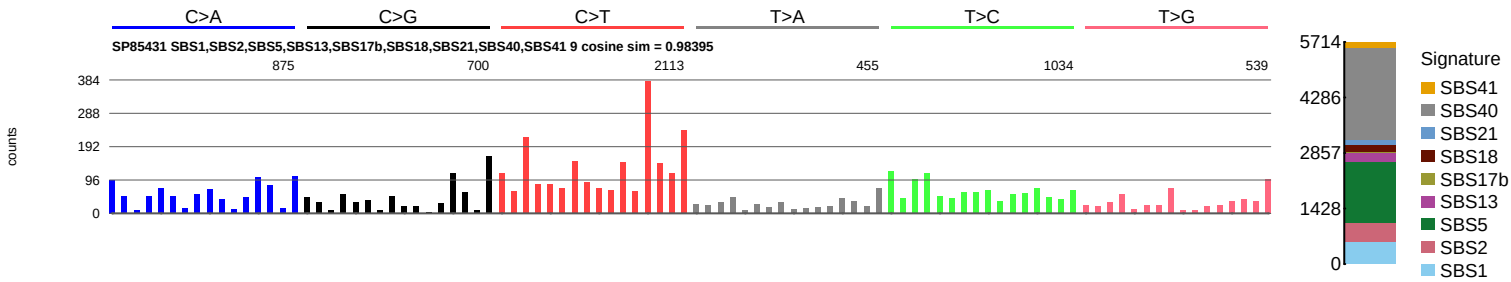
SBS5
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98395
875
700
2113
455
1034
539
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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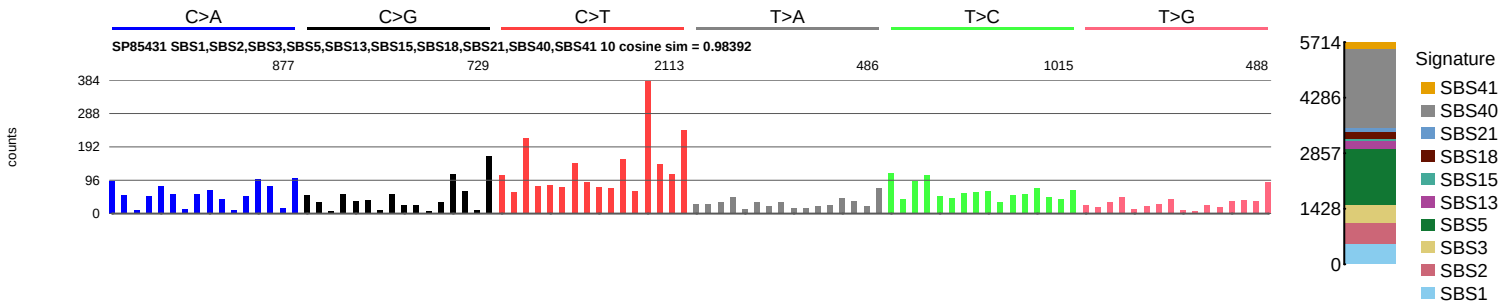
SBS2
SBS1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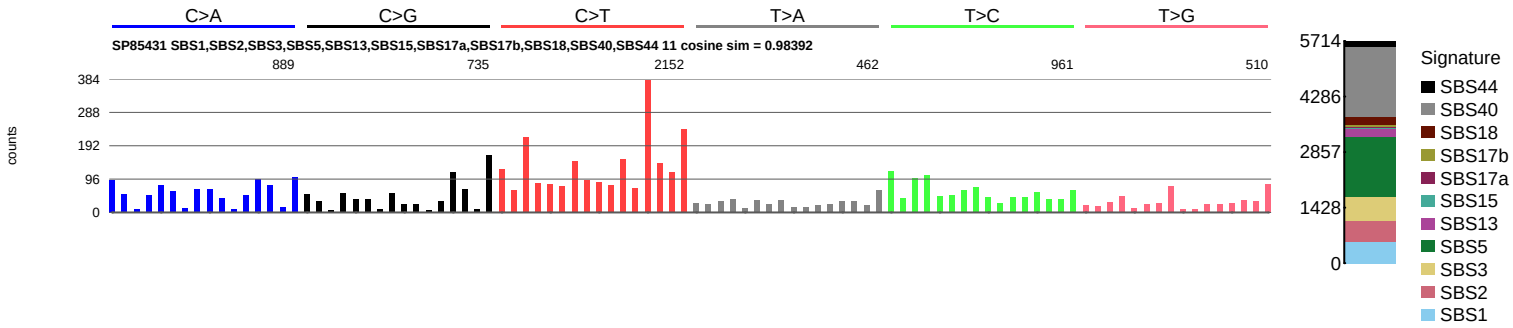
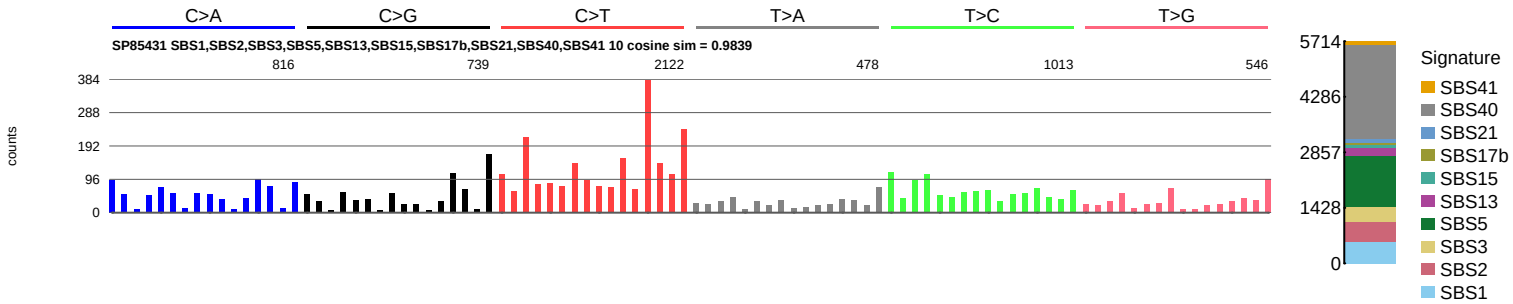
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98392
877
729
2113
486
1015
488
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 11 cosine sim = 0.98392
889
735
2152
462
961
510
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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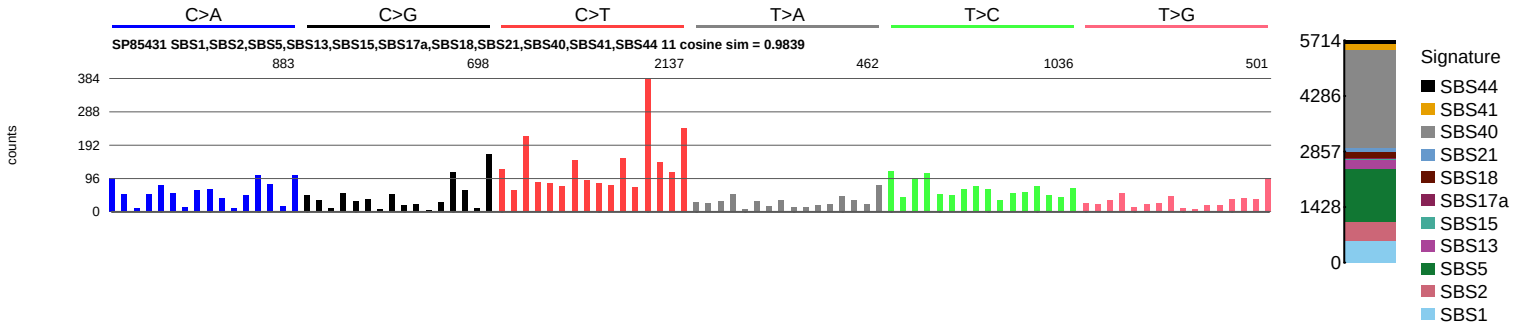
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.9839
816
739
2122
478
1013
546
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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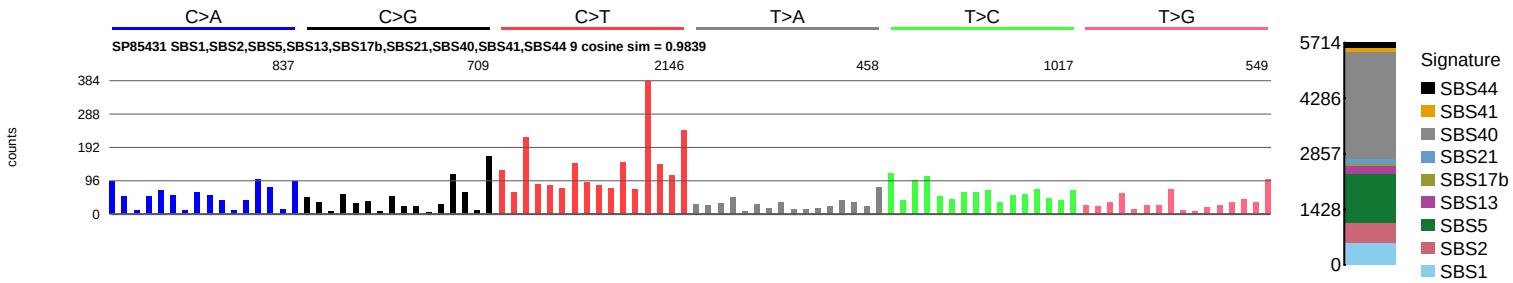
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.9839
883
698
2137
462
1036
501
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.9839
837
709
2146
458
1017
549
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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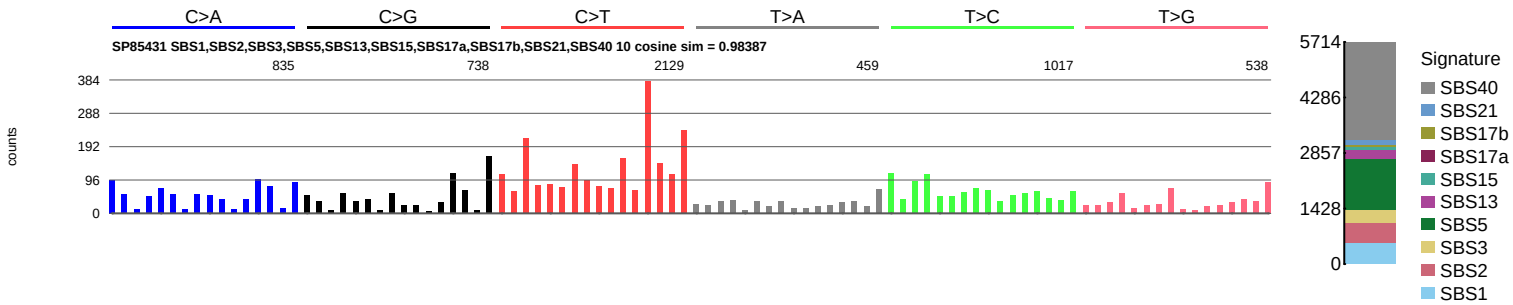
SBS2
SBS1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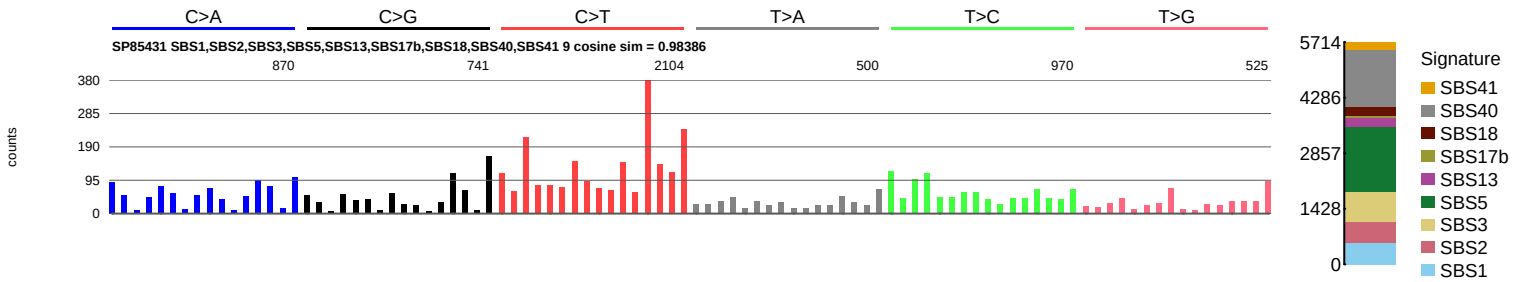
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 10 cosine sim = 0.98387
835
738
2129
459
1017
538
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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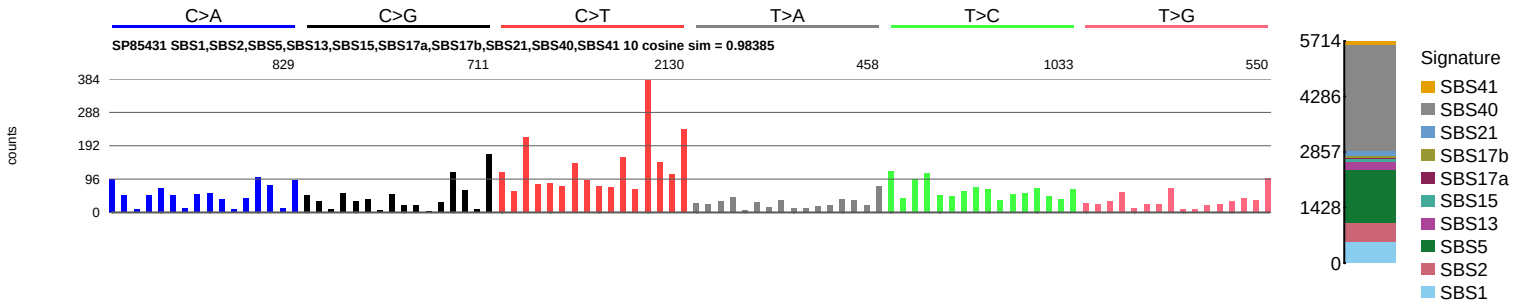
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98386
870
741
2104
500
970
525
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.98385
829
711
2130
458
1033
550
counts
384
288
192
96
0
5714
4286
2857
1428
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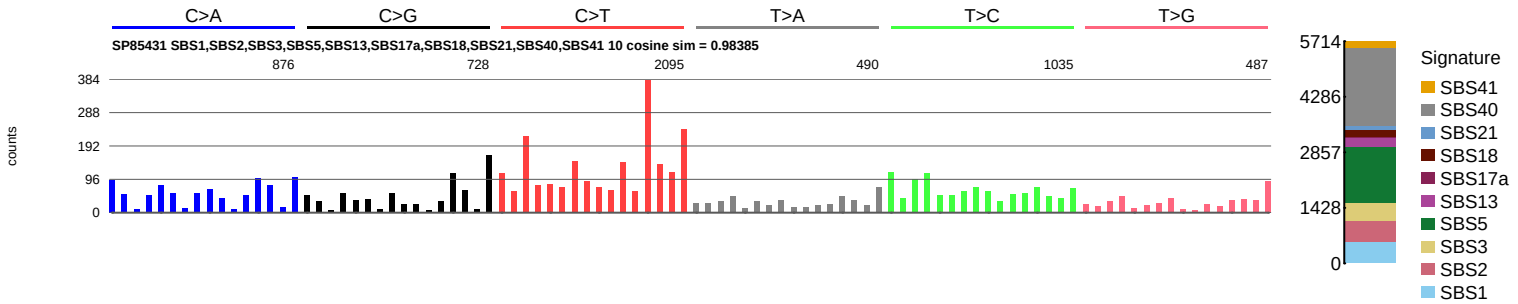
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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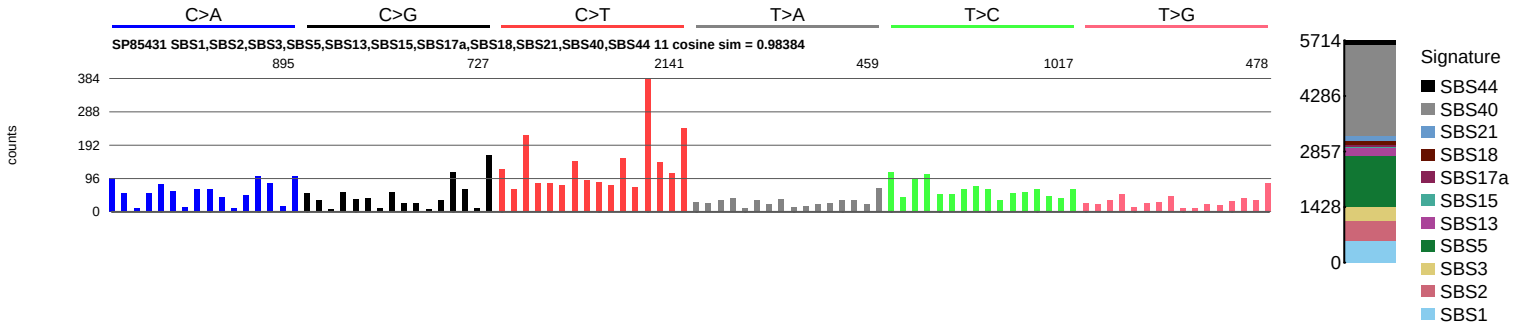
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98385
876
728
2095
490
1035
487
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98384
895
727
2141
459
1017
478
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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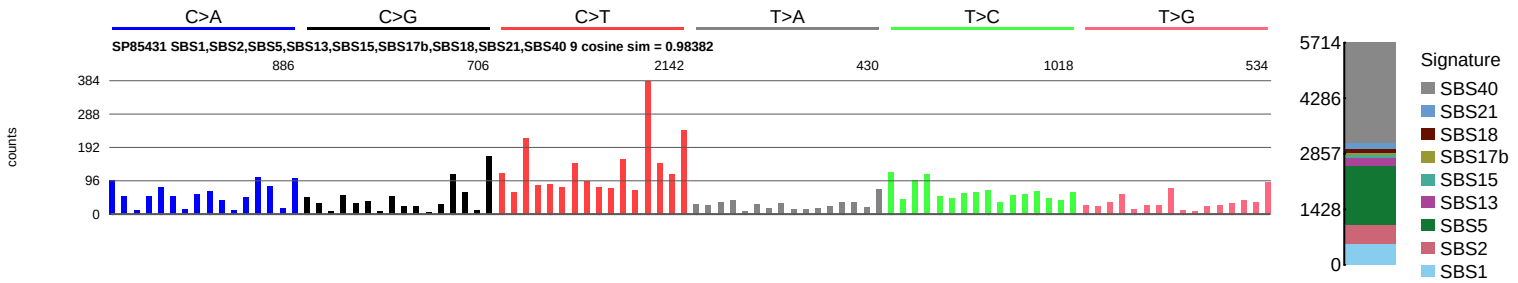
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98382
886
706
2142
430
1018
534
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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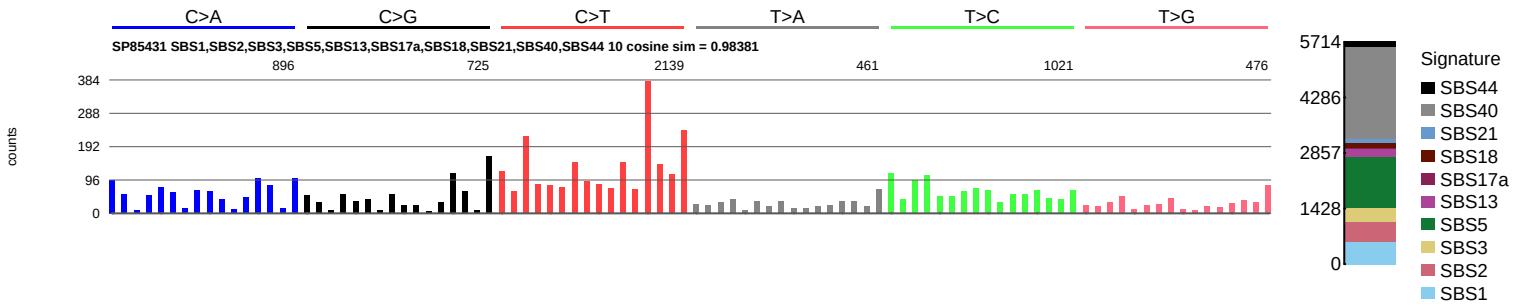
SBS2
SBS1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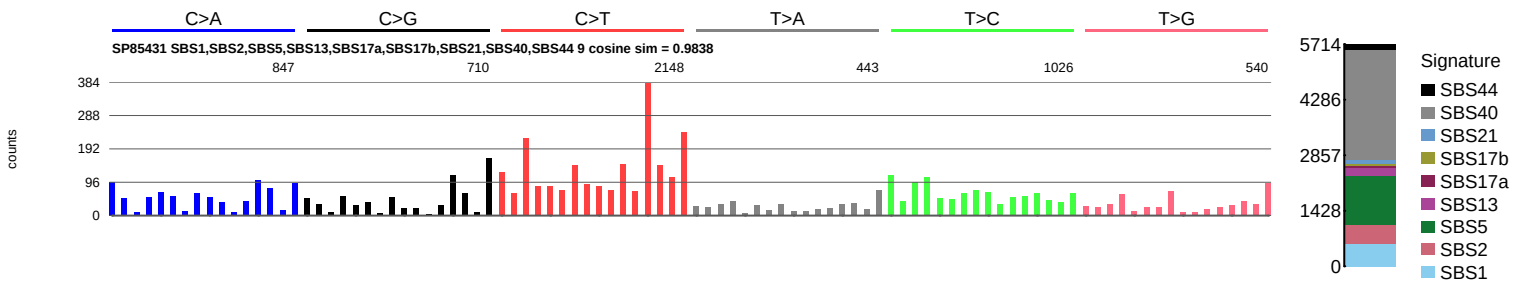
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98381
896
725
2139
461
1021
476
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.9838
847
710
2148
443
1026
540
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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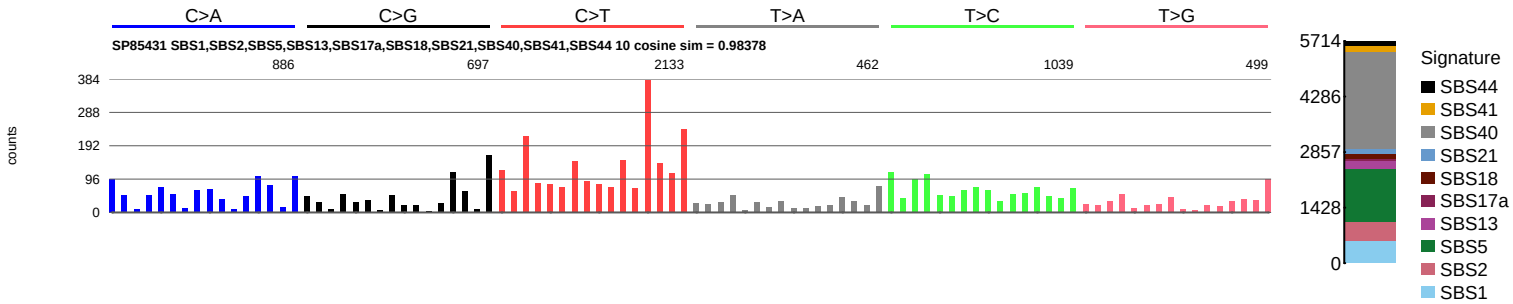
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98378
886
697
2133
462
1039
499
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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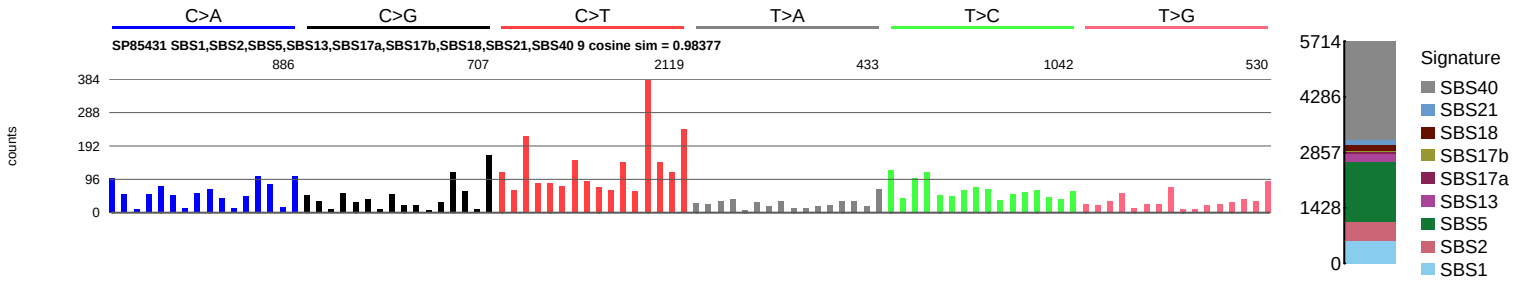
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98377
886
707
2119
433
1042
530
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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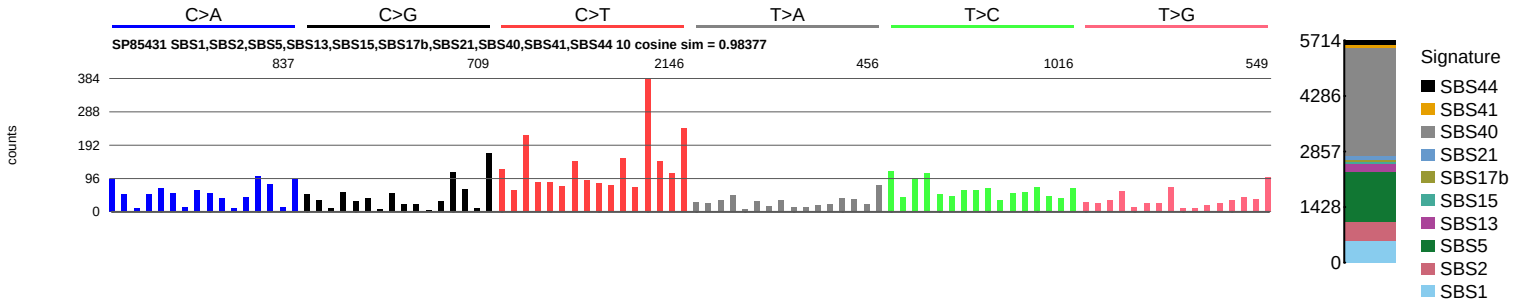
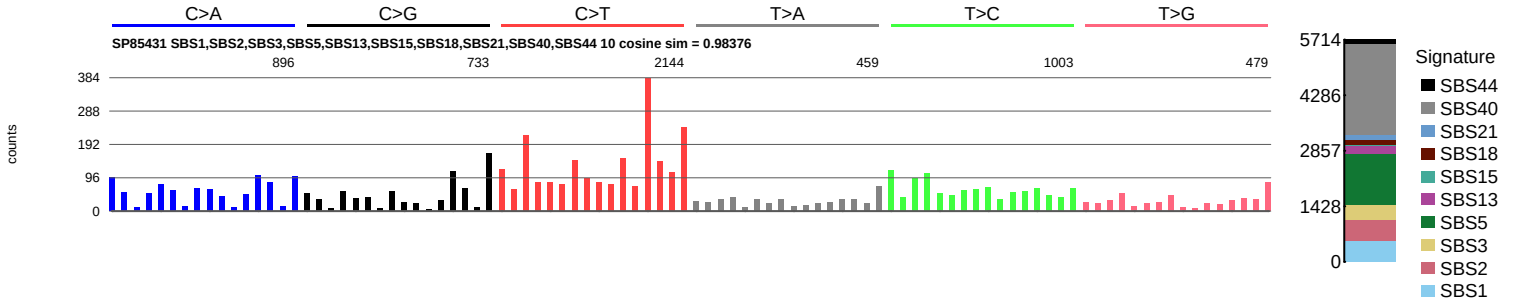
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98377
837
709
2146
456
1016
549
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98376
896
733
2144
459
1003
479
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
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SBS18
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SBS13
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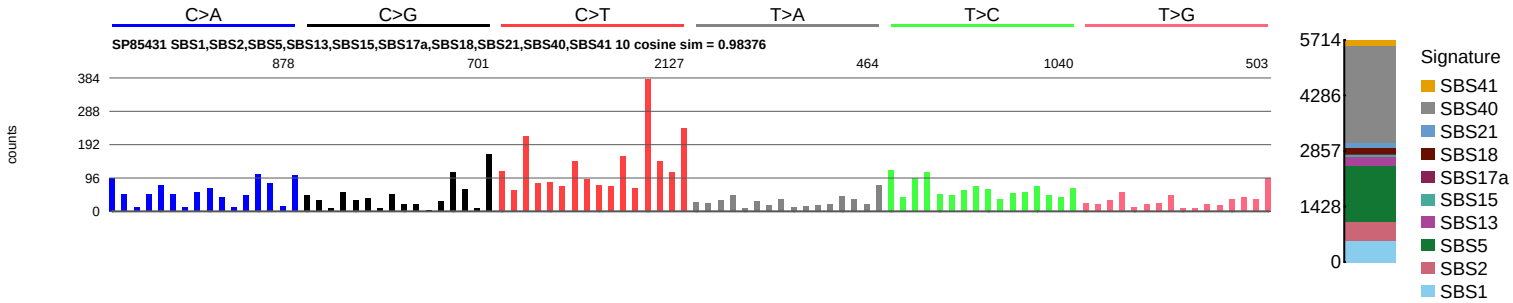
SBS2
SBS1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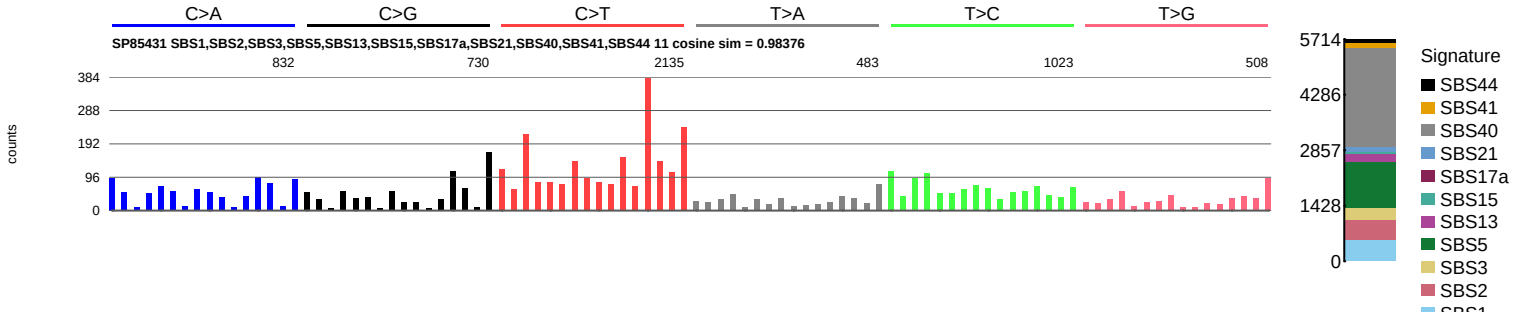
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98376
878
701
2127
464
1040
503
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98376
832
730
2135
483
1023
508
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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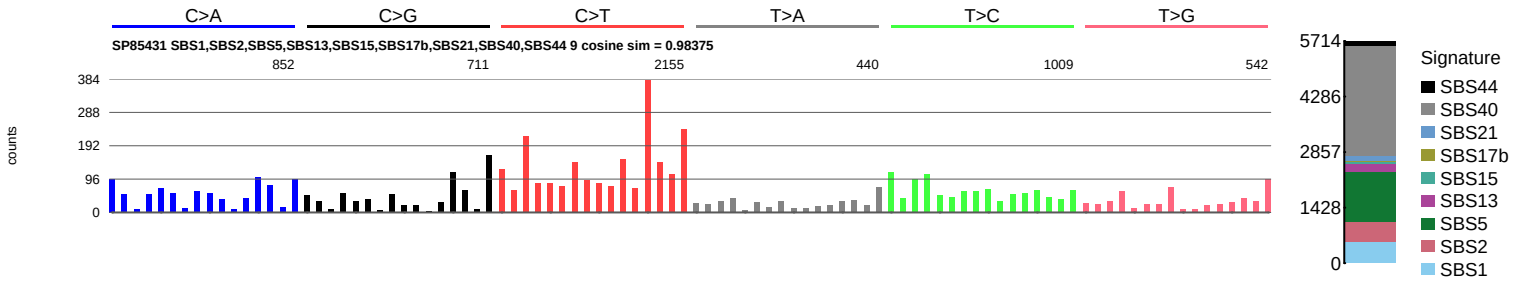
SBS40
SBS21
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2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SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98375
852
711
2155
440
1009
542
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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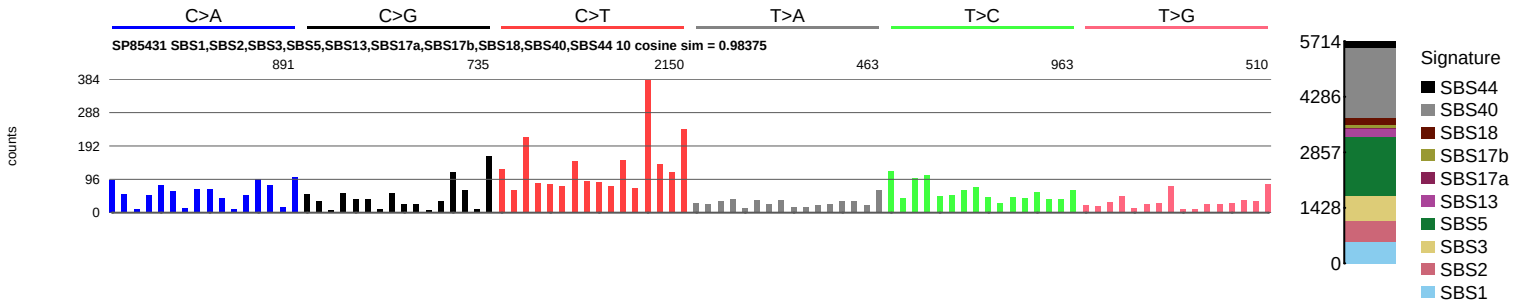
SBS2
SBS1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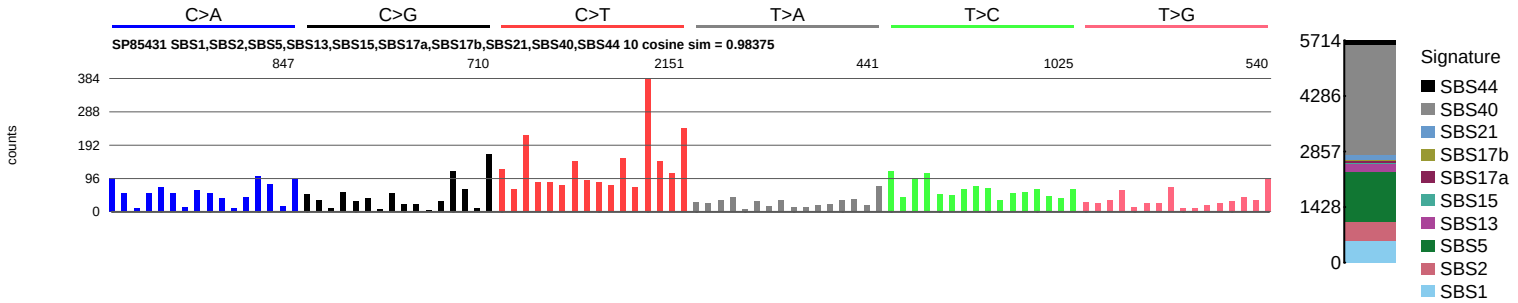
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98375
891
735
2150
463
963
510
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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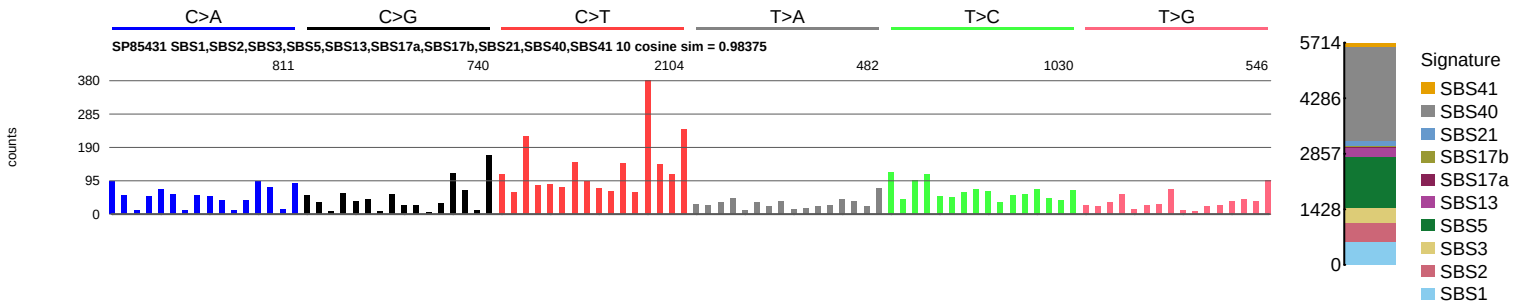
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 10 cosine sim = 0.98375
847
710
2151
441
1025
540
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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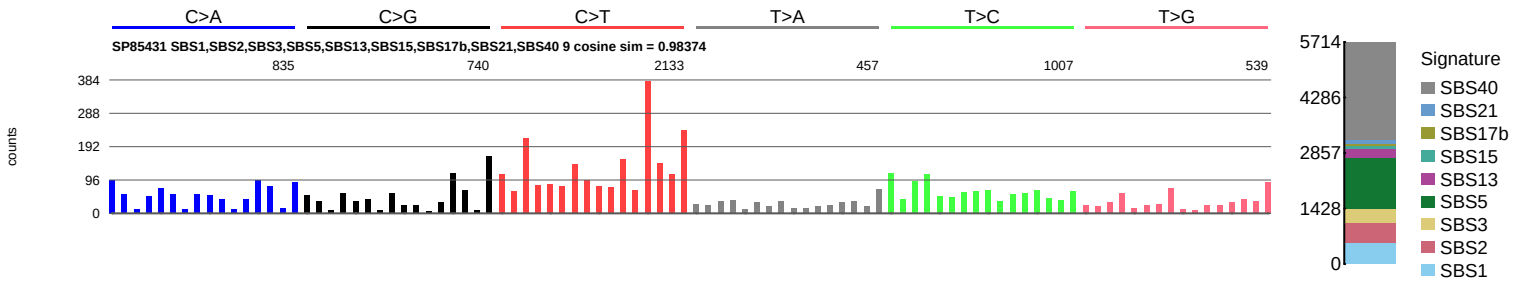
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.98375
811
740
2104
482
1030
546
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS21,SBS40 9 cosine sim = 0.98374
835
740
2133
457
1007
539
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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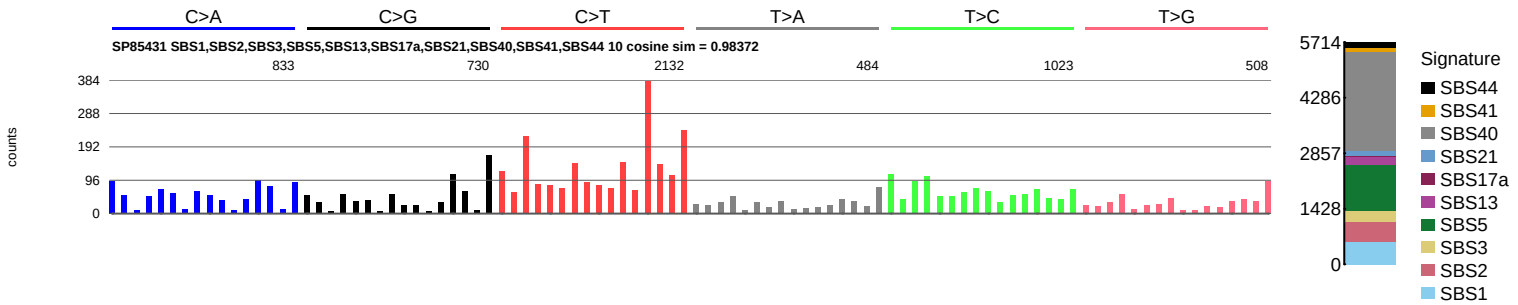
SBS2
SBS1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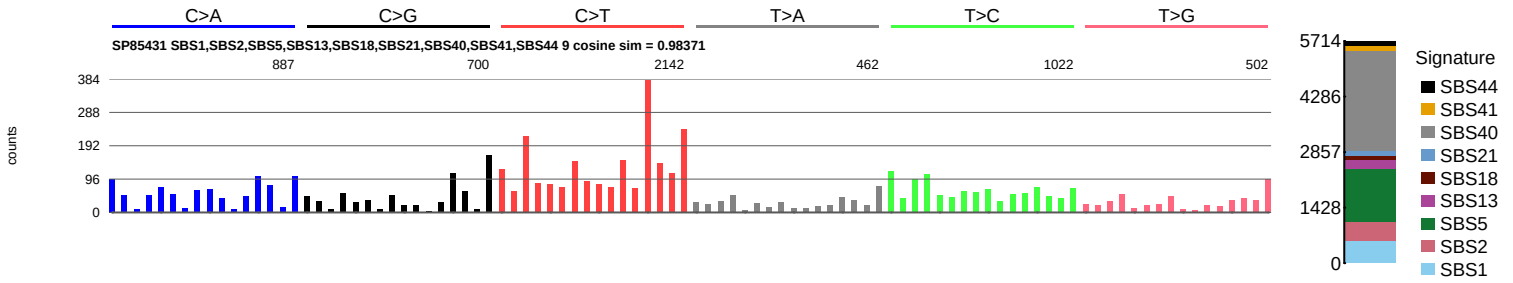
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98372
833
730
2132
484
1023
508
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98371
887
700
2142
462
1022
502
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
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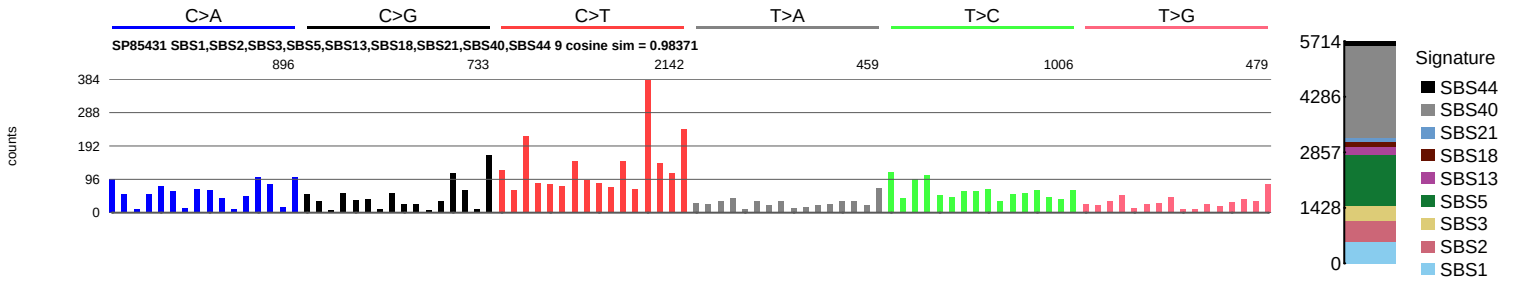
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98371
896
733
2142
459
1006
479
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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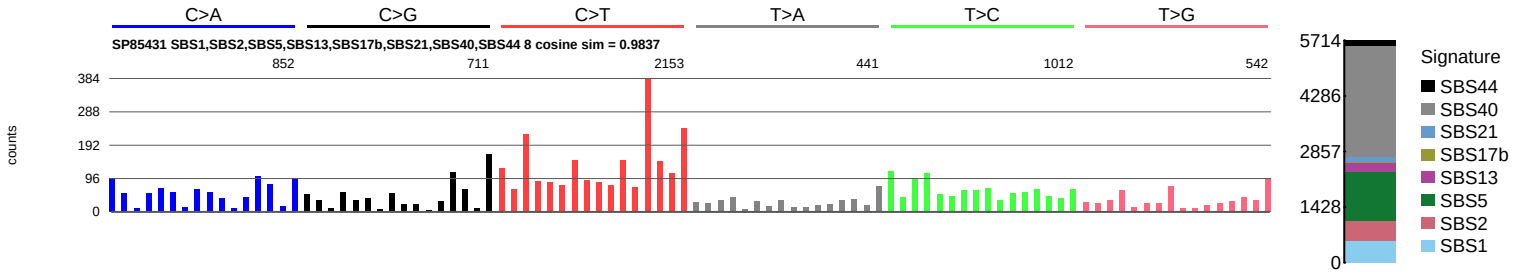
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.9837
852
711
2153
441
1012
542
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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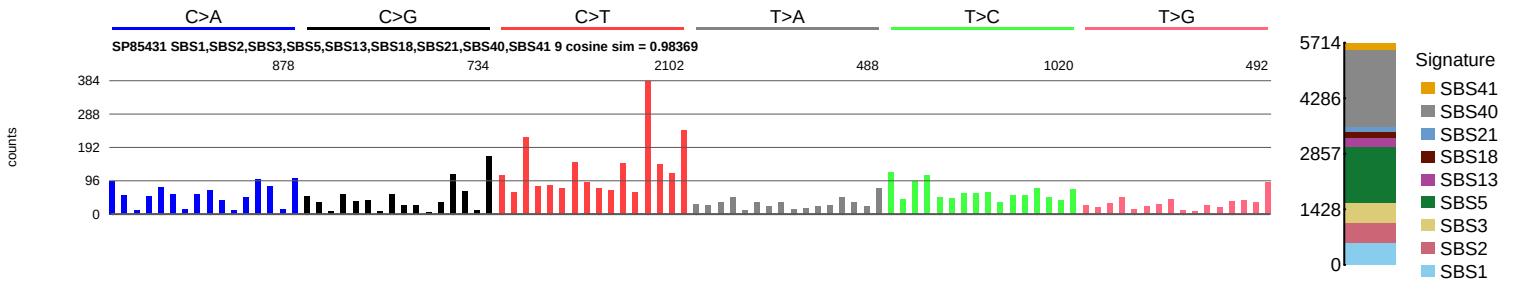
SBS2
SBS1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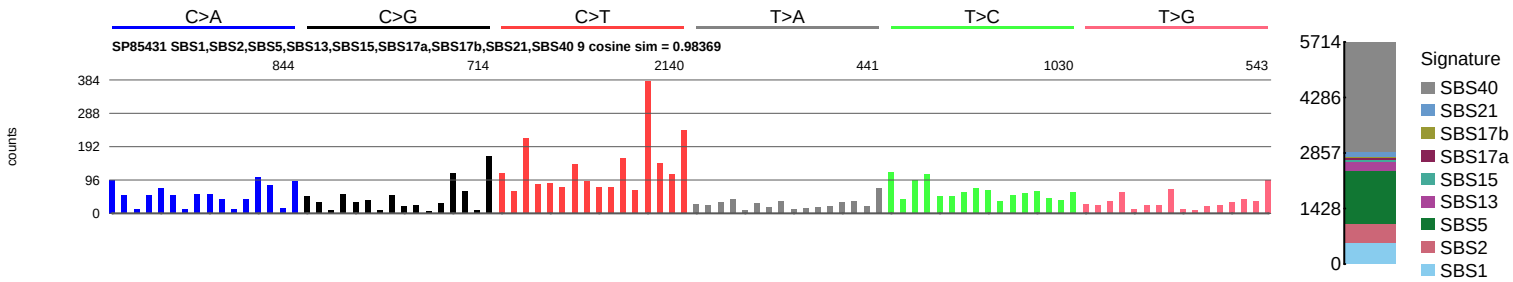
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98369
878
734
2102
488
1020
492
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 9 cosine sim = 0.98369
844
714
2140
441
1030
543
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
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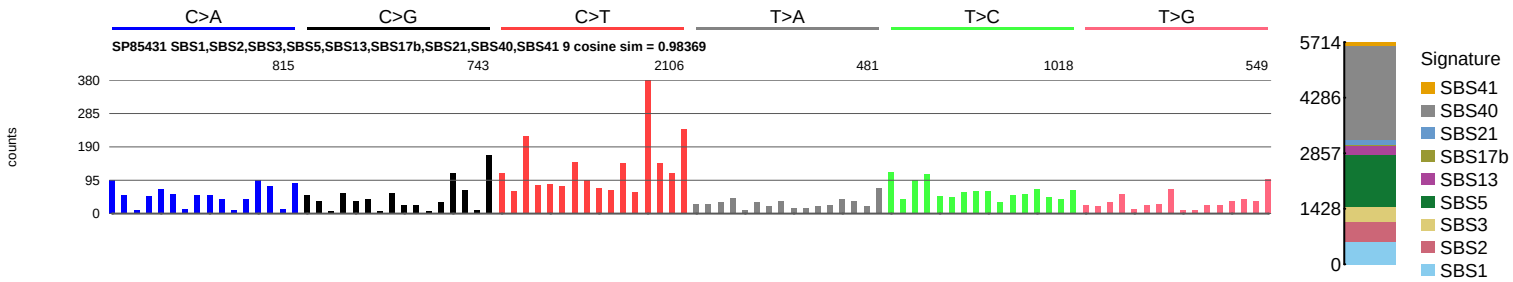
SBS5
SBS2
SBS1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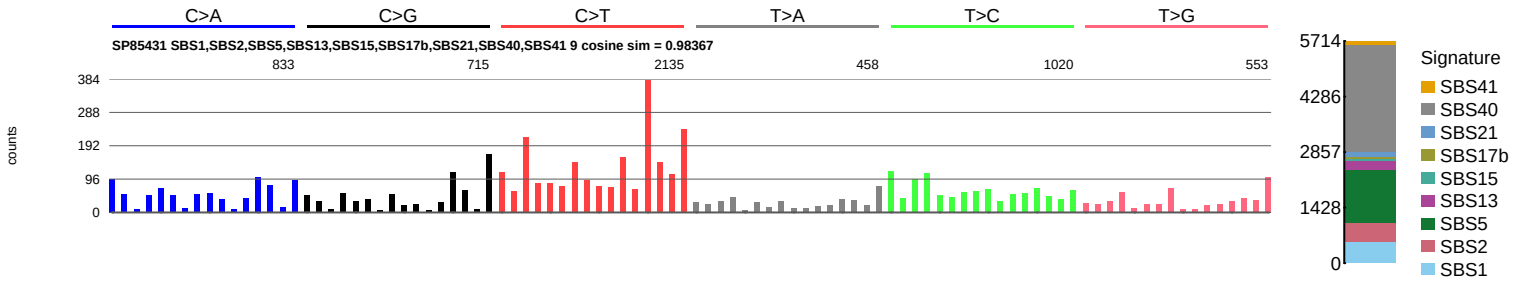
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98369
815
743
2106
481
1018
549
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98367
833
715
2135
458
1020
553
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
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SBS17b
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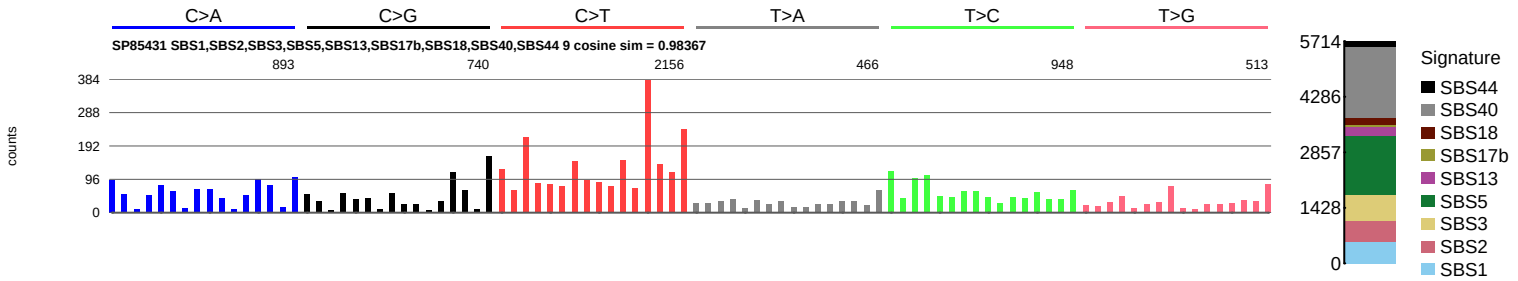
SBS5
SBS2
SBS1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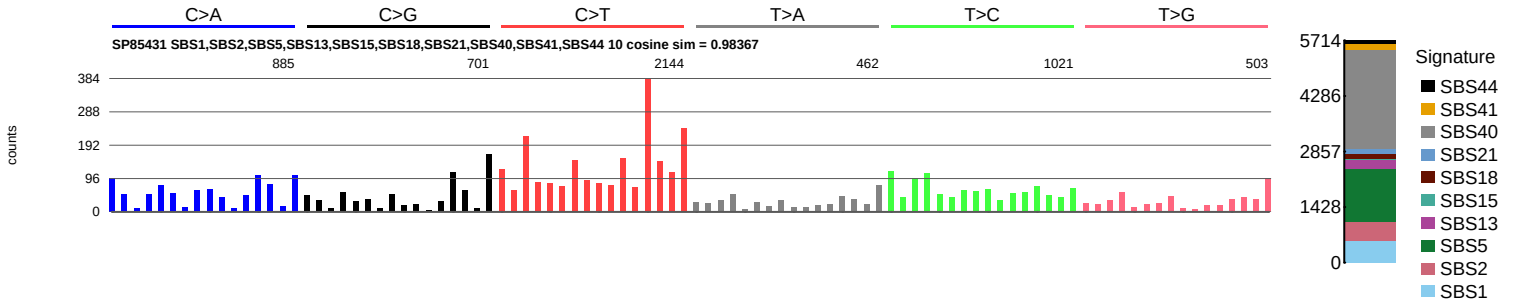
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98367
893
740
2156
466
948
513
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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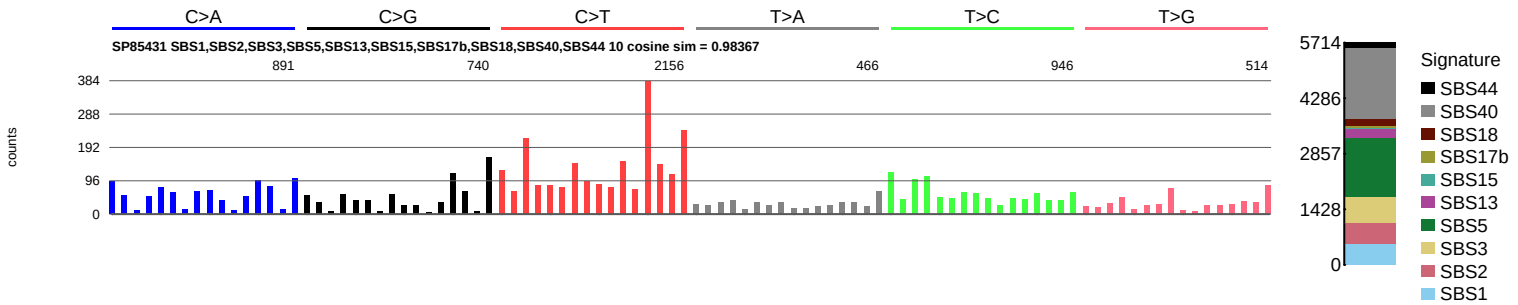
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98367
885
701
2144
462
1021
503
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1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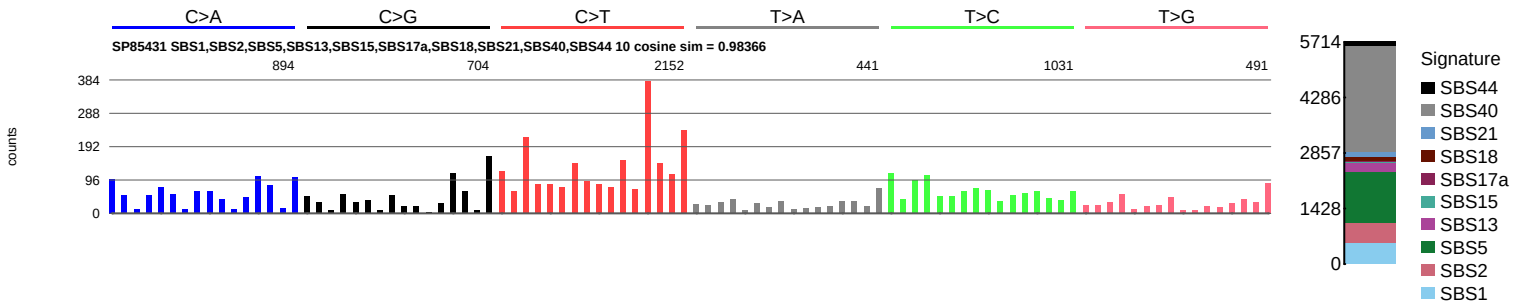
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98367
891
740
2156
466
946
514
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98366
894
704
2152
441
1031
491
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
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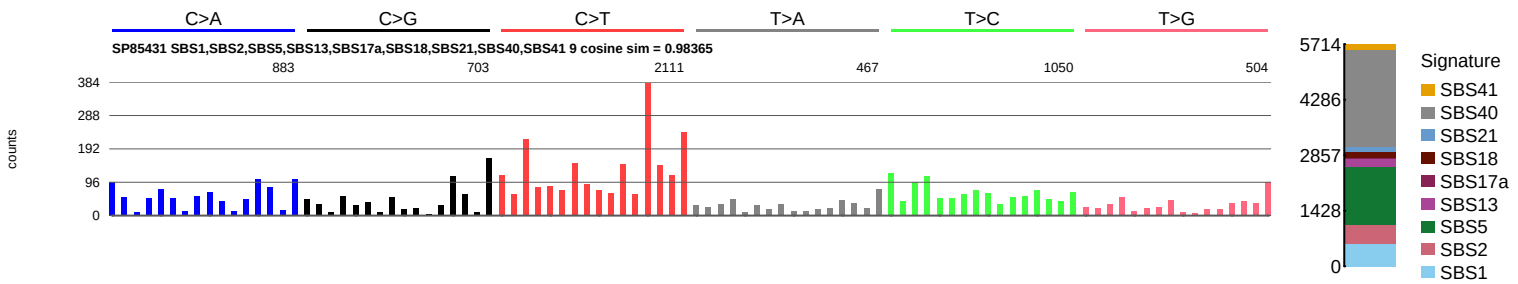
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98365
883
703
2111
467
1050
504
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
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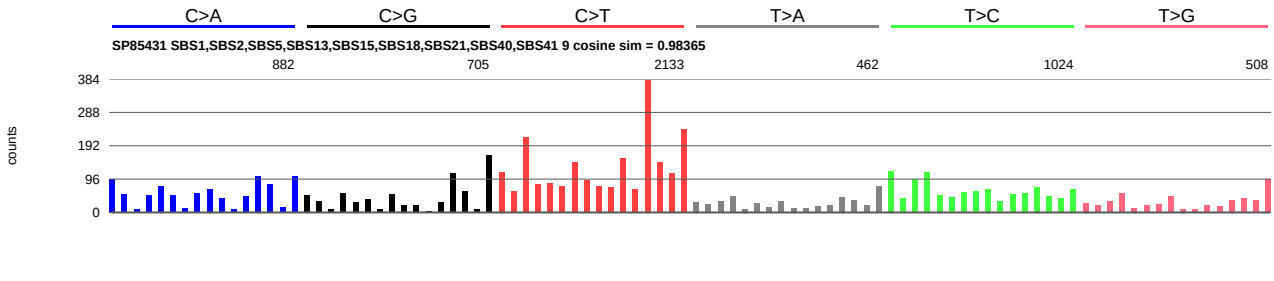
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98365
882
705
2133
462
1024
508
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
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SBS21
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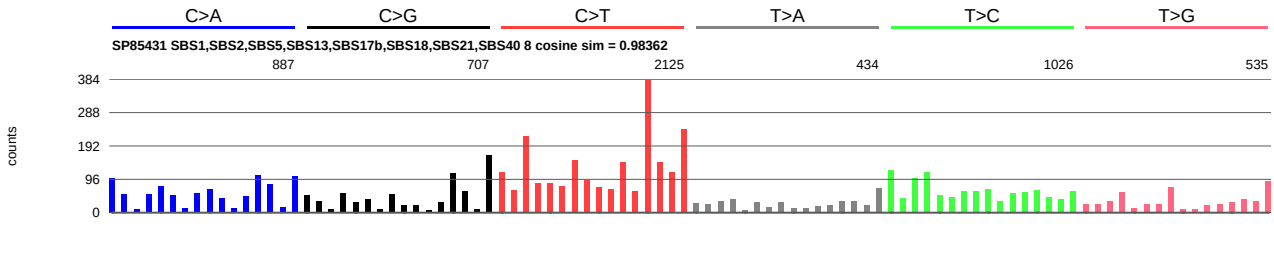
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98362
887
707
2125
434
1026
535
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS18
SBS17b
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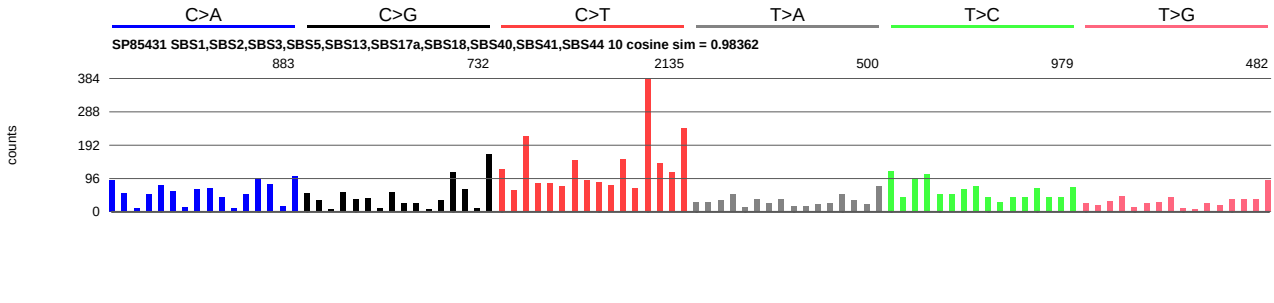
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98362
883
732
2135
500
979
482
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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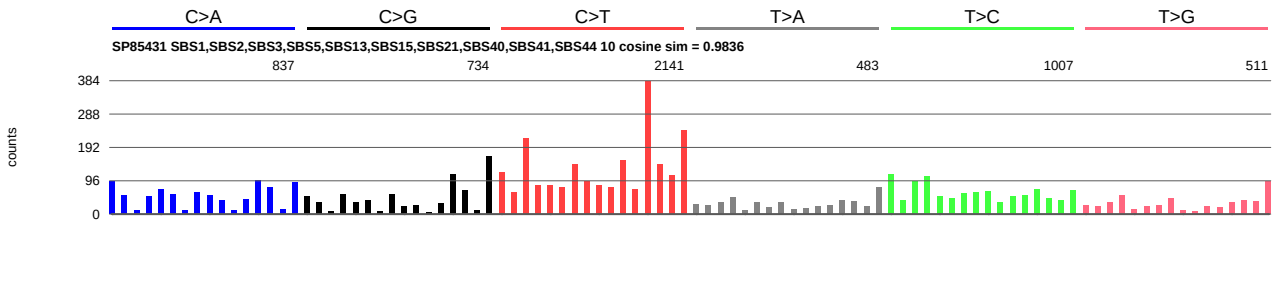
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.9836
837
734
2141
483
1007
511
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
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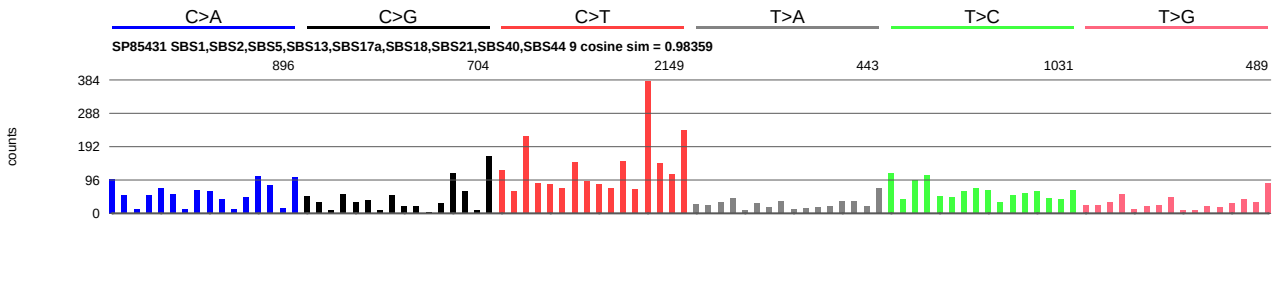
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98359
896
704
2149
443
1031
489
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS13
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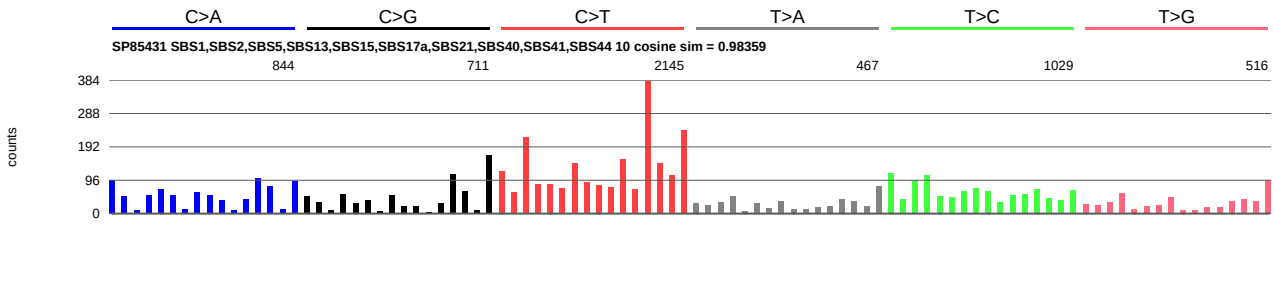
SBS2
SBS1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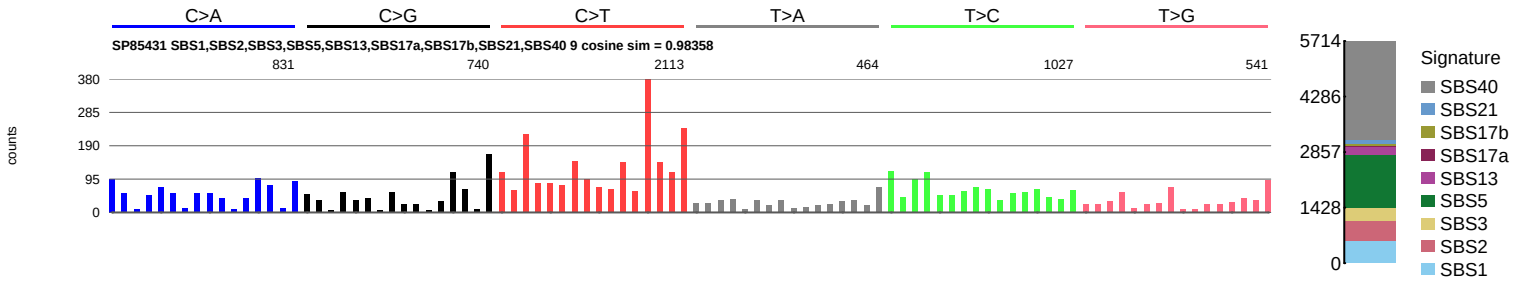
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98359
844
711
2145
467
1029
516
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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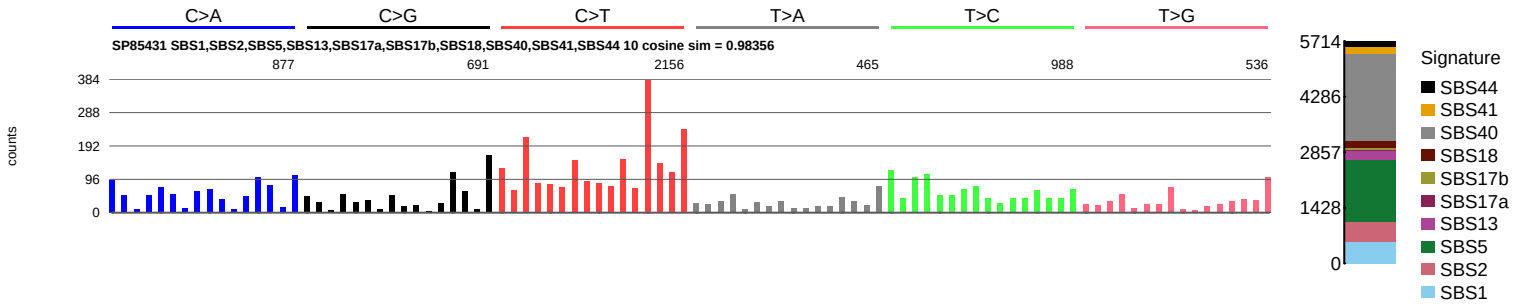
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40 9 cosine sim = 0.98358
831
740
2113
464
1027
541
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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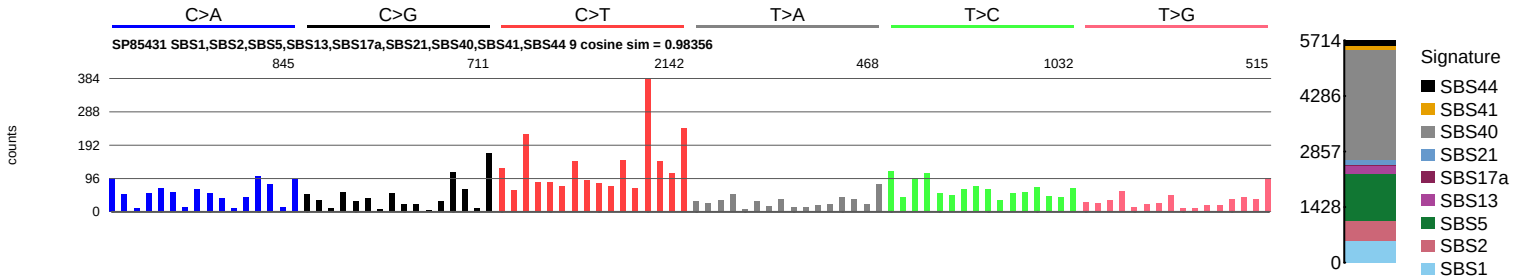
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98356
877
691
2156
465
988
536
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98356
845
711
2142
468
1032
515
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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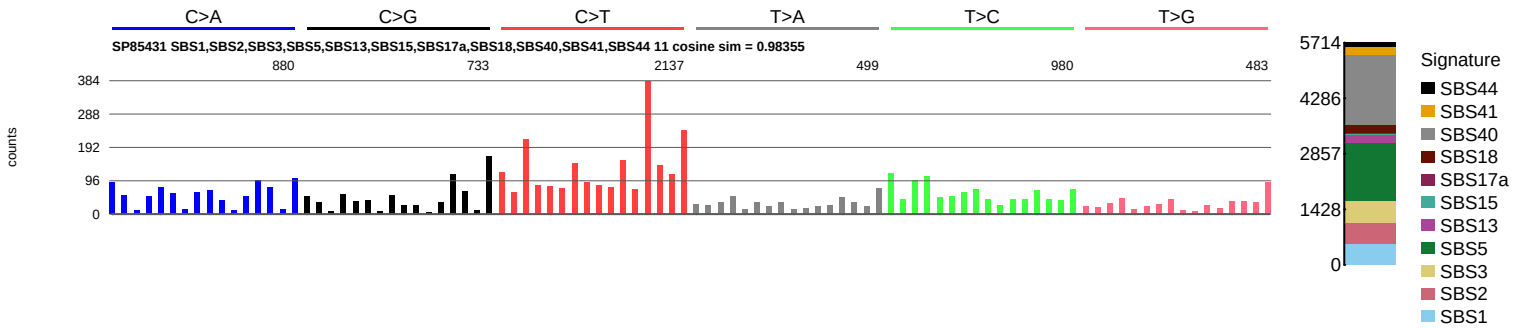
SBS2
SBS1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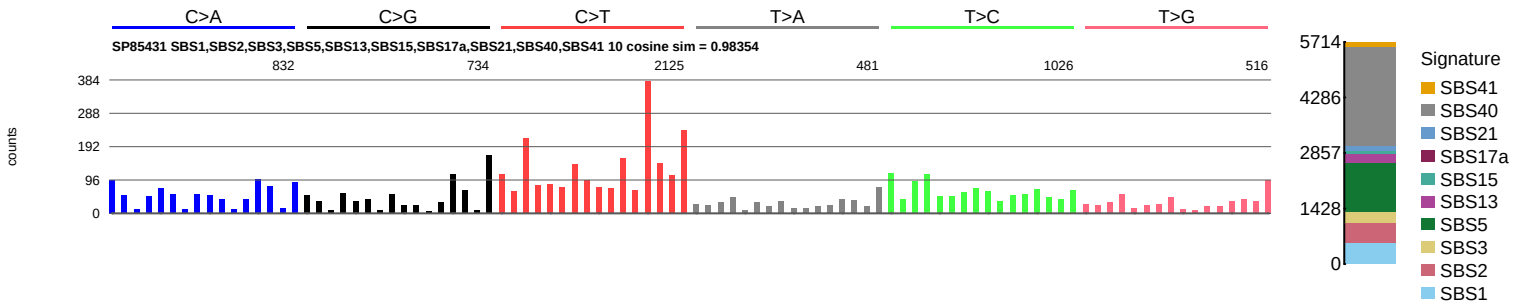
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98355
880
733
2137
499
980
483
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS21,SBS40,SBS41 10 cosine sim = 0.98354
832
734
2125
481
1026
516
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS21
SBS17a
SBS15
SBS13
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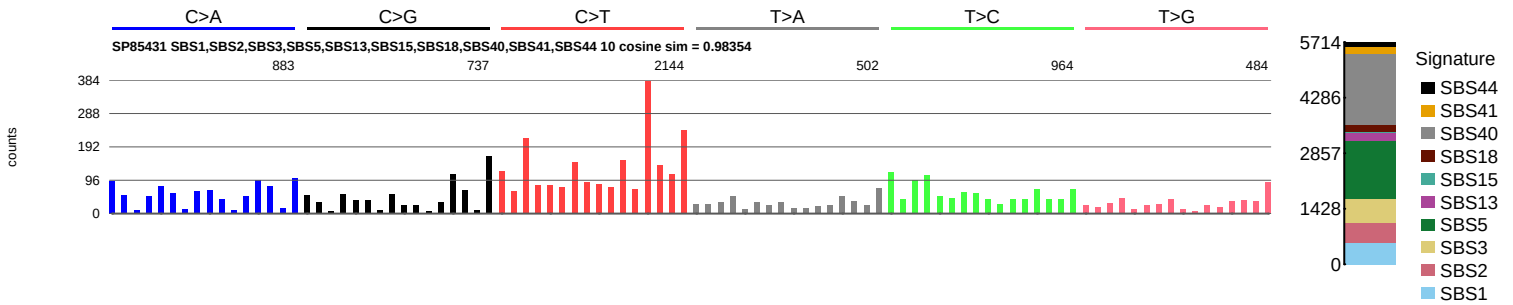
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98354
883
737
2144
502
964
484
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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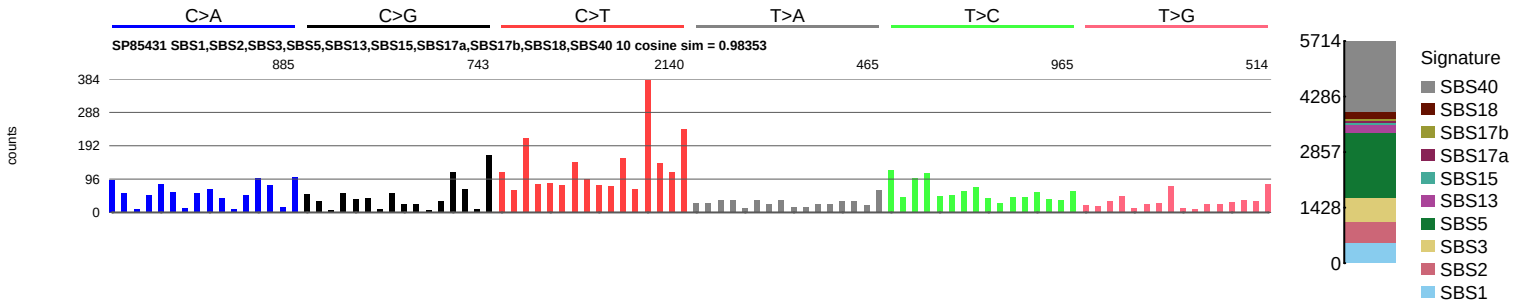
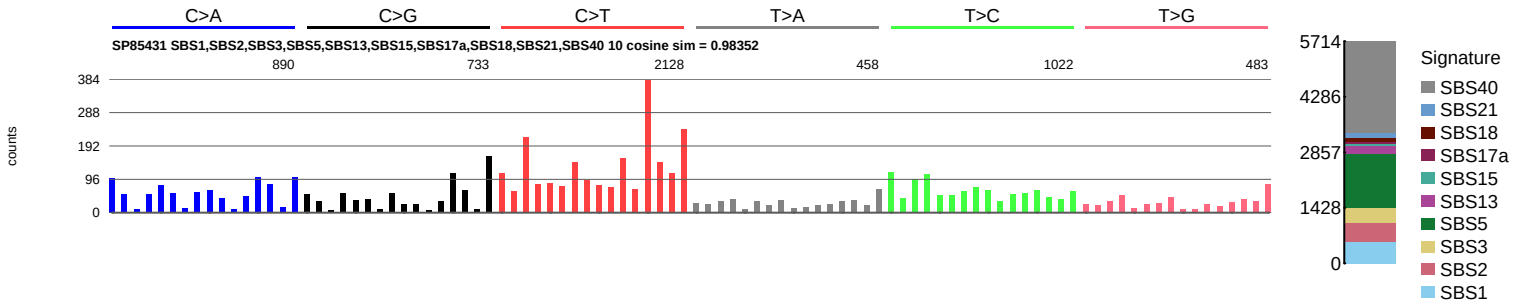
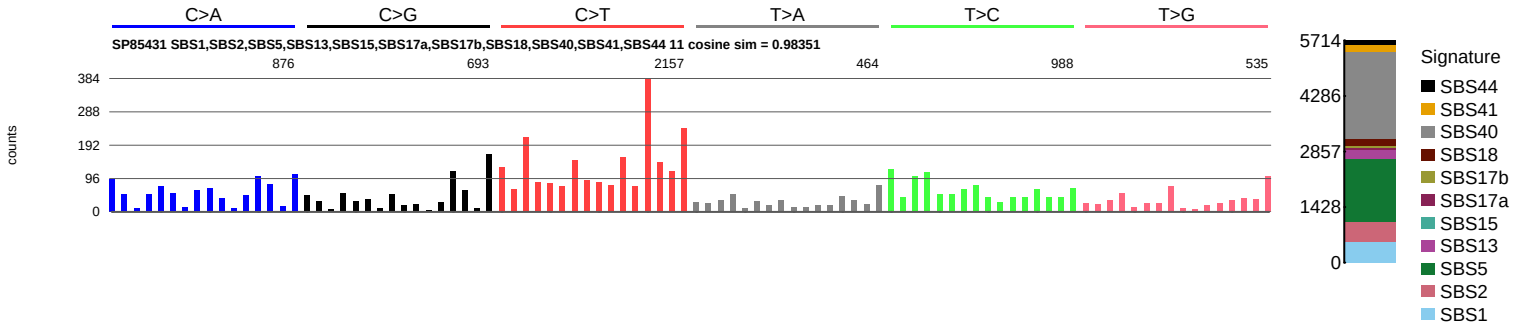
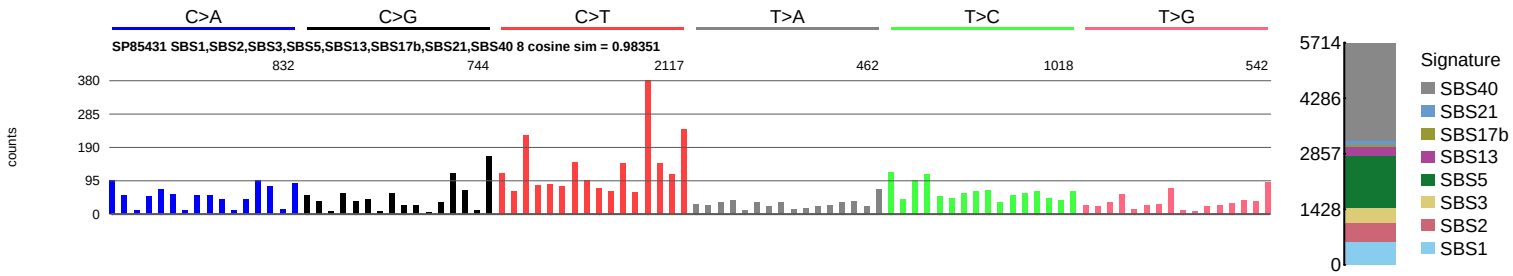
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 10 cosine sim = 0.98353
885
743
2140
465
965
514
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40 10 cosine sim = 0.98352
890
733
2128
458
1022
483
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98351
876
693
2157
464
988
535
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS21,SBS40 8 cosine sim = 0.98351
832
744
2117
462
1018
542
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
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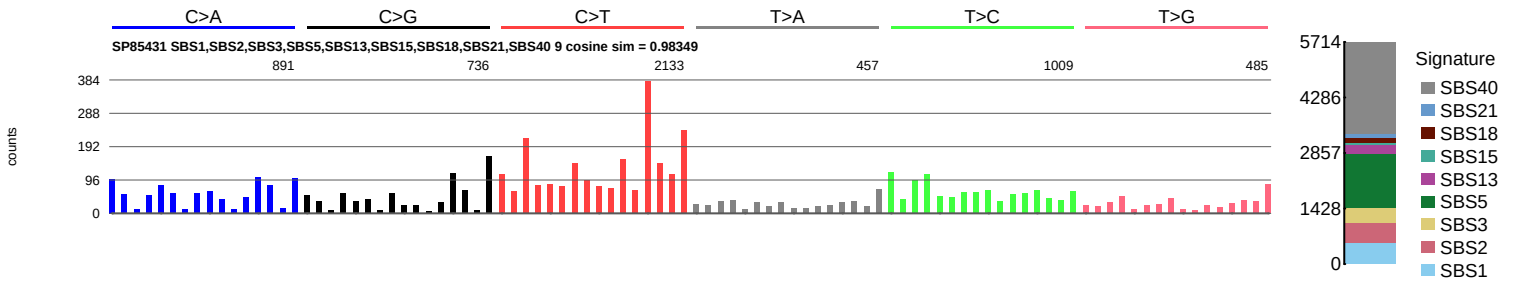
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40 9 cosine sim = 0.98349
891
736
2133
457
1009
485
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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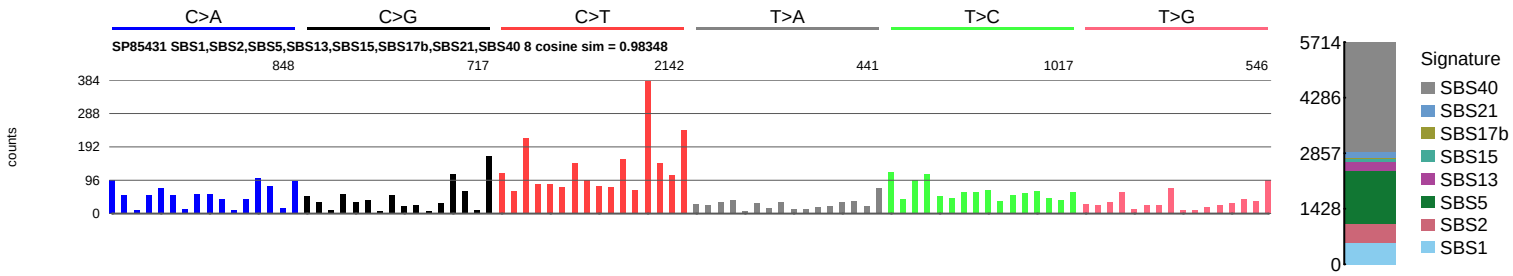
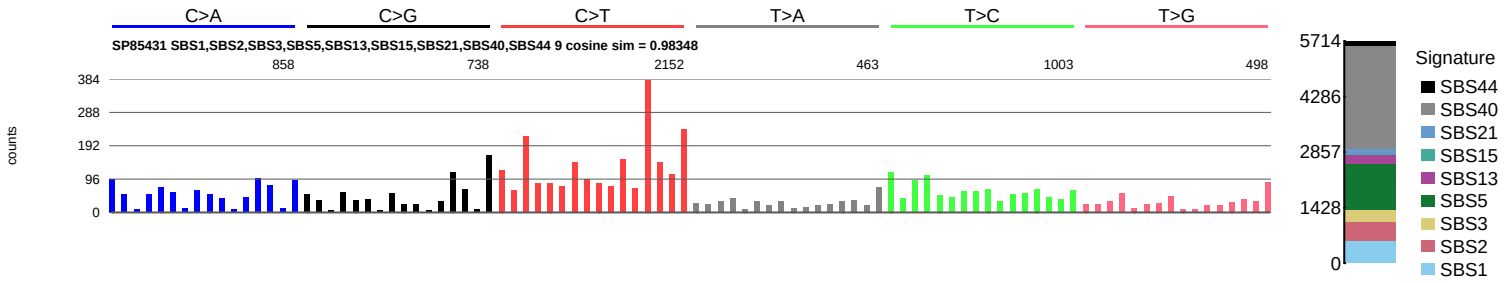
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS21,SBS40 8 cosine sim = 0.98348
848
717
2142
441
1017
546
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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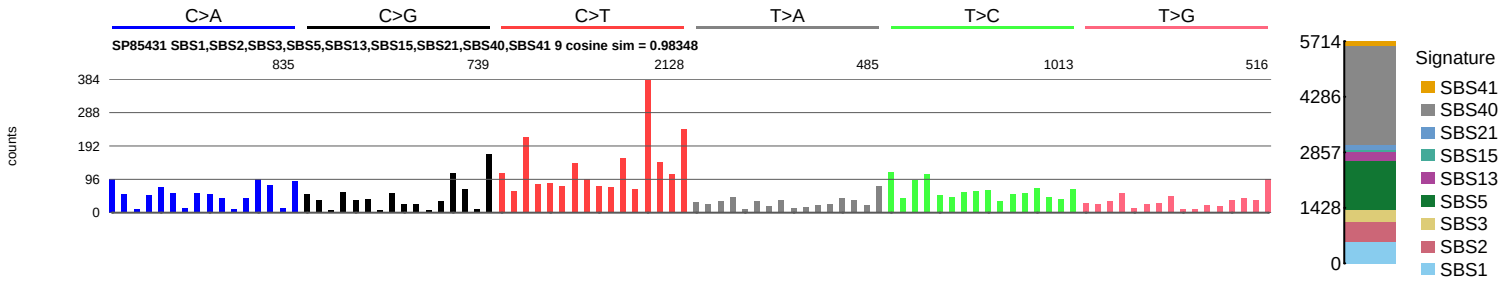
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS21,SBS40,SBS44 9 cosine sim = 0.98348
858
738
2152
463
1003
498
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS21
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS21,SBS40,SBS41 9 cosine sim = 0.98348
835
739
2128
485
1013
516
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
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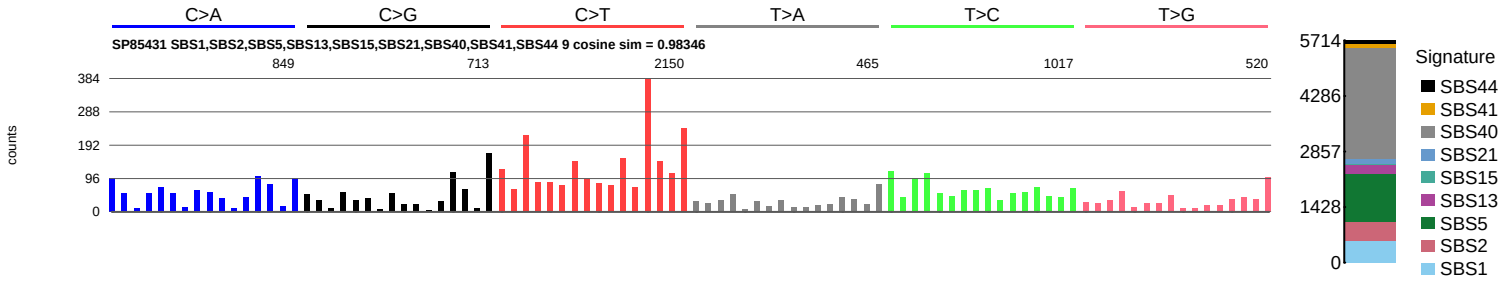
SBS2
SBS1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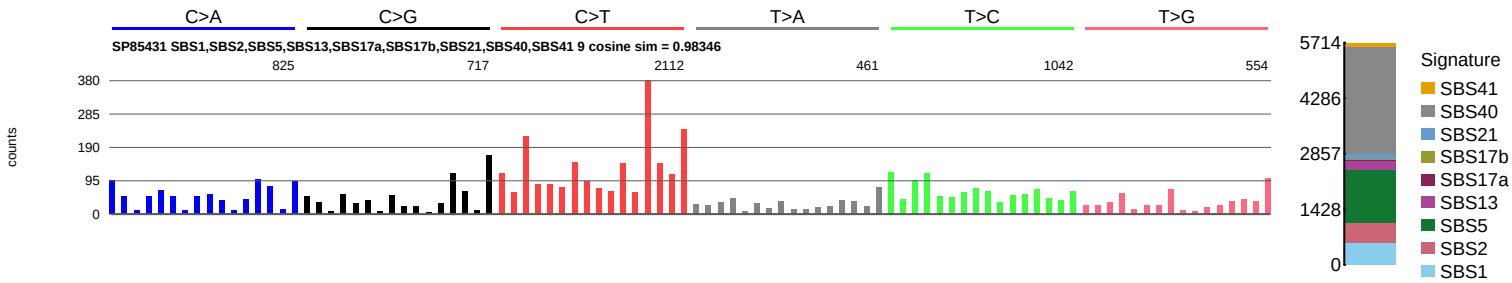
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98346
849
713
2150
465
1017
520
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98346
825
717
2112
461
1042
554
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
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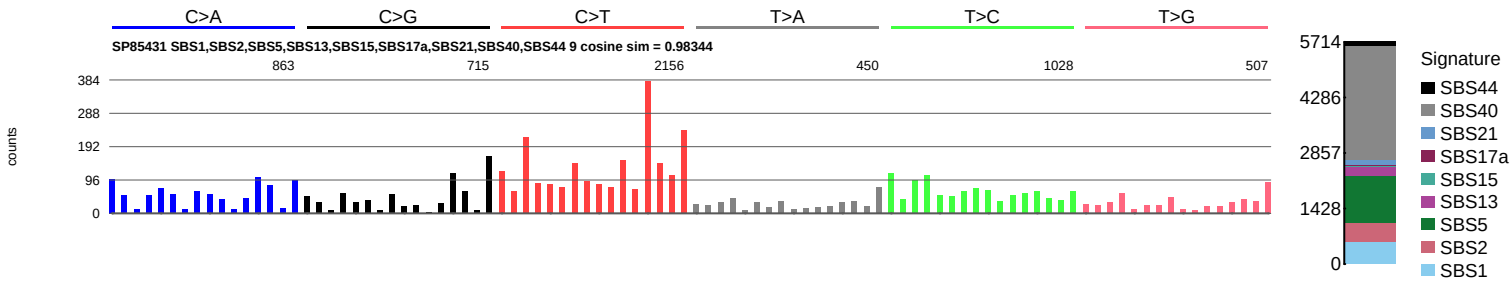
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS21,SBS40,SBS44 9 cosine sim = 0.98344
863
715
2156
450
1028
507
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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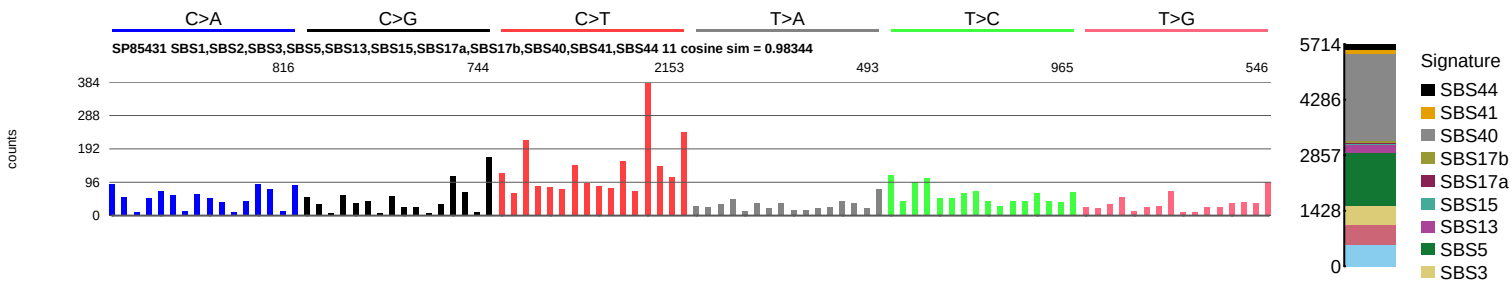
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 11 cosine sim = 0.98344
816
744
2153
493
965
546
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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SBS40
SBS17b
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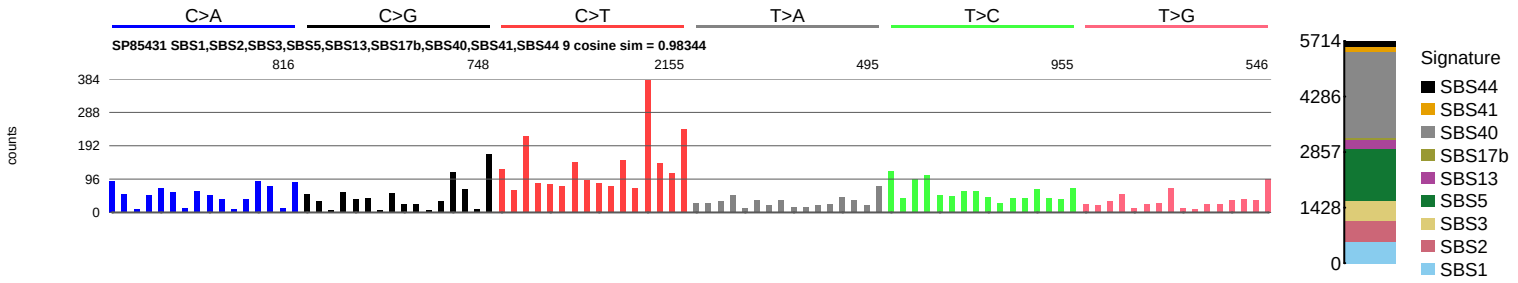
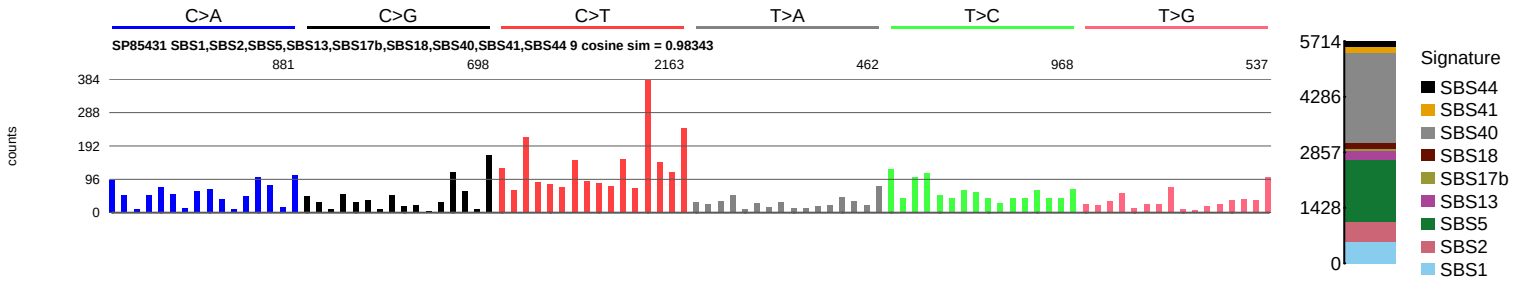
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98344
816
748
2155
495
955
546
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98343
881
698
2163
462
968
537
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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SBS40
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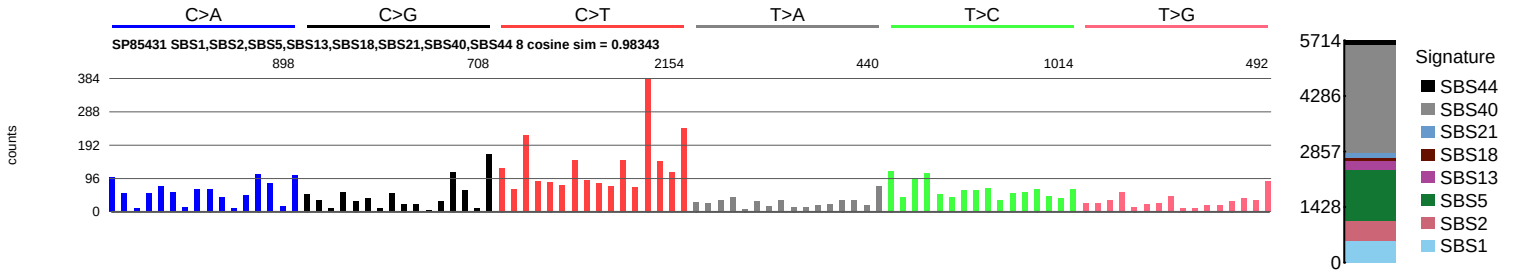
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98343
898
708
2154
440
1014
492
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
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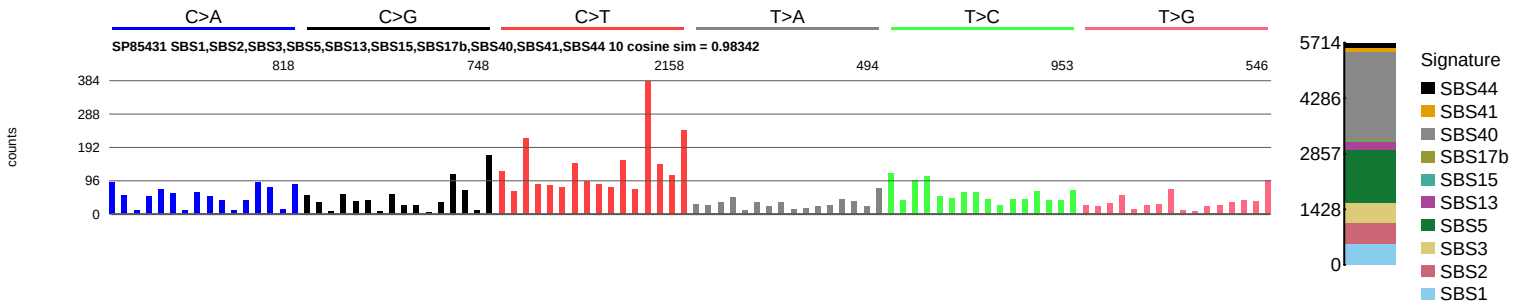
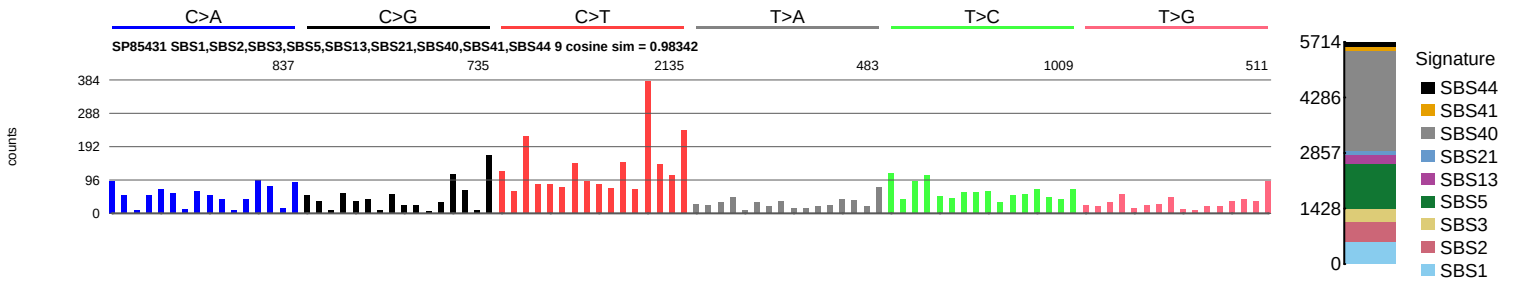
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS40,SBS41,SBS44 10 cosine sim = 0.98342
818
748
2158
494
953
546
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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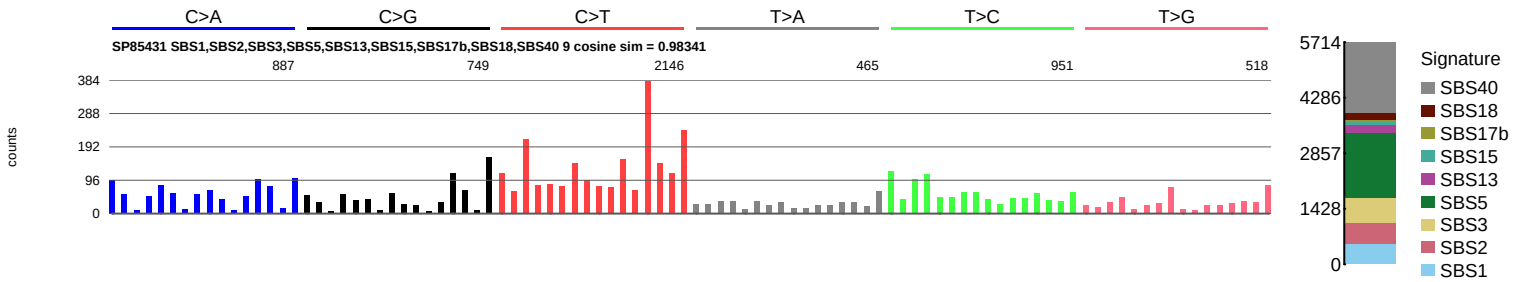
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98342
837
735
2135
483
1009
511
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS13
SBS5
SBS3
SBS2
SBS1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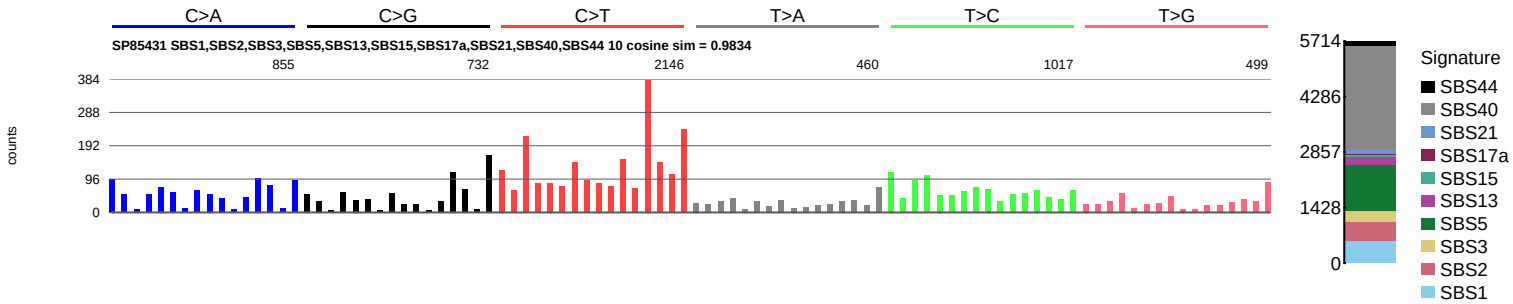
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 9 cosine sim = 0.98341
887
749
2146
465
951
518
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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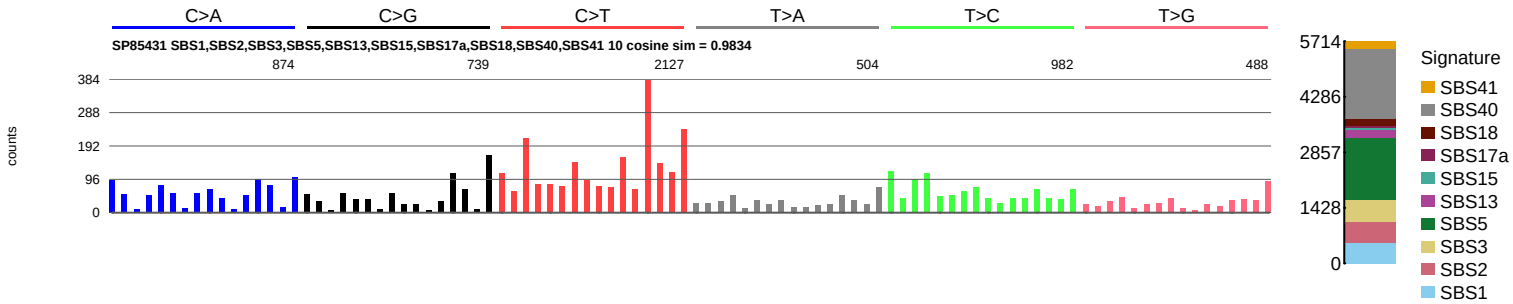
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS21,SBS40,SBS44 10 cosine sim = 0.9834
855
732
2146
460
1017
499
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS21
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>T
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T>C
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SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41 10 cosine sim = 0.9834
874
739
2127
504
982
488
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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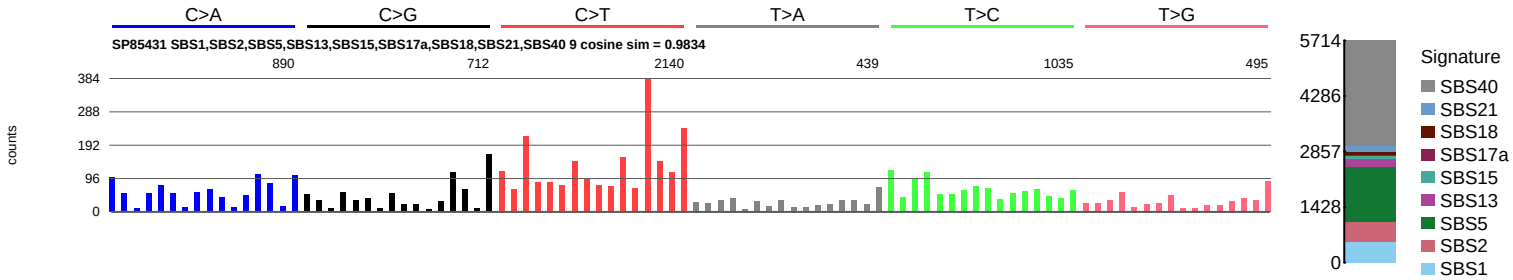
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40 9 cosine sim = 0.9834
890
712
2140
439
1035
495
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
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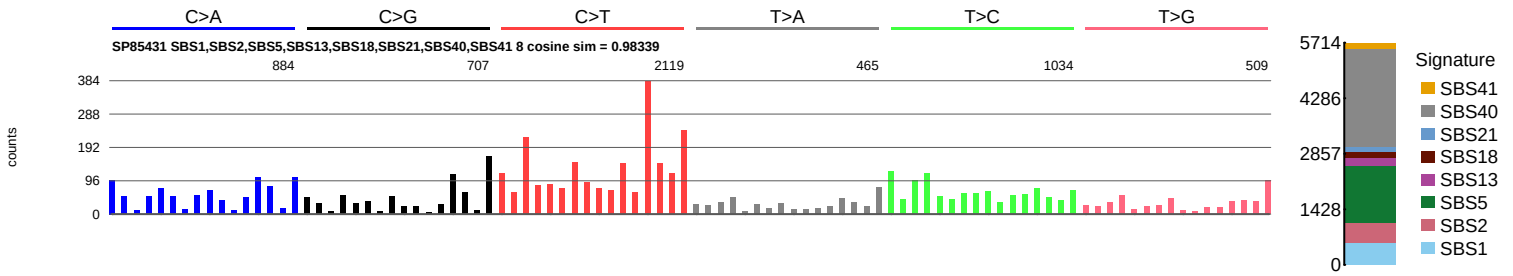
SBS2
SBS1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C>G
C>T
T>A
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T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98339
884
707
2119
465
1034
509
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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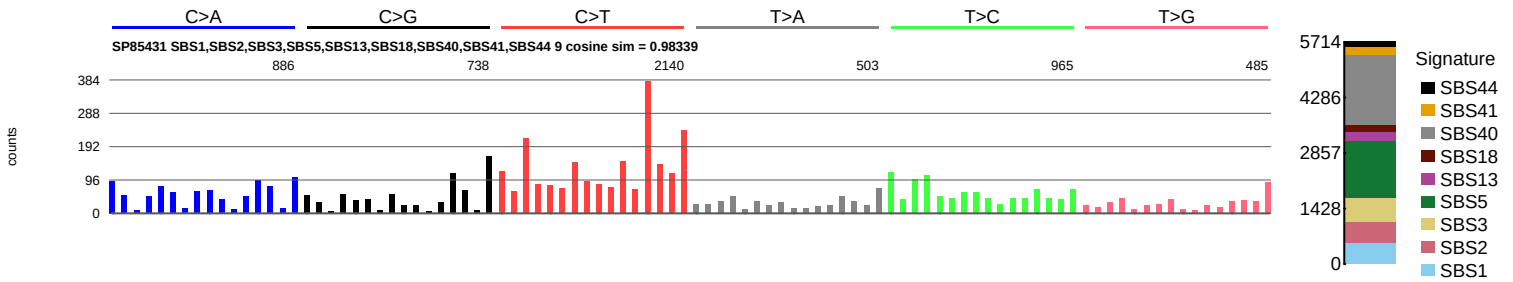
SBS2
SBS1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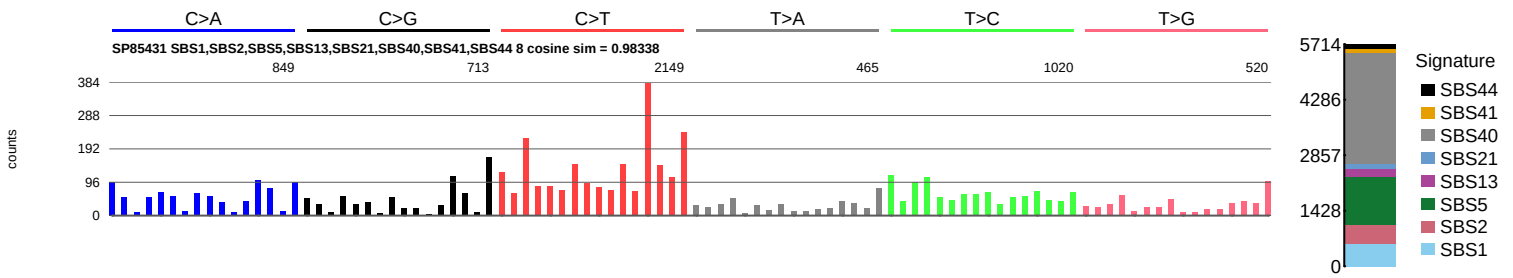
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98339
886
738
2140
503
965
485
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98338
849
713
2149
465
1020
520
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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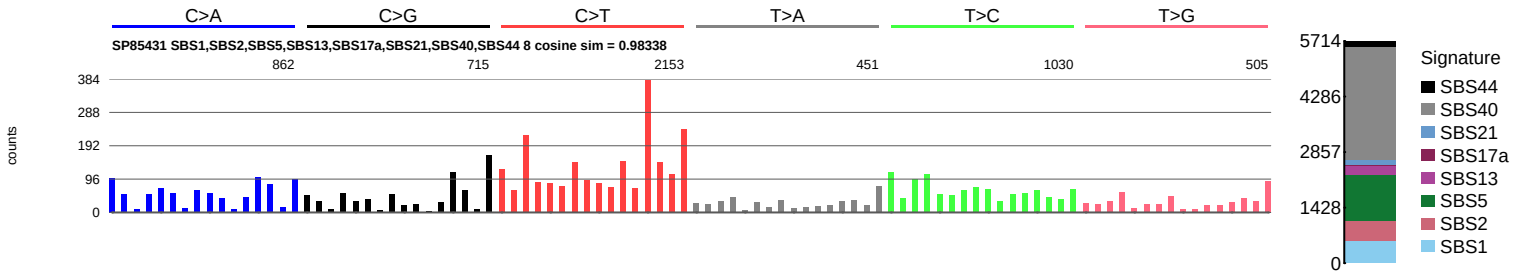
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS21,SBS40,SBS44 8 cosine sim = 0.98338
862
715
2153
451
1030
505
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS21
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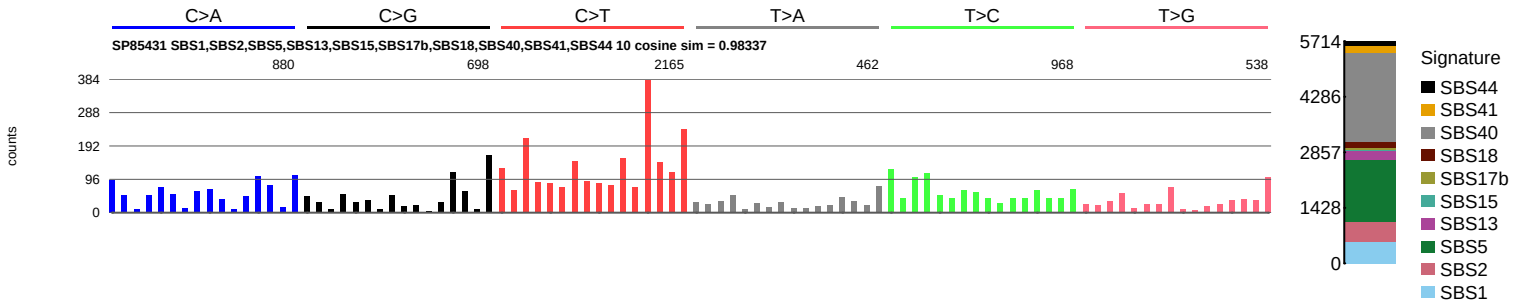
SBS2
SBS1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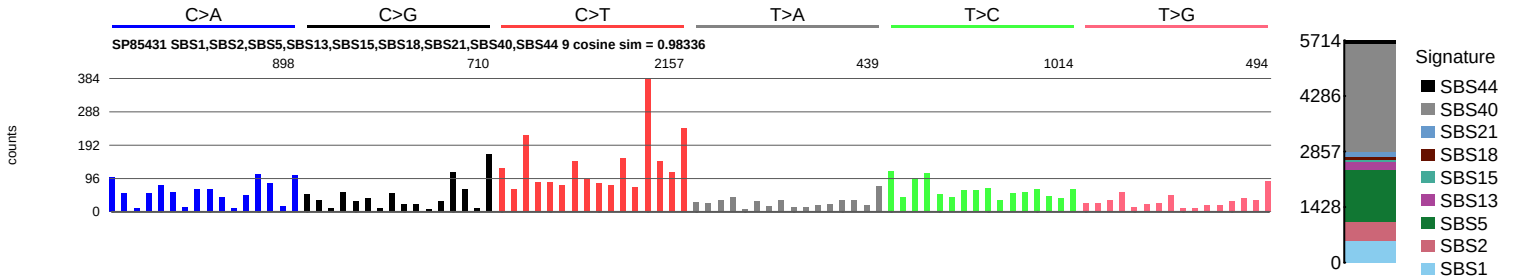
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98337
880
698
2165
462
968
538
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98336
898
710
2157
439
1014
494
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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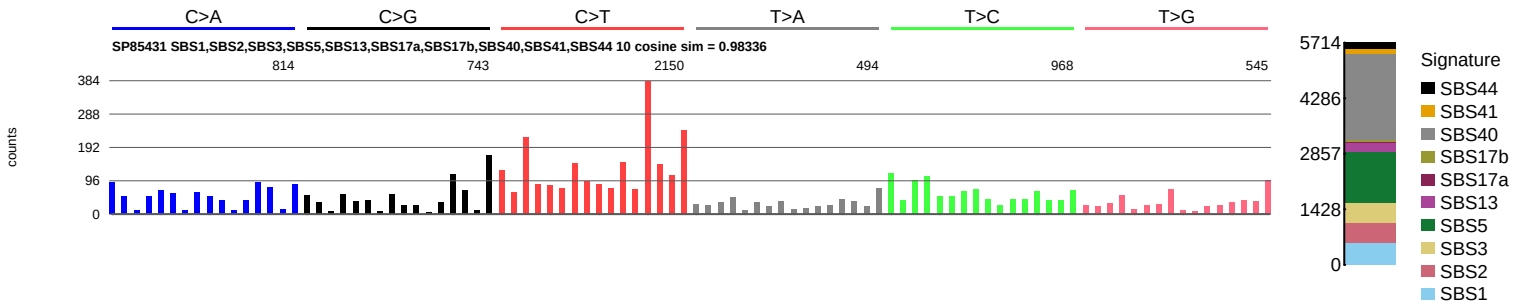
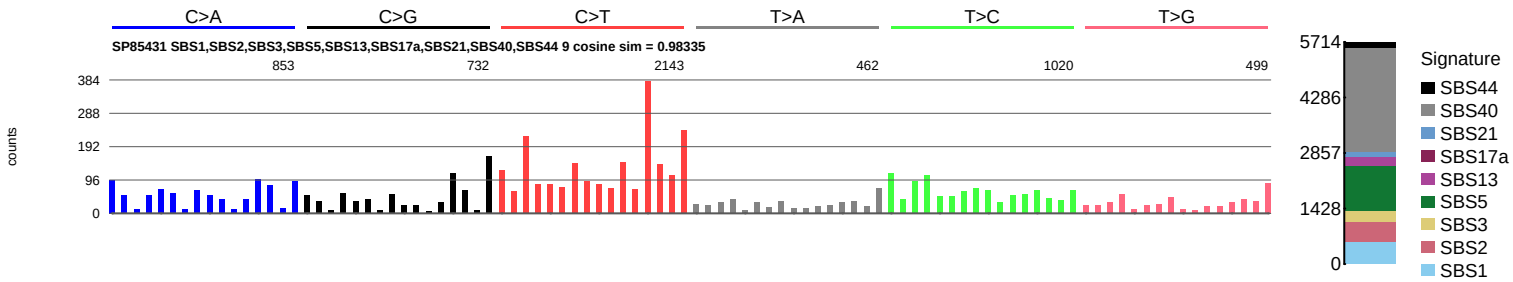
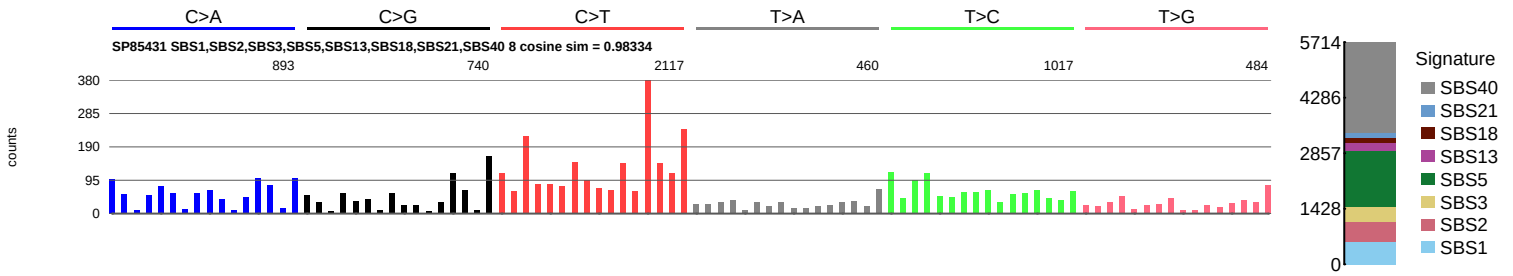
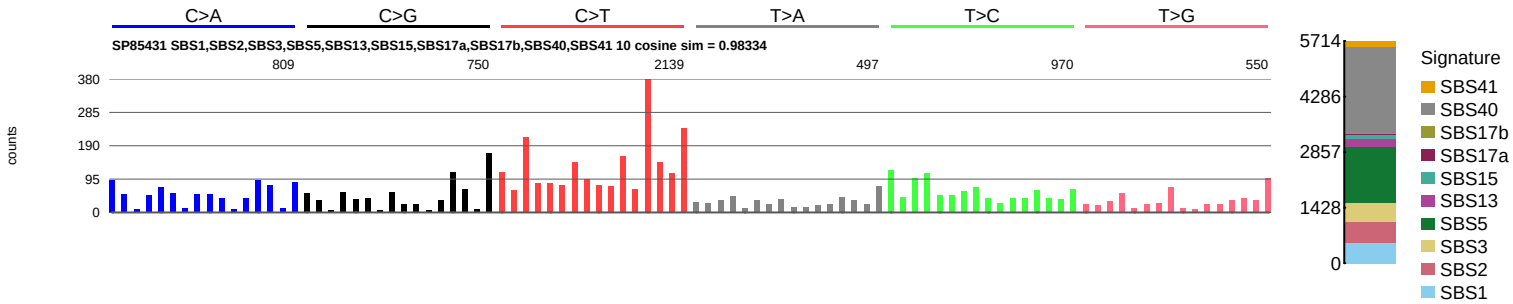
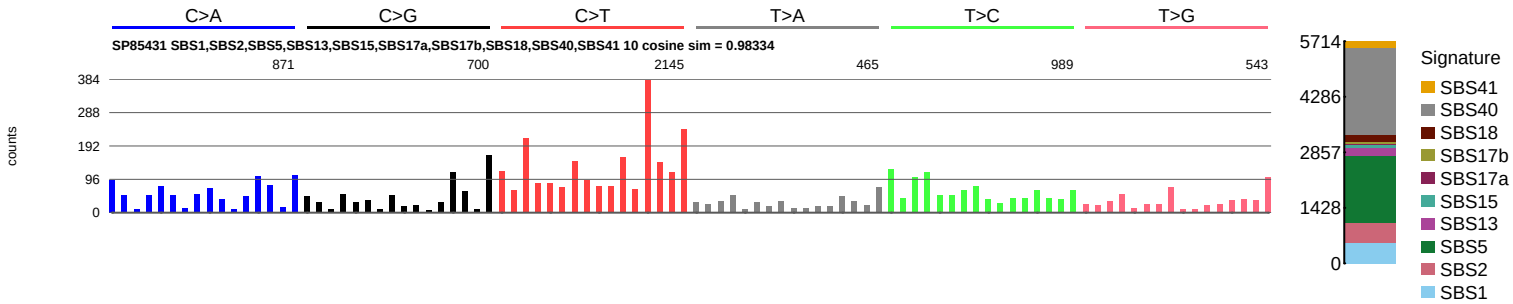
SBS2
SBS1

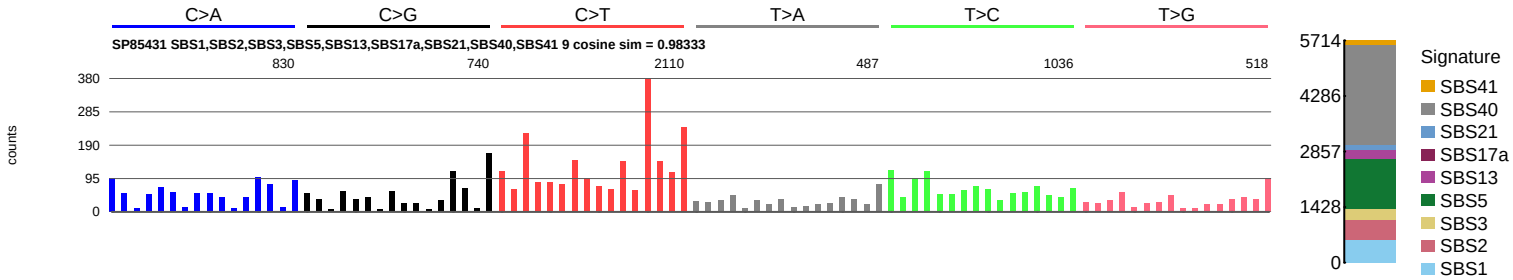

C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 10 cosine sim = 0.98336
814
743
2150
494
968
545
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS21,SBS40,SBS44 9 cosine sim = 0.98335
853
732
2143
462
1020
499
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS21
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS40 8 cosine sim = 0.98334
893
740
2117
460
1017
484
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 10 cosine sim = 0.98334
809
750
2139
497
970
550
380
285
190
95
0
counts
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98334
871
700
2145
465
989
543
384
288
192
96
0
counts
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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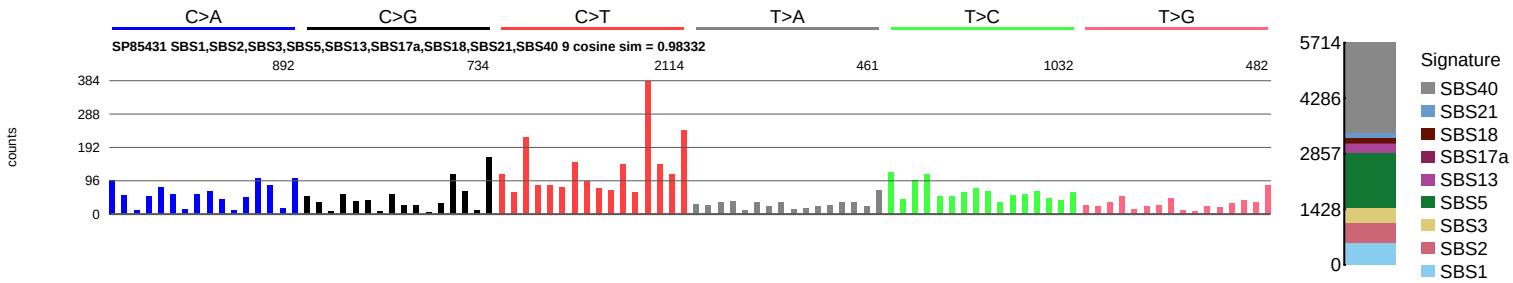
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS21,SBS40,SBS41 9 cosine sim = 0.98333
830
740
2110
487
1036
518
380
285
190
95
0
counts
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS21
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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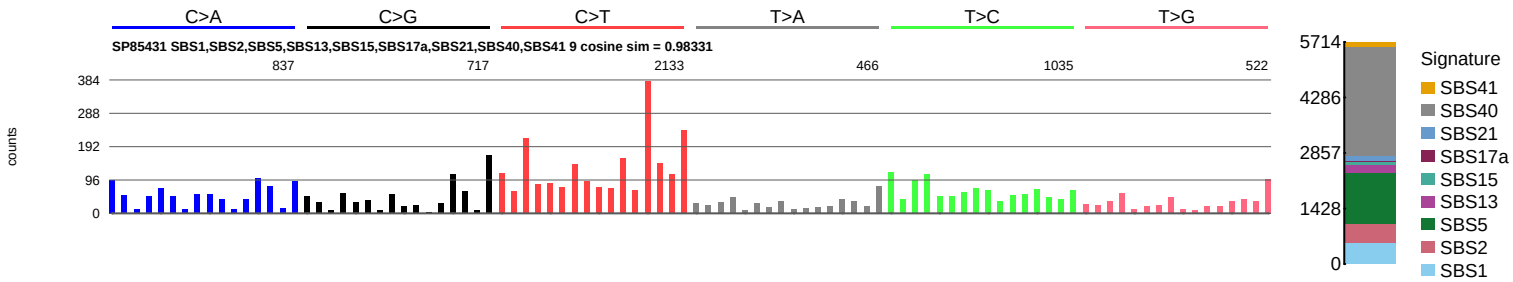
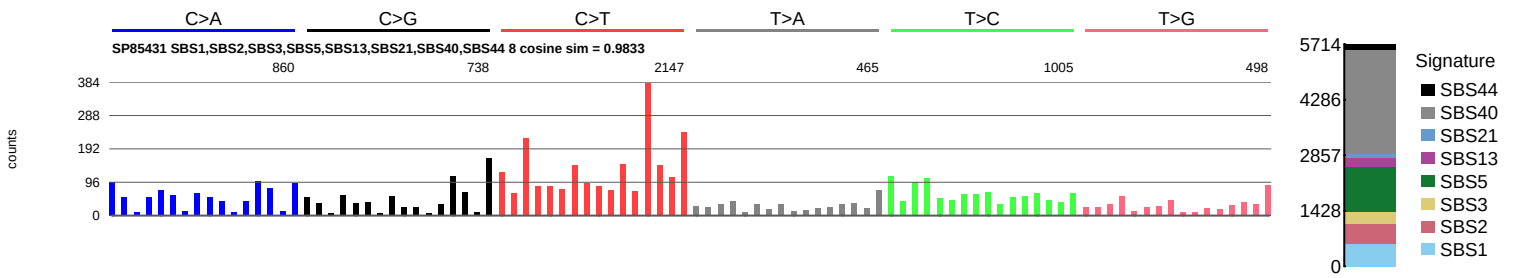
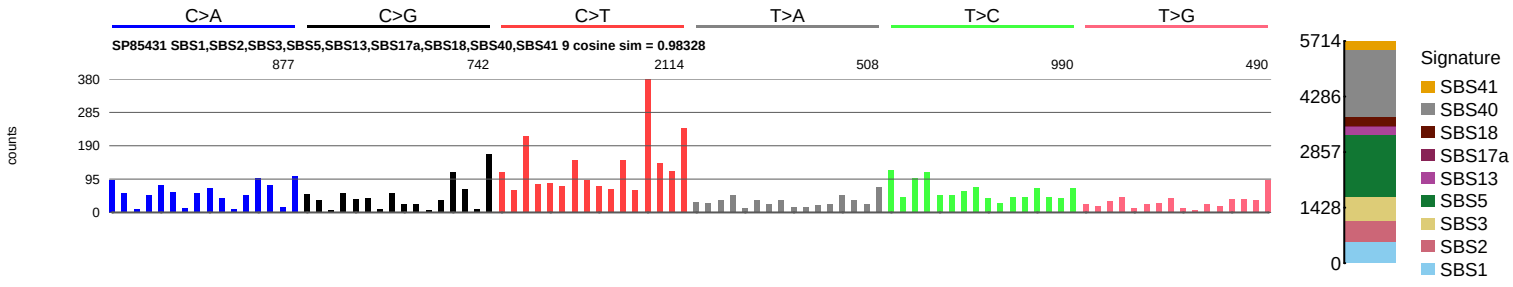
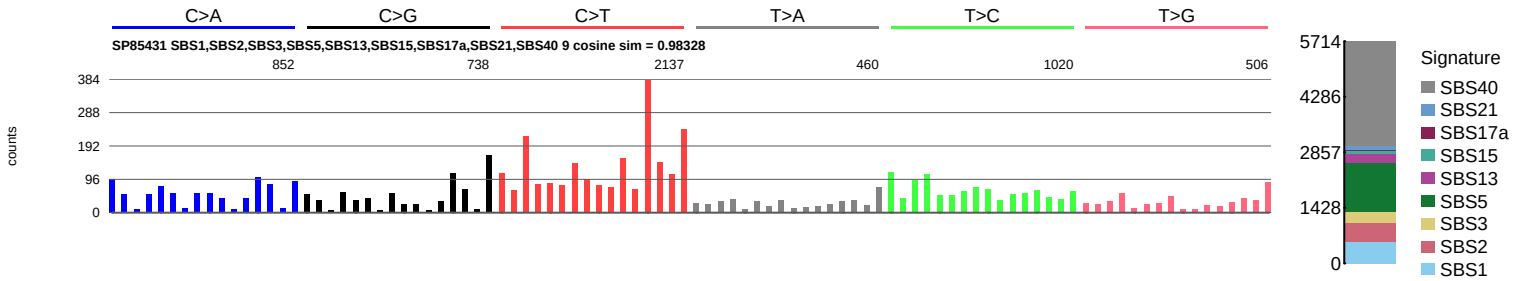
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40 9 cosine sim = 0.98332
892
734
2114
461
1032
482
384
288
192
96
0
counts
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS21,SBS40,SBS41 9 cosine sim = 0.98331
837
717
2133
466
1035
522
384
288
192
96
0
counts
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS21
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS21,SBS40,SBS44 8 cosine sim = 0.9833
860
738
2147
465
1005
498
384
288
192
96
0
counts
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS21
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41 9 cosine sim = 0.98328
877
742
2114
508
990
490
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS21,SBS40 9 cosine sim = 0.98328
852
738
2137
460
1020
506
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS17a
SBS15
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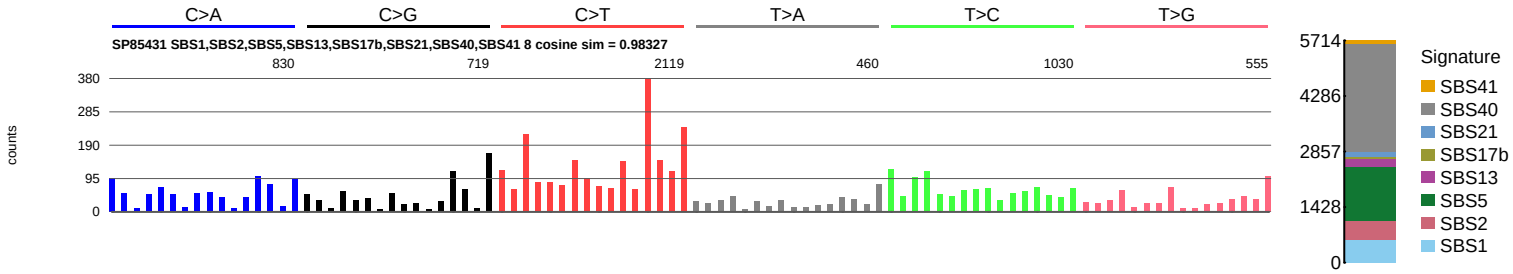
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98327
830
719
2119
460
1030
555
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS21
SBS17b
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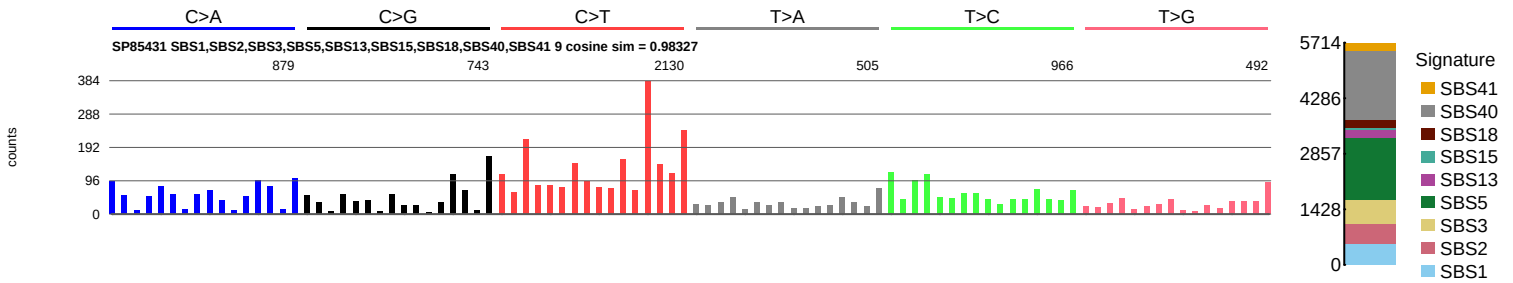
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS40,SBS41 9 cosine sim = 0.98327
879
743
2130
505
966
492
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
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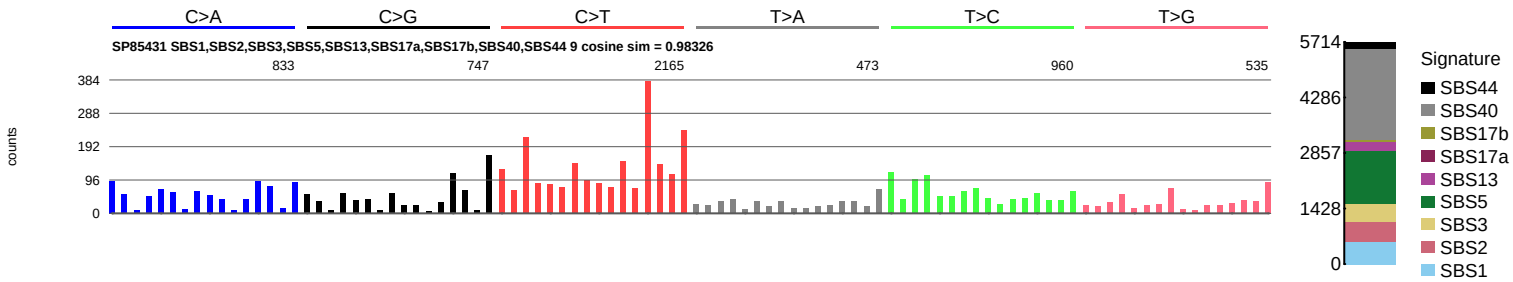
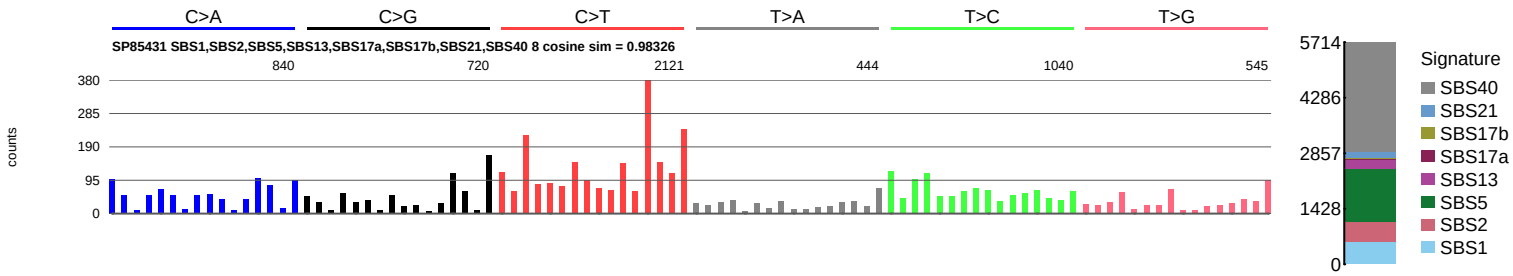
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 9 cosine sim = 0.98326
833
747
2165
473
960
535
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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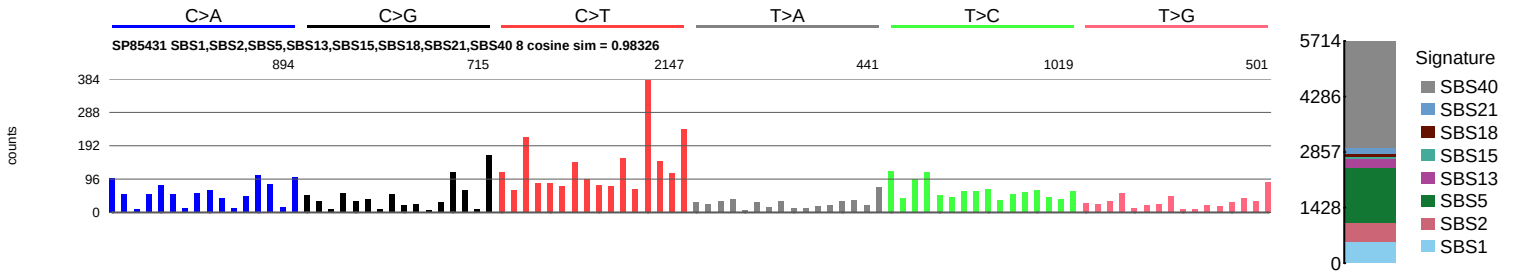
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98326
840
720
2121
444
1040
545
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40 8 cosine sim = 0.98326
894
715
2147
441
1019
501
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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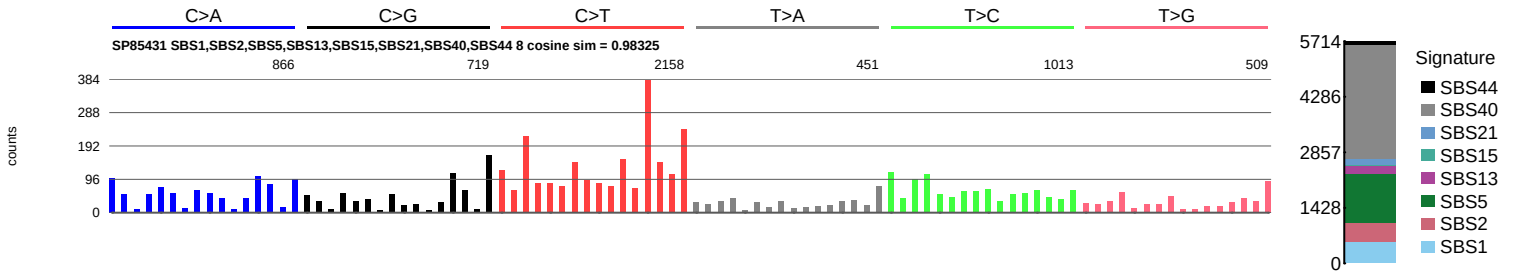
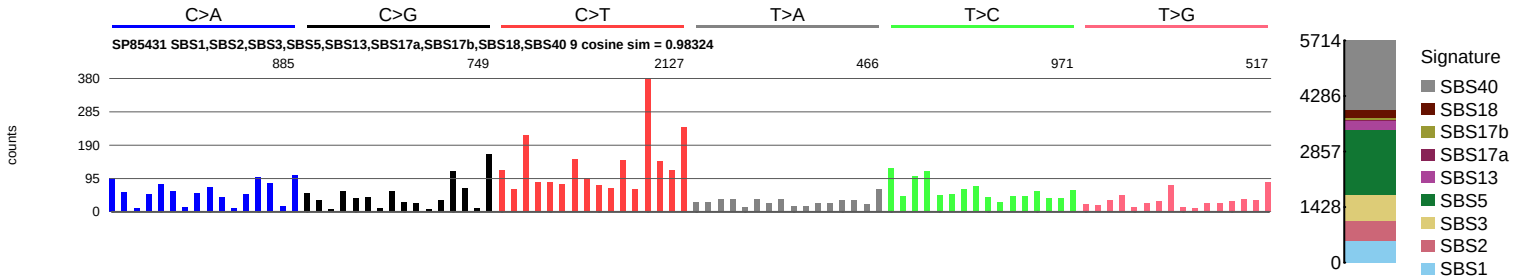
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS21,SBS40,SBS44 8 cosine sim = 0.98325
866
719
2158
451
1013
509
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS21
SBS15
SBS13
SBS5
SBS2
SBS1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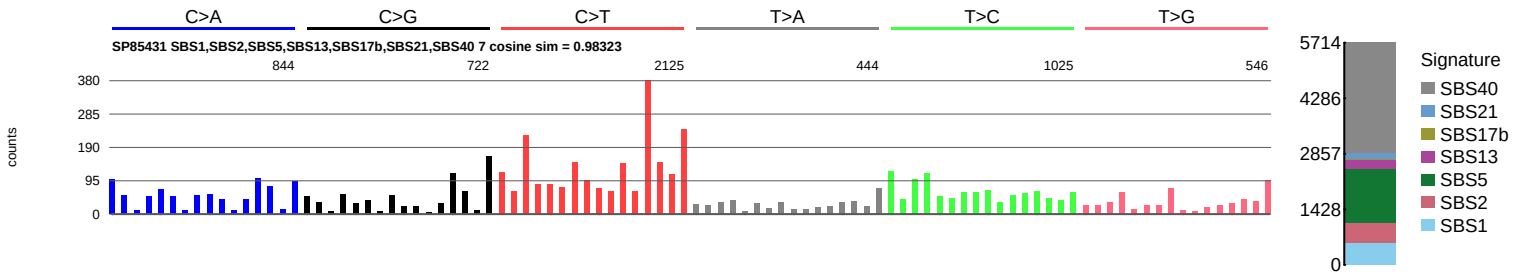
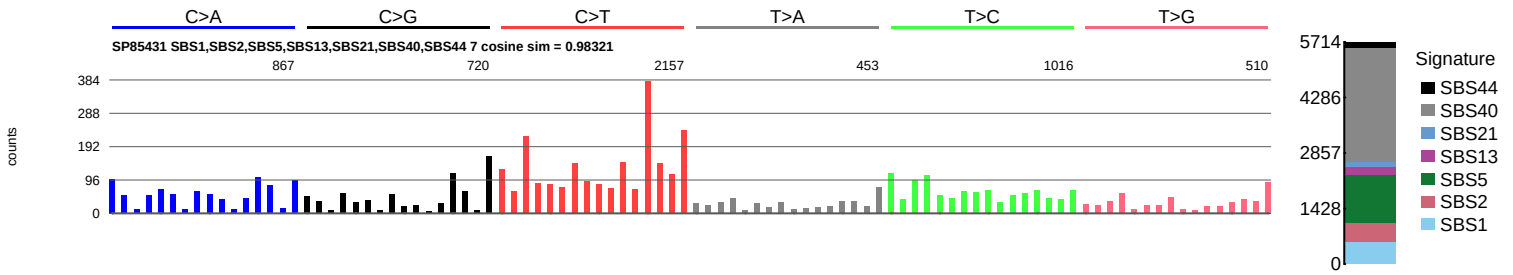
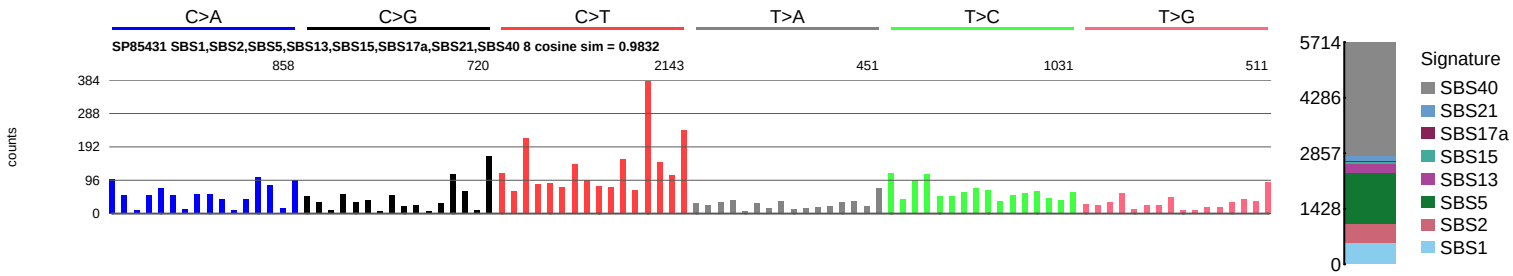
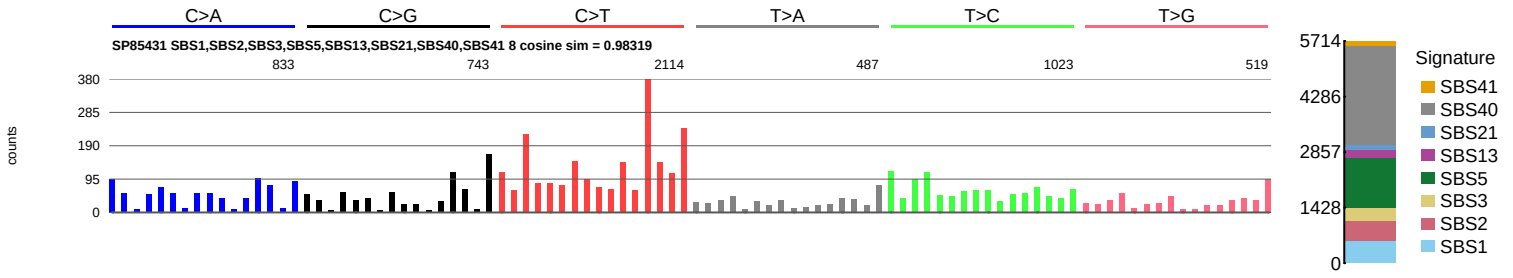
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98324
885
749
2127
466
971
517
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS21,SBS40 7 cosine sim = 0.98323
844
722
2125
444
1025
546
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS21,SBS40,SBS44 7 cosine sim = 0.98321
867
720
2157
453
1016
510
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS21
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS21,SBS40 8 cosine sim = 0.9832
858
720
2143
451
1031
511
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS21,SBS40,SBS41 8 cosine sim = 0.98319
833
743
2114
487
1023
519
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS21
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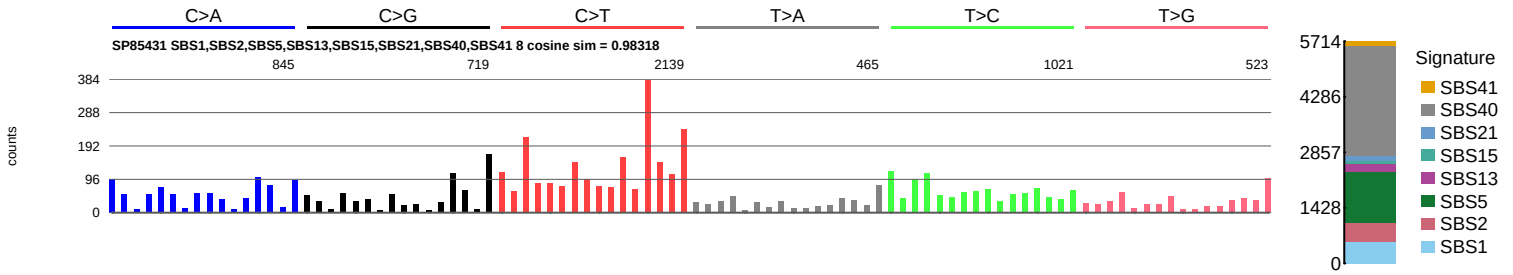
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS21,SBS40,SBS41 8 cosine sim = 0.98318
845
719
2139
465
1021
523
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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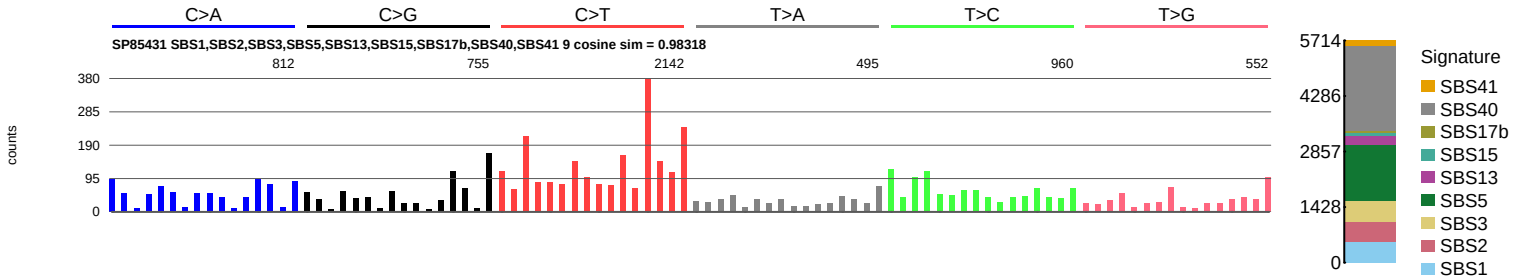
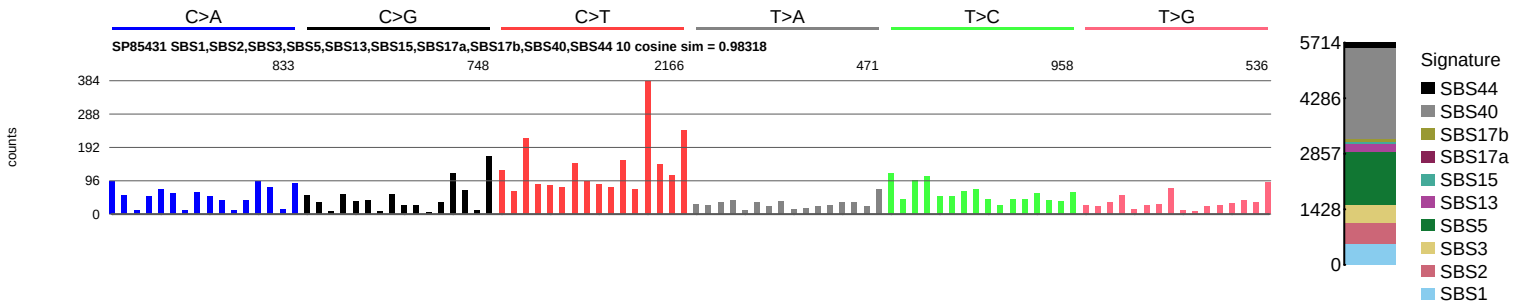
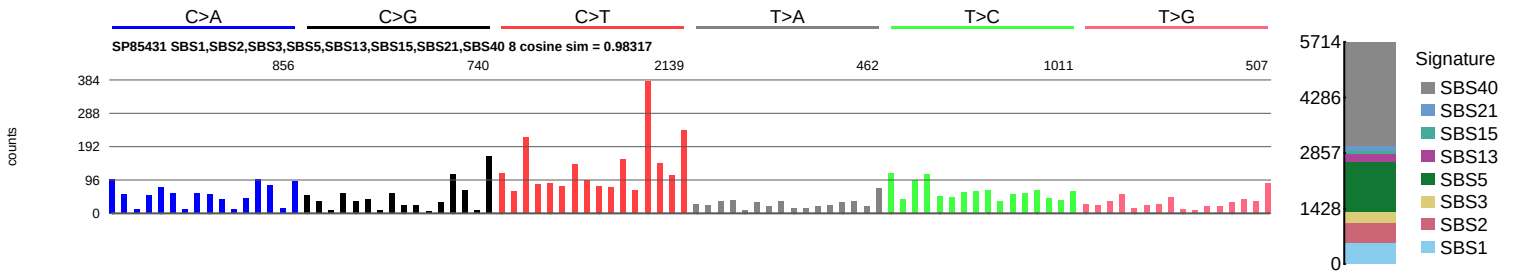
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS40,SBS41 9 cosine sim = 0.98318
812
755
2142
495
960
552
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 10 cosine sim = 0.98318
833
748
2166
471
958
536
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS21,SBS40 8 cosine sim = 0.98317
856
740
2139
462
1011
507
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS15
SBS13
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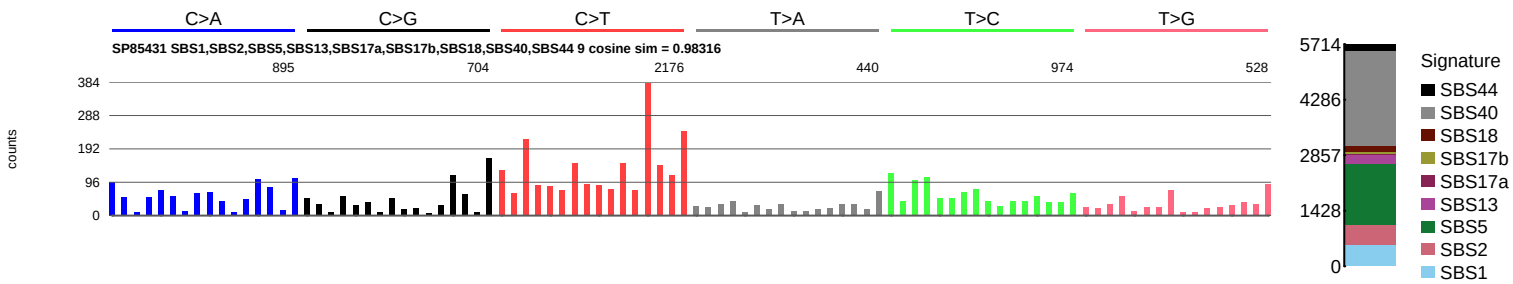
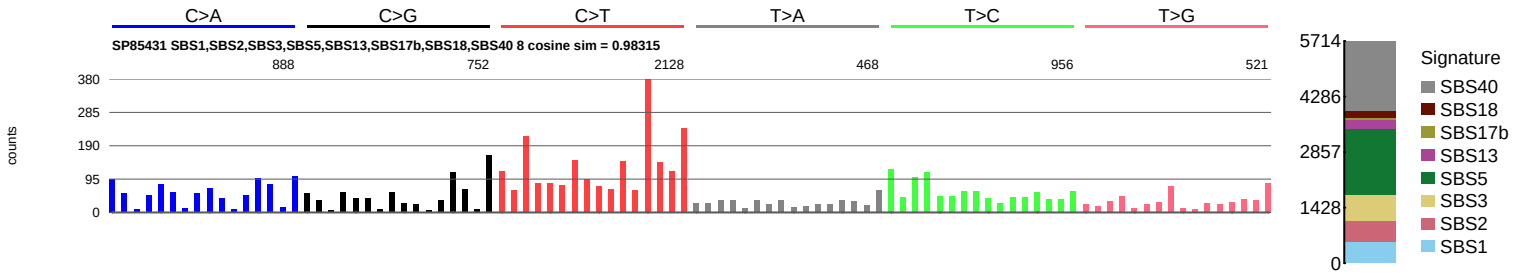
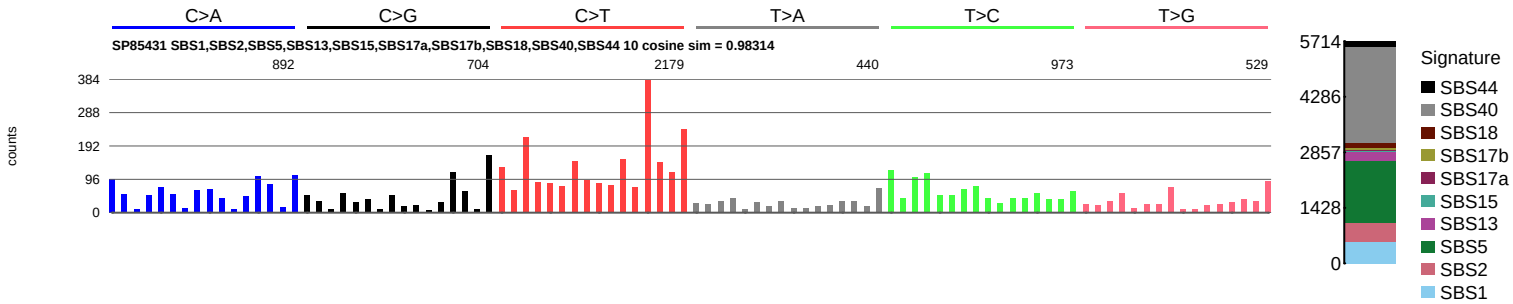
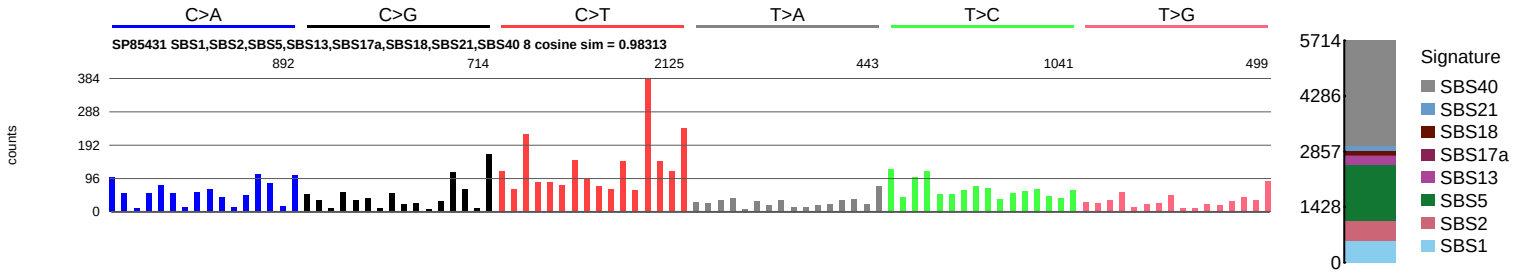
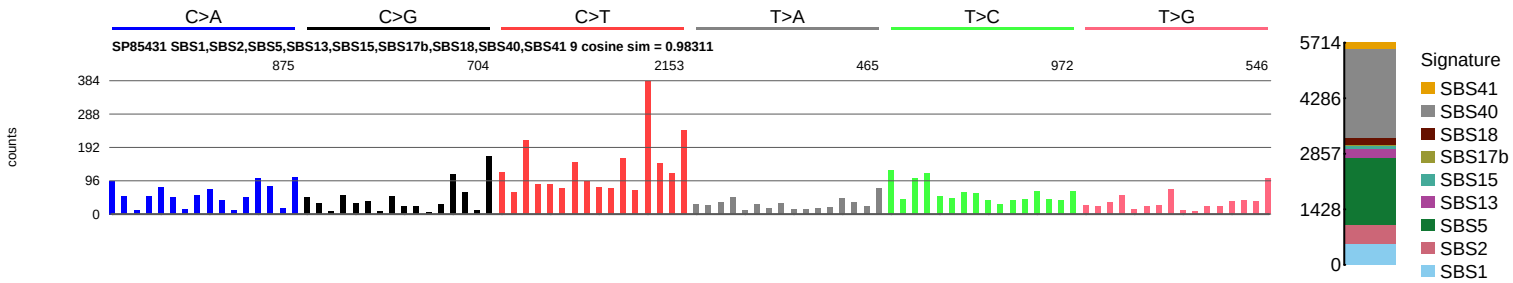
SBS3
SBS2
SBS1

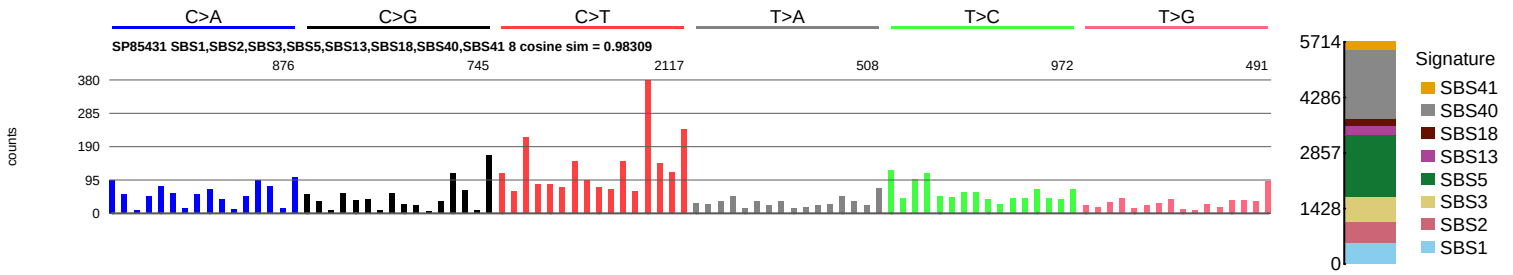

C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98316
895
704
2176
440
974
528
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

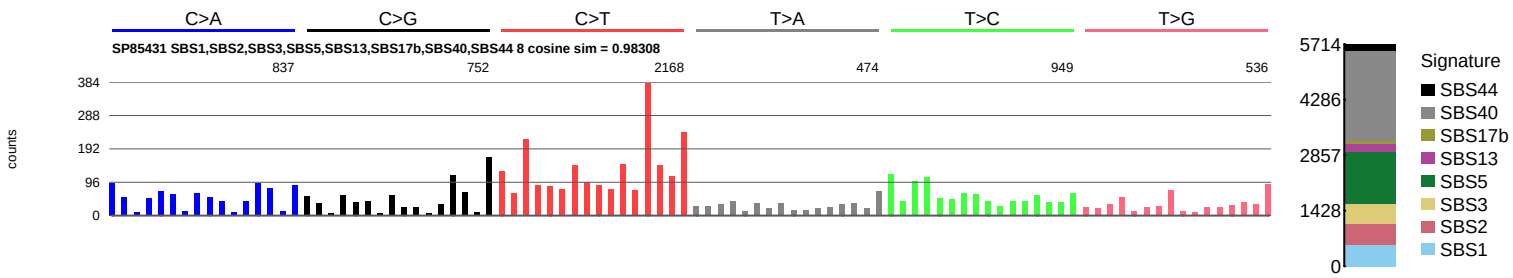

C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS40 8 cosine sim = 0.98315
888
752
2128
468
956
521
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98314
892
704
2179
440
973
529
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40 8 cosine sim = 0.98313
892
714
2125
443
1041
499
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98311
875
704
2153
465
972
546
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS40,SBS41 8 cosine sim = 0.98309
876
745
2117
508
972
491
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS40,SBS44 8 cosine sim = 0.98308
837
752
2168
474
949
536
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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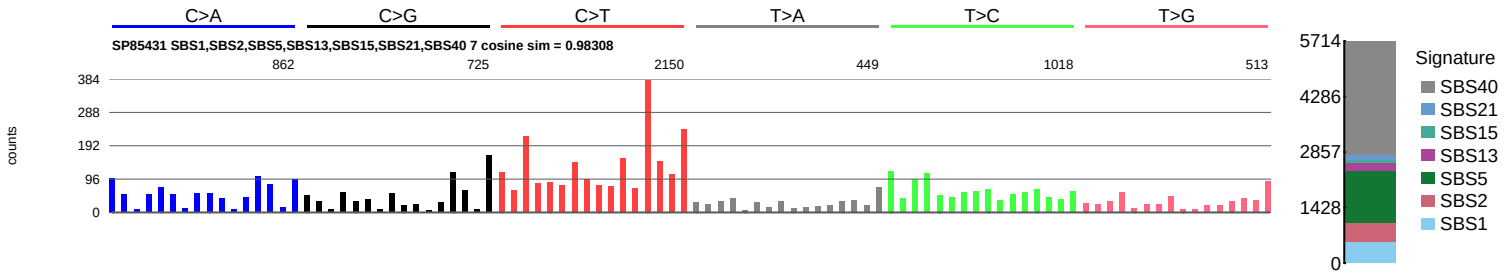
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS21,SBS40 7 cosine sim = 0.98308
862
725
2150
449
1018
513
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
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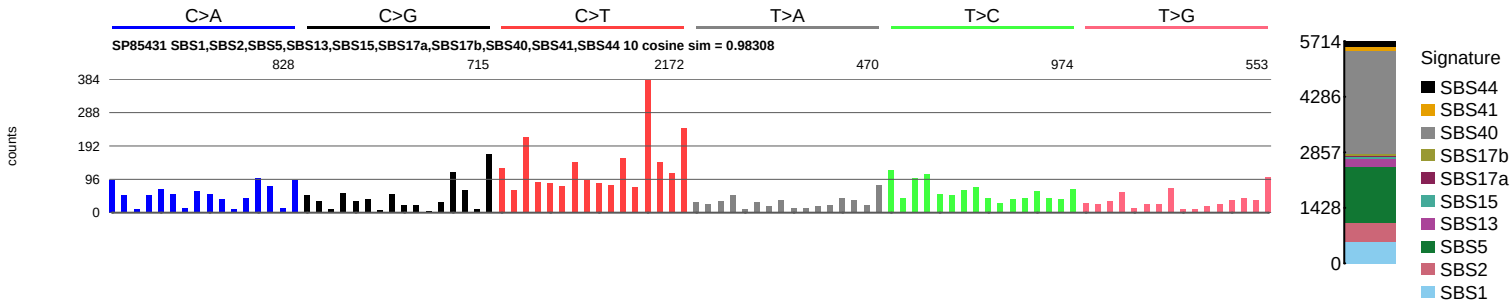
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 10 cosine sim = 0.98308
828
715
2172
470
974
553
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
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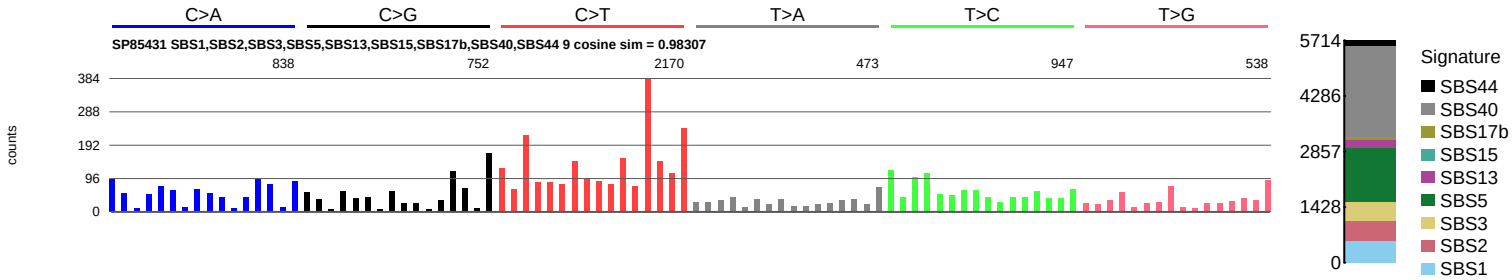
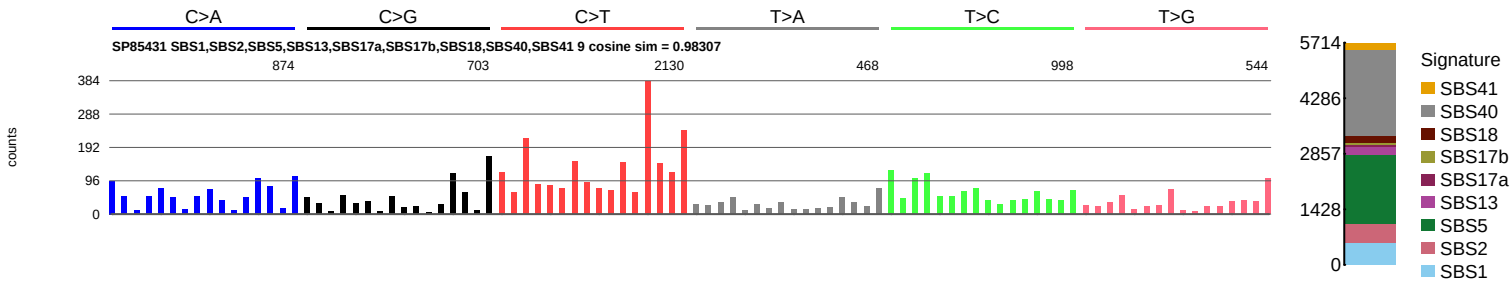
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS40,SBS44 9 cosine sim = 0.98307
838
752
2170
473
947
538
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98307
874
703
2130
468
998
544
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
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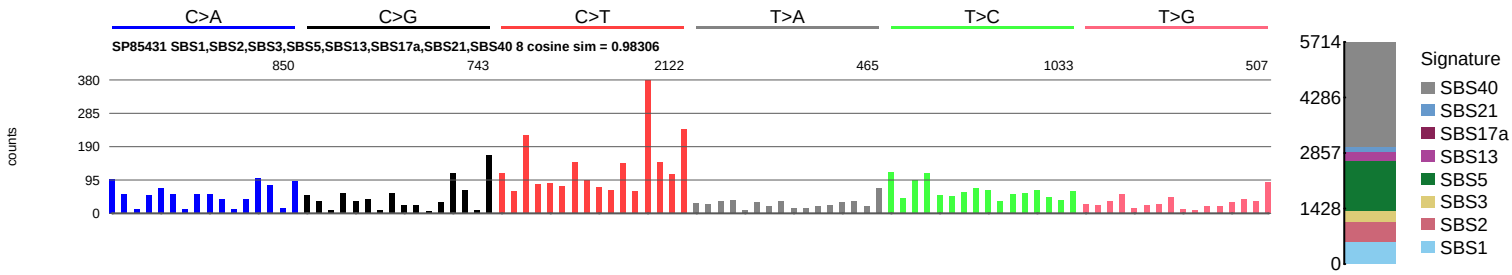
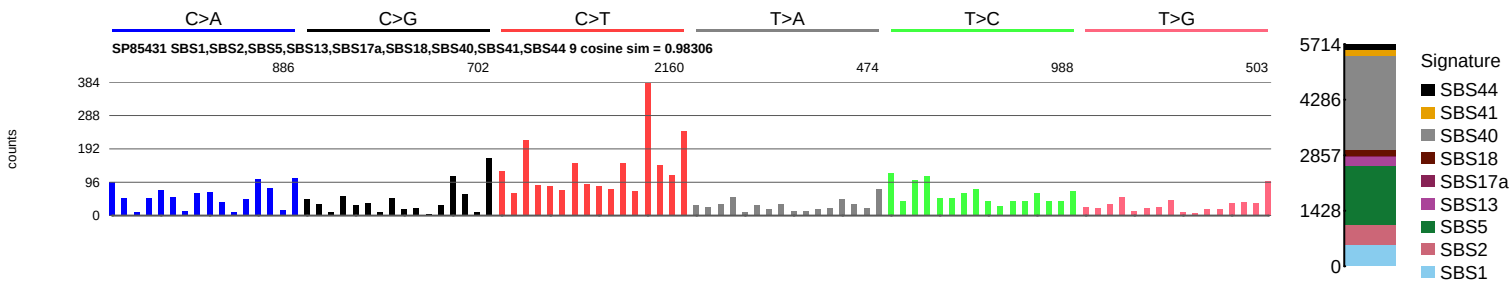
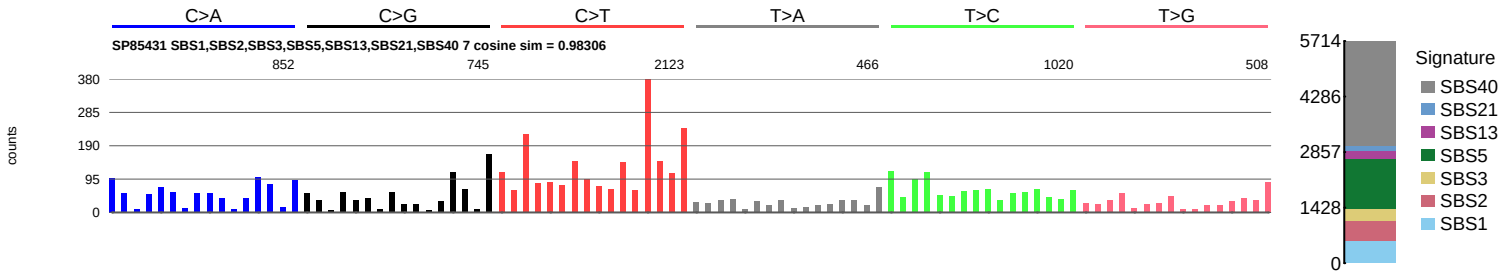
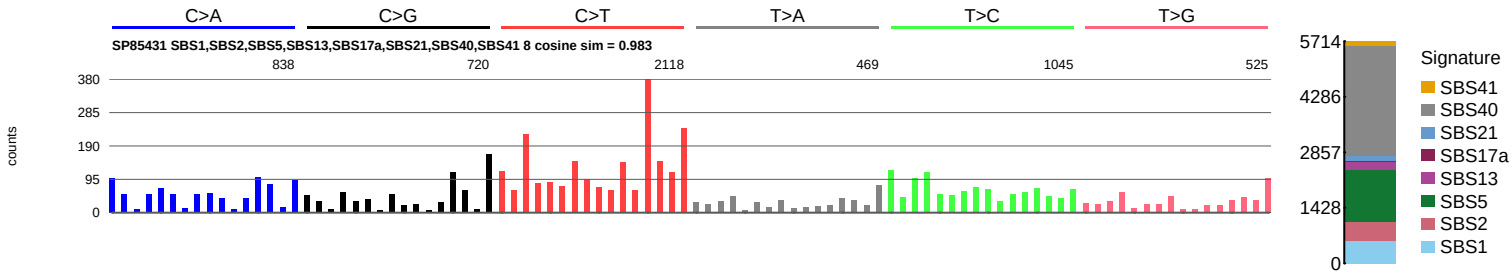
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS21,SBS40 8 cosine sim = 0.98306
850
743
2122
465
1033
507
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98306
886
702
2160
474
988
503
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS21,SBS40 7 cosine sim = 0.98306
852
745
2123
466
1020
508
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS13
SBS5
SBS3
SBS2
SBS1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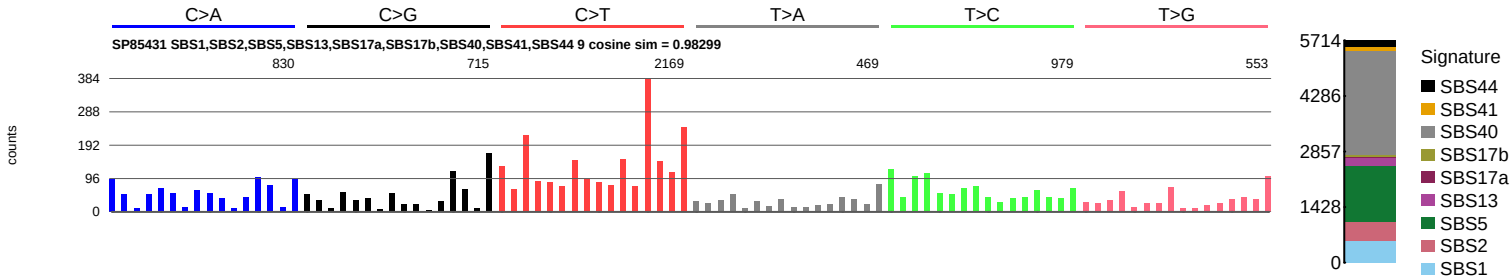
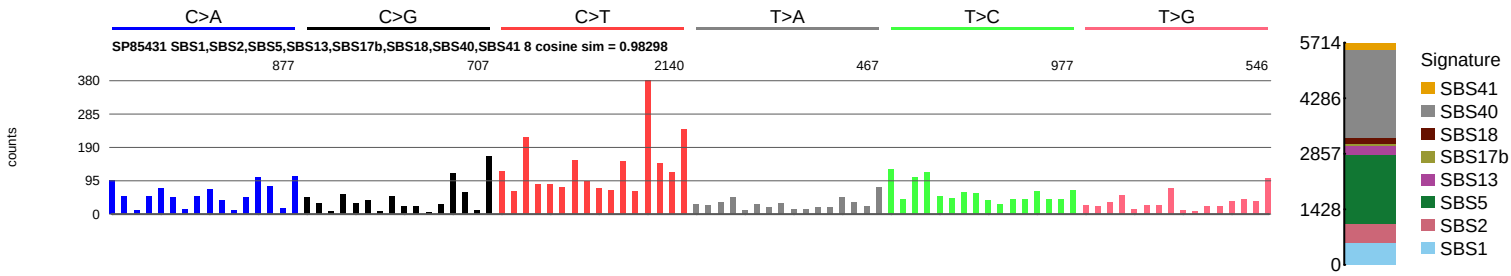
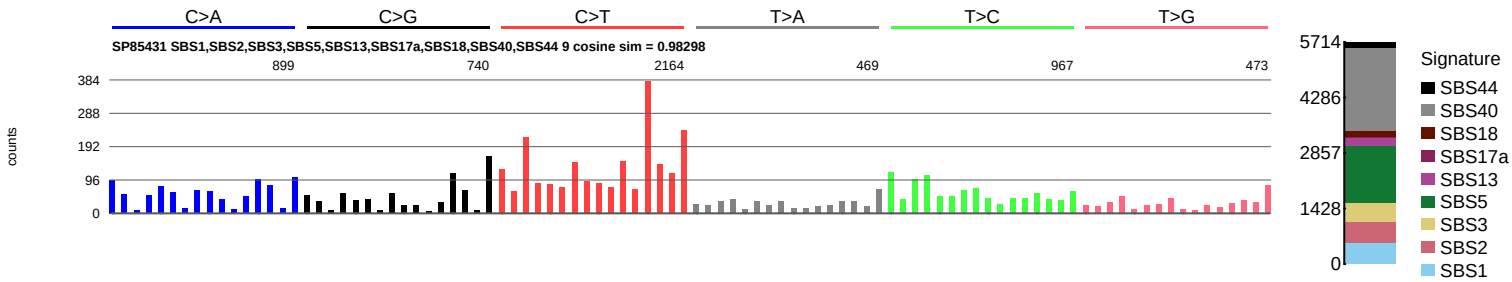
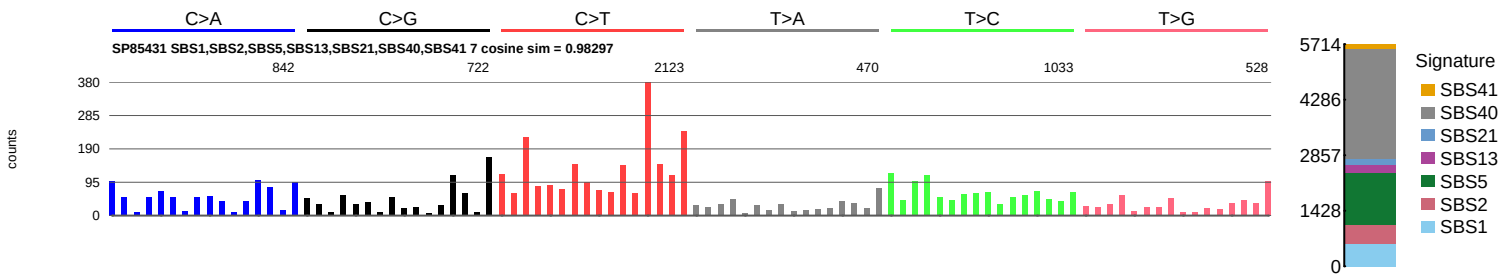
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS21,SBS40,SBS41 8 cosine sim = 0.983
838
720
2118
469
1045
525
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS21
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98299
830
715
2169
469
979
553
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98298
877
707
2140
467
977
546
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS44 9 cosine sim = 0.98298
899
740
2164
469
967
473
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS21,SBS40,SBS41 7 cosine sim = 0.98297
842
722
2123
470
1033
528
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS21
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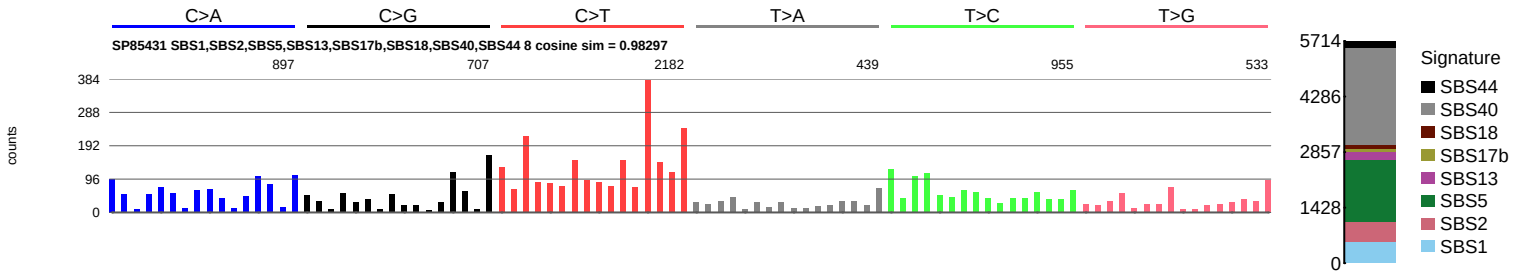
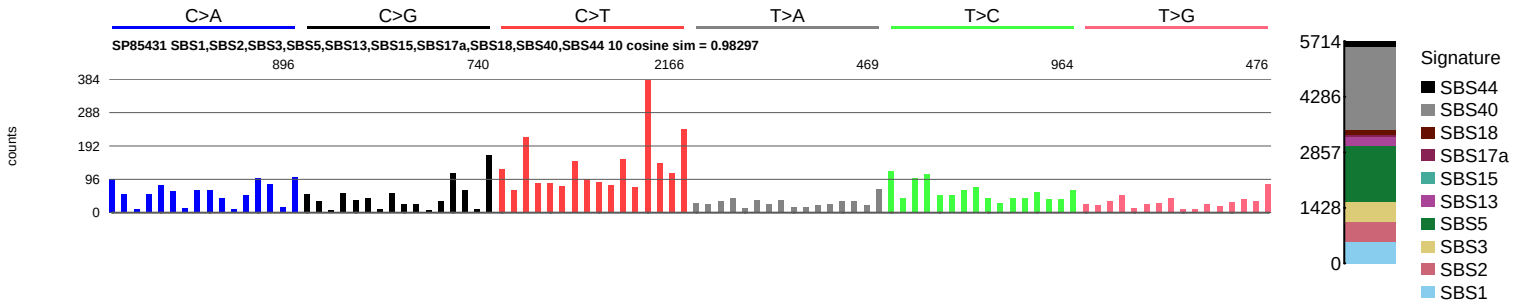
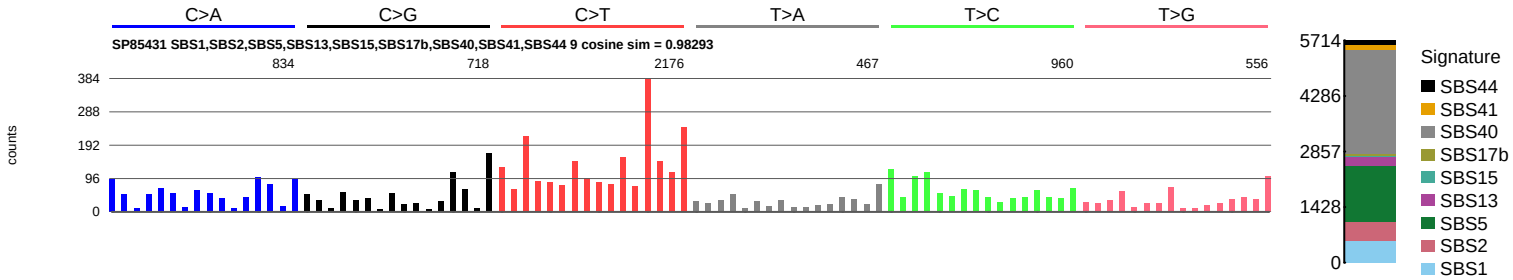
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98297
897
707
2182
439
955
533
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS44 10 cosine sim = 0.98297
896
740
2166
469
964
476
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98293
834
718
2176
467
960
556
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS17b
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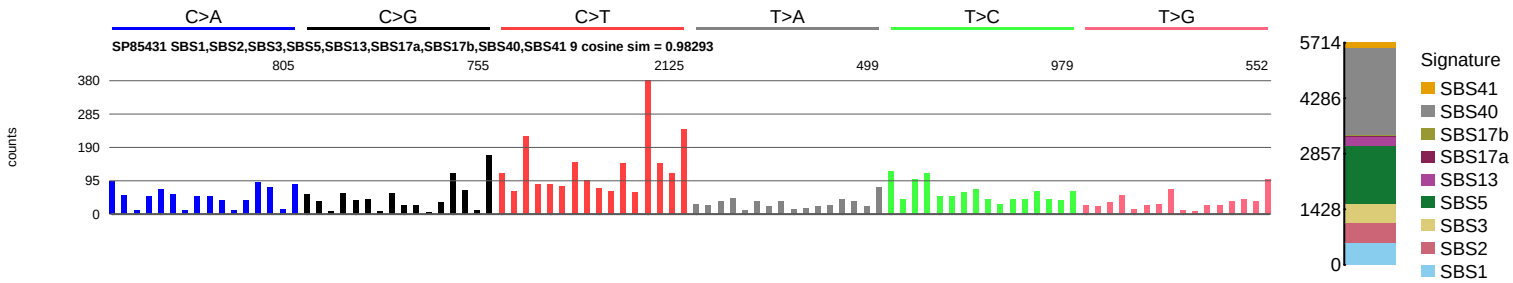
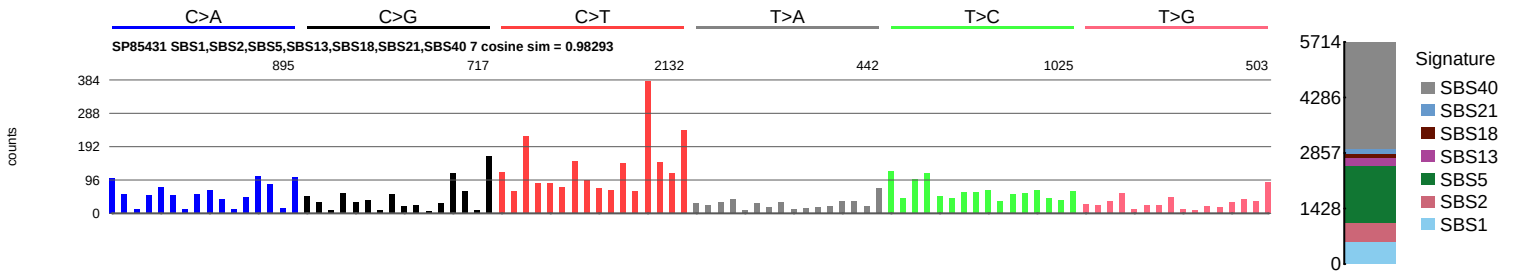
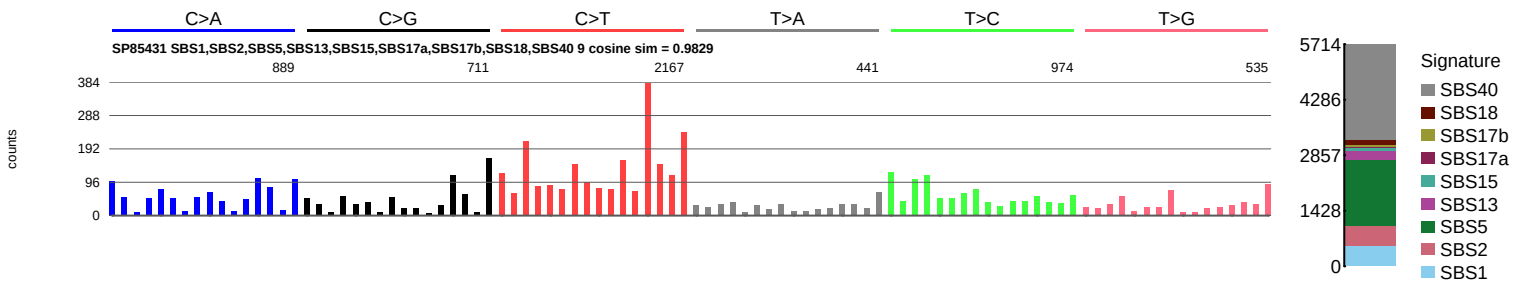
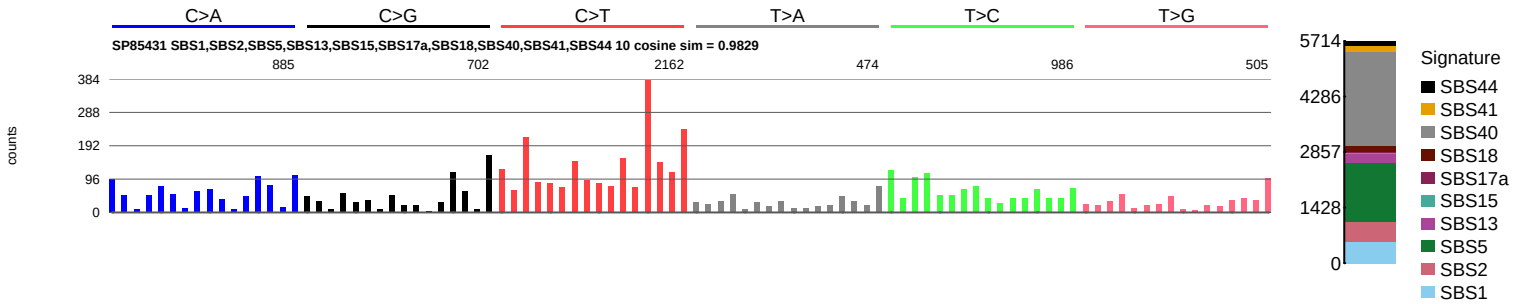
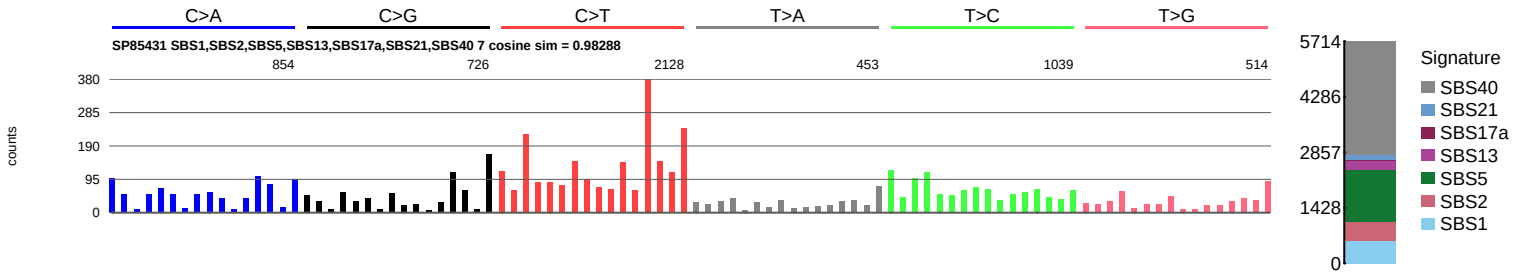
SBS5
SBS2
SBS1

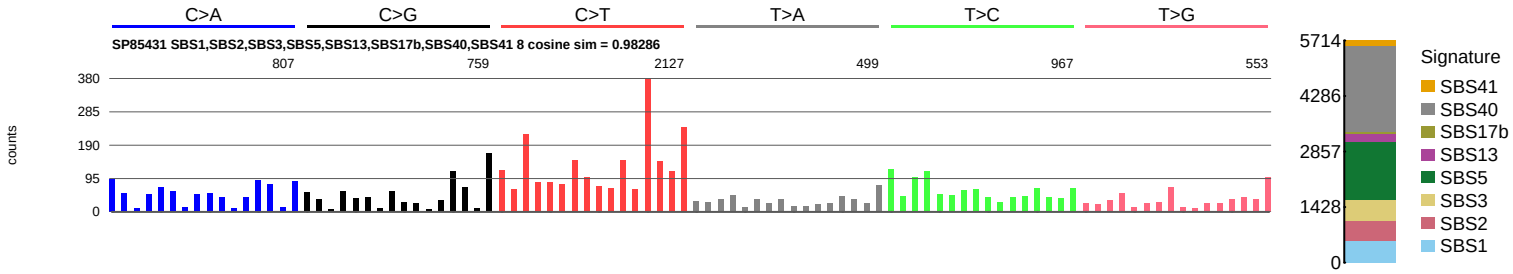

C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41 9 cosine sim = 0.98293
805
755
2125
499
979
552
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

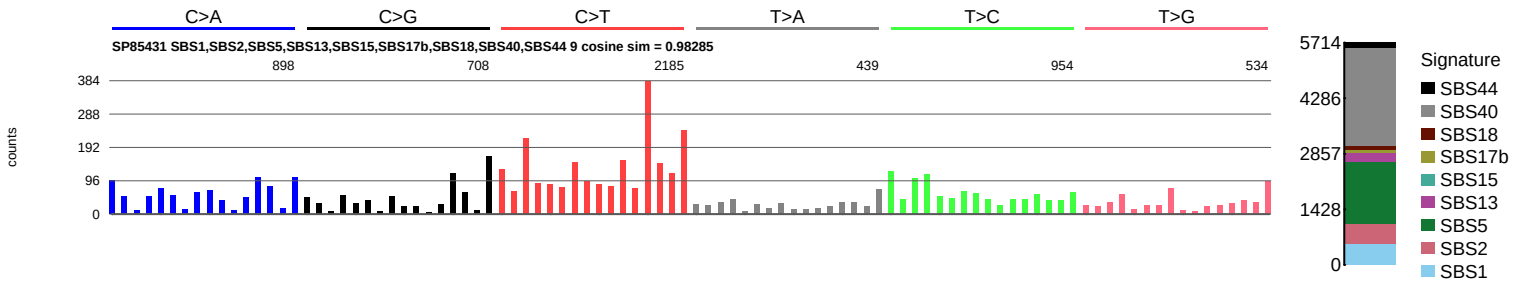

C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40 7 cosine sim = 0.98293
895
717
2132
442
1025
503
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1

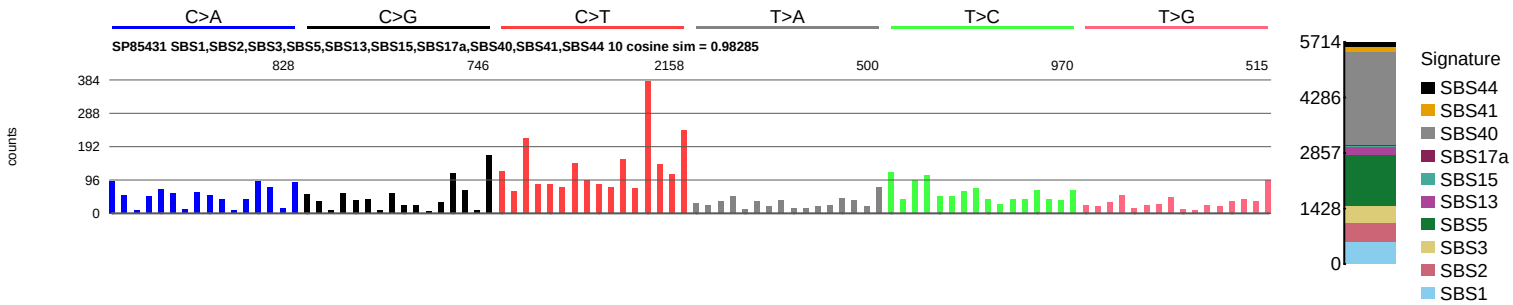

C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.9829
889
711
2167
441
974
535
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

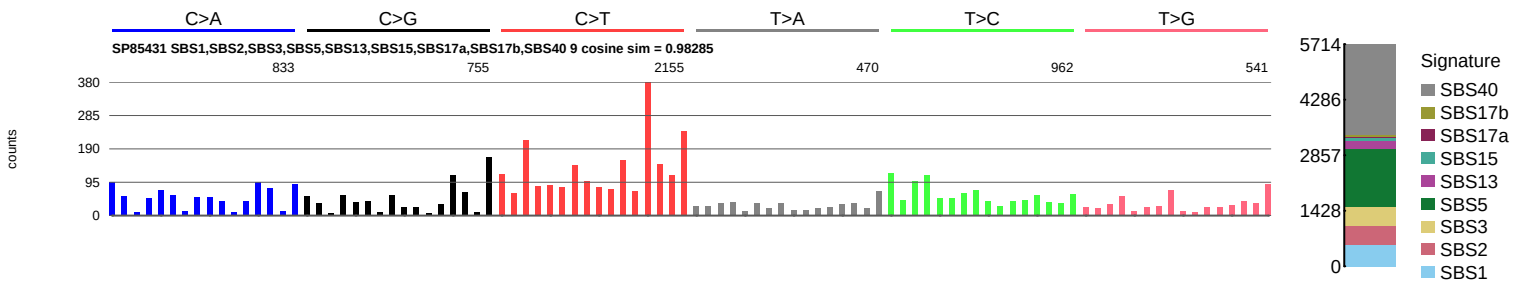

C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.9829
885
702
2162
474
986
505
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS21,SBS40 7 cosine sim = 0.98288
854
726
2128
453
1039
514
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS40,SBS41 8 cosine sim = 0.98286
807
759
2127
499
967
553
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98285
898
708
2185
439
954
534
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS40,SBS41,SBS44 10 cosine sim = 0.98285
828
746
2158
500
970
515
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 9 cosine sim = 0.98285
833
755
2155
470
962
541
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
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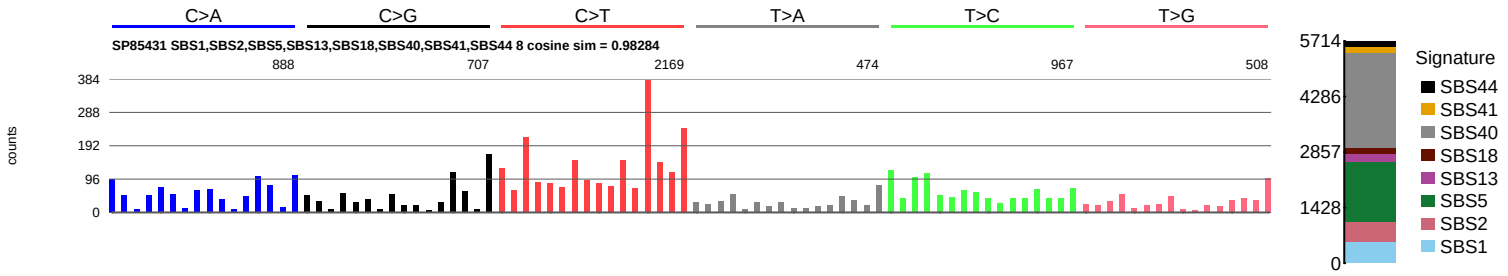
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98284
888
707
2169
474
967
508
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
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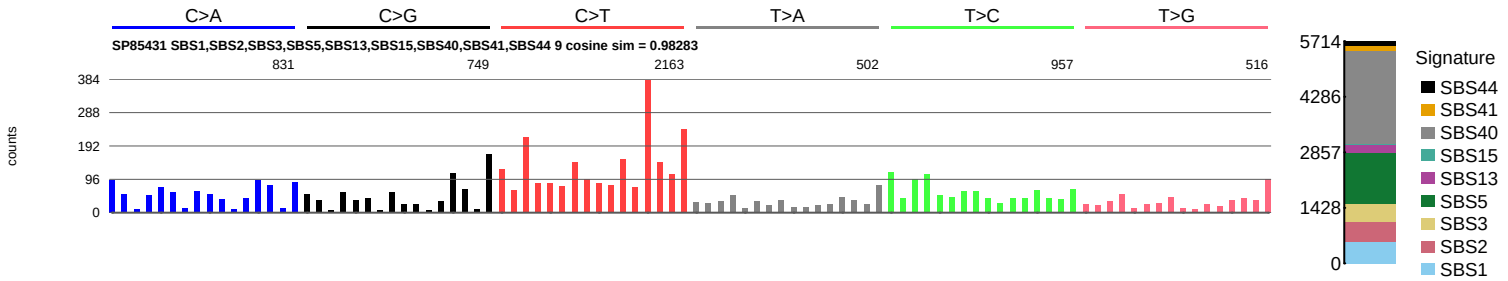
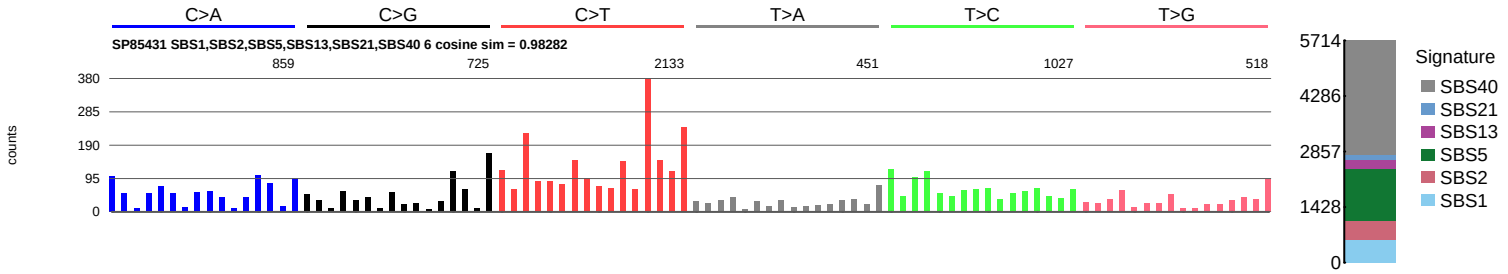
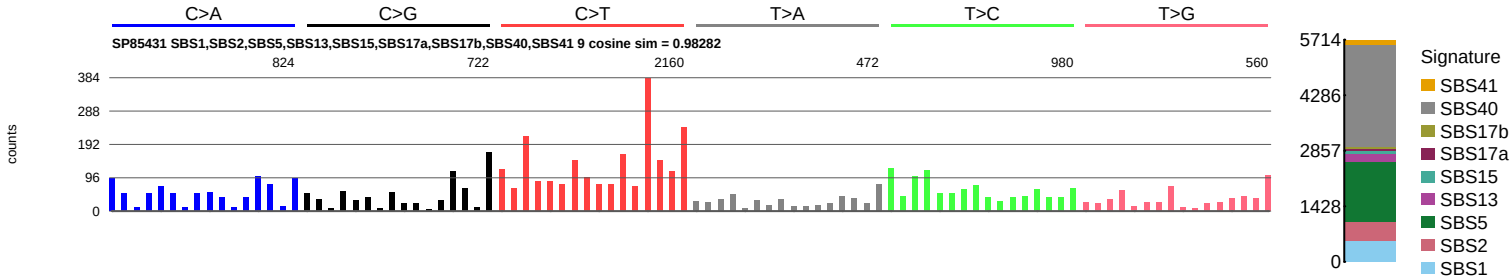
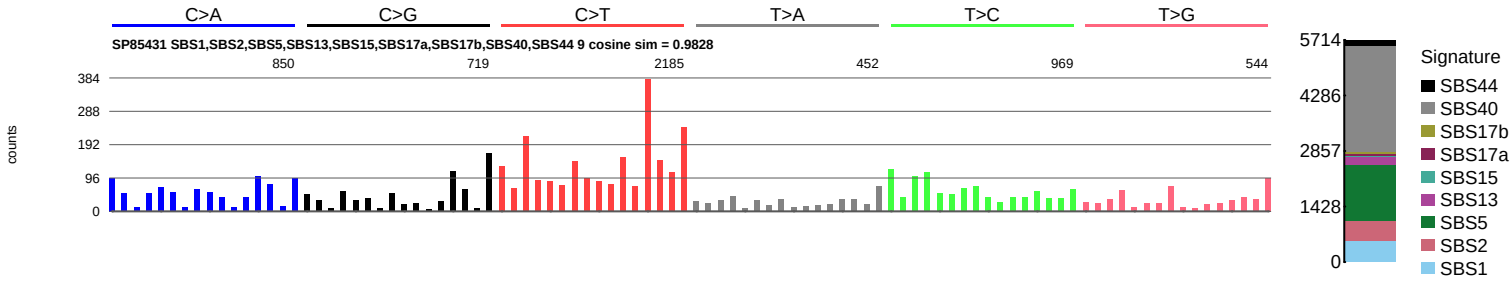
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS40,SBS41,SBS44 9 cosine sim = 0.98283
831
749
2163
502
957
516
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS21,SBS40 6 cosine sim = 0.98282
859
725
2133
451
1027
518
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS21
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 9 cosine sim = 0.98282
824
722
2160
472
980
560
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 9 cosine sim = 0.9828
850
719
2185
452
969
544
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS17b
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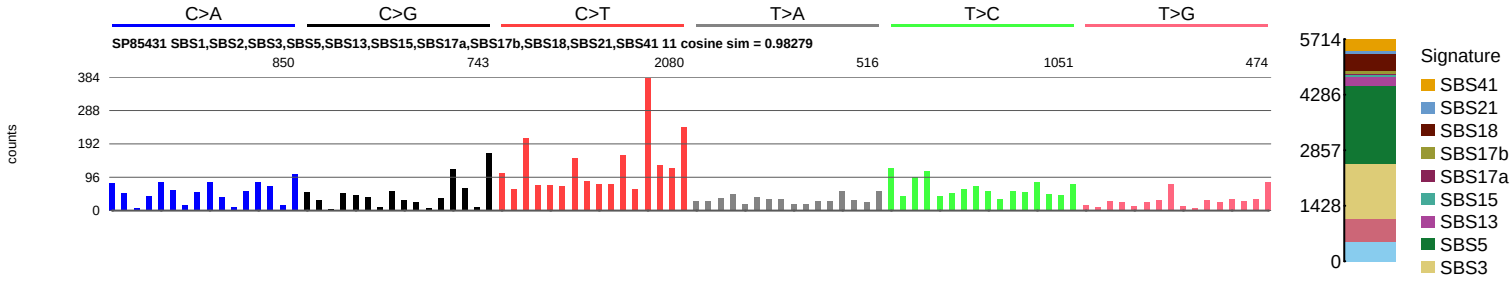
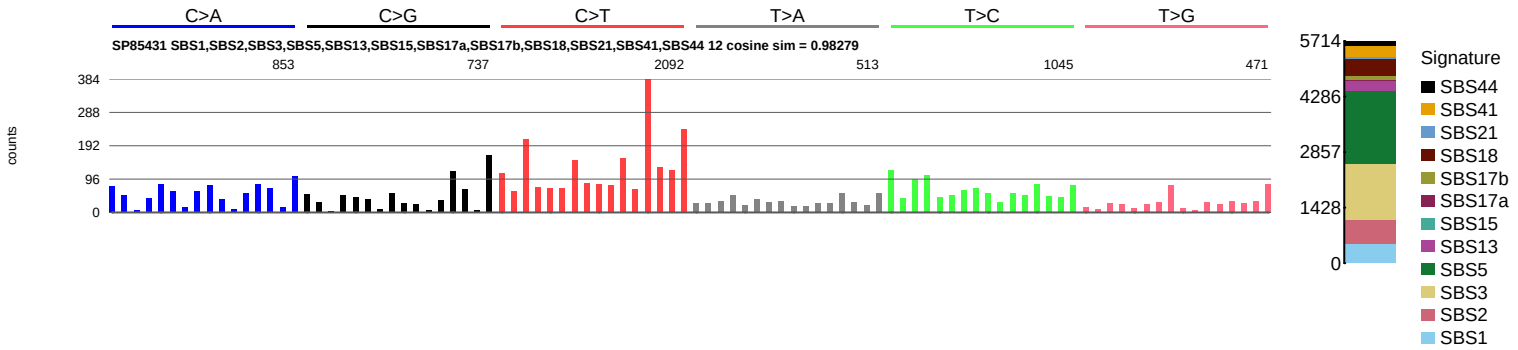
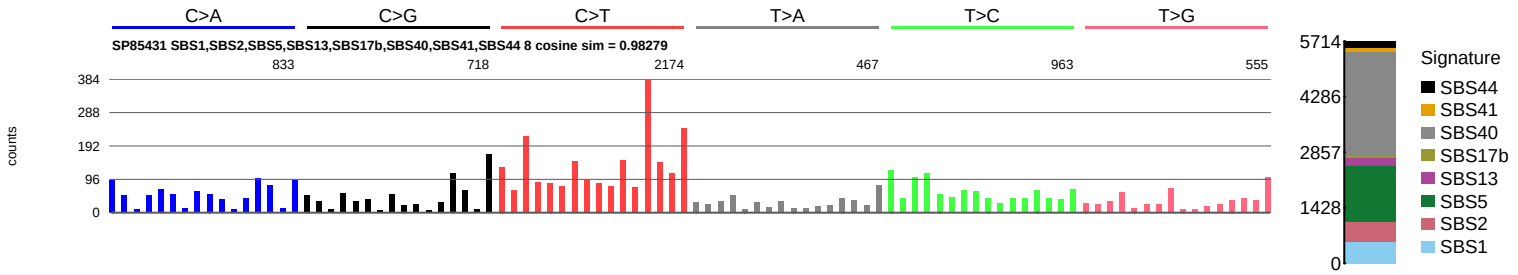
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 11 cosine sim = 0.98279
850
743
2080
516
1051
474
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 12 cosine sim = 0.98279
853
737
2092
513
1045
471
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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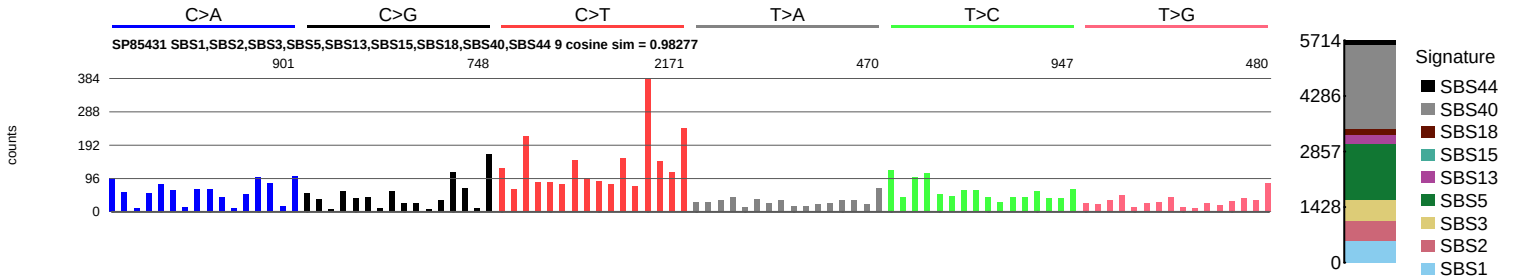
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98279
833
718
2174
467
963
555
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS13
SBS5
SBS2
SBS1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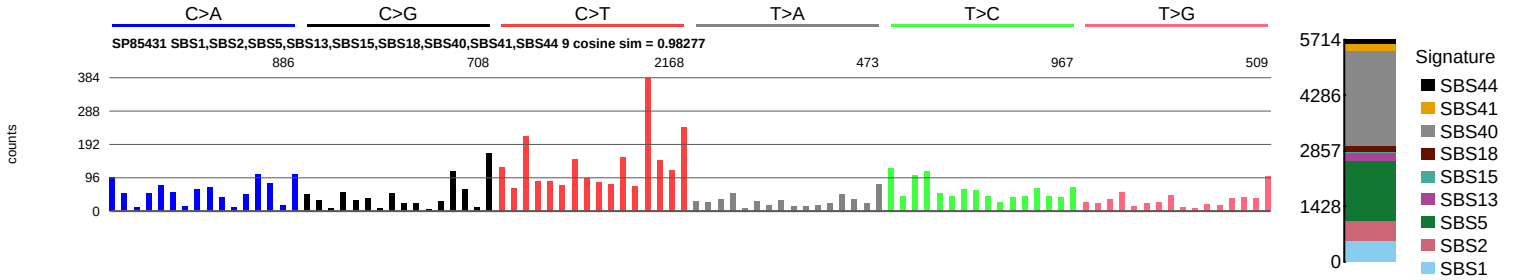
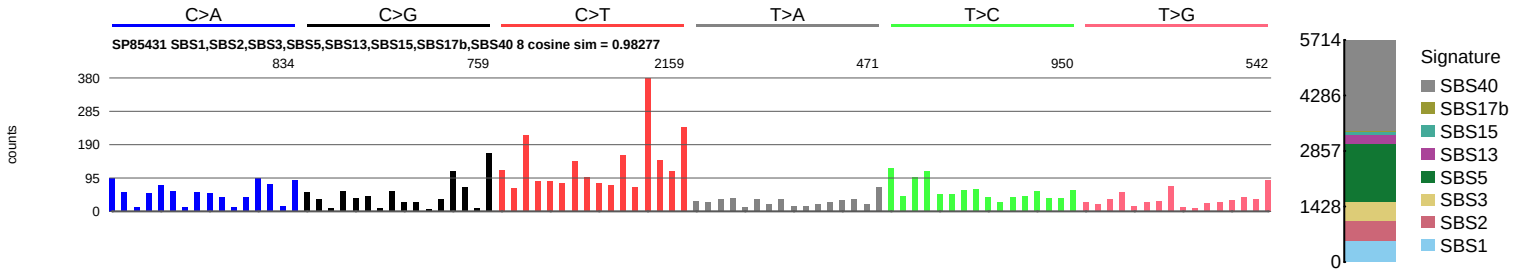
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS40,SBS44 9 cosine sim = 0.98277
901
748
2171
470
947
480
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98277
886
708
2168
473
967
509
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS40 8 cosine sim = 0.98277
834
759
2159
471
950
542
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
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SBS15
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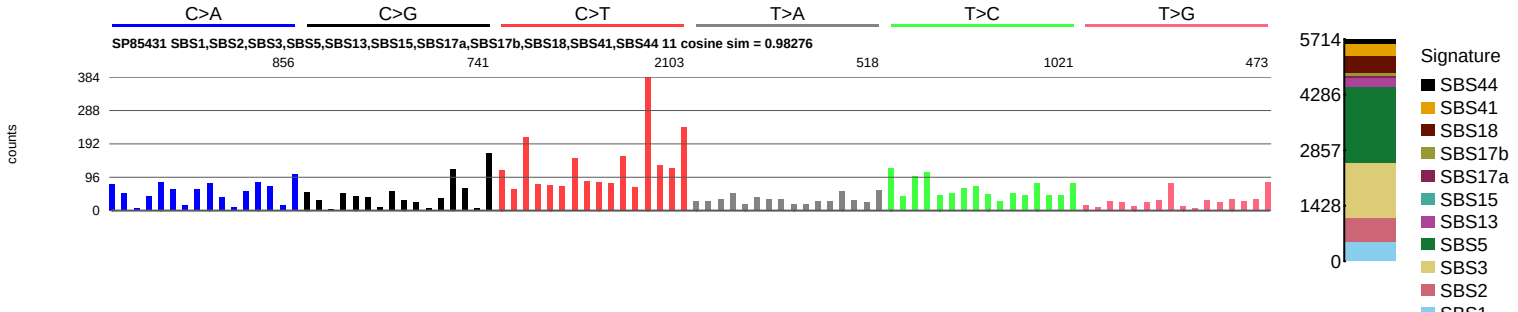
SBS2
SBS1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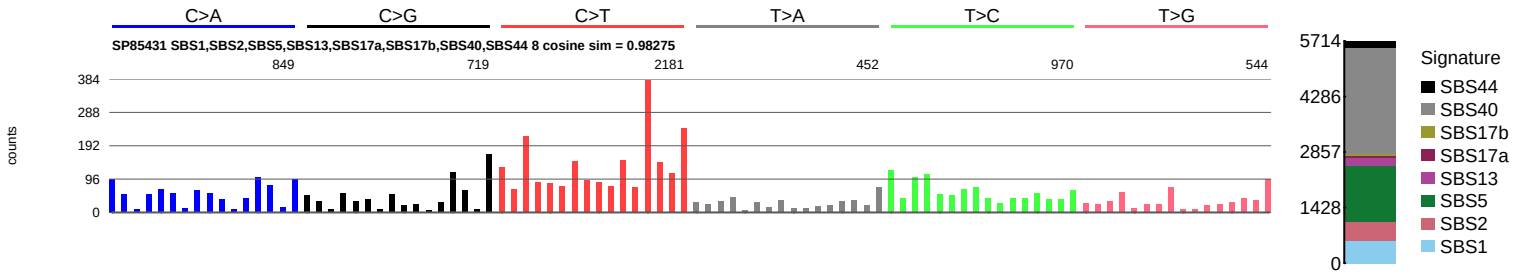
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 11 cosine sim = 0.98276
856
741
2103
518
1021
473
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2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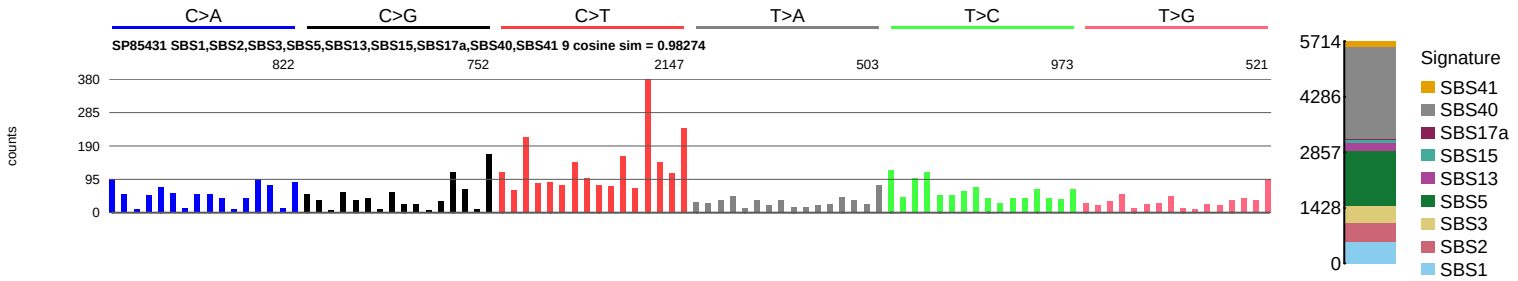
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.98275
849
719
2181
452
970
544
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS40,SBS41 9 cosine sim = 0.98274
822
752
2147
503
973
521
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS17a
SBS15
SBS13
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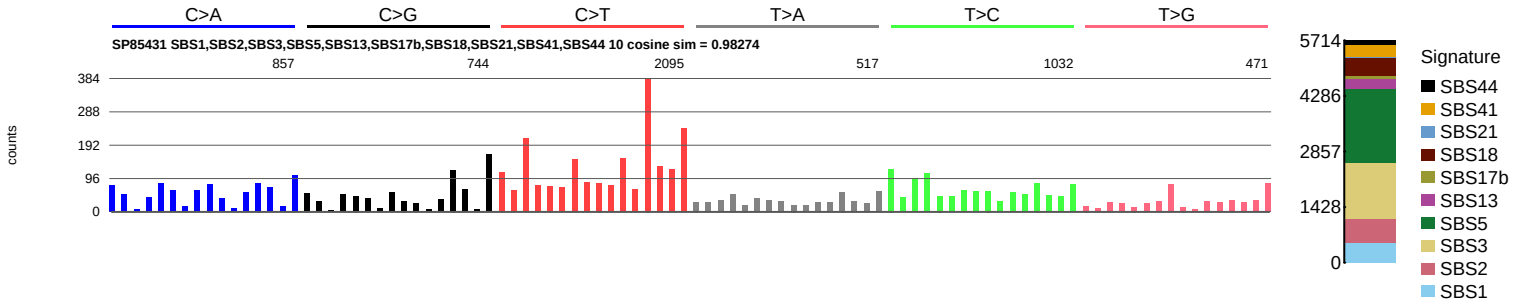
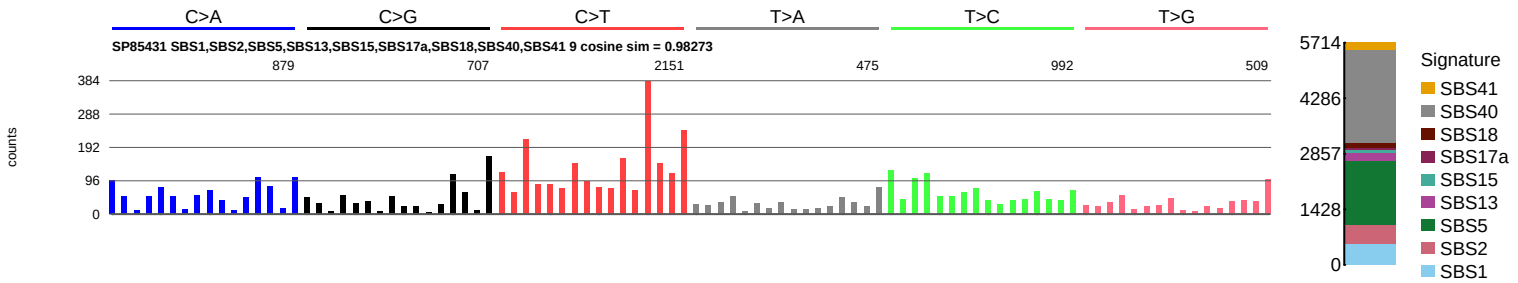
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98274
857
744
2095
517
1032
471
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41 9 cosine sim = 0.98273
879
707
2151
475
992
509
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
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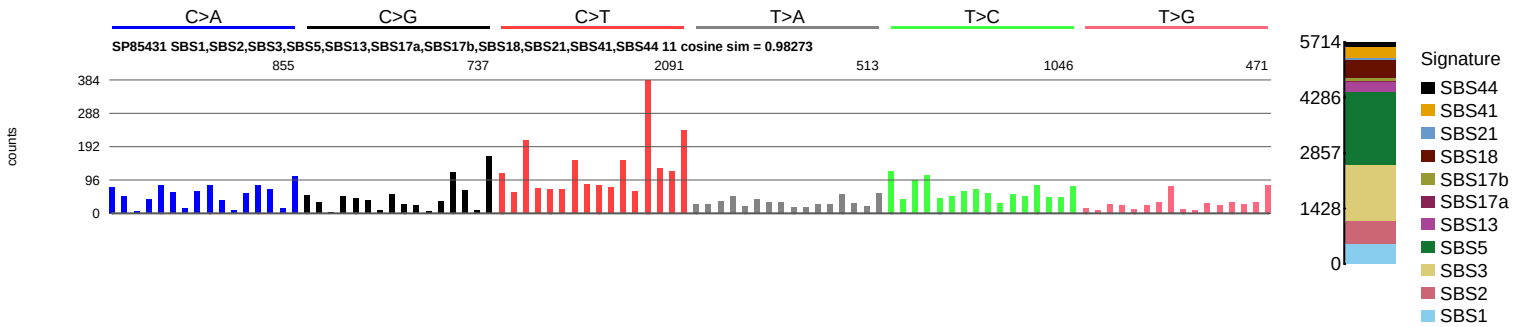
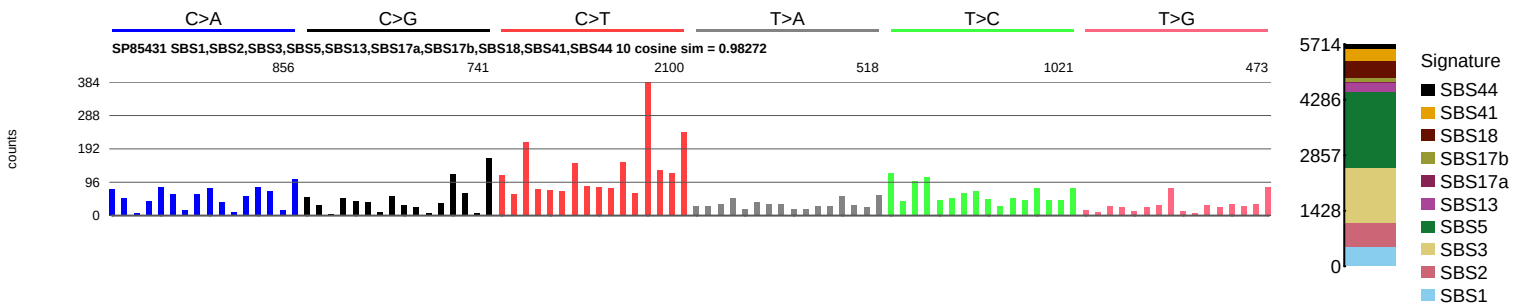
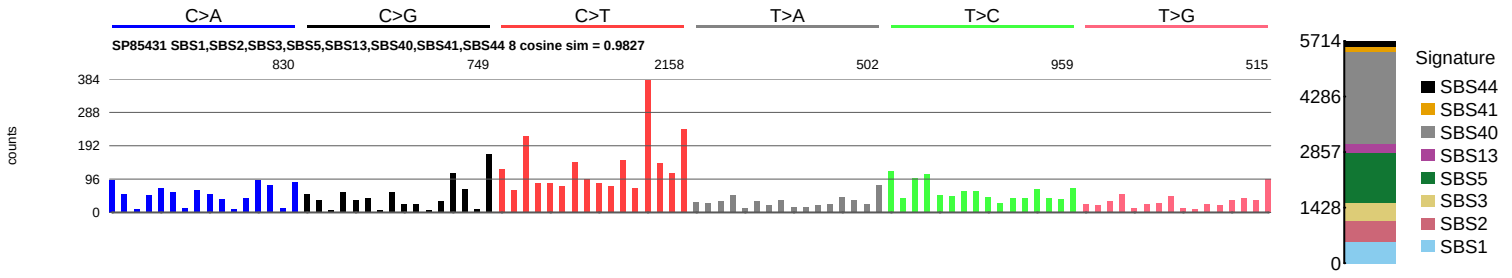
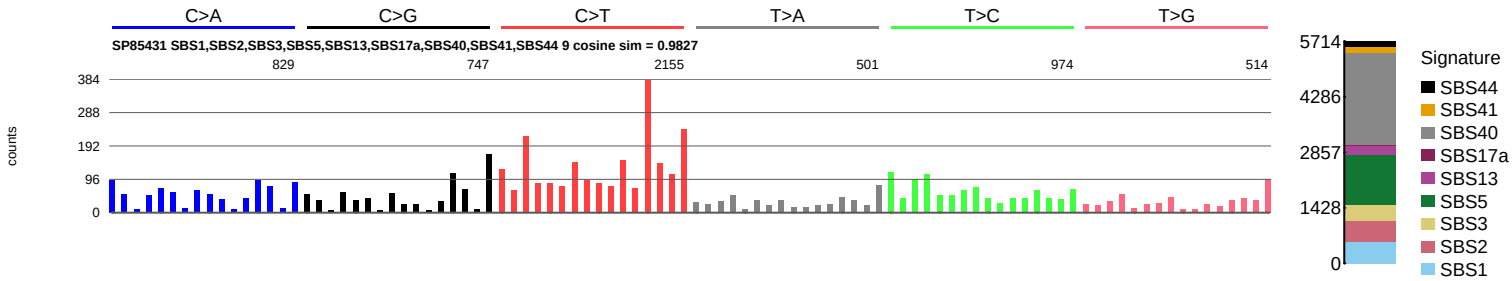
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98273
855
737
2091
513
1046
471
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98272
856
741
2100
518
1021
473
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS40,SBS41,SBS44 8 cosine sim = 0.9827
830
749
2158
502
959
515
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS40,SBS41,SBS44 9 cosine sim = 0.9827
829
747
2155
501
974
514
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS17a
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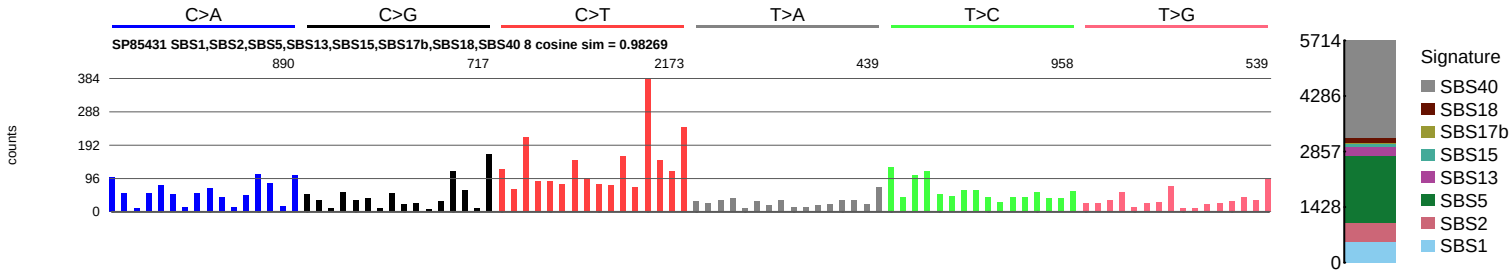
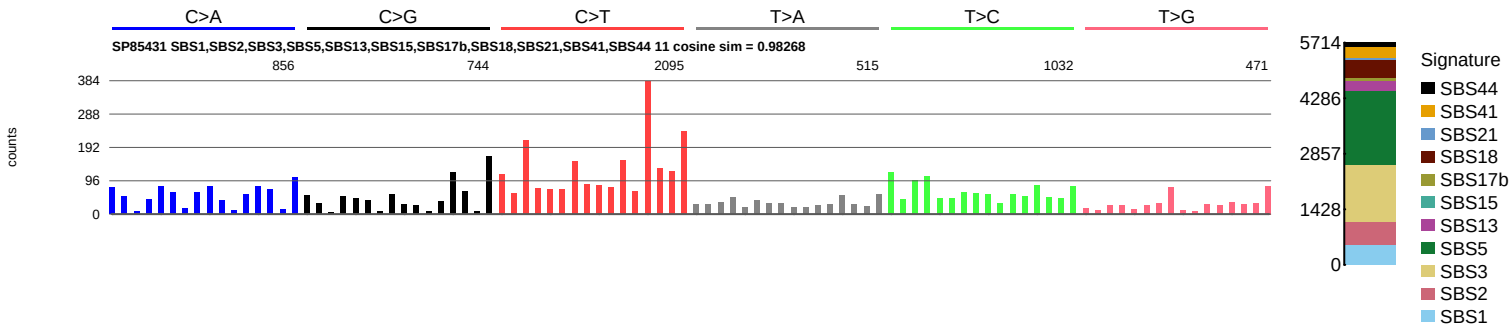
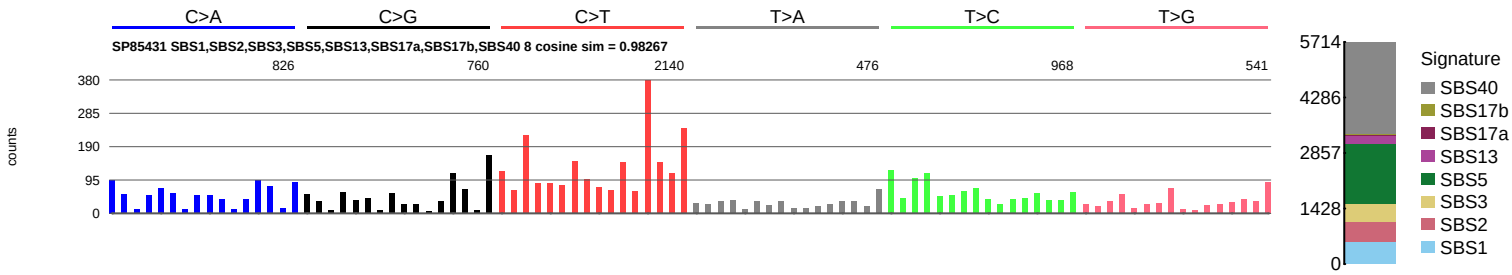
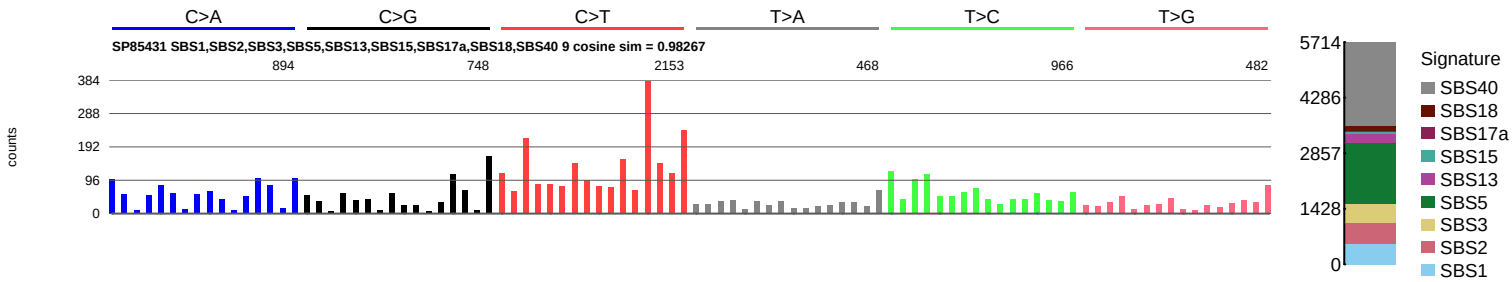
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 8 cosine sim = 0.98269
890
717
2173
439
958
539
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98268
856
744
2095
515
1032
471
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40 8 cosine sim = 0.98267
826
760
2140
476
968
541
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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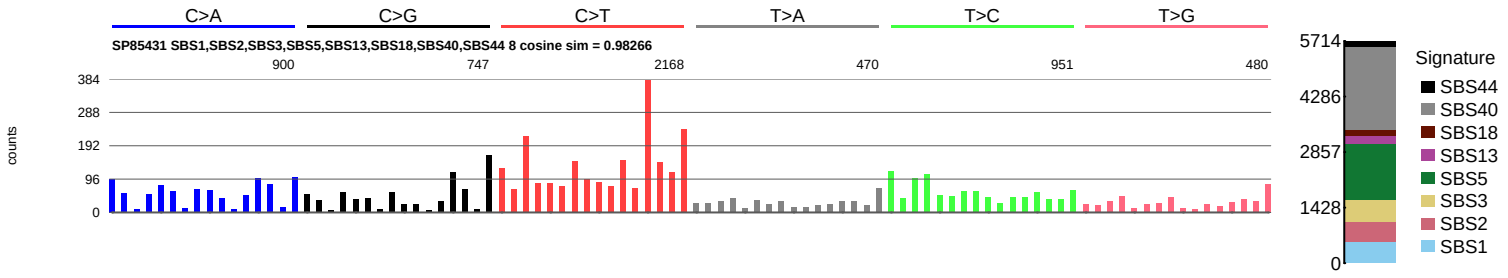
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40 9 cosine sim = 0.98267
894
748
2153
468
966
482
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS40,SBS44 8 cosine sim = 0.98266
900
747
2168
470
951
480
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS18
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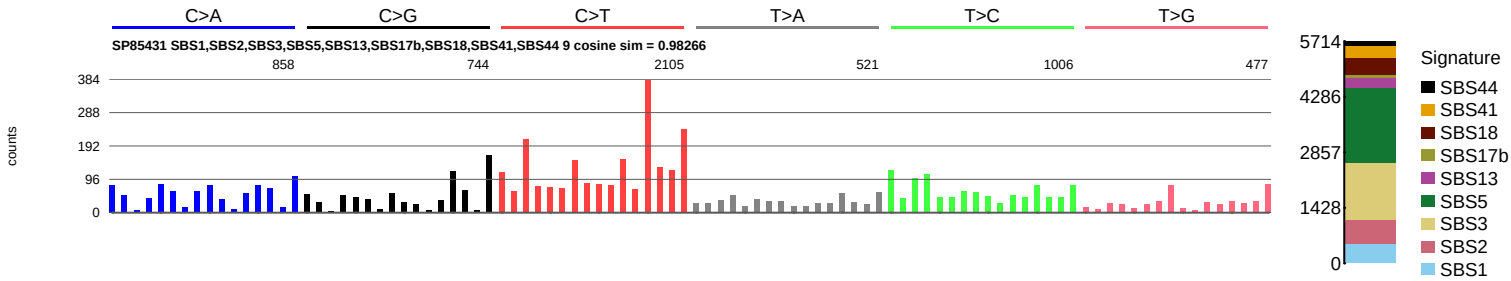
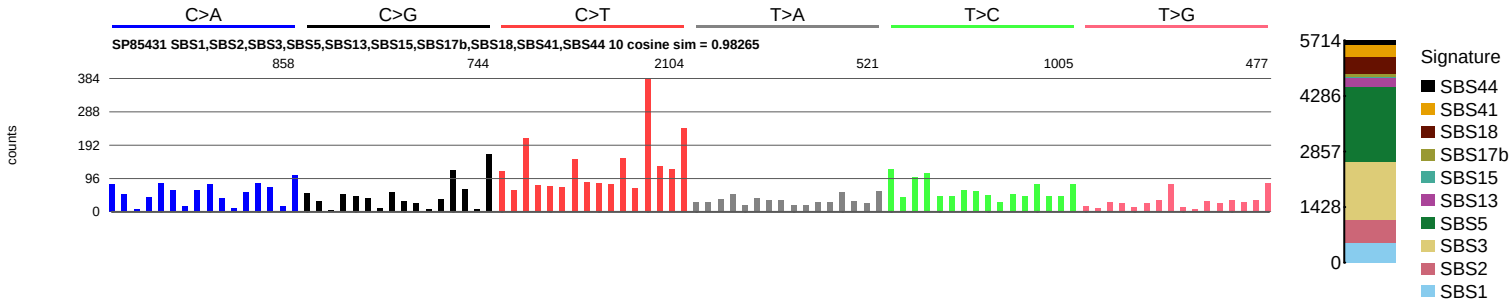
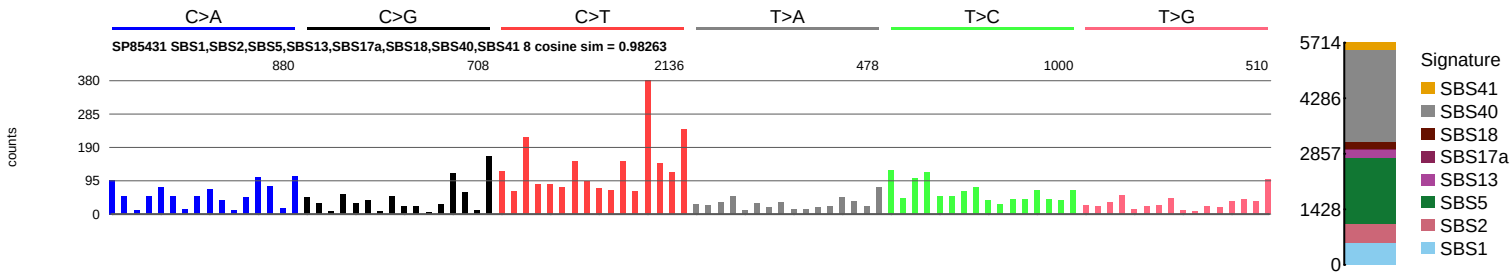
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98266
858
744
2105
521
1006
477
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98265
858
744
2104
521
1005
477
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.98263
880
708
2136
478
1000
510
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS18
SBS17a
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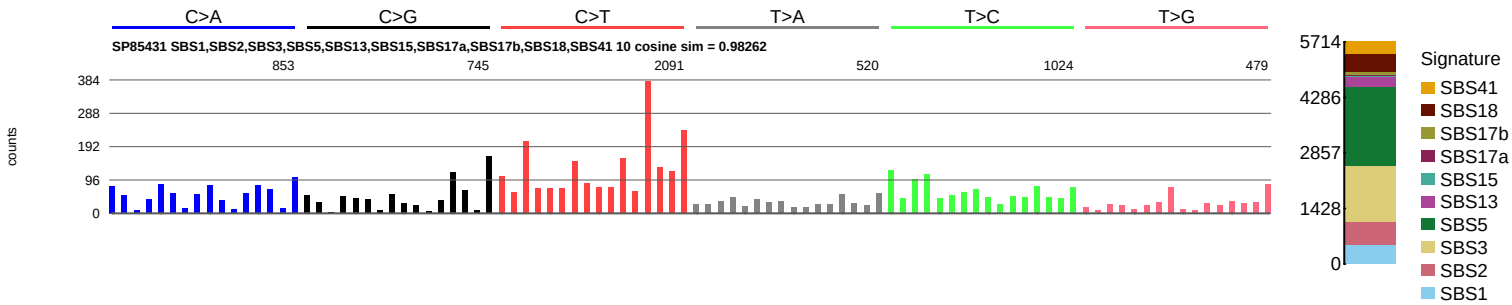
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.98262
853
745
2091
520
1024
479
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
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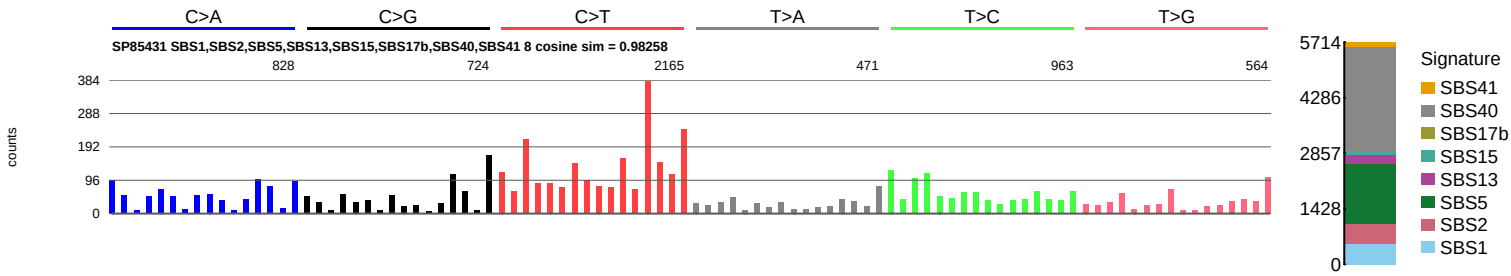
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS40,SBS41 8 cosine sim = 0.98258
828
724
2165
471
963
564
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS17b
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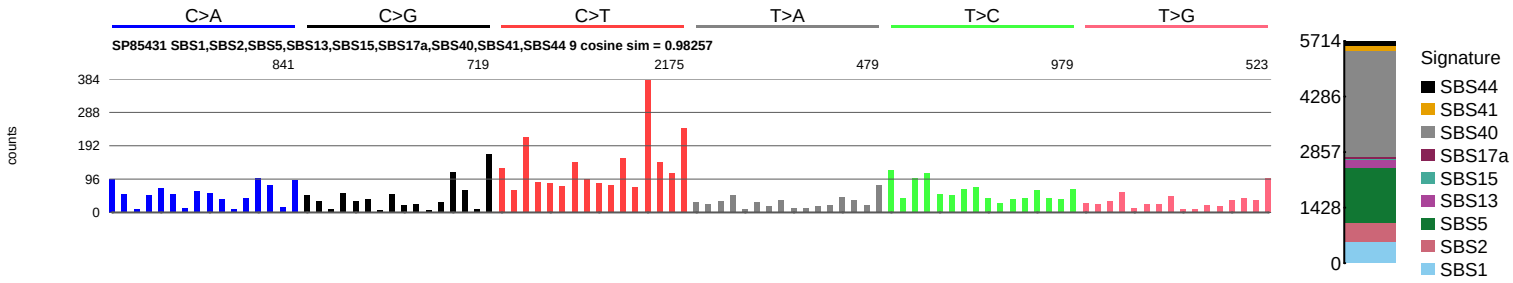
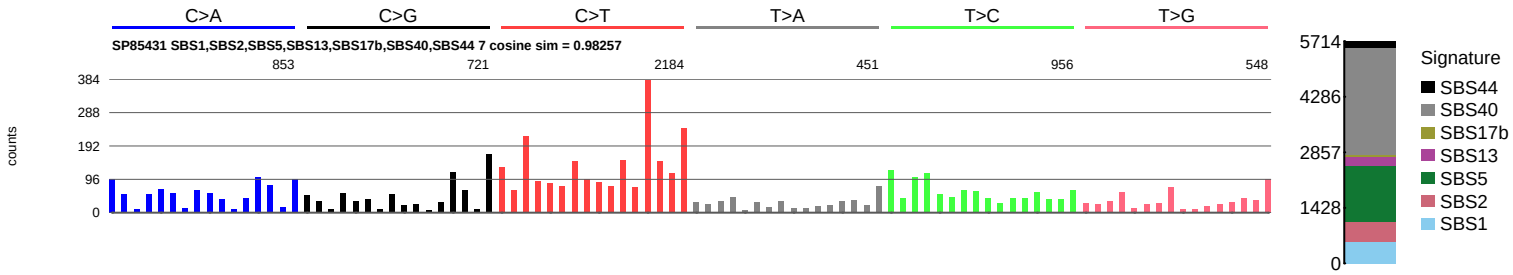
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS40,SBS41,SBS44 9 cosine sim = 0.98257
841
719
2175
479
979
523
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
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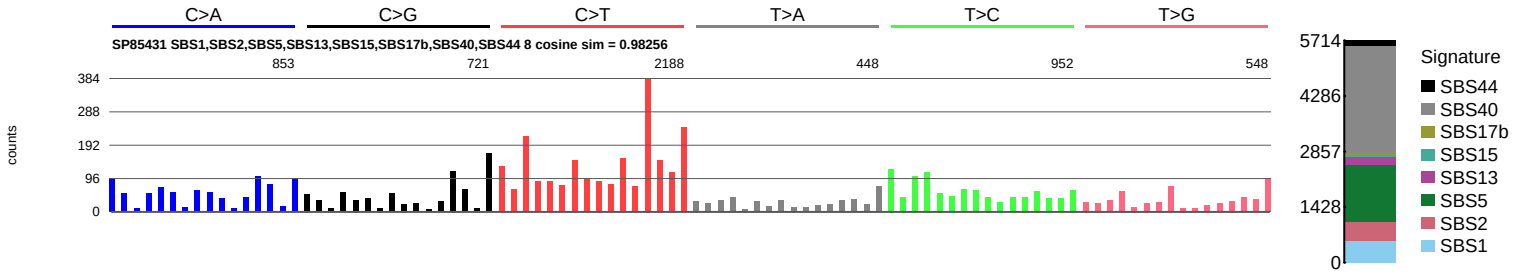
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS40,SBS44 7 cosine sim = 0.98257
853
721
2184
451
956
548
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS17b
SBS13
SBS5
SBS2
SBS1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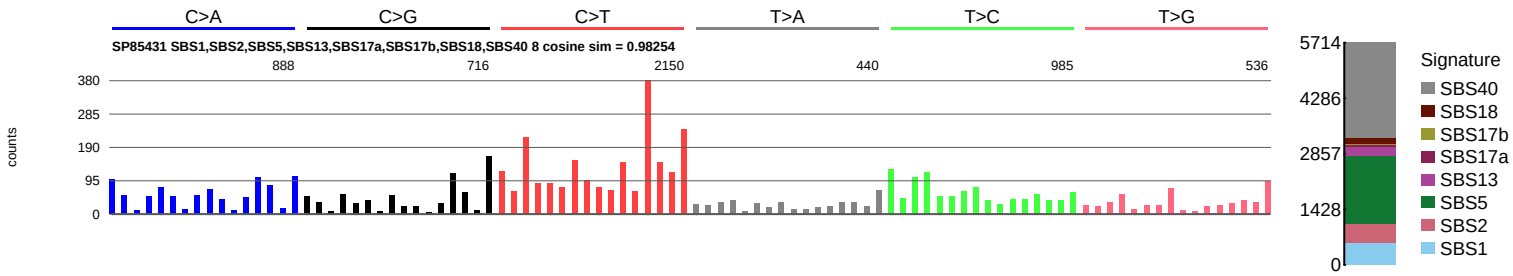
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS40,SBS44 8 cosine sim = 0.98256
853
721
2188
448
952
548
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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C>T
T>A
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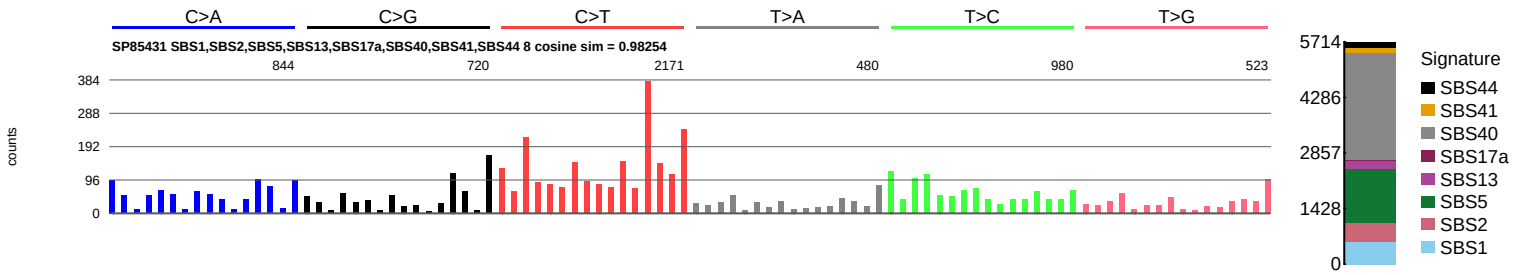
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98254
888
716
2150
440
985
536
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
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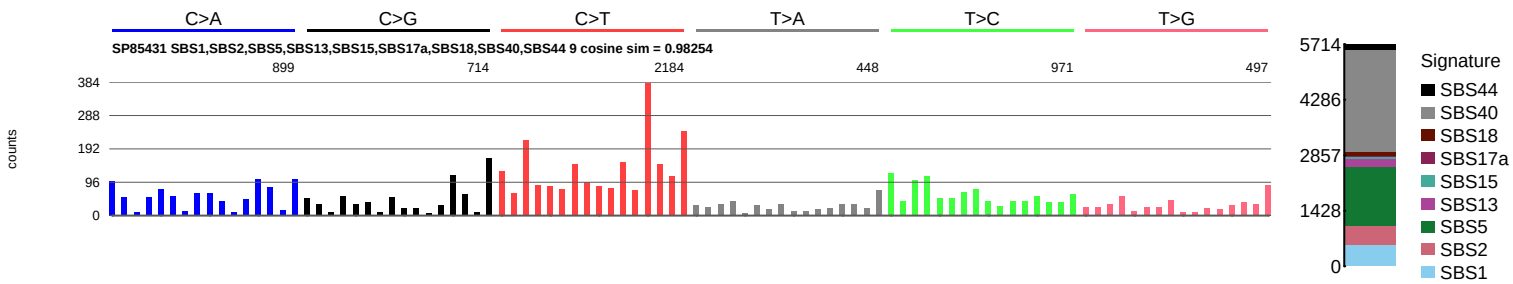
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS40,SBS41,SBS44 8 cosine sim = 0.98254
844
720
2171
480
980
523
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS44 9 cosine sim = 0.98254
899
714
2184
448
971
497
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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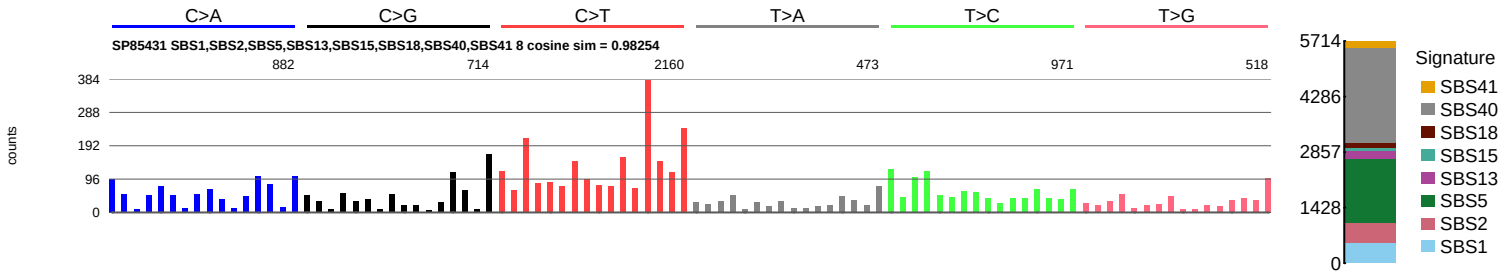
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS40,SBS41 8 cosine sim = 0.98254
882
714
2160
473
971
518
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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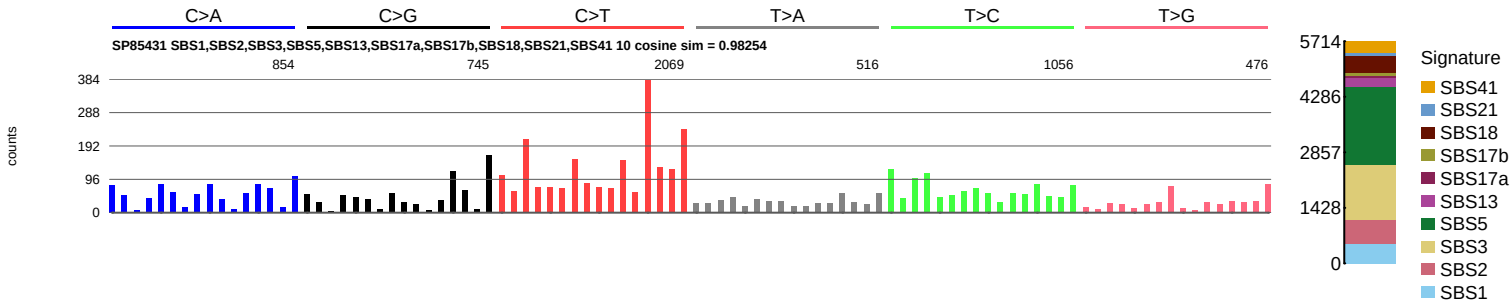
SBS2
SBS1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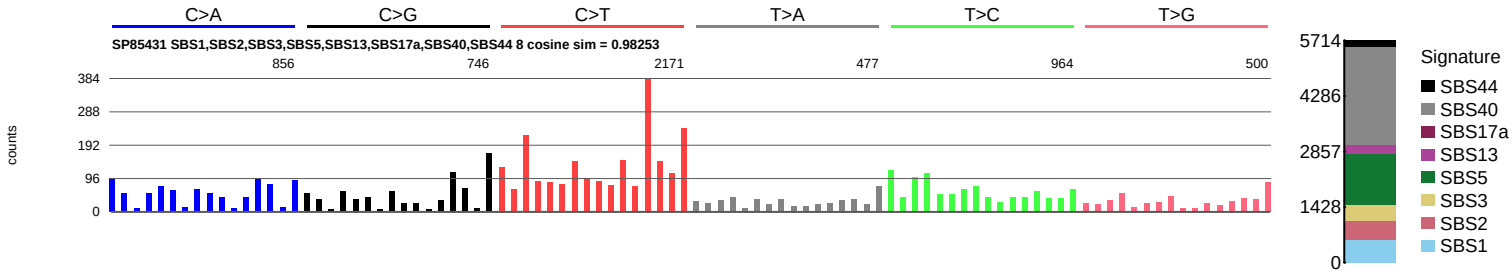
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98254
854
745
2069
516
1056
476
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS40,SBS44 8 cosine sim = 0.98253
856
746
2171
477
964
500
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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SBS17a
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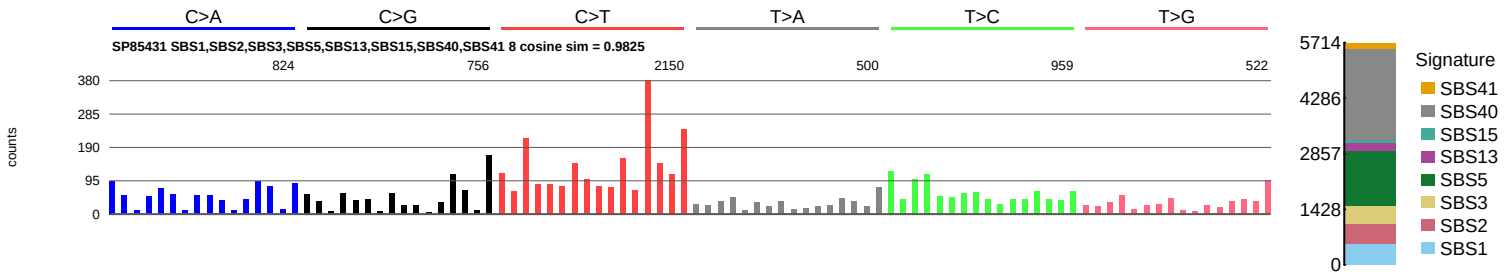
SBS2
SBS1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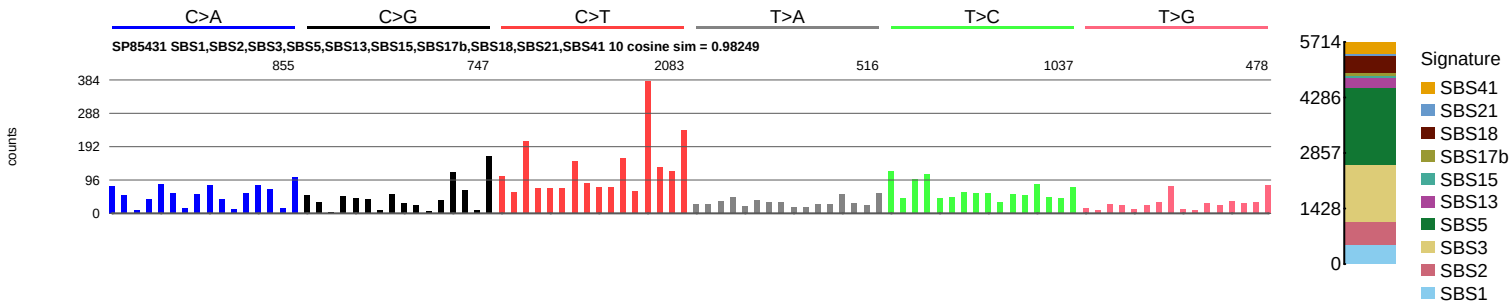
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS40,SBS41 8 cosine sim = 0.9825
824
756
2150
500
959
522
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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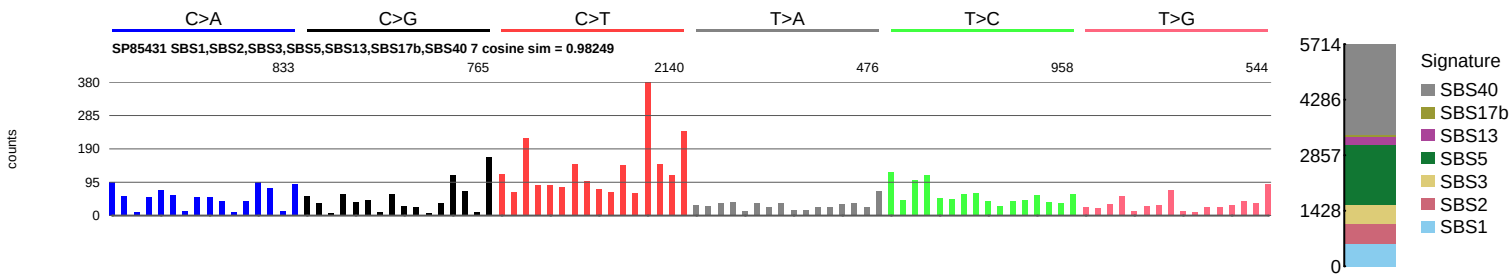
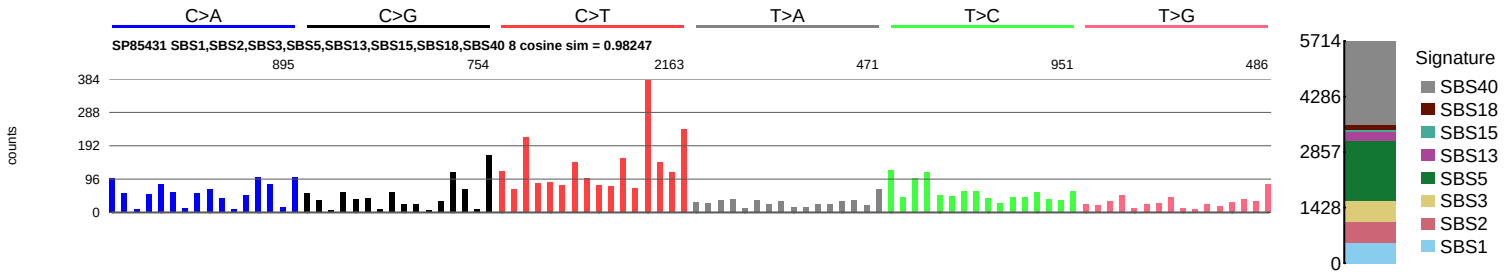
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98249
855
747
2083
516
1037
478
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS40 7 cosine sim = 0.98249
833
765
2140
476
958
544
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS40 8 cosine sim = 0.98247
895
754
2163
471
951
486
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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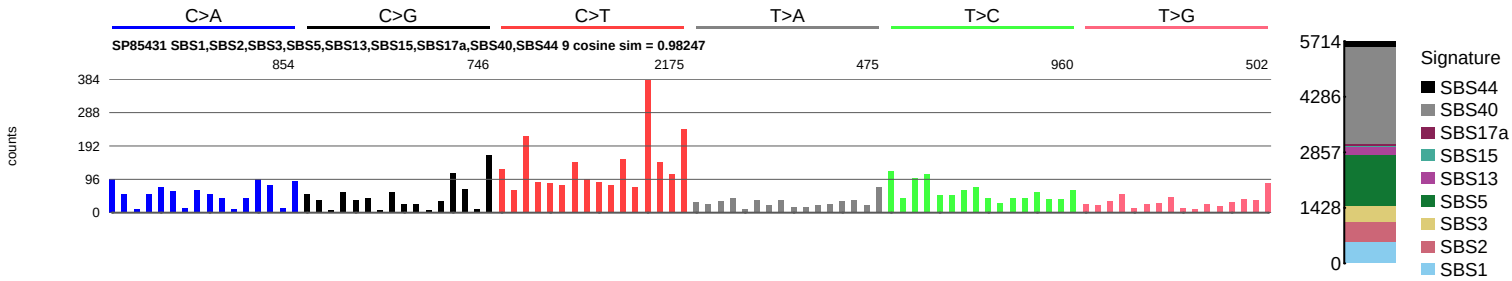
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS40,SBS44 9 cosine sim = 0.98247
854
746
2175
475
960
502
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS17a
SBS15
SBS13
SBS5
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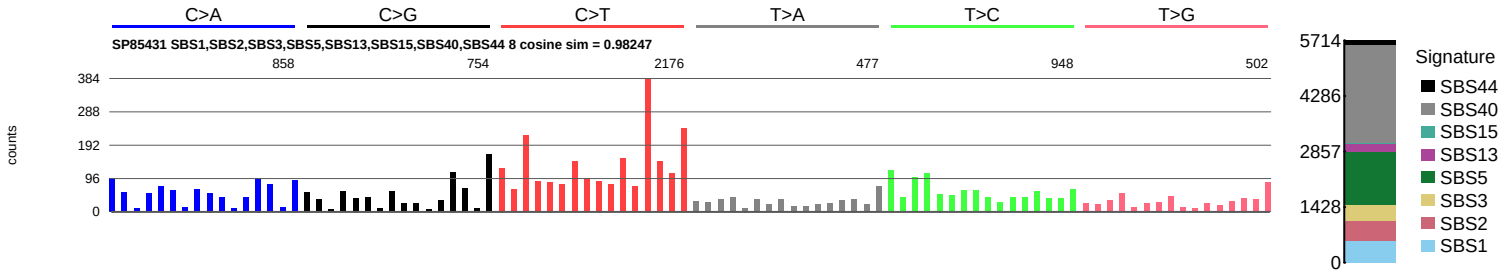
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS40,SBS44 8 cosine sim = 0.98247
858
754
2176
477
948
502
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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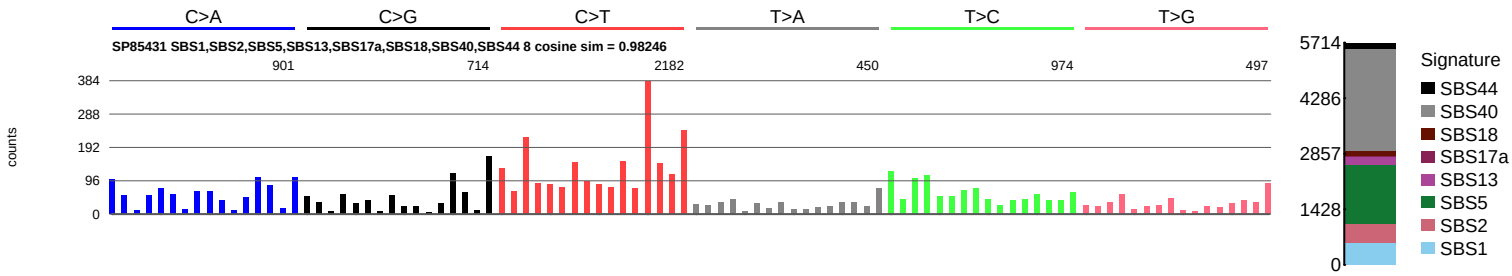
SBS2
SBS1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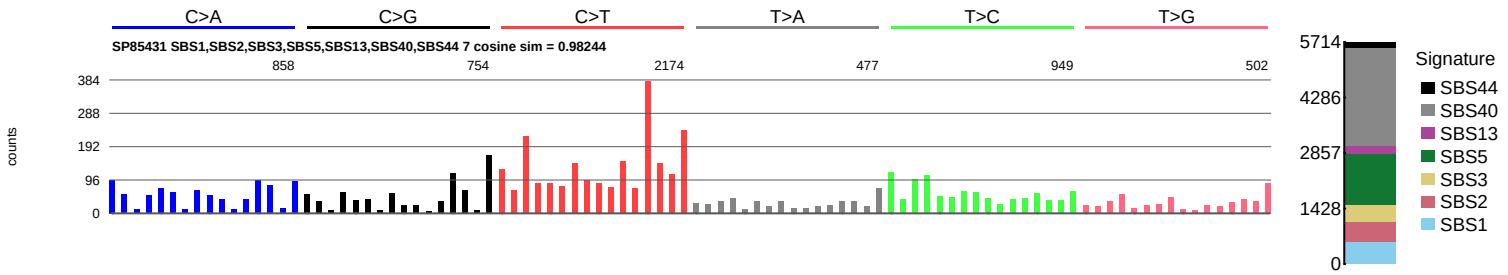
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS44 8 cosine sim = 0.98246
901
714
2182
450
974
497
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1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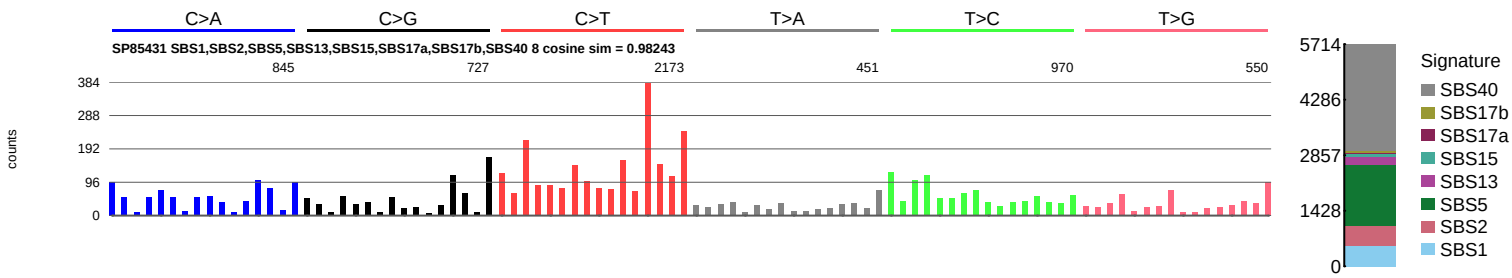
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS40,SBS44 7 cosine sim = 0.98244
858
754
2174
477
949
502
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS13
SBS5
SBS3
SBS2
SBS1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C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 8 cosine sim = 0.98243
845
727
2173
451
970
550
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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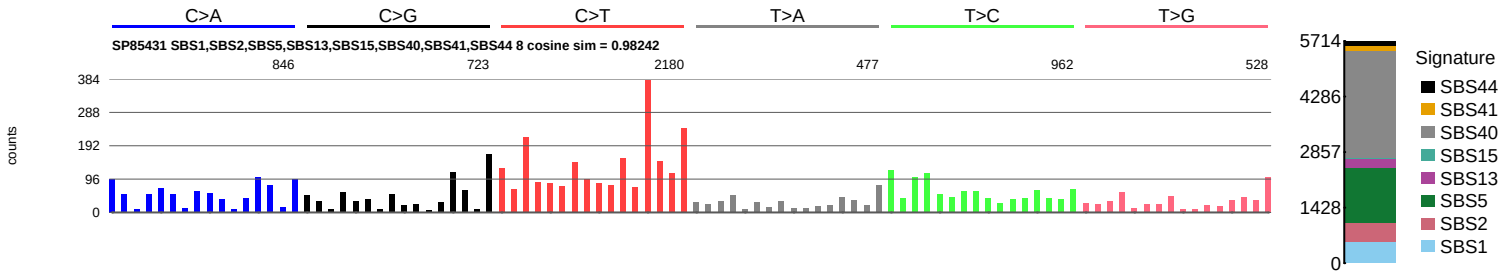
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS40,SBS41,SBS44 8 cosine sim = 0.98242
846
723
2180
477
962
528
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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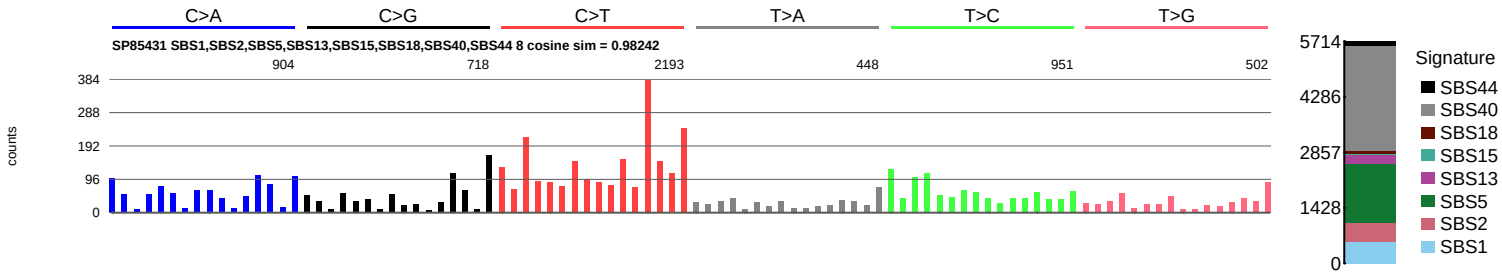
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS40,SBS44 8 cosine sim = 0.98242
904
718
2193
448
951
502
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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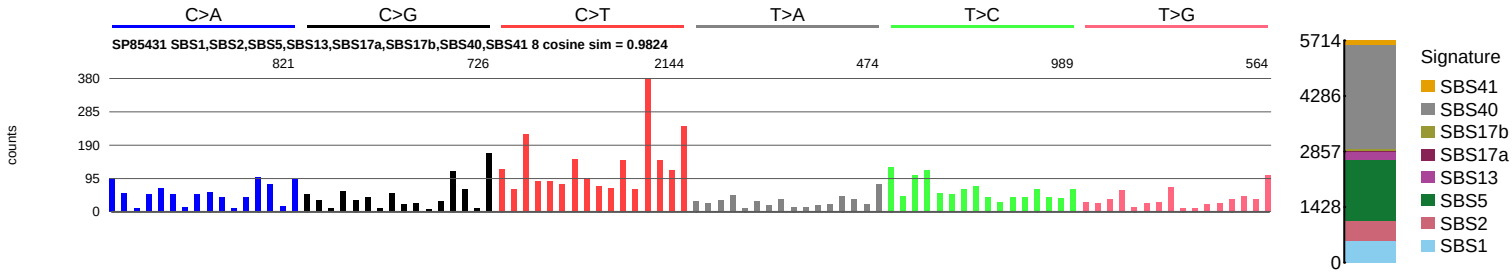
SBS2
SBS1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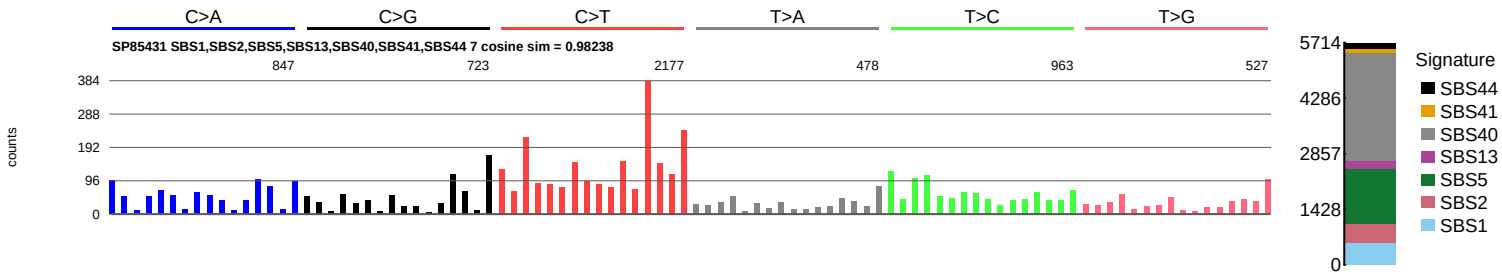
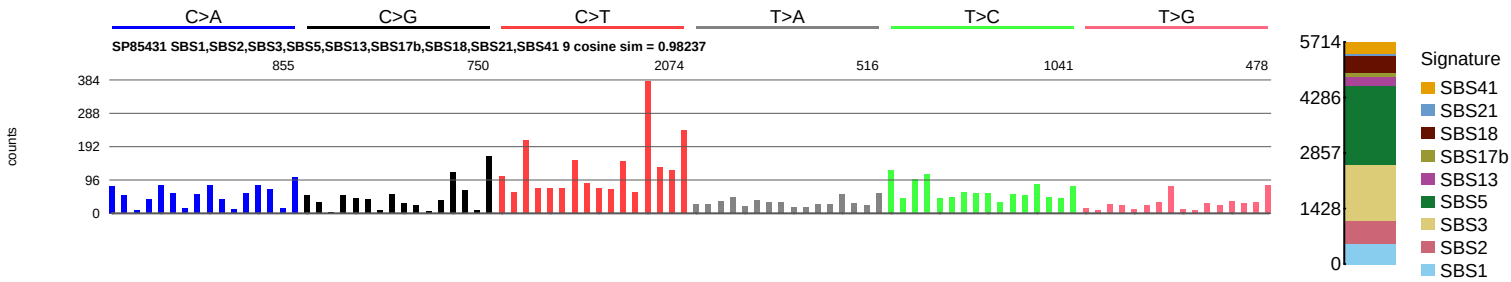
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.9824
821
726
2144
474
989
564
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS40,SBS41,SBS44 7 cosine sim = 0.98238
847
723
2177
478
963
527
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98237
855
750
2074
516
1041
478
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
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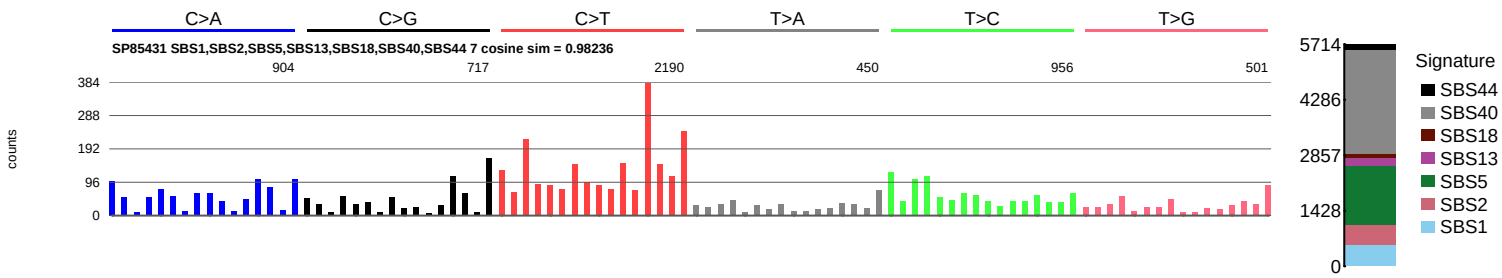
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40,SBS44 7 cosine sim = 0.98236
904
717
2190
450
956
501
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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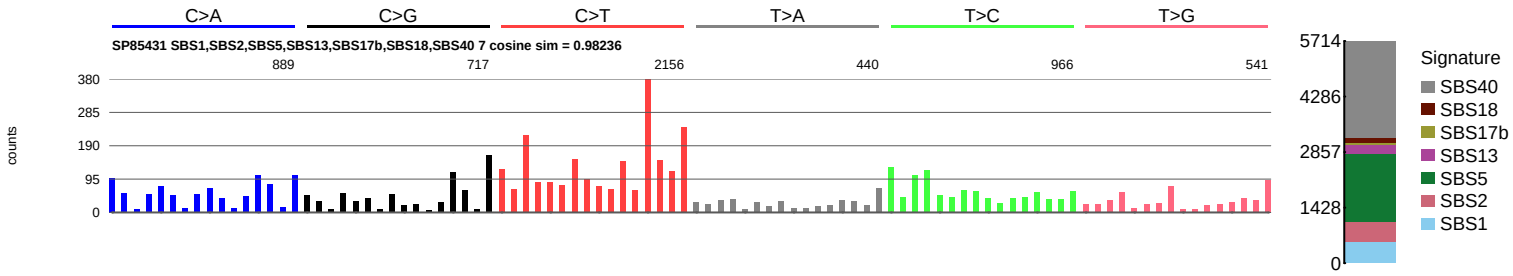
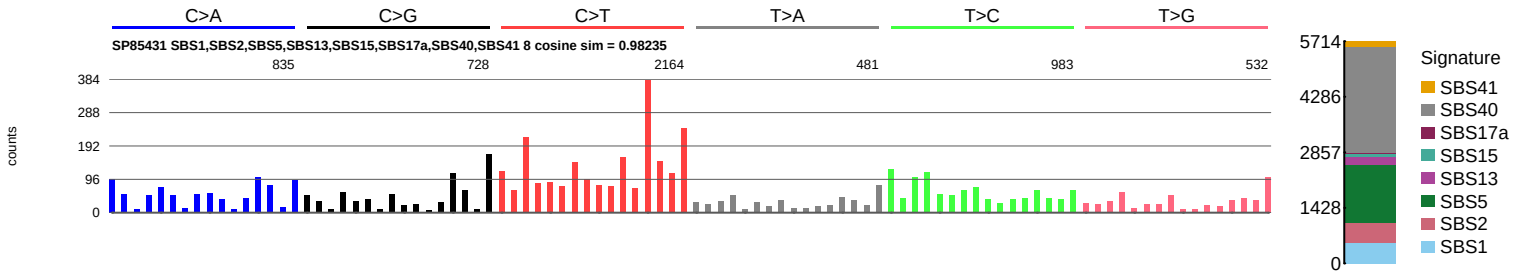
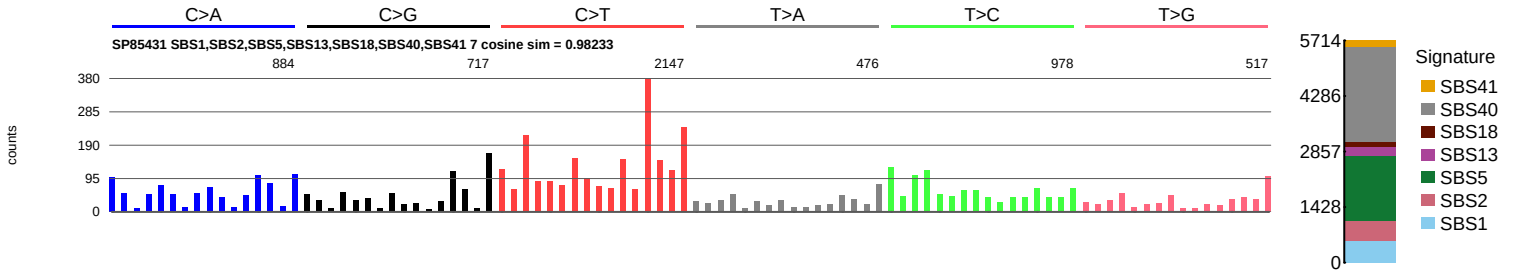
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40 7 cosine sim = 0.98236
889
717
2156
440
966
541
380
285
190
95
0
counts
5714
4286
2857
1428
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS40,SBS41 8 cosine sim = 0.98235
835
728
2164
481
983
532
384
288
192
96
0
counts
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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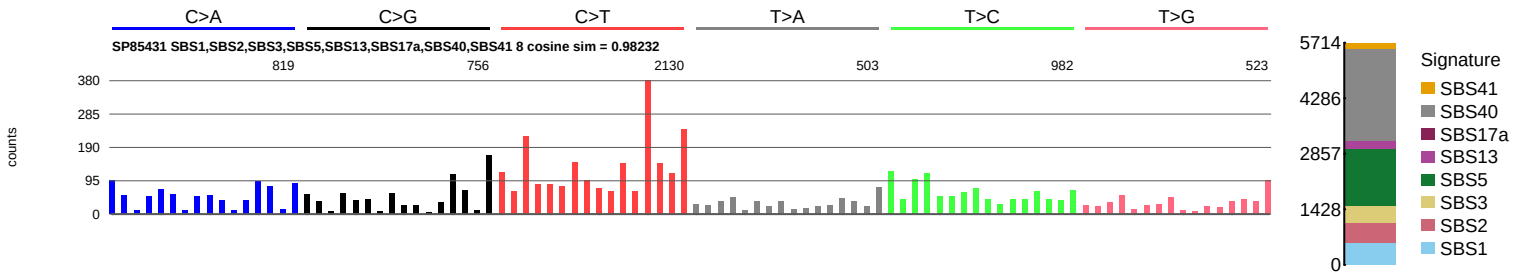
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40,SBS41 7 cosine sim = 0.98233
884
717
2147
476
978
517
380
285
190
95
0
counts
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS18
SBS13
SBS5
SBS2
SBS1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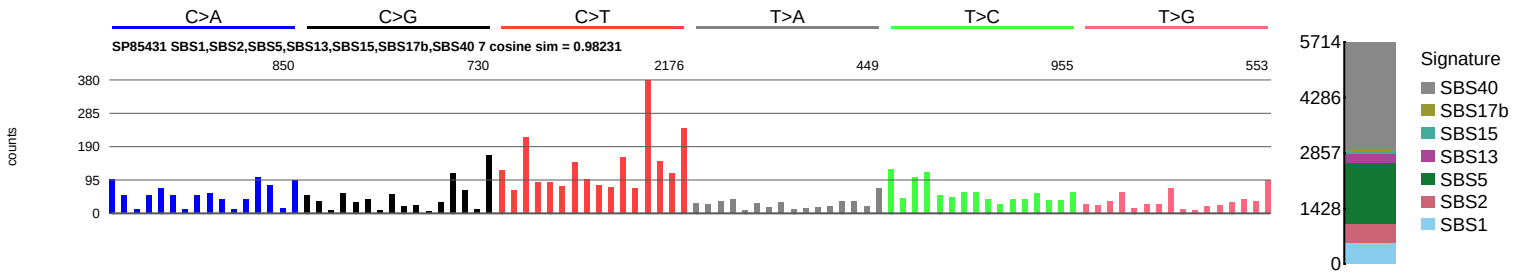
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS40,SBS41 8 cosine sim = 0.98232
819
756
2130
503
982
523
380
285
190
95
0
counts
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS40 7 cosine sim = 0.98231
850
730
2176
449
955
553
380
285
190
95
0
counts
5714
4286
2857
1428
0
Signature
SBS40
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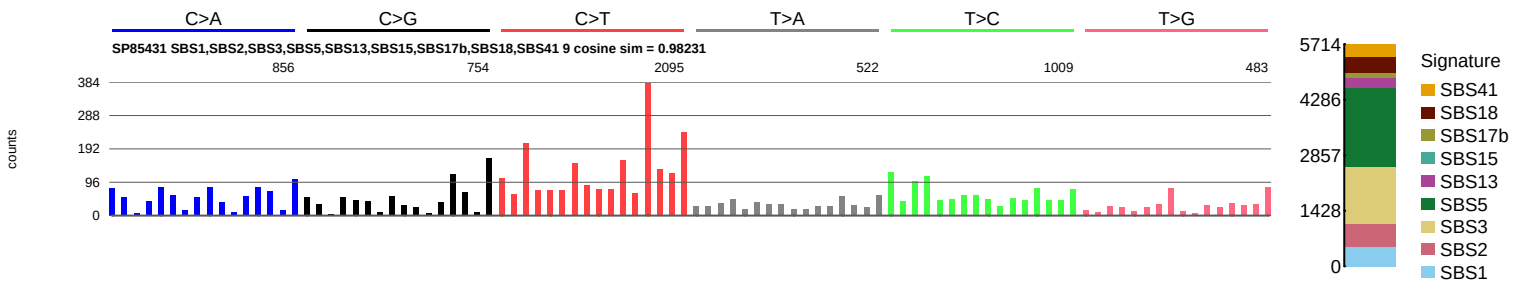
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 9 cosine sim = 0.98231
856
754
2095
522
1009
483
384
288
192
96
0
counts
5714
4286
2857
1428
0
Signature
SBS41
SBS18
SBS17b
SBS15
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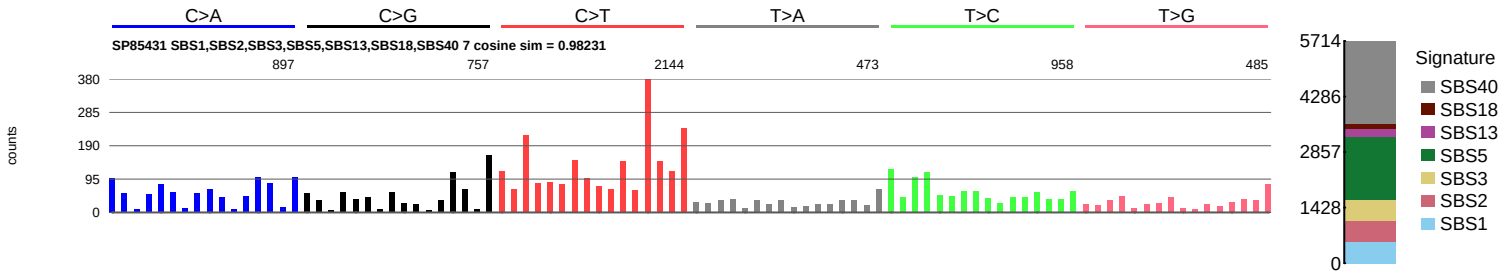
SBS3
SBS2
SBS1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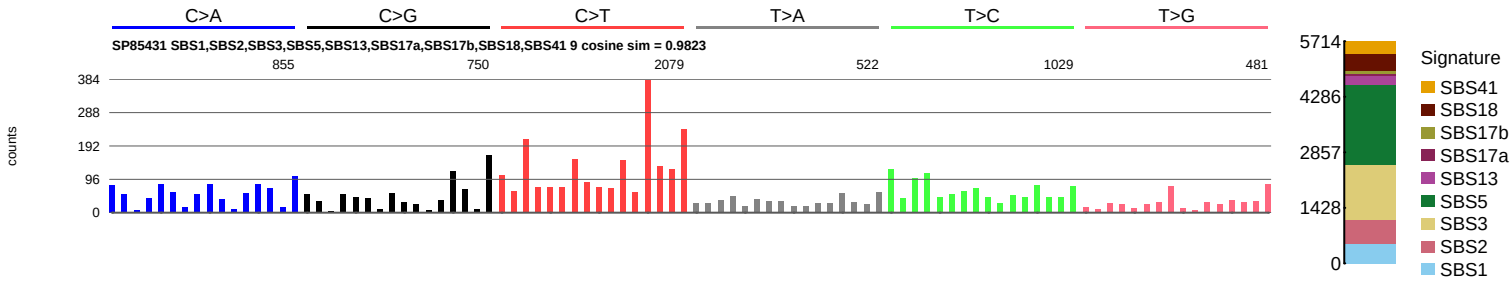
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS40 7 cosine sim = 0.98231
897
757
2144
473
958
485
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1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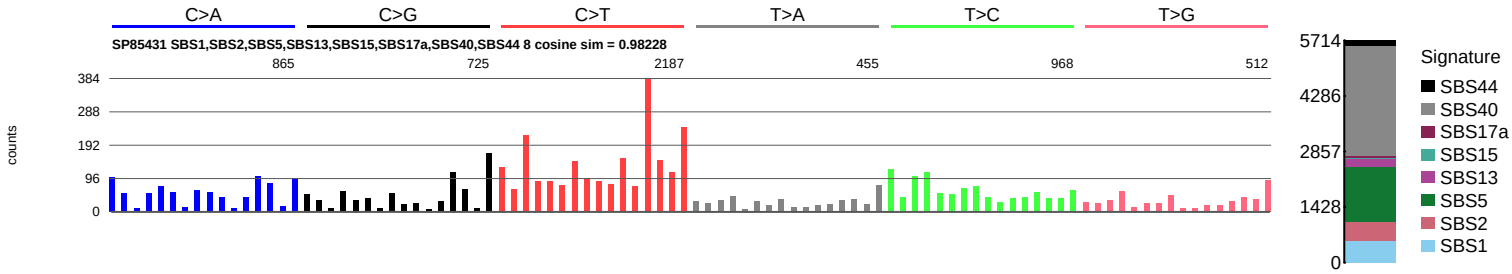
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.9823
855
750
2079
522
1029
481
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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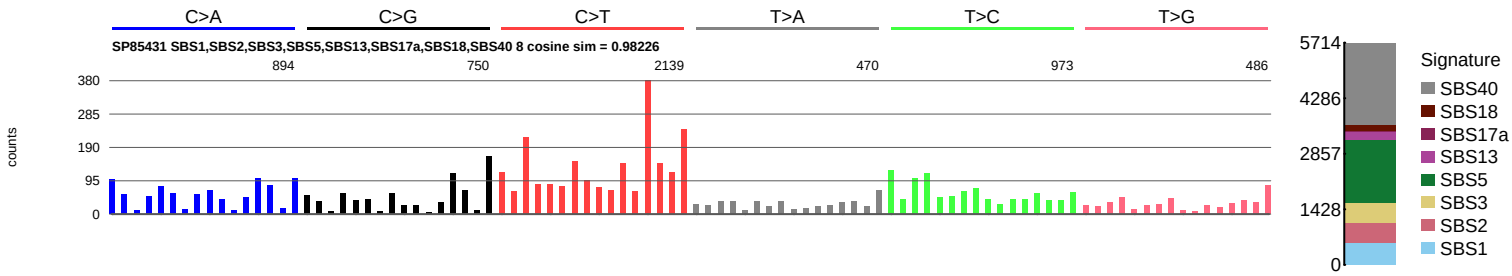
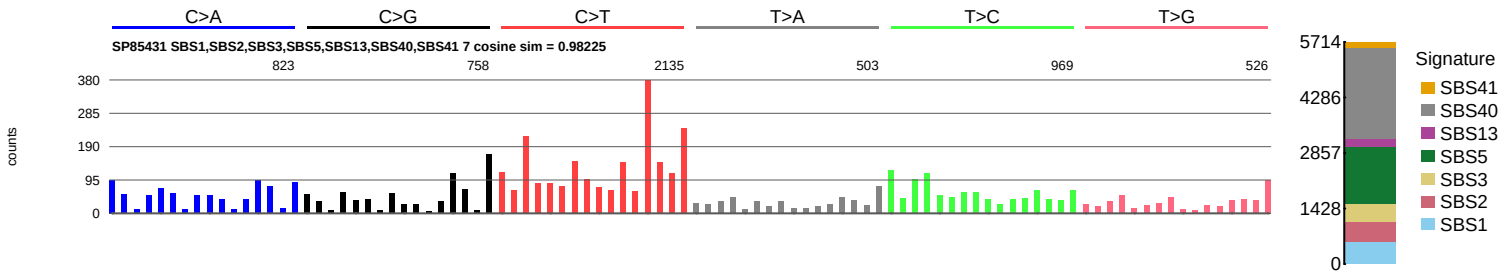
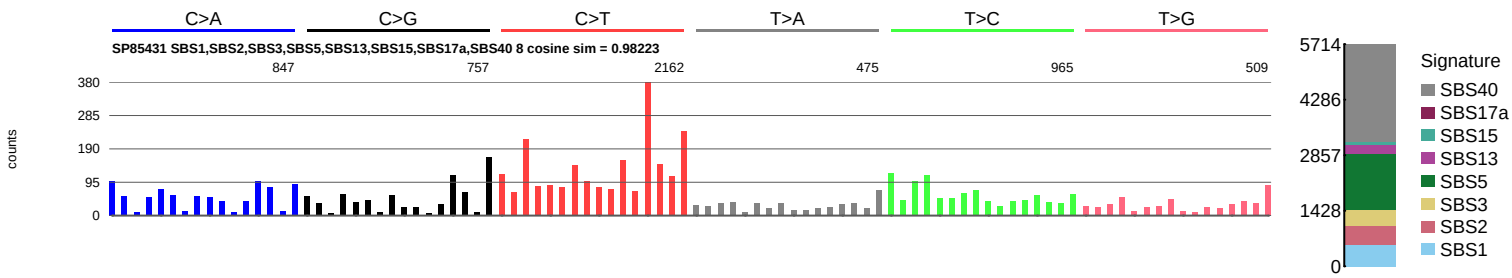
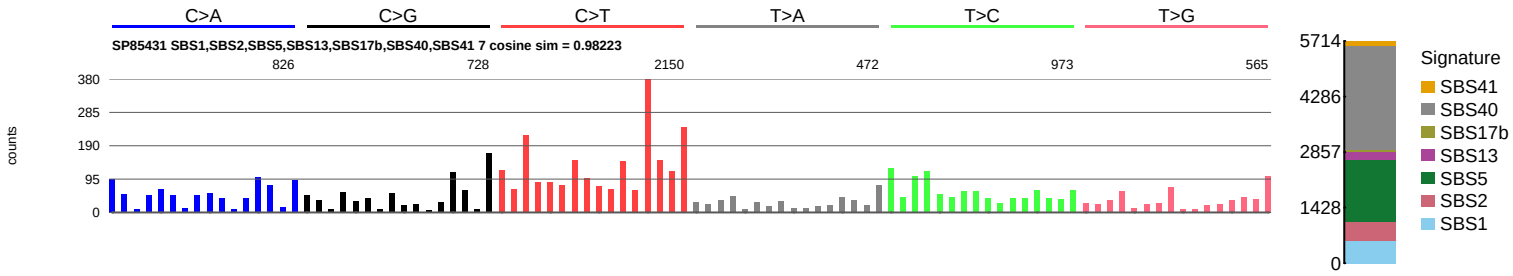
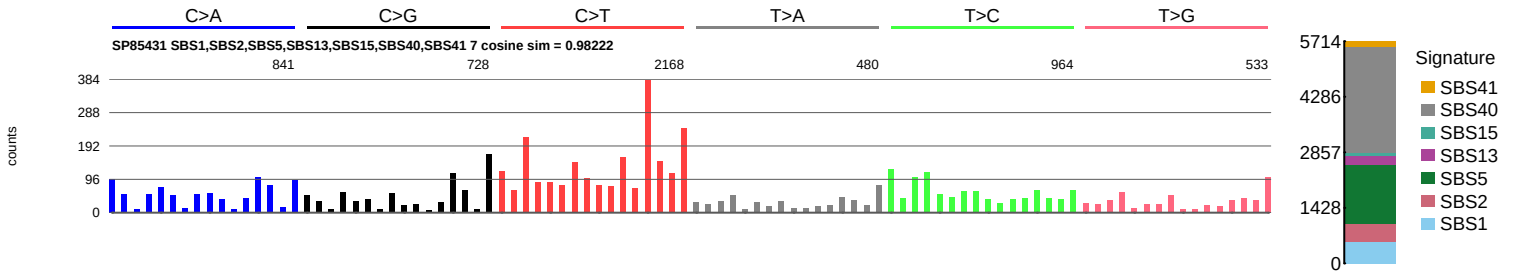
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS40,SBS44 8 cosine sim = 0.98228
865
725
2187
455
968
512
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

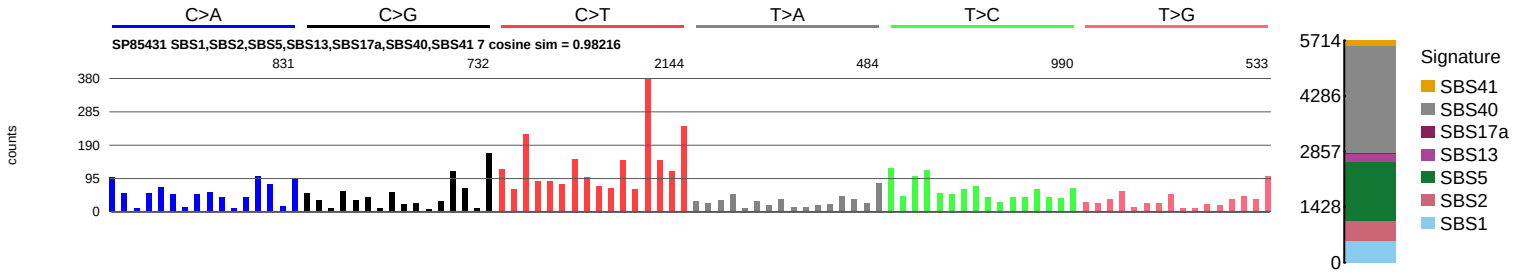

C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS40 8 cosine sim = 0.98226
894
750
2139
470
973
486
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

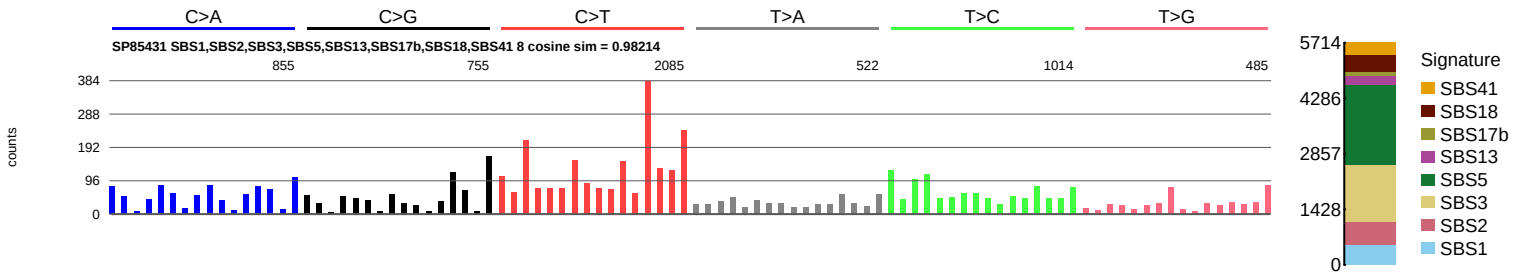

C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS40,SBS41 7 cosine sim = 0.98225
823
758
2135
503
969
526
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS40 8 cosine sim = 0.98223
847
757
2162
475
965
509
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS40,SBS41 7 cosine sim = 0.98223
826
728
2150
472
973
565
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS40,SBS41 7 cosine sim = 0.98222
841
728
2168
480
964
533
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS40,SBS41 7 cosine sim = 0.98216
831
732
2144
484
990
533
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41 8 cosine sim = 0.98214
855
755
2085
522
1014
485
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS18
SBS17b
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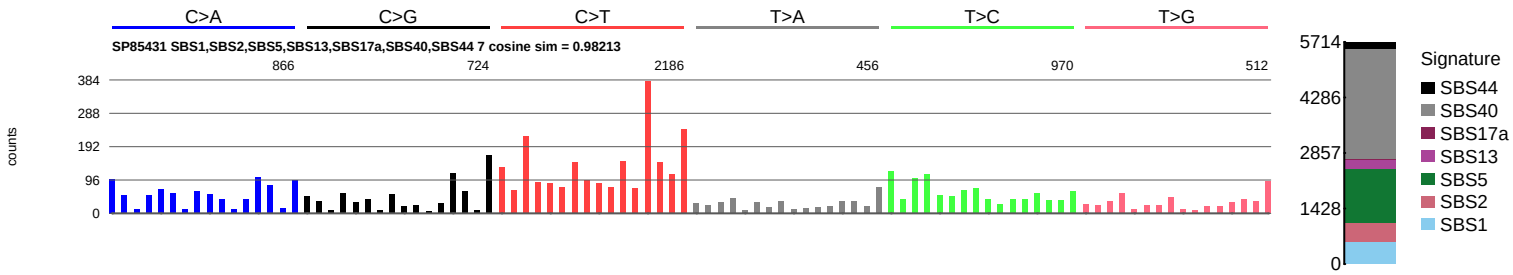
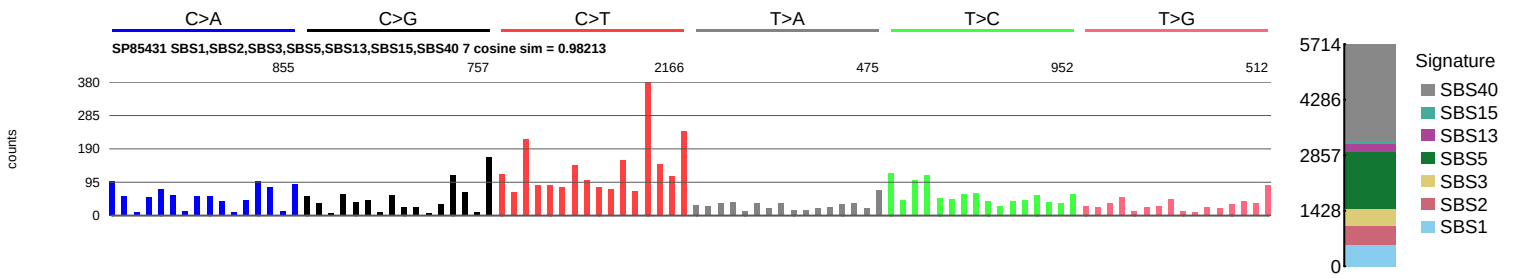
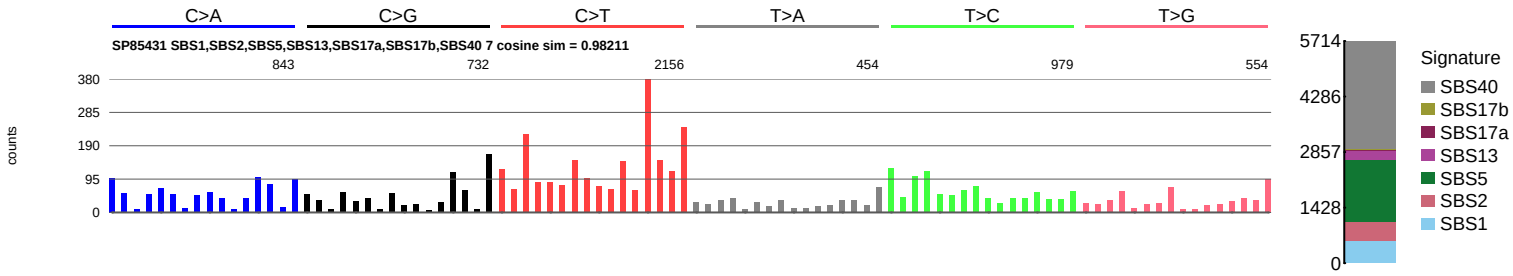
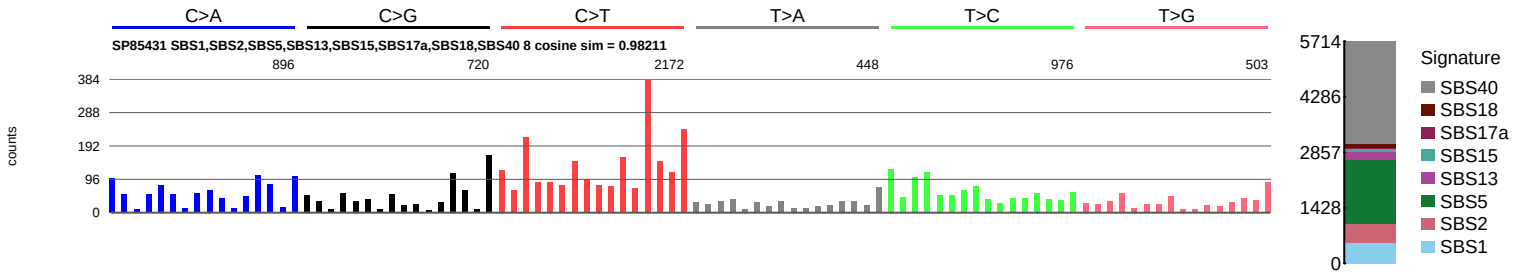
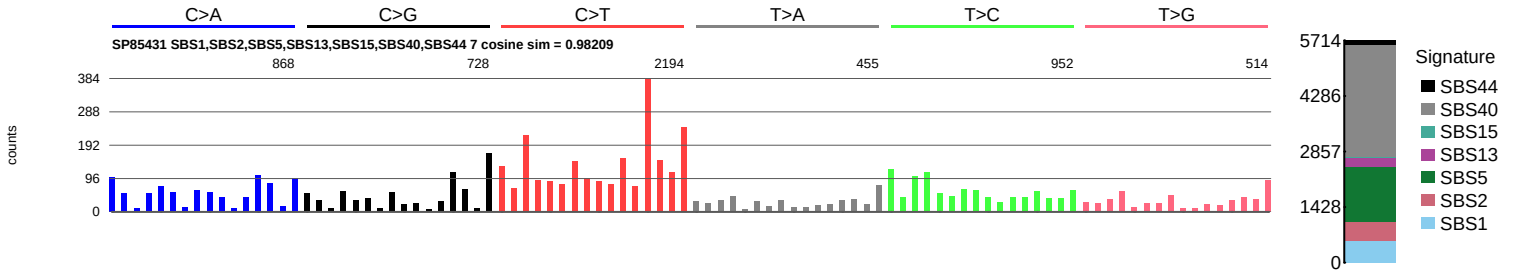
SBS3
SBS2
SBS1

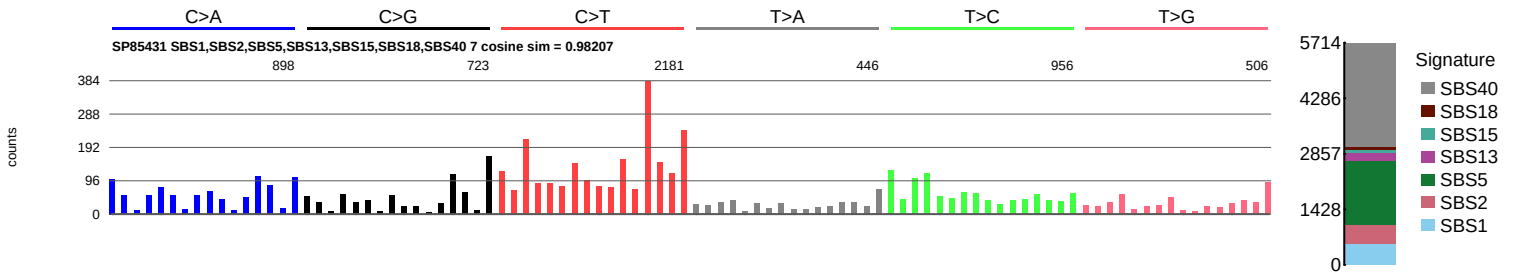

C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS40,SBS44 7 cosine sim = 0.98213
866
724
2186
456
970
512
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS40 7 cosine sim = 0.98213
855
757
2166
475
952
512
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98211
843
732
2156
454
979
554
380
285
190
95
0
counts
5714
4286
2857
1428
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40 8 cosine sim = 0.98211
896
720
2172
448
976
503
384
288
192
96
0
counts
5714
4286
2857
1428
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS40,SBS44 7 cosine sim = 0.98209
868
728
2194
455
952
514
384
288
192
96
0
counts
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS40 7 cosine sim = 0.98207
898
723
2181
446
956
506
384
288
192
96
0
counts
5714
4286
2857
1428
0
Signature
SBS40
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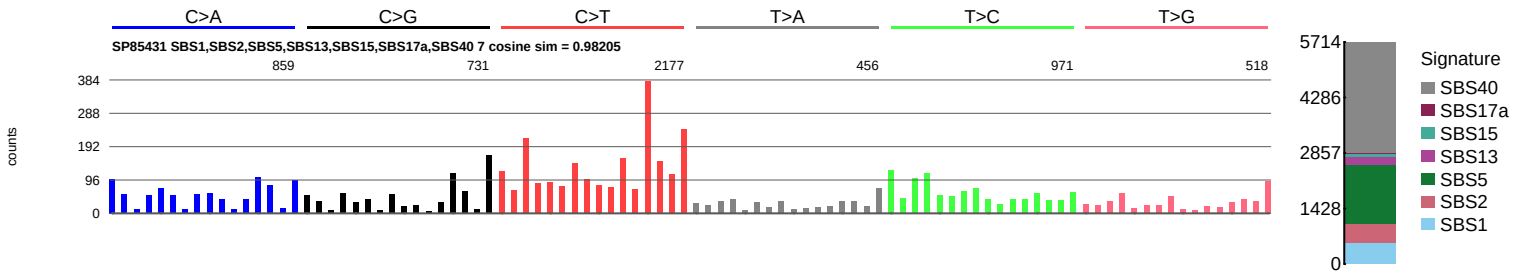
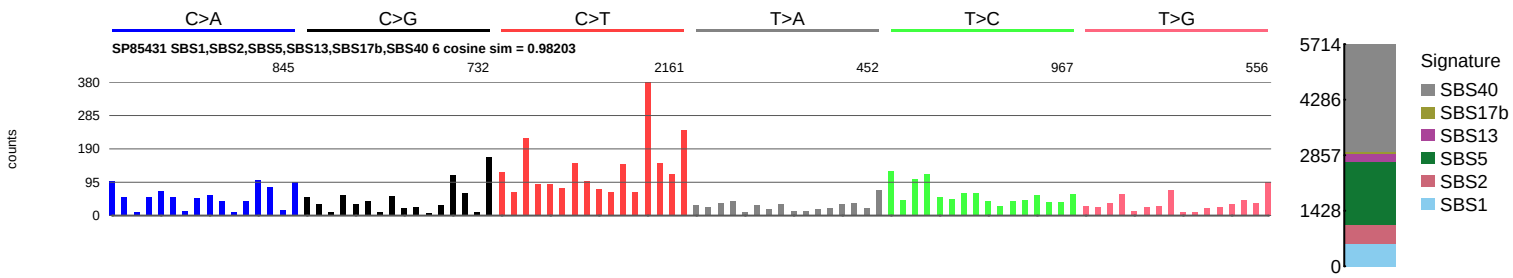
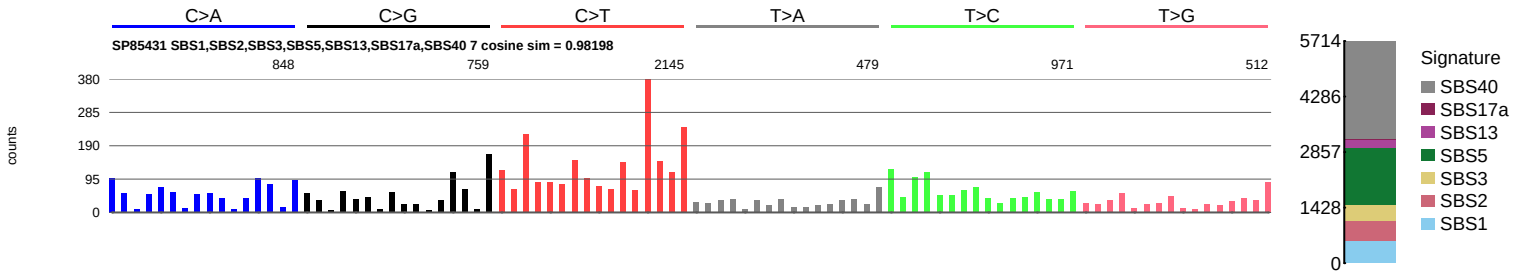
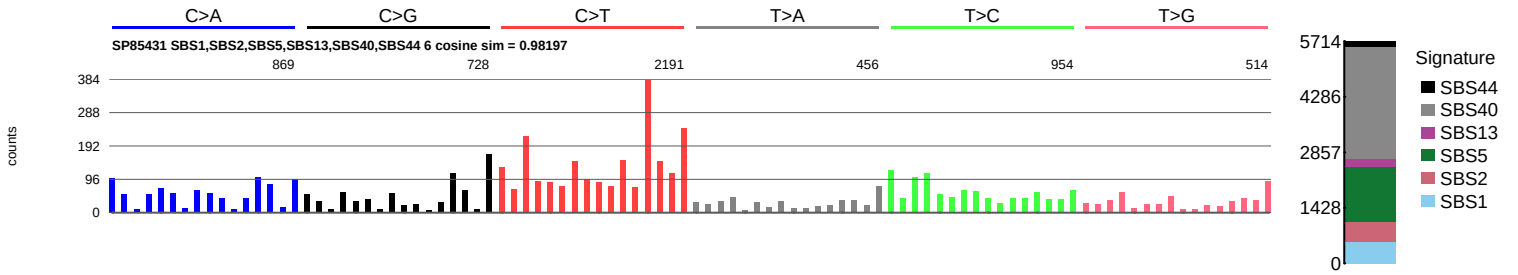
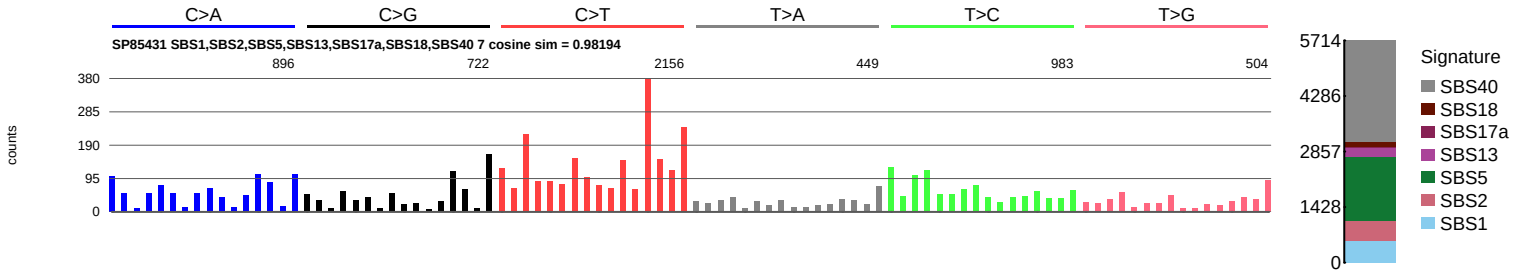
SBS2
SBS1

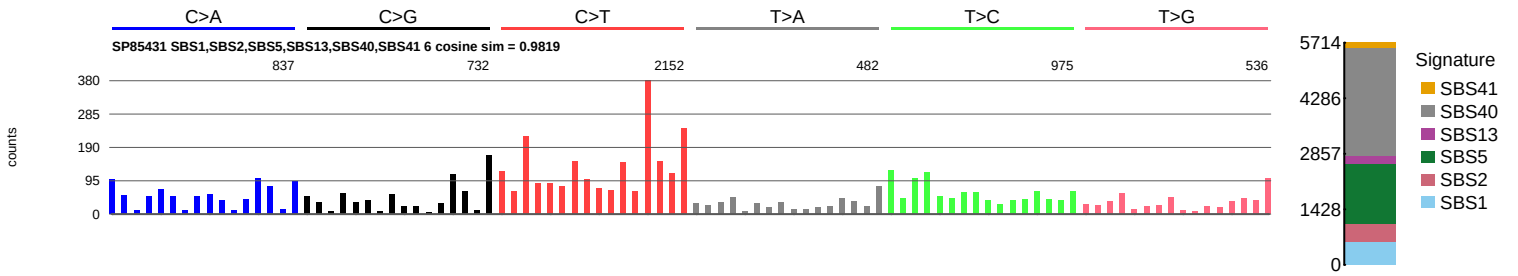

C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS40 7 cosine sim = 0.98205
859
731
2177
456
971
518
384
288
192
96
0
counts
5714
4286
2857
1428
0
Signature
SBS40
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

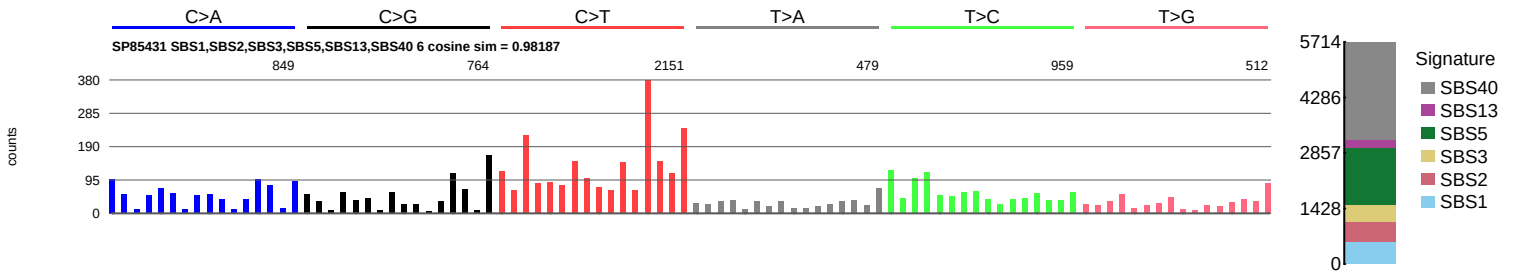

C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS40 6 cosine sim = 0.98203
845
732
2161
452
967
556
380
285
190
95
0
counts
5714
4286
2857
1428
0
Signature
SBS40
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS40 7 cosine sim = 0.98198
848
759
2145
479
971
512
380
285
190
95
0
counts
5714
4286
2857
1428
0
Signature
SBS40
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS40,SBS44 6 cosine sim = 0.98197
869
728
2191
456
954
514
384
288
192
96
0
counts
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40 7 cosine sim = 0.98194
896
722
2156
449
983
504
380
285
190
95
0
counts
5714
4286
2857
1428
0
Signature
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS40,SBS41 6 cosine sim = 0.9819
837
732
2152
482
975
536
380
285
190
95
0
counts
5714
4286
2857
1428
0
Signature
SBS41
SBS40
SBS13
SBS5
SBS2
SBS1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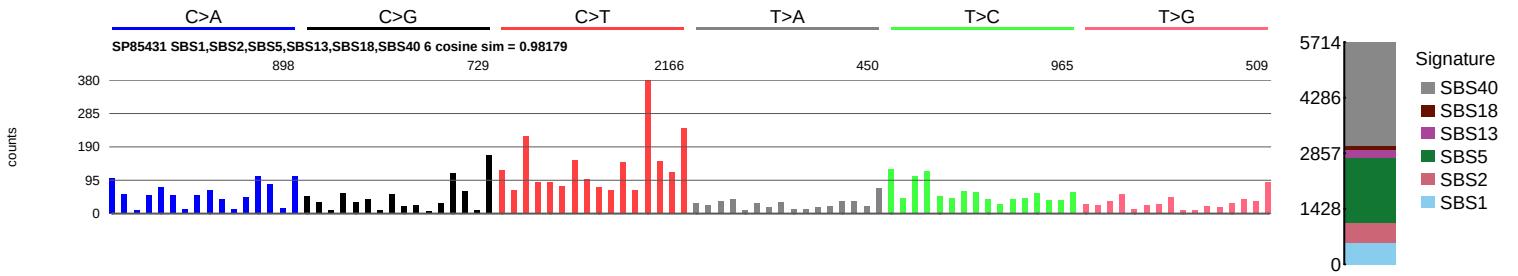
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS40 6 cosine sim = 0.98187
849
764
2151
479
959
512
380
285
190
95
0
counts
5714
4286
2857
1428
0
Signature
SBS40
SBS13
SBS5
SBS3
SBS2
SBS1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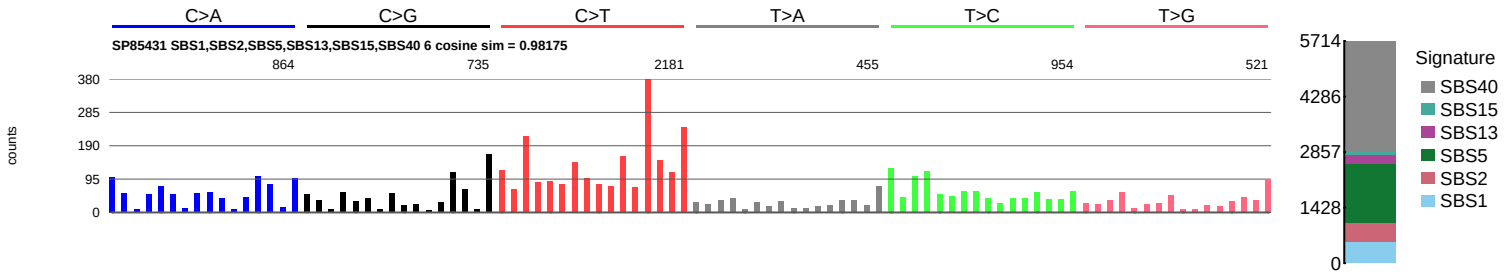
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40 6 cosine sim = 0.98179
898
729
2166
450
965
509
380
285
190
95
0
counts
5714
4286
2857
1428
0
Signature
SBS40
SBS18
SBS13
SBS5
SBS2
SBS1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C>G
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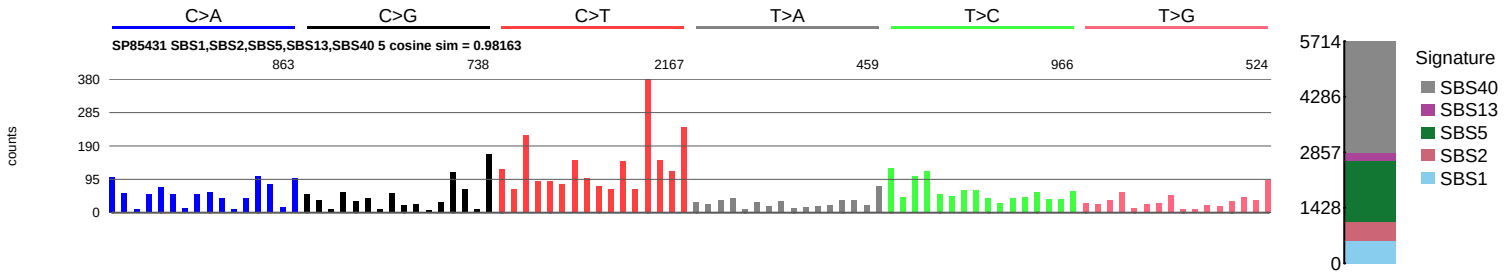
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS15,SBS40 6 cosine sim = 0.98175
864
735
2181
455
954
521
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
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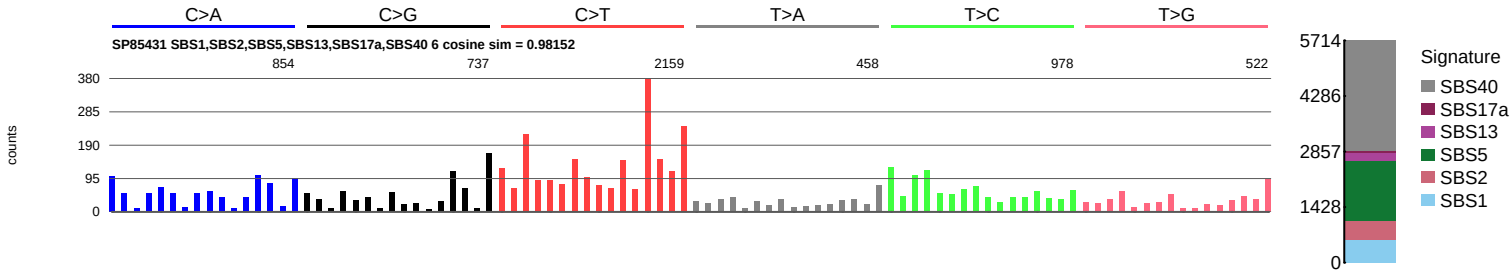
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS40 5 cosine sim = 0.98163
863
738
2167
459
966
524
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
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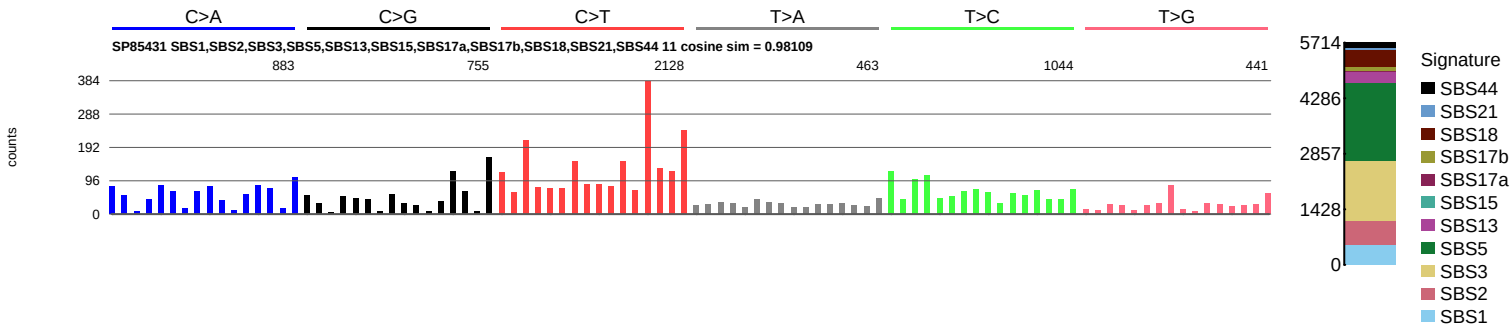
T>C
T>G
SP85431 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS40 6 cosine sim = 0.98152
854
737
2159
458
978
522
counts
380
285
190
95
0
5714
4286
2857
1428
0
Signature
SBS40
SBS17a
SBS13
SBS5
SBS2
SBS1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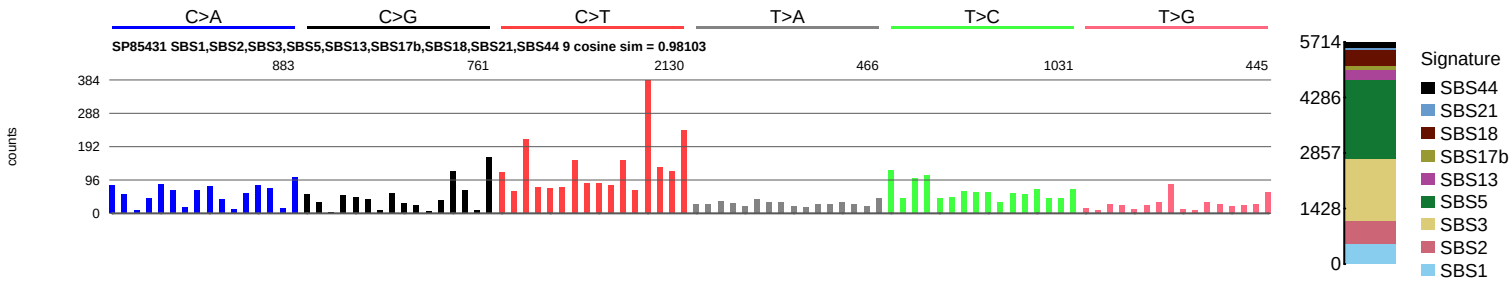
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 11 cosine sim = 0.98109
883
755
2128
463
1044
441
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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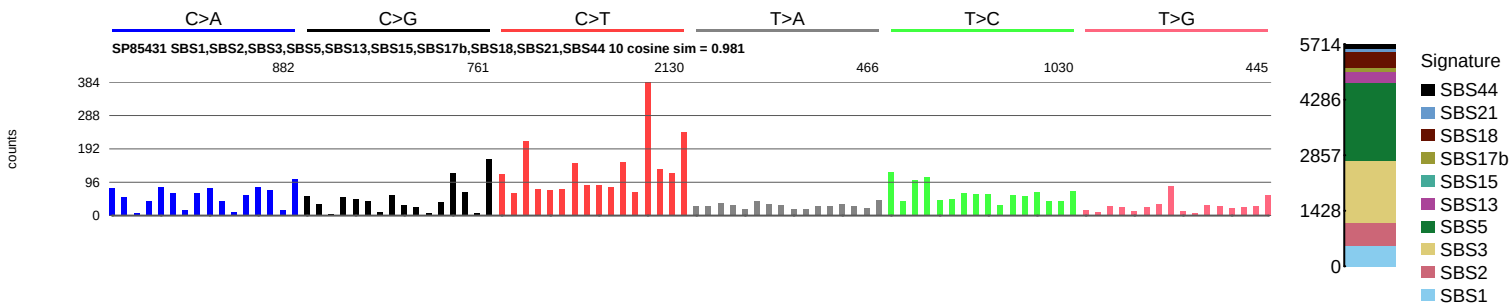
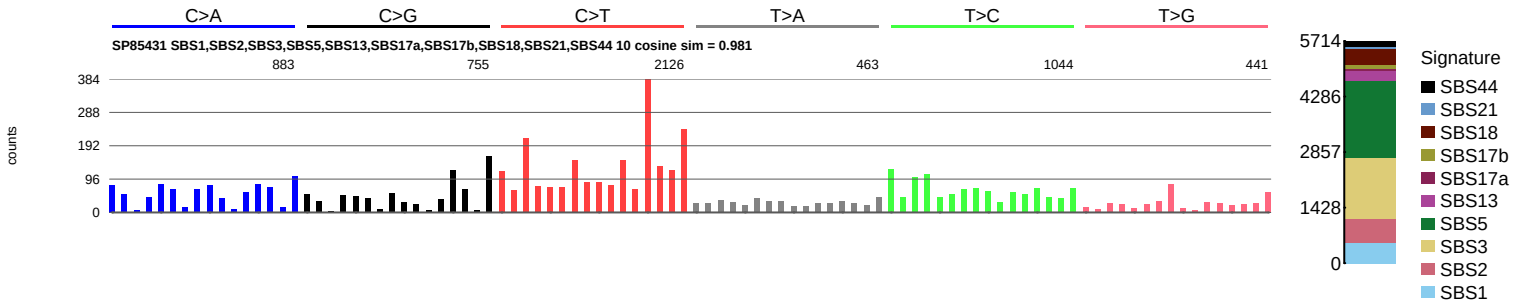
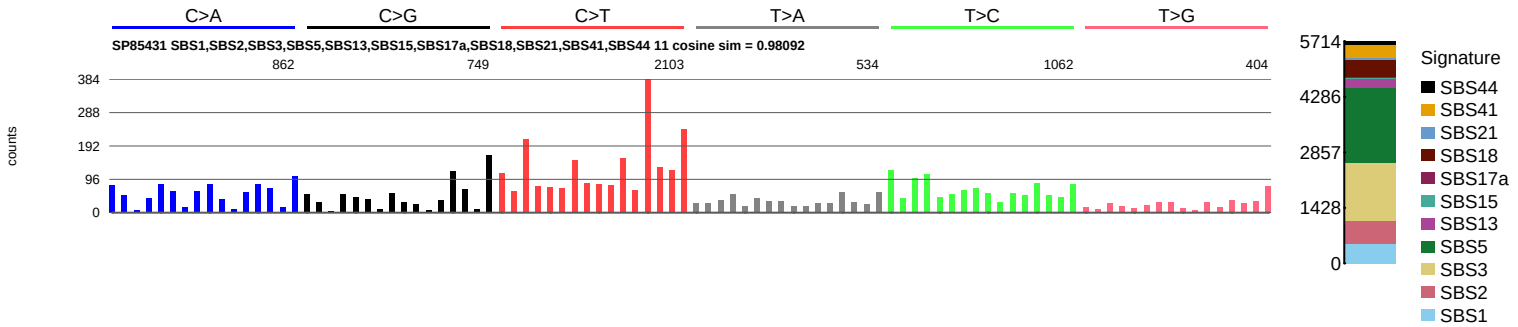
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98103
883
761
2130
466
1031
445
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.981
882
761
2130
466
1030
445
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.981
883
755
2126
463
1044
441
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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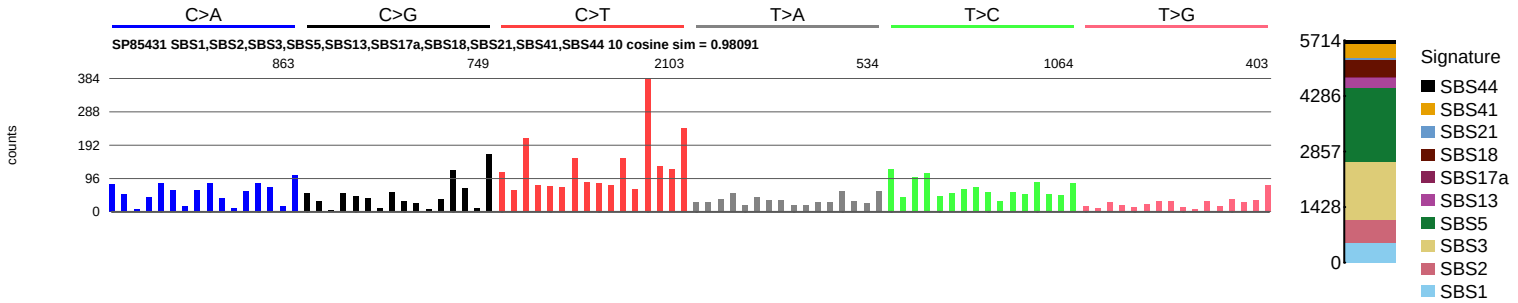
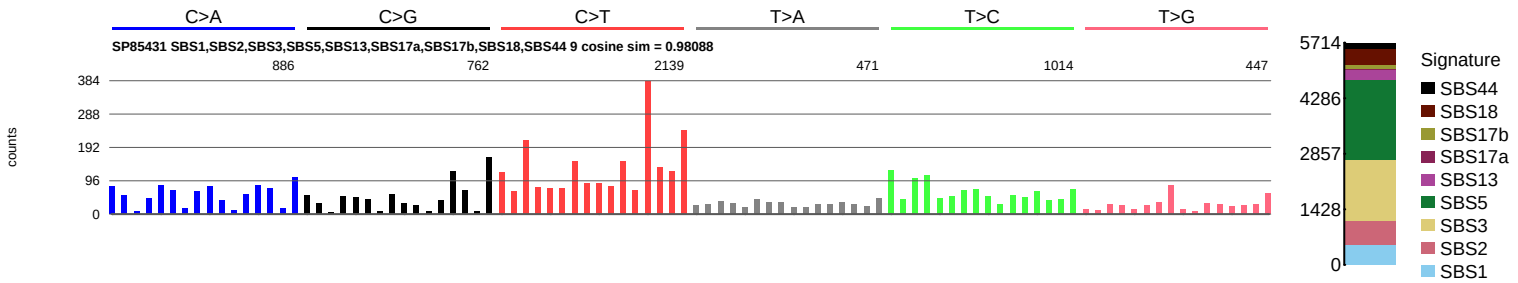
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98092
862
749
2103
534
1062
404
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98091
863
749
2103
534
1064
403
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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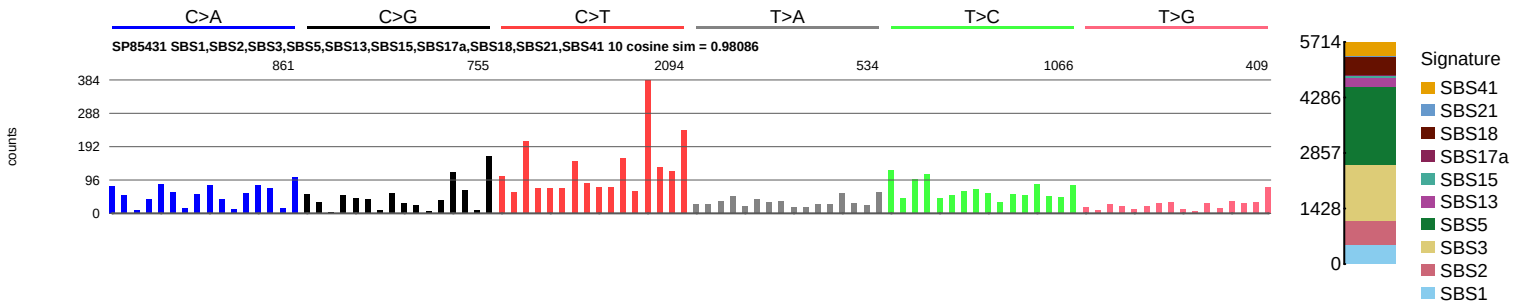
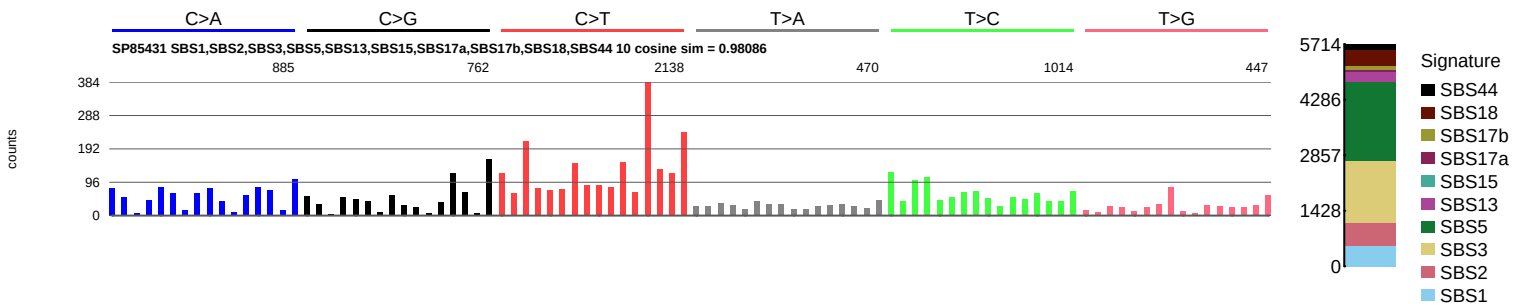
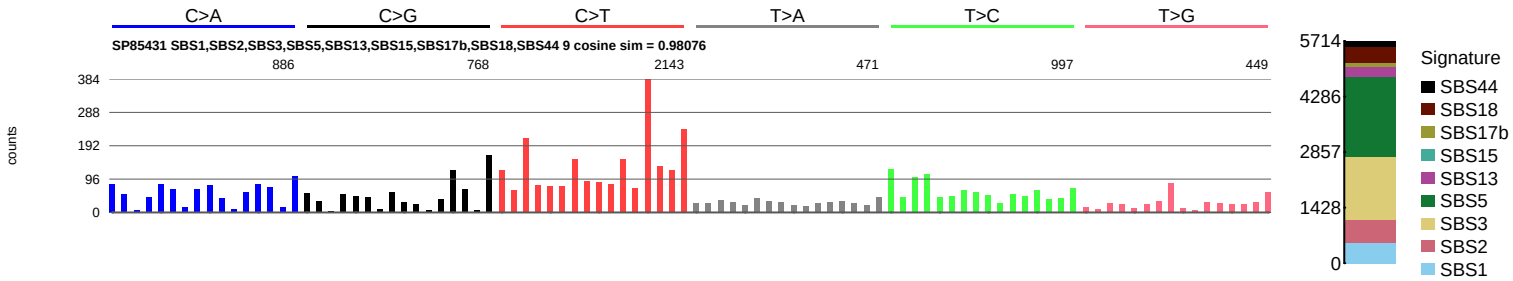
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98088
886
762
2139
471
1014
447
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS41 10 cosine sim = 0.98086
861
755
2094
534
1066
409
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 10 cosine sim = 0.98086
885
762
2138
470
1014
447
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS44 9 cosine sim = 0.98076
886
768
2143
471
997
449
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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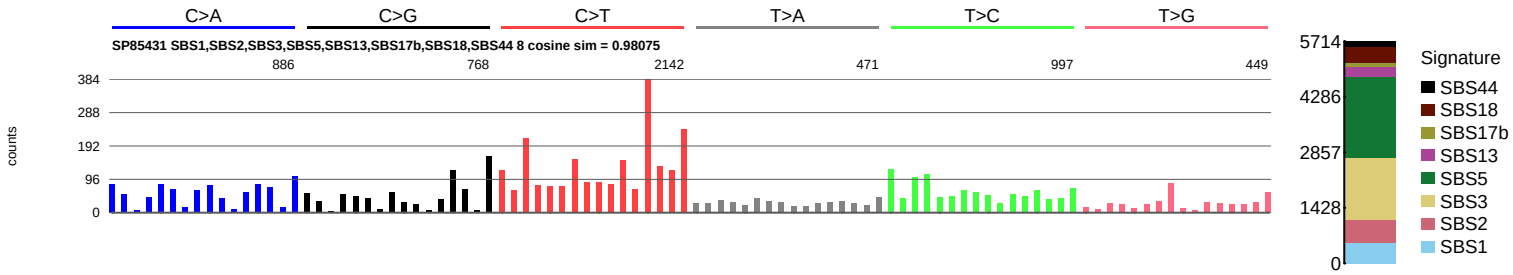
SBS2
SBS1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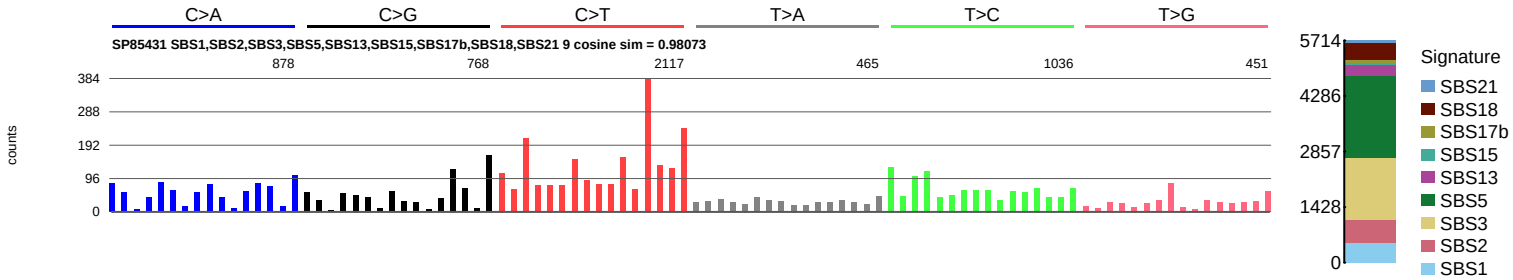
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS44 8 cosine sim = 0.98075
886
768
2142
471
997
449
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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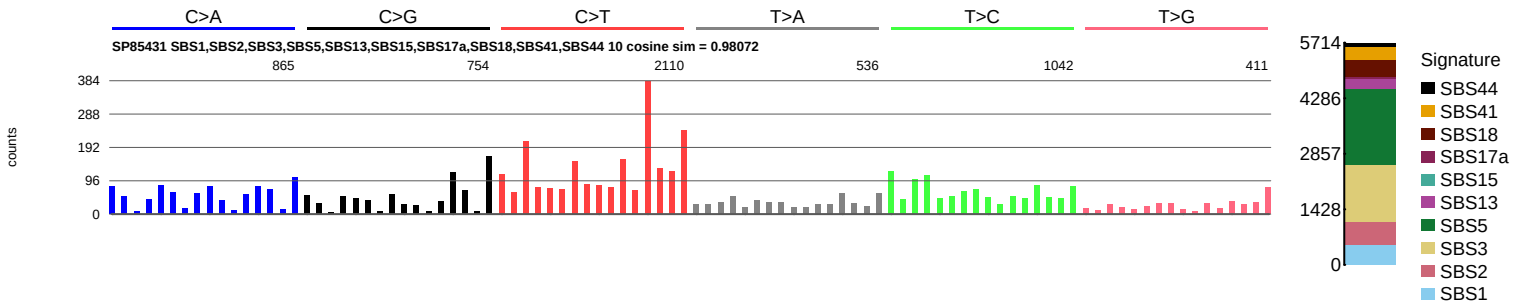
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21 9 cosine sim = 0.98073
878
768
2117
465
1036
451
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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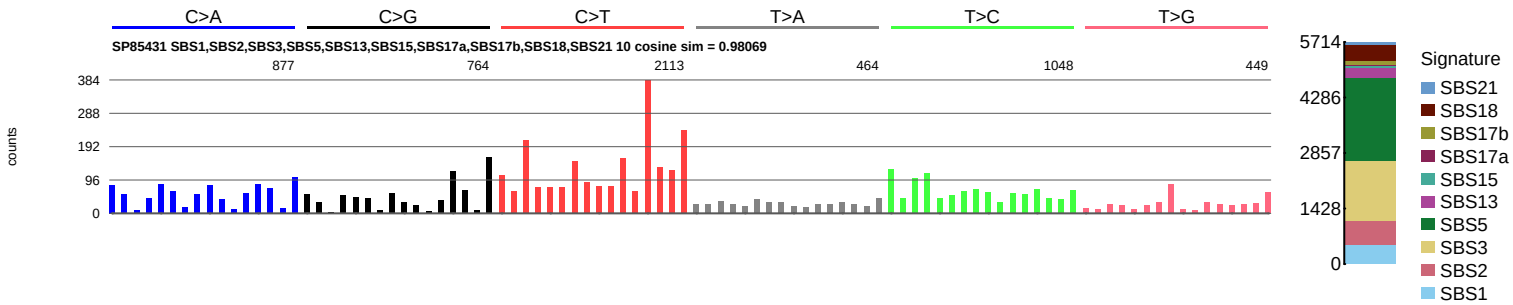
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41,SBS44 10 cosine sim = 0.98072
865
754
2110
536
1042
411
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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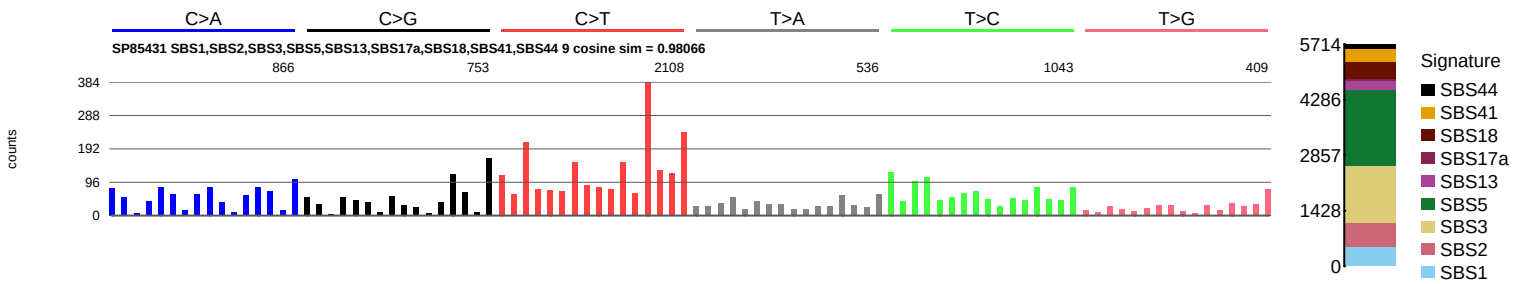
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.98069
877
764
2113
464
1048
449
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS41,SBS44 9 cosine sim = 0.98066
866
753
2108
536
1043
409
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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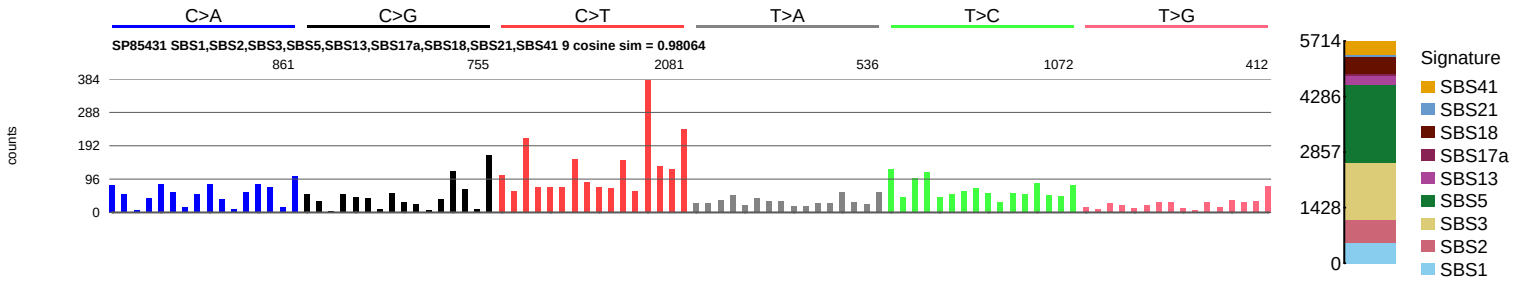
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS41 9 cosine sim = 0.98064
861
755
2081
536
1072
412
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
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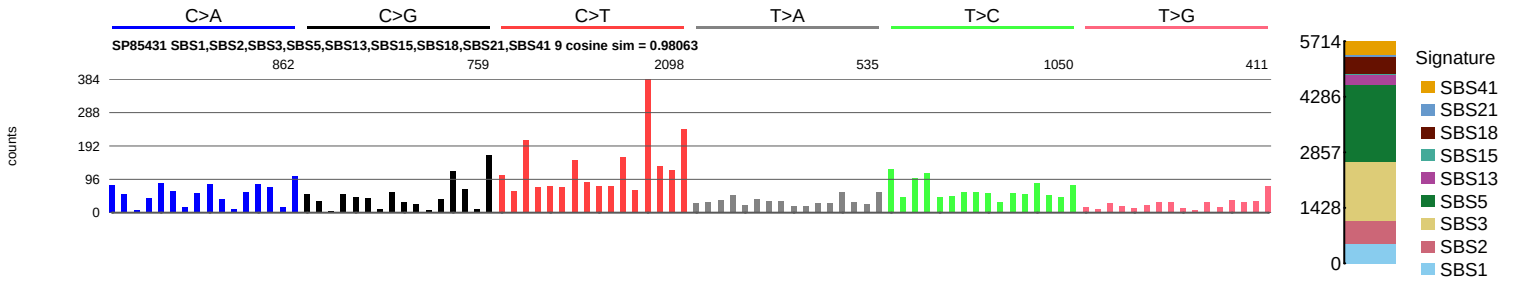
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS41 9 cosine sim = 0.98063
862
759
2098
535
1050
411
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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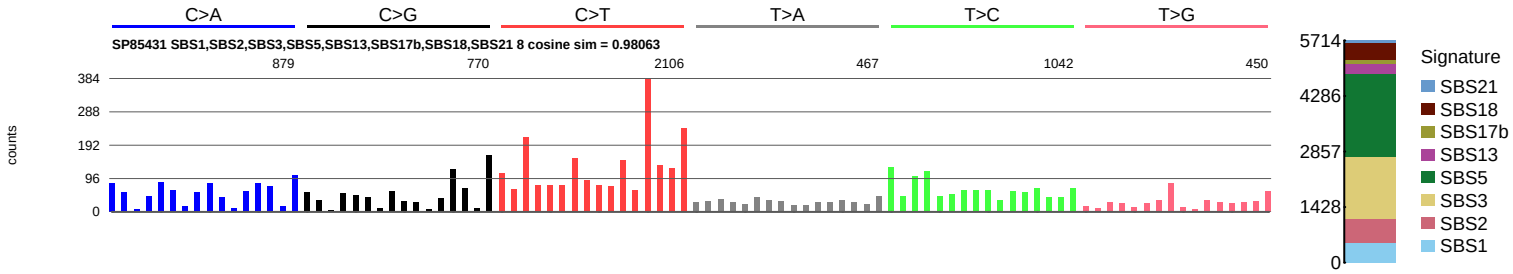
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21 8 cosine sim = 0.98063
879
770
2106
467
1042
450
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS21
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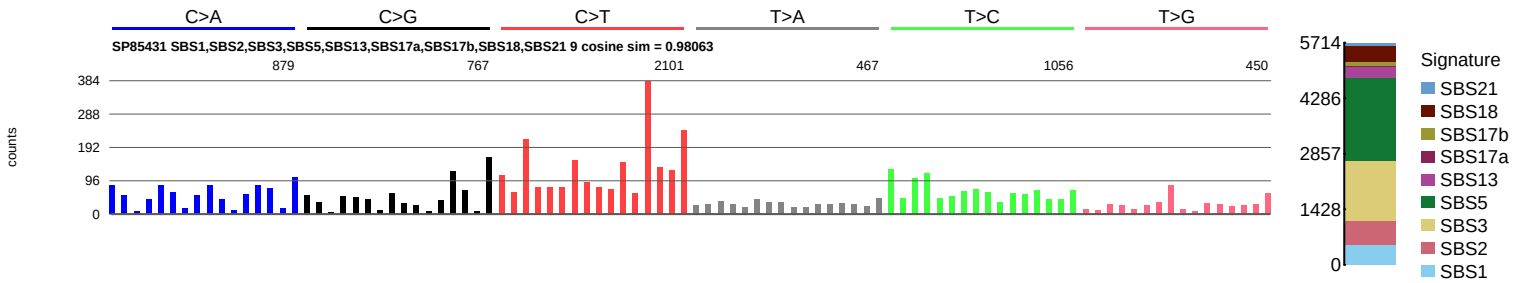
SBS2
SBS1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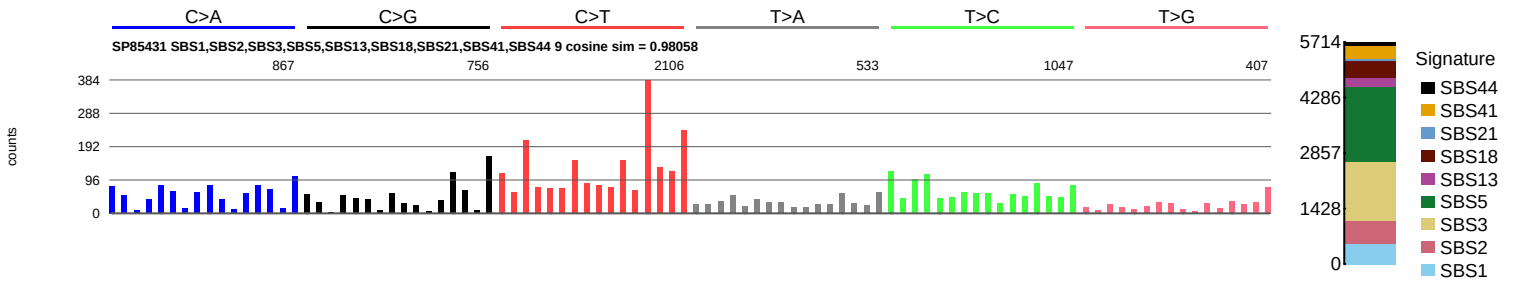
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98063
879
767
2101
467
1056
450
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98058
867
756
2106
533
1047
407
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
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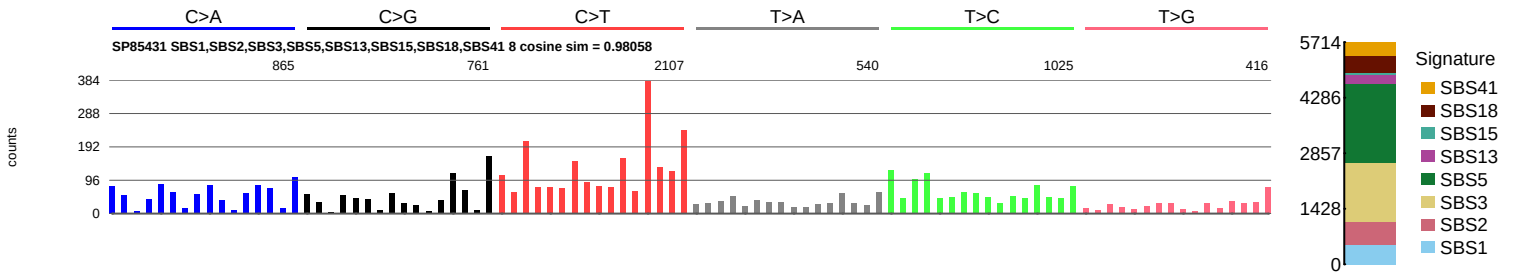
SBS1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SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS41 8 cosine sim = 0.98058
865
761
2107
540
1025
416
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS18
SBS15
SBS13
SBS5
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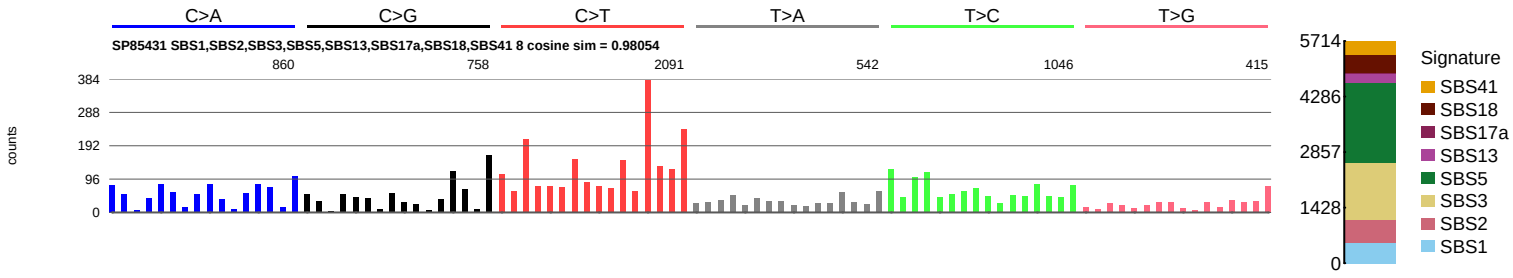
SBS2
SBS1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C>T
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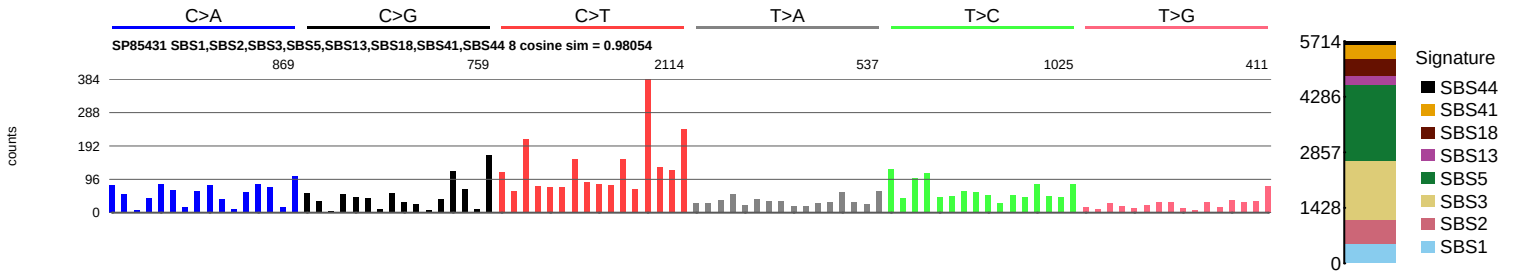
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS41 8 cosine sim = 0.98054
860
758
2091
542
1046
415
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>T
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T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS41,SBS44 8 cosine sim = 0.98054
869
759
2114
537
1025
411
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
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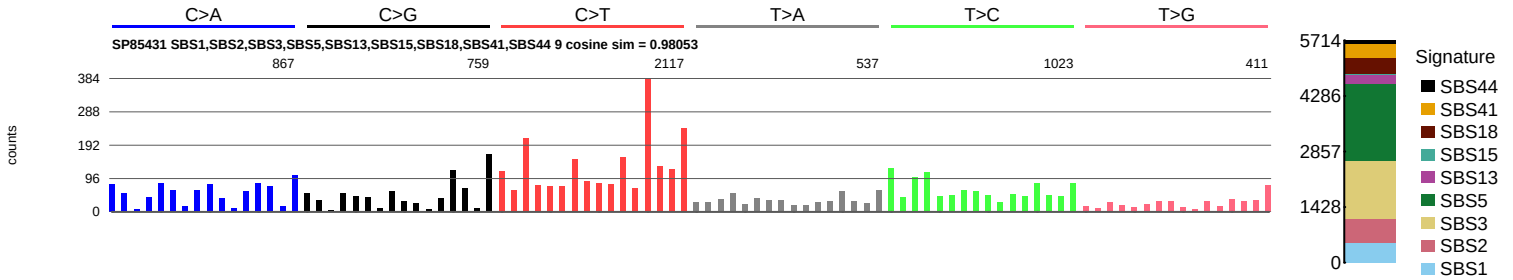
SBS13
SBS5
SBS3
SBS2
SBS1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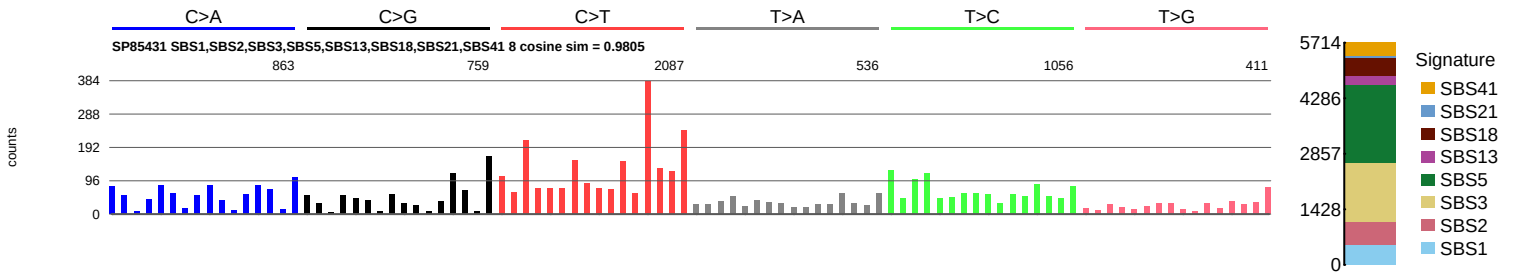
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS41,SBS44 9 cosine sim = 0.98053
867
759
2117
537
1023
411
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS21,SBS41 8 cosine sim = 0.9805
863
759
2087
536
1056
411
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
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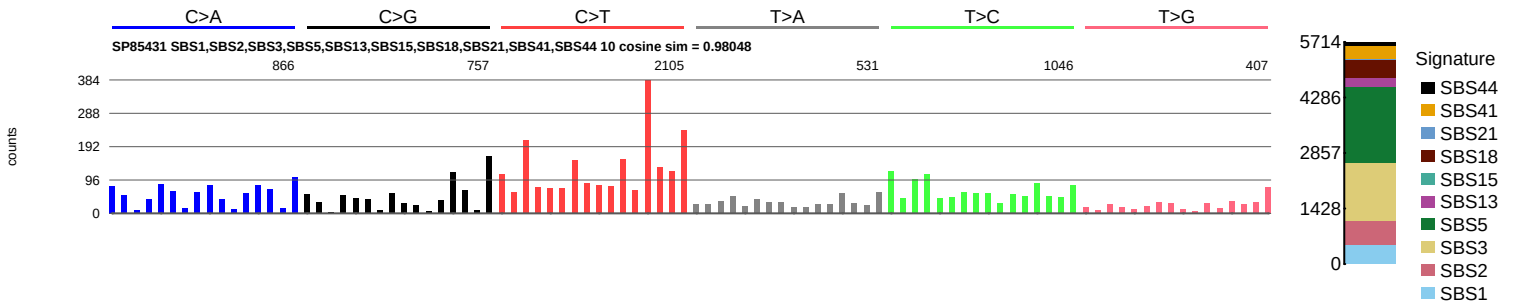
SBS2
SBS1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C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98048
866
757
2105
531
1046
407
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS21
SBS18
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SBS13
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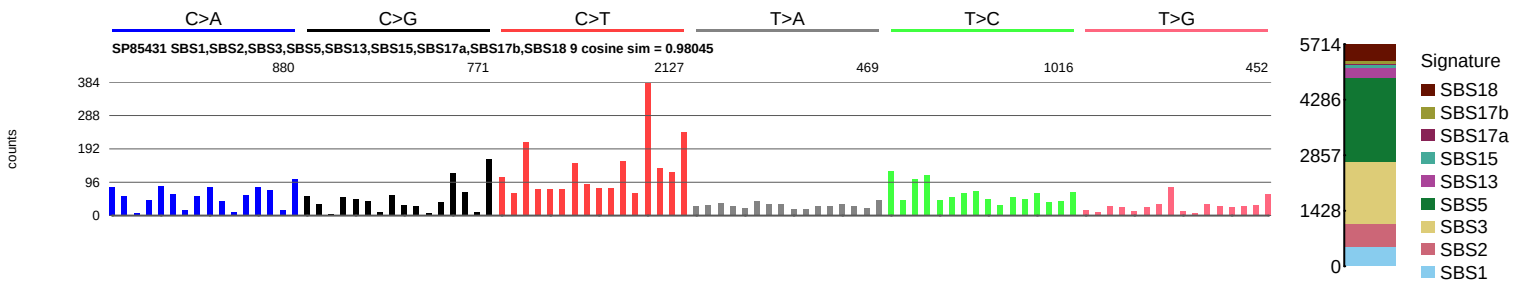
SBS3
SBS2
SBS1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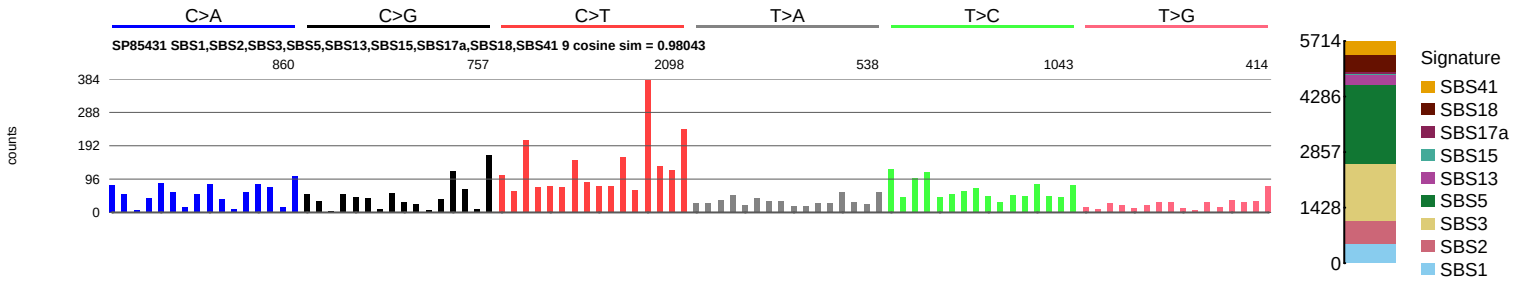
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.98045
880
771
2127
469
1016
452
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41 9 cosine sim = 0.98043
860
757
2098
538
1043
414
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
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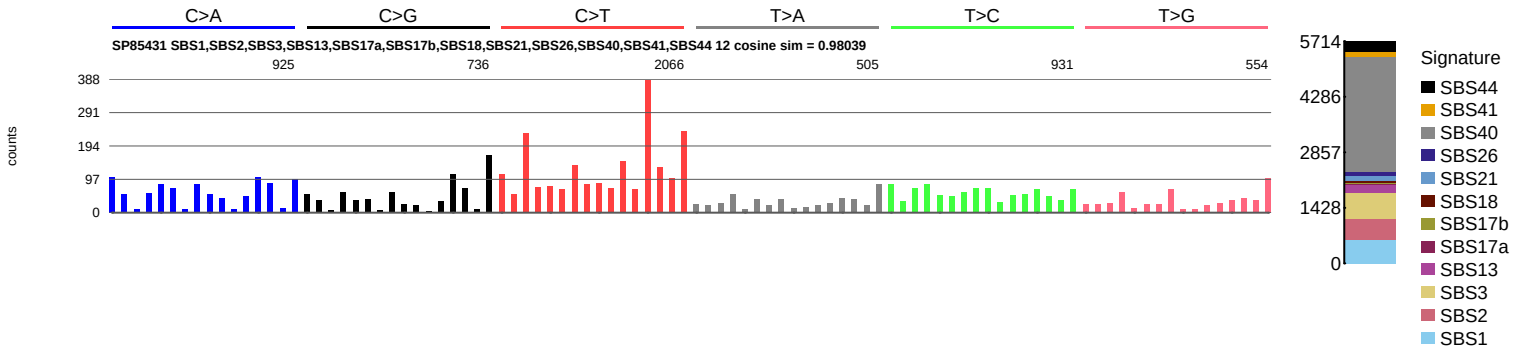
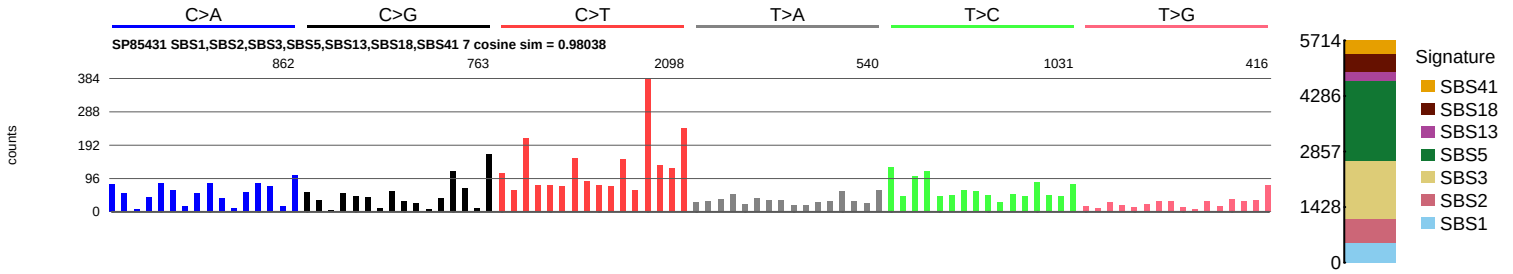
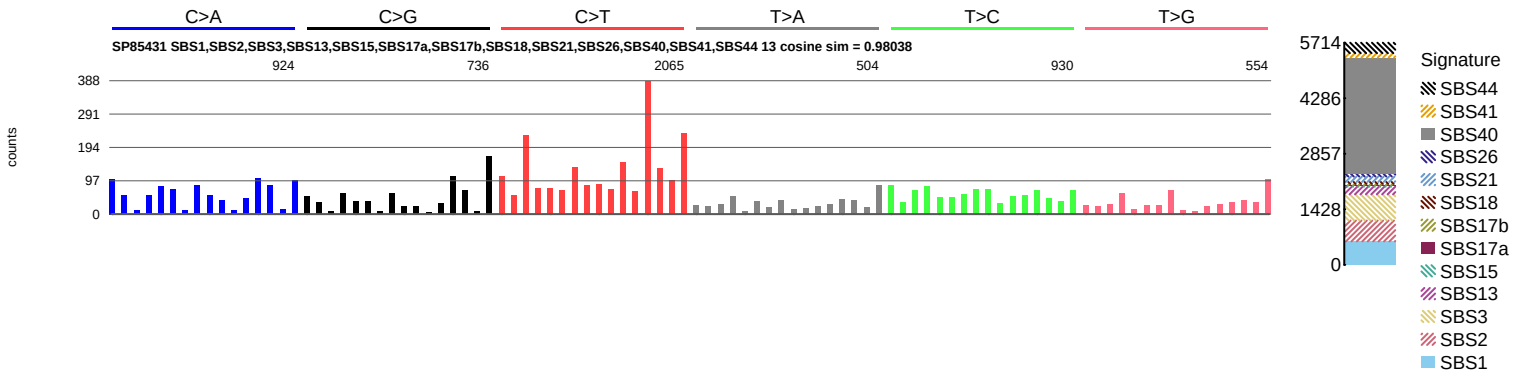
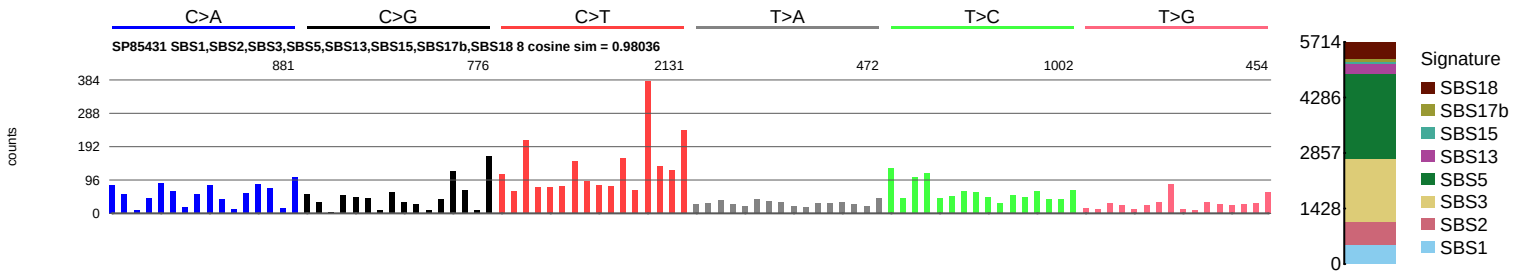
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98039
925
736
2066
505
931
554
counts
388
291
194
97
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS41 7 cosine sim = 0.98038
862
763
2098
540
1031
416
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS41
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 13 cosine sim = 0.98038
924
736
2065
504
930
554
counts
388
291
194
97
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18 8 cosine sim = 0.98036
881
776
2131
472
1002
454
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS18
SBS17b
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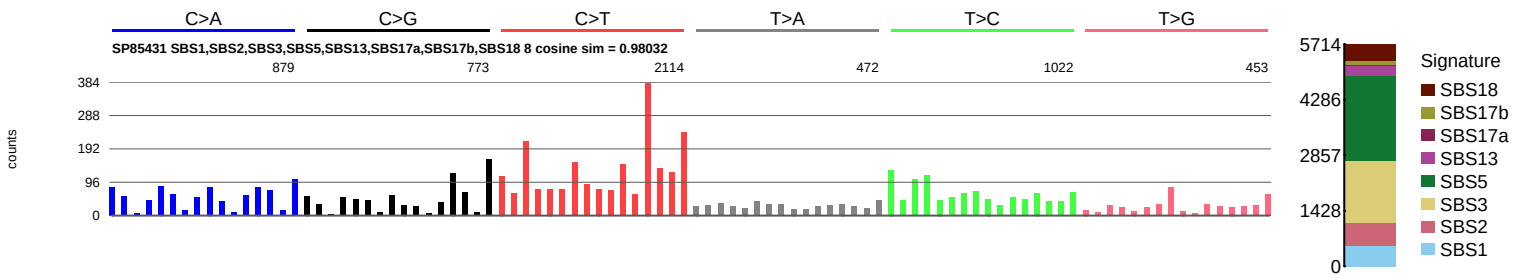
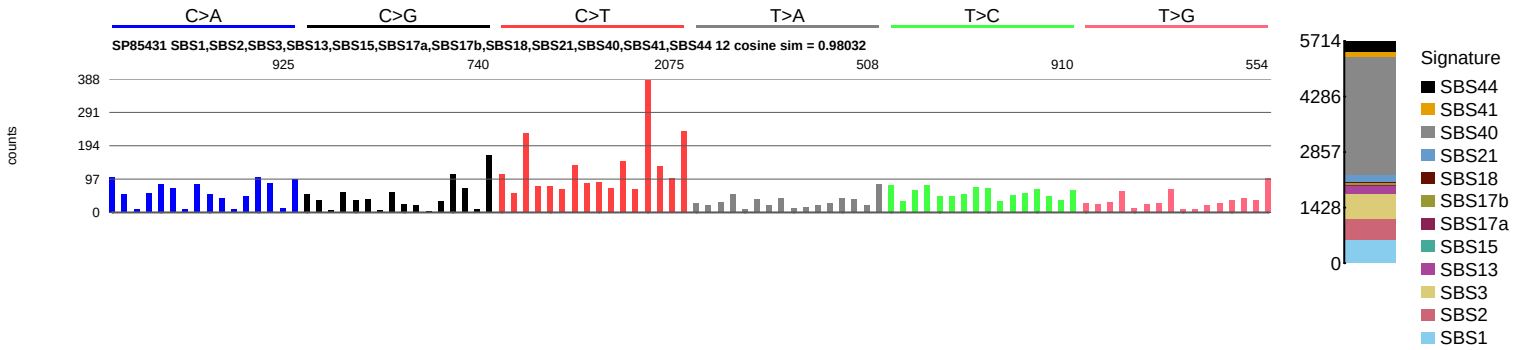
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98032
879
773
2114
472
1022
453
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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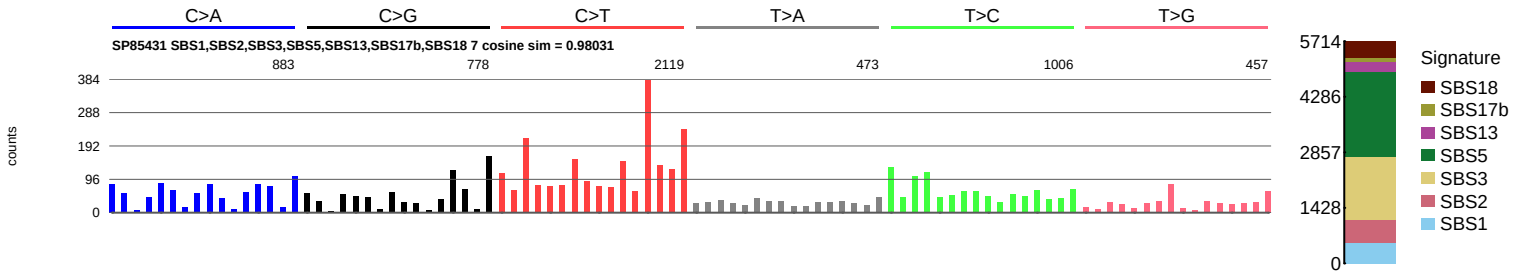
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98032
925
740
2075
508
910
554
counts
388
291
194
97
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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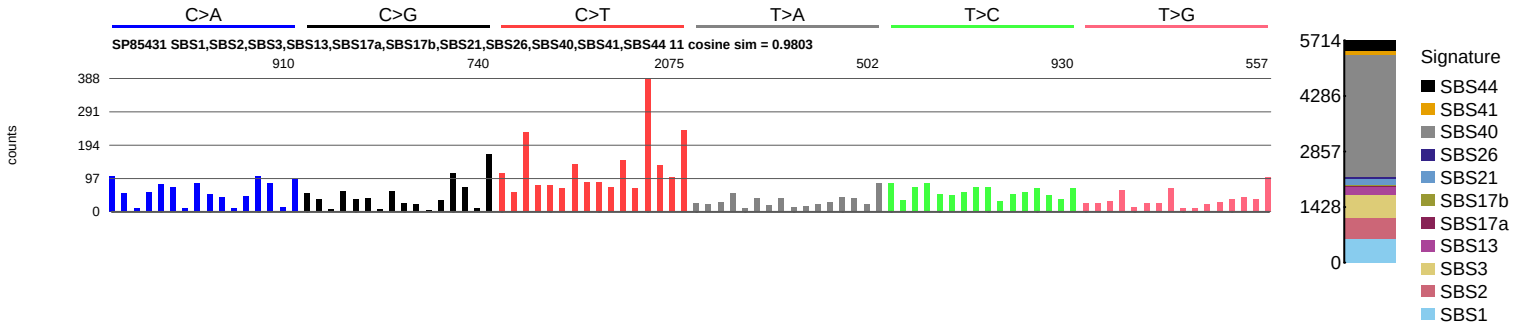
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18 7 cosine sim = 0.98031
883
778
2119
473
1006
457
counts
384
288
192
96
0
5714
4286
2857
1428
0
Signature
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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T>C
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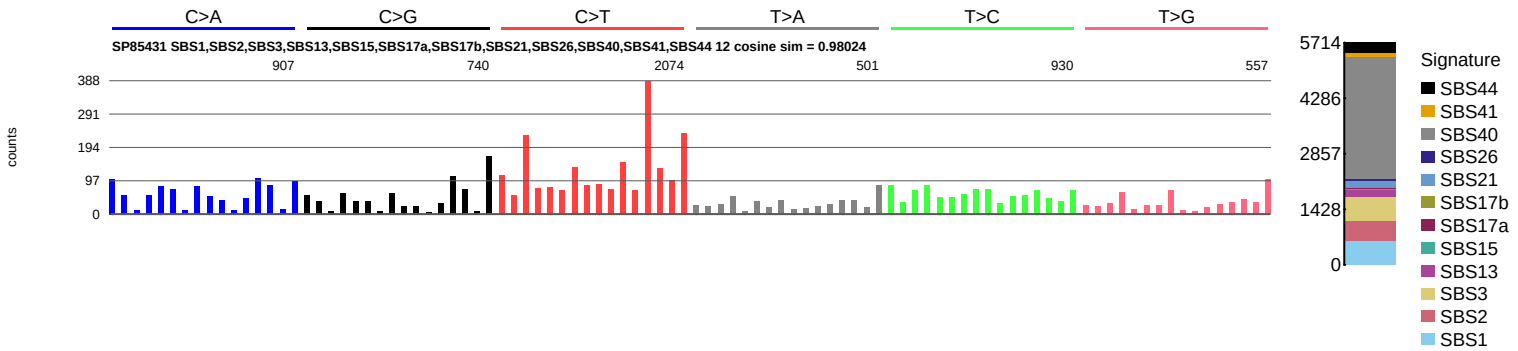
SP85431 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.9803
910
740
2075
502
930
557
counts
388
291
194
97
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98024
907
740
2074
501
930
557
counts
388
291
194
97
0
5714
4286
2857
1428
0
Signature
SBS44
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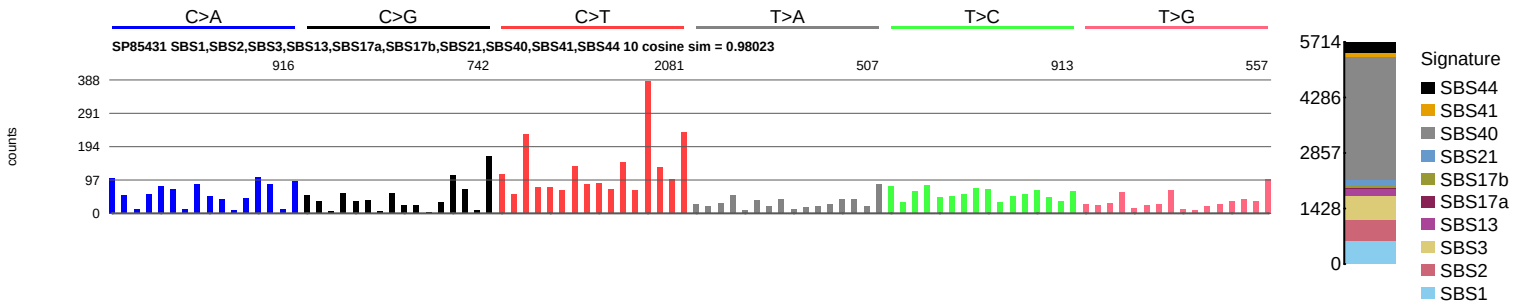
SBS2
SBS1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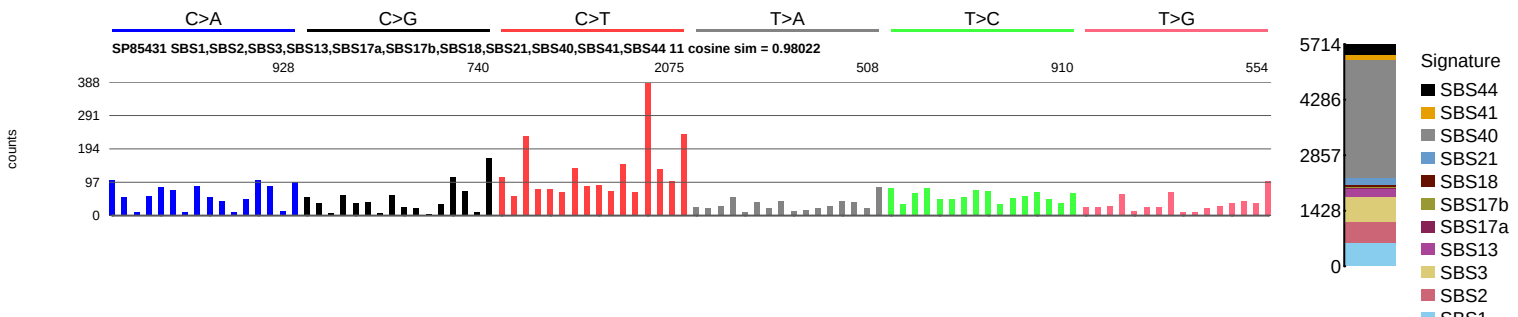
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98023
916
742
2081
507
913
557
counts
388
291
194
97
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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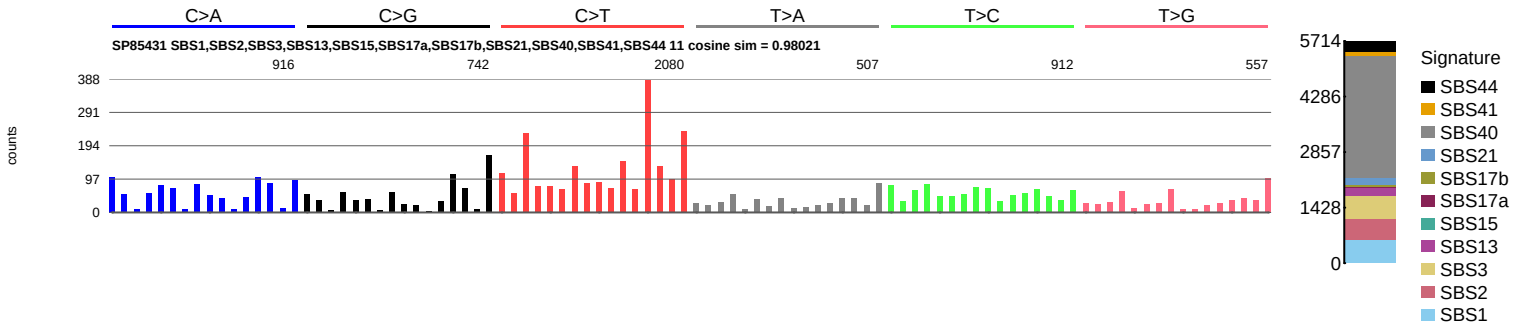
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98022
928
740
2075
508
910
554
counts
388
291
194
97
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98021
916
742
2080
507
912
557
counts
388
291
194
97
0
5714
4286
2857
1428
0
Signature
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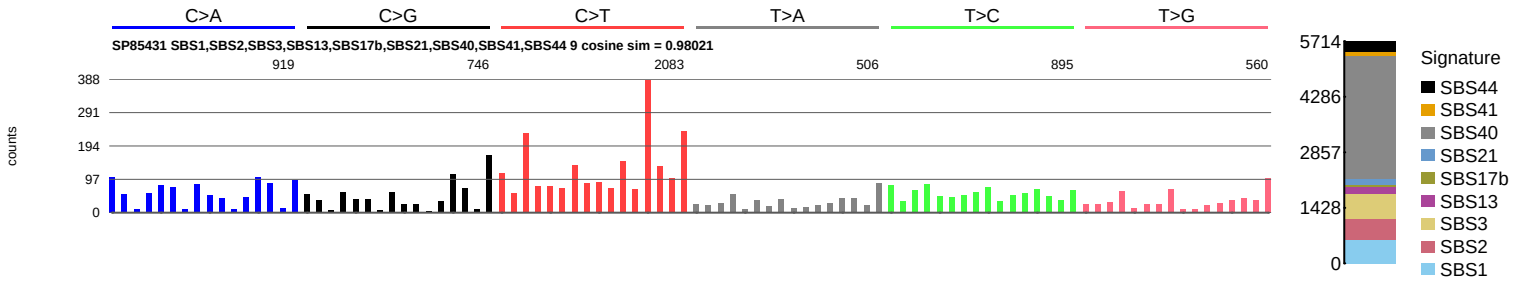
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98021
919
746
2083
506
895
560
counts
388
291
194
97
0
5714
4286
2857
1428
0
Signature
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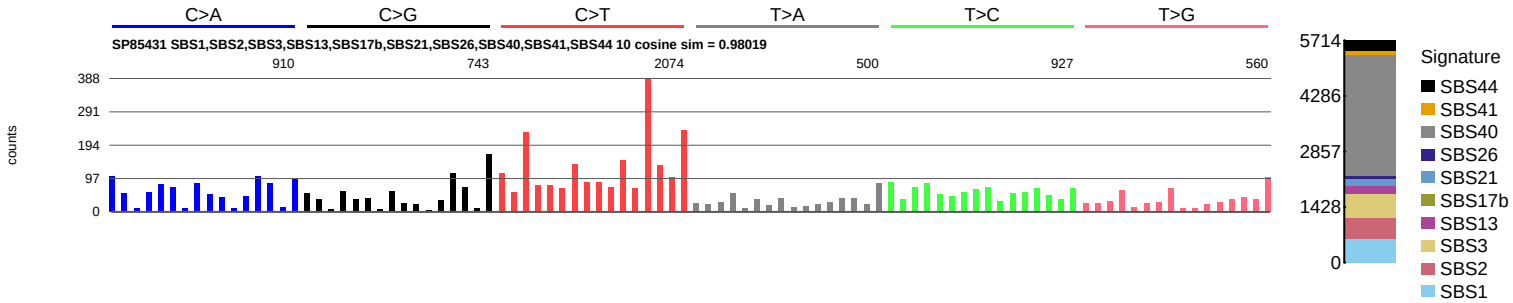
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98019
910
743
2074
500
927
560
counts
388
291
194
97
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
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SBS26
SBS21
SBS17b
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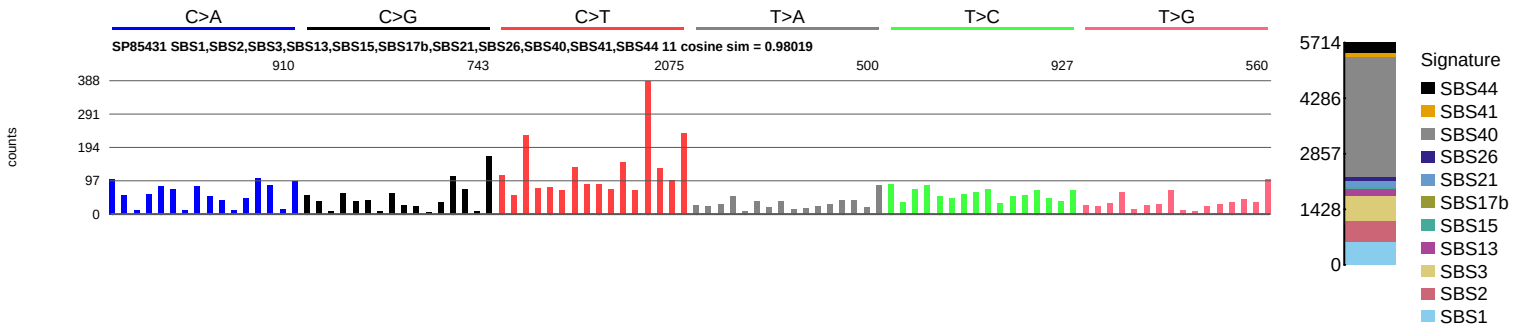
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98019
910
743
2075
500
927
560
counts
388
291
194
97
0
5714
4286
2857
1428
0
Signature
SBS44
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SBS26
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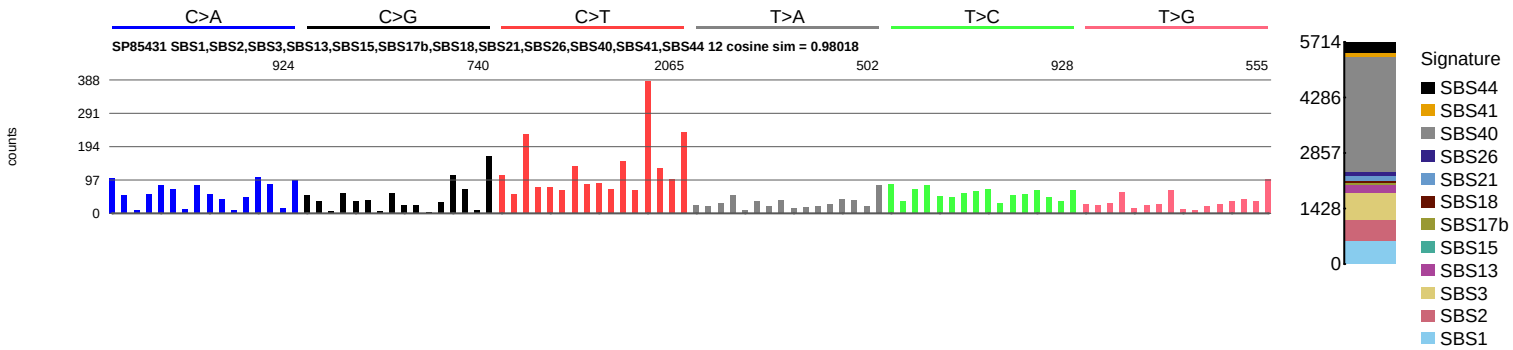
SBS2
SBS1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C>G
C>T
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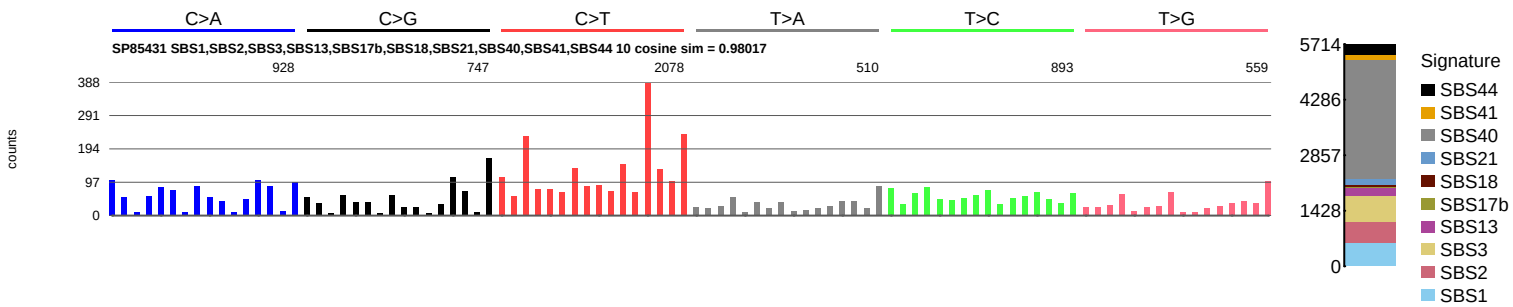
T>C
T>G
SP85431 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98018
924
740
2065
502
928
555
counts
388
291
194
97
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1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SP85431 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98017
928
747
2078
510
893
559
counts
388
291
194
97
0
5714
4286
2857
1428
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Signature
SBS44
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SBS40
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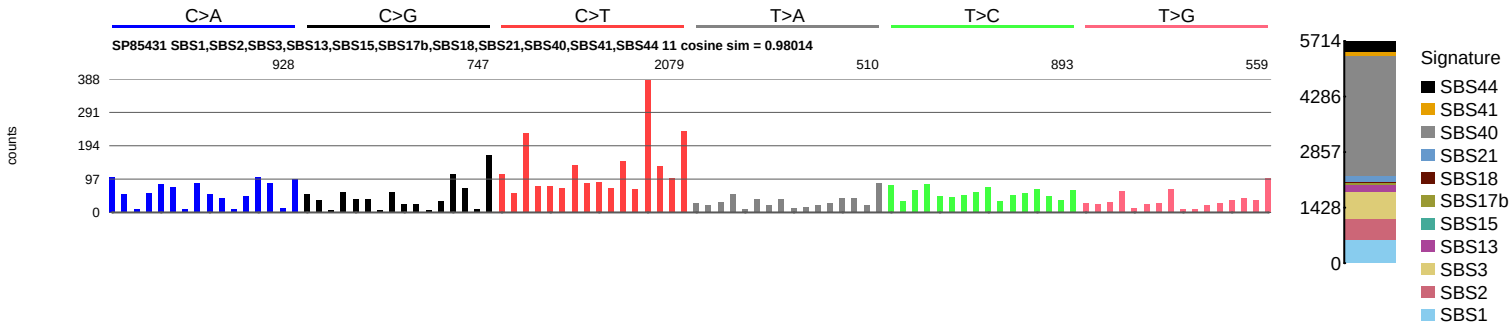
SBS1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928
747
2079
510
893
559
counts
388
291
194
97
0
5714
4286
2857
1428
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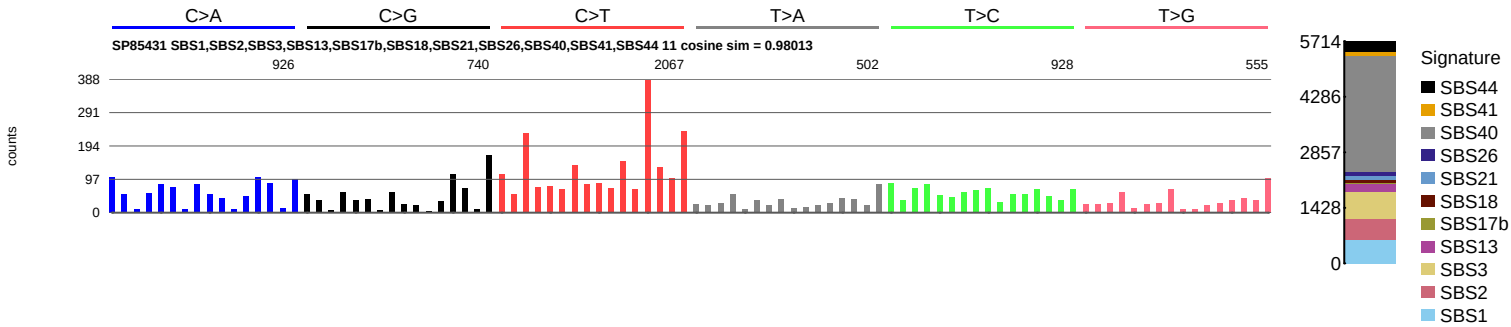
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SP85431 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98013
926
740
2067
502
928
555
counts
388
291
194
97
0
5714
4286
2857
1428
0
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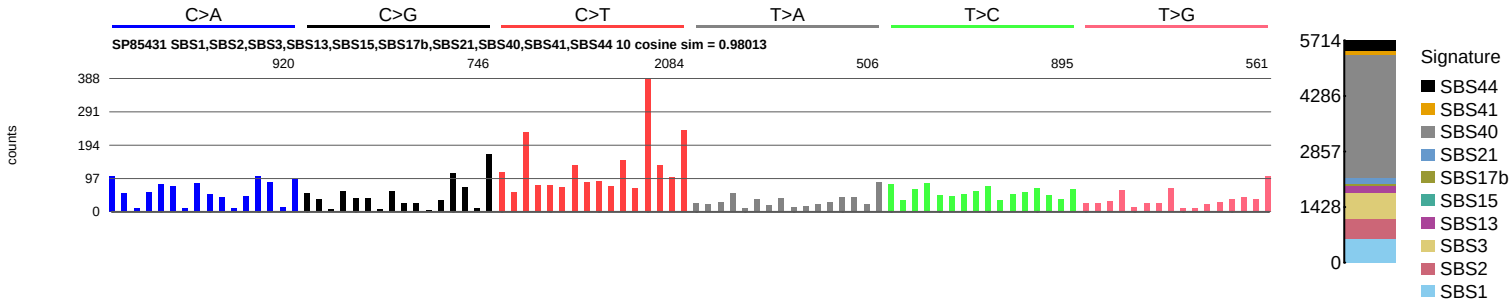
SBS13
SBS3
SBS2
SBS1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SP85431 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98013
920
746
2084
506
895
561
counts
388
291
194
97
0
5714
4286
2857
1428
0
Signature
SBS44
SBS41
SBS40
SBS21
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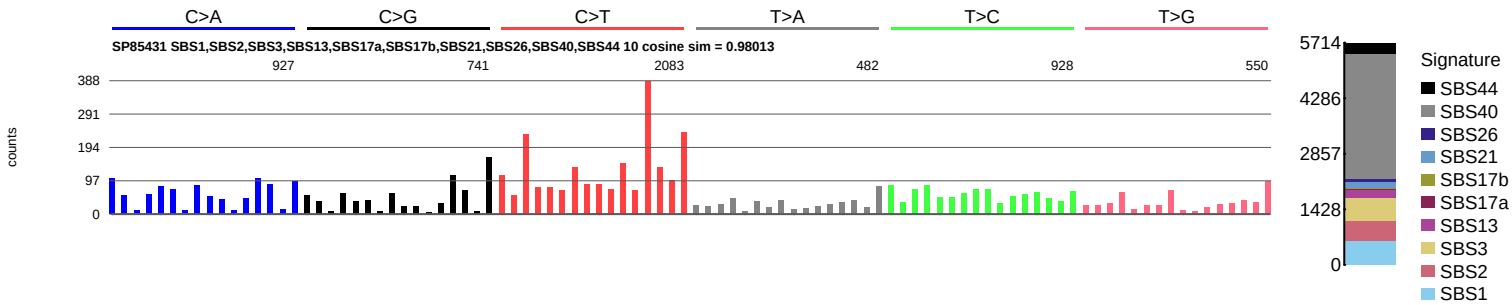
SBS15
SBS13
SBS3
SBS2
SBS1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T>G
SP85431 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98013
927
741
2083
482
928
550
counts
388
291
194
97
0
5714
4286
2857
1428
0
Signature
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SBS17b
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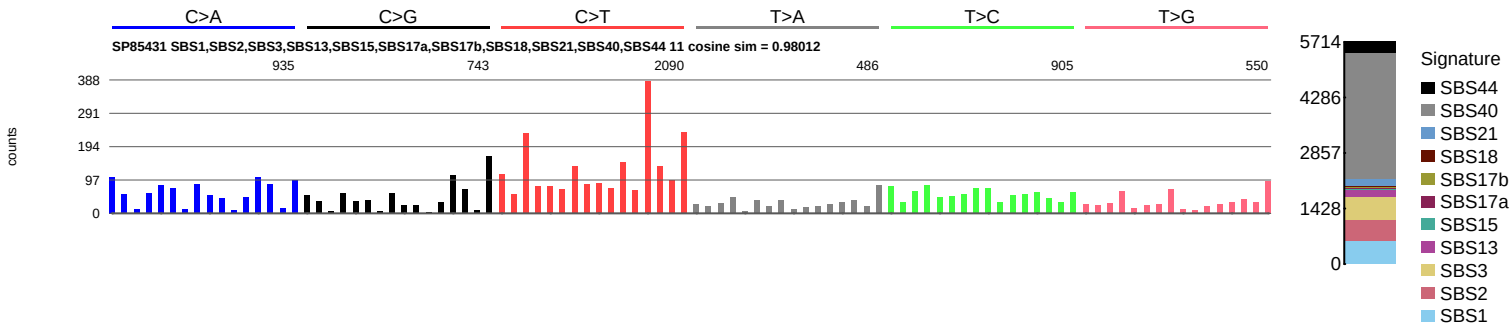
SBS13
SBS3
SBS2
SBS1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935
743
2090
486
905
550
counts
388
291
194
97
0
5714
4286
2857
1428
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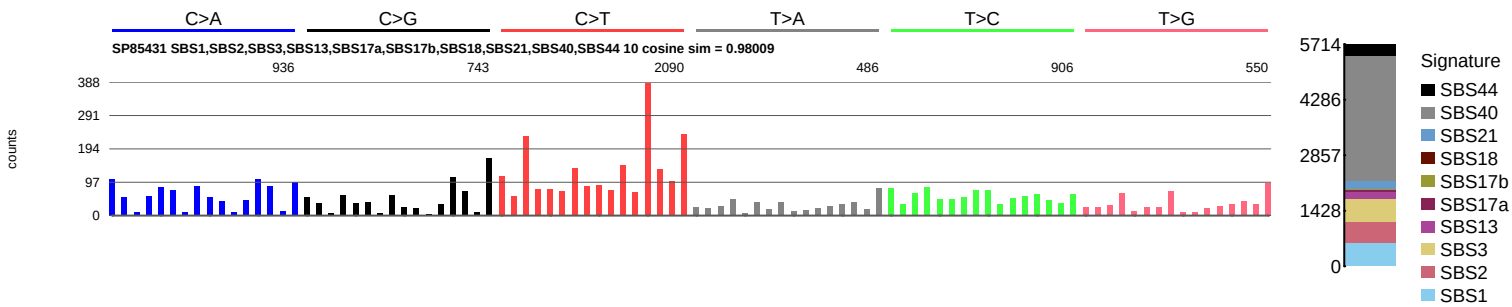
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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T>C
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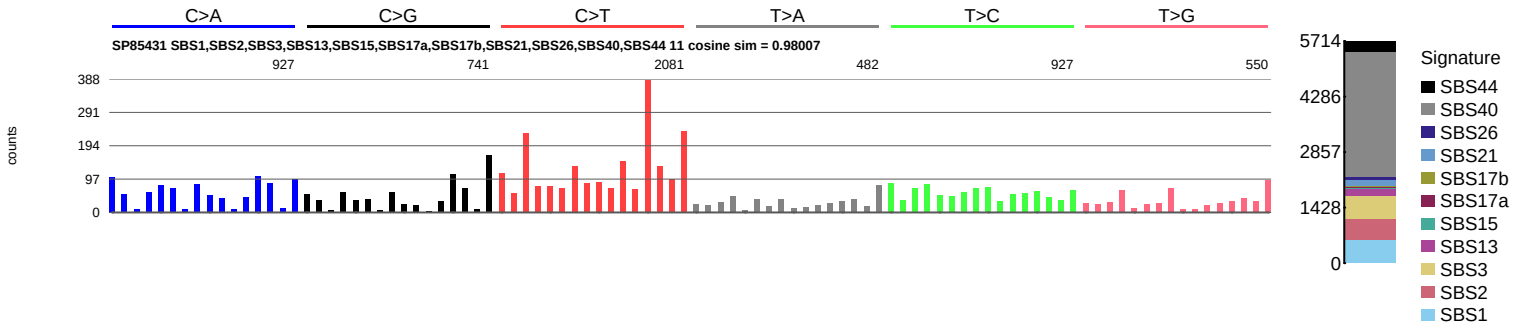
SP85431 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98009
936
743
2090
486
906
550
counts
388
291
194
97
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
C>T
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T>C
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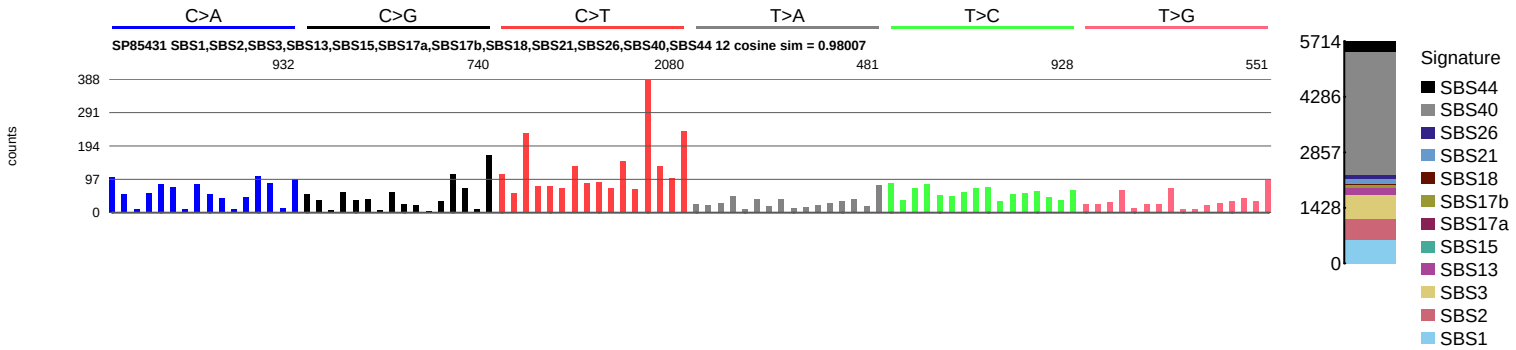
SP85431 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98007
927
741
2081
482
927
550
counts
388
291
194
97
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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C>G
C>T
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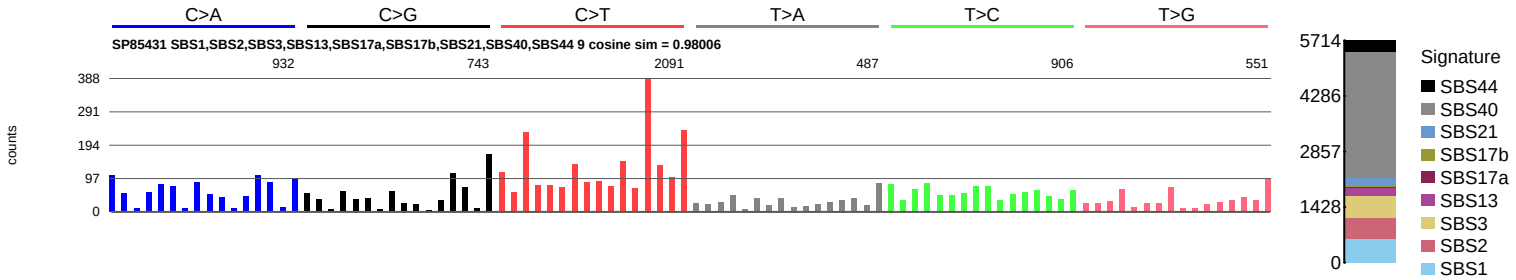
SP85431 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 12 cosine sim = 0.98007
932
740
2080
481
928
551
counts
388
291
194
97
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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C>G
C>T
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T>C
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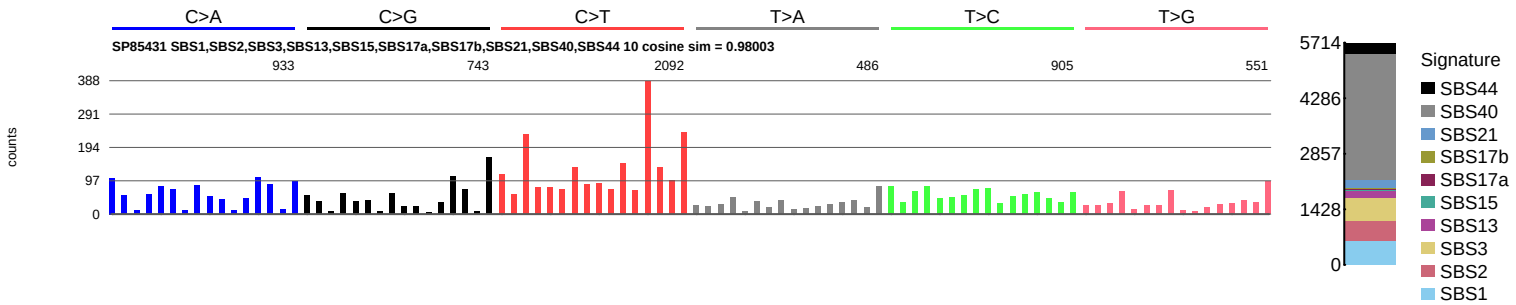
SP85431 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98006
932
743
2091
487
906
551
counts
388
291
194
97
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
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SP85431 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 10 cosine sim = 0.98003
933
743
2092
486
905
551
counts
388
291
194
97
0
5714
4286
2857
1428
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1